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Hamilton–Jacobi Theories in k -Contact Geometry and Their Applications

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Abstract

This PhD thesis develops geometric Hamilton–Jacobi theories for non-conservative classical field theories within the framework of k -contact geometry. After reviewing the geometric structures required for the formulation of k -contact Hamiltonian systems, particular attention is devoted to their Hamilton–De Donder–Weyl equations and their geometric properties.

Two complementary Hamilton–Jacobi formulations are developed. In the first, the dissipative variables are not retained in the base of the projected dynamics, whereas in the second, they are incorporated explicitly. In each approach, both the Hamiltonian and evolution k -contact formalisms are analysed, and the geometric structures underlying their solutions, including Legendrian and maximally coisotropic submanifolds, are investigated. Complete solutions and the corresponding notions of integrability are also discussed. Several notions from contact dynamics are extended to the k -contact setting, while classical results are recovered as particular cases.

The resulting formalism is applied to a broad class of non-conservative partial differential equations. These include dissipative wave and telegrapher-type equations, nonlinear field equations, and models motivated by non-equilibrium thermodynamics. The examples illustrate how k -contact geometry provides a systematic geometric framework for describing dissipation, attenuation, source terms, and balance laws in classical field theories. These results significantly extend the range of applications of k -contact geometry found in the existing literature.

Keywords:

k -contact geometry, Hamilton–Jacobi theory, contact geometry, Hamilton–De Donder–Weyl equations, non-conservative field theories, dissipative partial differential equations, geometric field theory, non-equilibrium thermodynamics.

Streszczenie

Niniejsza rozprawa doktorska rozwija geometryczne teorie Hamiltona–Jacobiego dla niekonserwatywnych klasycznych teorii pola w ramach geometrii k -kontaktowej. Po przedstawieniu struktur geometrycznych niezbędnych do sformułowania k -kontaktowych układów Hamiltona szczególną uwagę poświęcono odpowiadającym im równaniom Hamiltona–De Dondera–Weyla oraz ich własnościom geometrycznym.

W pracy rozwinięto dwa komplementarne sformułowania teorii Hamiltona–Jacobiego. W pierwszym z nich zmienne dyssypacyjne nie są uwzględniane w bazie rzutowanej dynamiki, natomiast w drugim są one jawnie włączone do tej bazy. W obu podejściach analizowane są zarówno Hamiltonowski, jak i ewolucyjny formalizm k -kontaktowy, a także struktury geometryczne związane z ich rozwiązaniami, w tym podrozmaitości legendrowskie i maksymalnie koizotropowe. Omówiono również rozwiązania zupełne oraz odpowiadające im pojęcia całkowalności. Kilka pojęć z dynamiki kontaktowej zostaje rozszerzonych na przypadek k -kontaktowy, podczas gdy klasyczne wyniki są odzyskiwane jako przypadki szczególne.

Otrzymany formalizm zastosowano do szerokiej klasy niekonserwatywnych równań różniczkowych cząstkowych. Obejmują one m.in. dyssypacyjne równania falowe i równania typu telegraficznego, nieliniowe równania pola oraz modele inspirowane termodynamiką nierównowagową. Przedstawione przykłady pokazują, że geometria k -kontaktowa dostarcza systematycznych narzędzi geometrycznych do opisu dyssypacji, tłumienia, członów źródłowych oraz praw bilansu w klasycznych teoriach pola. Wyniki te znacząco poszerzają zakres zastosowań geometrii k -kontaktowej przedstawionych w istniejącej literaturze.

Słowa kluczowe

geometria k -kontaktowa, teoria Hamiltona–Jacobiego, geometria kontaktowa, równania Hamiltona–De Dondera–Weyla, niekonserwatywne teorie pola, dyssypacyjne równania różniczkowe cząstkowe, geometryczna teoria pola, termodynamika nierównowagowa.

Tytuł pracy w języku polskim

Teorie Hamiltona–Jacobiego w geometrii k -kontaktowej i ich zastosowania

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Introduction

Geometric methods provide a natural framework for the formulation and analysis of physical systems [1, 7, 89]. In classical Hamiltonian mechanics, symplectic geometry gives an intrinsic description of conservative dynamics, while contact geometry extends this viewpoint to systems involving dissipation, thermodynamic variables, or more general non-conservative effects (see [80] and references therein). In field theory, the presence of several independent variables leads to the use of further geometric structures, such as k -symplectic, k -cosymplectic, multisymplectic, and, more recently, k -contact geometries [52, 72, 110]. In fact, k -contact geometry appeared as a generalisation of contact geometry to deal with non-conservative field theories [110].

Among the classical tools of Hamiltonian mechanics, the Hamilton–Jacobi theory plays a distinguished role [1, 7, 25, 64, 65, 75, 113]. Besides providing a method for integrating Hamiltonian systems, it establishes a deep relation between dynamics, geometry, and first-order partial differential equations. Its geometric formulation in terms of Lagrangian submanifolds and projected Hamiltonian dynamics has motivated extensions to time-dependent systems, classical field theories, contact Hamiltonian systems, and other generalised geometrical settings [61]. This topic has attracted considerable interest in recent years. To mention just a few representative contributions, Grabowska and Grabowski developed a contact Hamilton–Jacobi theory based on homogeneous symplectic geometry and line bundles [37]. De León, Lainz and Muñoz-Brea formulated a Hamilton–Jacobi theory for contact Hamiltonian systems [61], while de León, Lainz, López-Gordón and Rivas further studied Hamilton–Jacobi theory and integrability for both autonomous and non-autonomous contact systems [60]. Hamilton–Jacobi theory has also been extended to classical field theories within several geometric frameworks. Multisymplectic formulations were developed in [63, 71], while the theory was formulated in the k -symplectic and k -cosymplectic settings in [64, 75]. Further extensions include Hamilton–Jacobi theory for gauge field theories [76]. Nevertheless, a systematic Hamilton–Jacobi theory for non-conservative field theories within the k -contact framework remained largely undeveloped. This approach is particularly relevant since k -contact geometry provides an appropriate framework for describing dissipation.

The use of k -contact geometry in this thesis is not motivated by the search for a further formal generalisation of symplectic or contact geometry. Rather, it arises from a structural requirement of non-conservative field theories. While k -symplectic and k -cosymplectic formalisms provide natural geometric descriptions of conservative classical fields, they do not intrinsically encode dissipation, source terms, or balance laws. In the k -contact setting, these effects are incorporated directly into the geometry through the contact variables and the associated Reeb directions. Thus, dissipation is not introduced as an external perturbation of a conservative model, but becomes part of the geometric dynamics itself. At the same time, k -contact geometry retains the structure required by field theories and reduces to ordinary contact Hamiltonian mechanics when $k = 1$. In this sense, it provides a natural field-theoretic counterpart of contact mechanics rather than merely an additional layer of generality.

This feature is particularly relevant for Hamilton–Jacobi theory. In conservative field theories, the Hamilton–Jacobi equation is naturally tied to the underlying symplectic or multisymplectic structure. For non-conservative systems, however, the corresponding geometric formulation must also account for the evolution of the dissipative variables and for the modification of the usual conservation laws. The

k -contact framework provides precisely the additional geometric structure needed to formulate these effects intrinsically, while preserving the geometric interpretation of Hamilton–Jacobi solutions in terms of Legendrian or maximally coisotropic structures compatible with the corresponding dynamics.

The main purpose of this Doctoral Dissertation is to develop Hamilton–Jacobi theories for k -contact Hamiltonian systems and to investigate their application to non-conservative field equations. Particular attention is paid to the geometric mechanisms by which dissipation and balance laws can be encoded through the additional contact variables, as well as to the relation between Hamilton–Jacobi equations and the corresponding k -contact Hamilton–De Donder–Weyl equations. The dissertation also studies a broad range of physically motivated examples, showing that k -contact geometry provides a flexible language for the geometric treatment of dissipative partial differential equations. It also recovers or provides a new interpretation of some previous Hamilton–Jacobi theories, including those for contact and k -symplectic systems [61, 64].

The core of this Doctoral Dissertation is based on two research papers:

- J. de Lucas, J. Lange, X. Rivas, C. Sardón, “*Hamilton–Jacobi theory for non-conservative field theories in the k -contact framework*”. *Journal of Physics A: Mathematical and Theoretical*, 2026. DOI:10.1088/1751-8121/ae9b8c. Preprint: [arXiv:2604.27670](https://arxiv.org/abs/2604.27670).

This work develops a Hamilton–Jacobi theory for non-conservative classical field theories in the framework of co-oriented k -contact geometry. A central contribution is the introduction of evolution k -contact k -vector fields and the corresponding evolution Hamilton–De Donder–Weyl equations, providing a field-theoretical counterpart of the evolution formalism appearing in ordinary contact Hamiltonian mechanics [62].

Two different families of Hamilton–Jacobi theories are developed. The first one is a z -independent, or action-independent, formulation in which the dynamics is projected onto the configuration manifold. The second is a z -dependent, or action-dependent, formulation in which the dissipative variables are retained as part of the base manifold of the projected dynamics. Both approaches are studied for the Hamiltonian and evolution k -contact descriptions. The distinction between them leads to different geometric objects: in the canonical k -contact model, the z -independent formulation is naturally related to Legendrian submanifolds, whereas maximally coisotropic submanifolds appear in the z -dependent theory.

The article also analyses several structural properties specific to k -contact Hamiltonian systems. In particular, for $k > 1$, a Hamiltonian function does not uniquely determine its associated Hamiltonian k -vector field. This freedom, which has to be handled with technical conditions, is interpreted geometrically and incorporated into the formulation of the Hamilton–Jacobi problem. Complete solutions and integrable k -contact Hamiltonian systems are introduced in terms of appropriate foliations, extending ideas familiar from symplectic and contact Hamilton–Jacobi theories. The relation with previous contact and k -symplectic approaches to Hamilton–Jacobi theories is analysed.

The resulting formalism is illustrated through several non-conservative partial differential equations. These include the damped telegrapher/Klein–Gordon equation, a dissipative Hunter–Saxton equation, a non-regular dissipative first-order field model, and a covariant model inspired by extended irreversible thermodynamics. In the latter case, the contact variables can be interpreted as components of an entropy current, and a Maxwell–Cattaneo model is recovered together with a non-negative entropy production law. The applications also show that Hamiltonians with nonlinear, and in particular quadratic, dependence on the dissipative variables substantially enlarge the range of systems that can be described within the k -contact framework.

- J. de Lucas, J. Lange, M. Krych, “*A Guide to Applications of k -Contact Geometry in Dissipative Field Equations*”. Accepted for publication in *Journal of Physics A: Mathematical and Theoretical*, 2026. Preprint: [arXiv:2605.13313](https://arxiv.org/abs/2605.13313).

This work develops an application-oriented study of k -contact geometry for dissipative field equations. Its purpose is to demonstrate that the k -contact Hamiltonian formalism is not restricted to a small number of specially designed examples, but provides a systematic geometric framework for constructing and analysing broad classes of non-conservative partial differential equations.

A variety of dissipative field models are formulated as k -contact Hamiltonian systems, with particular attention to the construction of the corresponding Hamiltonians and to the role played by the dissipative variables. The analysis shows how different forms of dependence on the dissipative variables generate different types of damping, attenuation, source terms, and balance laws. In this way, the article provides a catalogue of models on which the Hamilton–Jacobi techniques developed in the previous work can be applied. The analysis also considers additional terms in the Hamiltonian that are not linear in the dissipative variables and shows how such nonlinear terms allow for the description of non-autonomous systems of partial differential equations via k -contact geometry.

Our new examples also clarify the relation between the geometric data of a k -contact Hamiltonian system and the physical partial differential equations obtained after eliminating the momentum variables. Consequently, the work provides both a collection of relevant applications and a methodological guide for identifying new physical systems admitting a k -contact description.

The two papers are strongly complementary. The first develops the geometric k -contact Hamilton–Jacobi theory and establishes its fundamental structural properties, while the second demonstrates the applicability of the k -contact framework through a systematic collection of non-conservative field equations. Together, they show that k -contact geometry can be used not merely as an abstract generalisation of contact geometry, but as an effective geometric language for dissipative classical field theories.

A primary objective of this research is therefore to extend the Hamilton–Jacobi theory from conservative Hamiltonian systems to field theories in which dissipation or other non-conservative effects form an intrinsic part of the dynamics. This requires several conceptual modifications. In contrast with ordinary contact mechanics, k -contact Hamiltonian systems involve families of vector fields rather than a single Hamiltonian vector field, and the associated Hamiltonian k -vector fields are generally non-unique. Moreover, different choices of projection lead naturally to different Hamilton–Jacobi problems. Understanding these phenomena and incorporating them into a coherent geometric theory constitutes one of the main contributions of this dissertation and involves several technically demanding proofs.

An equally important aim is to connect the abstract geometry with explicit differential equations of physical relevance. The examples studied throughout this work show that k -contact Hamiltonian systems can reproduce dissipative wave equations, nonlinear field equations, transmission-line models, and thermodynamic balance laws for field theories. In particular, the thermodynamic applications suggest that the additional dissipative variables can acquire a direct physical interpretation as entropy-related quantities rather than being merely auxiliary geometrical variables. Moreover, the theory considers Hamiltonians with additional terms that are not merely linear in the dissipative variables and explains the relevance of this more general dependence.

Several directions for further research emerge naturally from the results of this dissertation. One important problem is the development of a more complete theory of integrability for k -contact Hamiltonian systems, relating complete Hamilton–Jacobi solutions, invariant foliations, and reconstruction of integral sections. It would also be useful to establish intrinsic criteria characterising when the Hamiltonian and evolution k -contact descriptions yield equivalent projected dynamics.

Another natural direction concerns the thermodynamic models developed in this dissertation. In these models, the interpretation of the dissipative variables as entropy density and entropy flux provides a particularly promising setting for further investigations.

Further extensions may involve higher-order field theories, more general k -contact structures beyond the canonical models considered here, singular and constrained systems, and the relationship with multisymplectic and jet-bundle descriptions of classical field theory. In this regard, the theory of [82] could

be used to extend our formalism to study more general PDEs.

In summary, this Doctoral Dissertation develops a Hamilton–Jacobi framework for non-conservative field theories based on k -contact geometry and demonstrates its applicability to a broad family of dissipative physical models. The results contribute to the interaction between differential geometry, geometric mechanics, partial differential equations, and mathematical physics.

There are also other papers related to the scientific development underlying this Doctoral Dissertation, although they are not part of its principal content:

1. J. Lange, B.M. Zawora, “*Reduction of exact symplectic manifolds and energy hypersurfaces*”. Proceedings of the 7th International Conference on Geometric Science of Information, Lecture Notes in Computer Science, vol. 16035, pp. 328–336, Springer, 2025.
2. J. de Lucas, J. Lange, X. Rivas, “*A symplectic approach to Schrödinger equations in the infinite-dimensional unbounded setting*”. *AIMS Mathematics* **9** (2024), 27998–28043.

A brief description of these complementary works is as follows.

The article “Reduction of exact symplectic manifolds and energy hypersurfaces” develops two reduction procedures for Hamiltonian systems defined on exact symplectic manifolds endowed with Lie group symmetries. One approach is formulated directly at the level of the exact symplectic manifold, whereas the second is based on the contact geometry naturally induced on appropriate energy hypersurfaces. The equivalence between both reduction procedures is established by means of suitable versions of the Marsden–Meyer–Weinstein reduction theorem. The constructions are illustrated through explicit examples and clarify the relationship between exact symplectic reduction and contact geometry.

This work received one of the eight prizes awarded at the 7th International Conference on Geometric Science of Information (GSI 2025), which had around two hundred participants. The Prize Committee highlighted, in particular, the mathematical content connecting symplectic reduction with contact structures and the clarity of its presentation. Besides its independent interest, the work is conceptually related to the present dissertation through the close interaction between symplectic and contact geometries that underlies many of the k -contact constructions considered here.

The article “A symplectic approach to Schrödinger equations in the infinite-dimensional unbounded setting” concerns the geometric formulation of Schrödinger equations using infinite-dimensional symplectic geometry. In particular, it studies the difficulties caused by the presence of unbounded operators and develops a rigorous symplectic approach appropriate to this infinite-dimensional setting.

Although these works are not directly concerned with k -contact geometry, they form part of the broader geometric background of this dissertation and reflect the application of geometric methods to dynamical equations beyond the standard finite-dimensional framework.

These complementary works reflect additional results obtained during the scientific formation associated with this Doctoral Dissertation. They illustrate the broader role of symplectic and contact methods in geometric mechanics and mathematical physics and provide useful perspectives for possible future extensions of the k -contact formalism.

The dissertation is divided into four main chapters.

Chapter 1, entitled *On k -contact geometries*, introduces the geometric structures required in the subsequent developments. It begins with the symplectic, Poisson, and contact frameworks arising in classical mechanics and then reviews the k -polysymplectic and k -polycosymplectic descriptions of classical field theories. These structures motivate the introduction of k -contact geometry, its Reeb vector fields, canonical models, adapted coordinates, Hamiltonian k -vector fields, and the corresponding Hamilton–De Donder–Weyl equations. This chapter fixes the notation and provides the geometric foundations used throughout the dissertation.

Chapter 2, entitled *Classical Hamilton–Jacobi theory*, reviews the classical Hamilton–Jacobi theory and its geometric formulation in symplectic mechanics. It introduces the Hamilton–Jacobi equation,

complete solutions, and their relation with canonical transformations and the integration of Hamiltonian systems. Particular attention is paid to the interpretation of Hamilton–Jacobi solutions in terms of Lagrangian submanifolds and projected dynamics. This chapter establishes the concepts and notation that are subsequently extended to the k -contact setting.

Chapter 3, entitled *Hamilton–Jacobi equations in the k -contact setting*, constitutes the theoretical core of the dissertation. It first analyses regular k -contact Hamiltonian systems and their relation with higher-order systems of partial differential equations. It then develops the z -independent and z -dependent Hamilton–Jacobi theories, both for the Hamiltonian and evolution k -contact formulations. Particular attention is devoted to the geometric characterisation of admissible sections, the non-uniqueness of Hamiltonian k -vector fields, complete Hamilton–Jacobi solutions, and the corresponding notions of integrability.

Finally, Chapter 4, entitled *New k -contact physical models*, is devoted to applications of the k -contact geometric framework to physically relevant non-conservative field equations. The examples include dissipative wave and Klein–Gordon-type equations, telegrapher-type models, nonlinear dissipative equations, and thermodynamic systems. These applications illustrate how the abstract k -contact formalism produces explicit partial differential equations and how the dissipative variables encode attenuation, source terms, and balance laws. They also demonstrate the practical scope of the k -contact Hamiltonian formalism and the geometric methods developed in the previous chapter.

Chapter 1

On k -contact geometries

Geometric formulations of physical theories have consistently uncovered fundamental structural features that are less apparent in purely analytic approaches [1, 6, 7, 41, 72]. For example, in classical mechanics, symplectic geometry provides the natural language for Hamiltonian systems, encoding conservation laws and phase space dynamics in a coordinate-free manner [41]. However, many physically relevant systems fall outside the scope of strictly symplectic frameworks. These include, in particular, systems involving dissipation or thermodynamic variables, which are naturally described within contact geometry, a framework that captures non-conservative behaviour [29, 73].

In recent years, further generalisations of contact geometry have been developed to address increasingly complex settings. Among these, k -contact geometry provides a natural extension of classical contact geometry to the study of systems with several interacting contact-type structures [110]. This framework is particularly well suited to the geometric description of field theories, multi-time dynamical systems, and geometric generalisations of Hamilton–Jacobi theory [83].

Roughly speaking, a k -contact manifold can be understood as a smooth manifold equipped with a family of k one-forms satisfying compatibility and non-degeneracy conditions that generalise the standard geometric contact form [32]. While classical contact geometry is governed by a single maximally non-integrable distribution described as the kernel of a contact form, the k -contact setting replaces this with a collection of such structures, leading to a richer and more intricate geometric behaviour. This additional complexity allows the simultaneous encoding of multiple parameters or interacting subsystems within a unified geometric framework.

The main purpose of this chapter is to introduce the fundamental concepts and notions regarding k -contact geometry that will be used throughout the development of Hamilton–Jacobi theory in this generalised setting. A natural way to motivate the structure of this chapter is to present k -contact geometry as the culmination of a sequence of increasingly rich geometric frameworks arising from classical mechanics and field theory. Accordingly, the chapter begins with a review of symplectic and contact geometry, since these provide the classical geometric language for conservative autonomous, conservative non-autonomous, and dissipative dynamical systems, respectively. In particular, many of the constructions and intuitions appearing later in the k -contact setting are generalisations of structures first encountered in these geometries.

The next natural step is the geometric formulation of classical field theories. For this reason, we briefly introduce k -polysymplectic and k -polycosymplectic manifolds. These structures provide the appropriate geometric setting for first-order classical field theories and establish the conceptual and technical framework required for the introduction of k -contact geometry.

Finally, k -contact geometry is presented as a natural counterpart of contact geometry within the k -symplectic and k -cosymplectic formalisms [29, 110, 111, 127]. Introducing the preceding structures first allows the fundamental ideas, notation, and geometric mechanisms to emerge progressively, making the transition to the k -contact setting more transparent. Moreover, many properties of k -contact man-

ifolds are most naturally understood through comparison with their k -cosymplectic and polysymplectic counterparts.

Special attention is devoted to the role of Reeb vector fields in the k -contact setting, as well as to the construction of adapted coordinate systems, which are essential for applications and explicit computations [110]. The relationships between k -contact geometry and other geometric frameworks, such as k -cosymplectic, polysymplectic, and multisymplectic geometries, are also discussed.

Throughout this text, the following assumptions are adopted unless stated otherwise. All geometric objects are assumed to be smooth and defined over the real numbers. Every manifold considered here is connected, finite-dimensional, Hausdorff, and paracompact. In particular, these conditions guarantee the availability of partitions of unity and ensure that locally constant functions are globally constant. We also assume that structures are defined globally on the underlying manifolds. These assumptions are introduced primarily for clarity and convenience, allowing us to concentrate on the central geometric ideas without being distracted by technical subtleties that are not essential to the discussion or applications. Finally, for a manifold M , we write $\Omega^k(M)$ for the space of differential k -forms and $\mathfrak{X}(M)$ for the space of vector fields on M .

1.1 Fundamental geometric structures

This section recalls the fundamental geometric structures from classical mechanics that will be used throughout this dissertation. We first introduce the basic notions of symplectic and Poisson geometry and then turn to contact geometry, emphasising the concepts and notation required for the subsequent development of the k -contact framework.

1.1.1 Symplectic and Poisson geometry

Symplectic geometry provides the natural geometric framework for many areas of mathematics and physics, most notably classical Hamiltonian mechanics [1, 35, 91]. The importance of symplectic geometry in the present context stems from its close relationship with contact geometry, which will be discussed in subsequent sections. We begin by recalling the basic notions and fundamental properties of symplectic manifolds. We also establish the notation and conventions used throughout the thesis.

Definition 1.1.1. A *symplectic manifold* is a pair (P, ω) , where P is a manifold and $\omega \in \Omega^2(P)$ is a closed and *non-degenerate* differential two-form, called a *symplectic form*.

The non-degeneracy condition for $\omega \in \Omega^2(P)$ means that the bilinear form $\omega_p : T_p P \times T_p P \rightarrow \mathbb{R}$ is non-degenerate for every $p \in P$. This implies that the tangent space $T_p P$ is even-dimensional at each point $p \in P$. Consequently, every symplectic manifold has even dimension, say $\dim P = 2n$, for $n \in \mathbb{N}$. Moreover, ω induces a vector bundle isomorphism $\omega^\flat : TP \rightarrow T^*P$, given by $\omega^\flat(v_p) := \omega_p(v_p, \cdot) \in T_p^*P$ for every $p \in P$ and each $v_p \in T_p P$.

Another consequence of the non-degeneracy of a symplectic form is that every symplectic manifold is canonically orientable. Let (P, ω) be a symplectic manifold of dimension $2n$. Since ω is a two-form, its n -fold exterior product $\omega^n = \omega \wedge \cdots \wedge \omega$ (n factors) is a differential $2n$ -form on P . The non-degeneracy of ω is equivalent to the fact that ω^n is nowhere vanishing. Consequently, $\mu := \omega^n/n!$ defines a volume form on P , called the *symplectic volume form*. The volume form μ determines a canonical orientation on P , known as the *symplectic orientation*.

Let (P, ω) be a symplectic manifold, and consider a subspace $V_p \subset T_p P$, for $p \in P$. We define a *symplectic orthogonal complement* of V_p by

$$V_p^{\perp \omega} := \{w_p \in T_p P \mid \omega_p(w_p, v_p) = 0, \forall v_p \in V_p\}.$$

Then, V_p is called *isotropic* if $V_p \subset (V_p)^\perp$. Meanwhile, we say that V_p is *coisotropic* if $(V_p)^\perp \subset V_p$. On the other hand, if $(V_p)^\perp = V_p$, then V_p is *Lagrangian*. A submanifold $N \subset P$ is called (co)isotropic (resp. Lagrangian) if $T_p N \subset T_p P$ is a (co)isotropic (resp. Lagrangian) subspace for all $p \in N$.

Now, given a submanifold $N \subset P$, we consider the subspace $T_p N \subset T_p P$ at each point $p \in N$. Using the symplectic orthogonal complement of $T_p N$, we classify submanifolds of a symplectic manifold as follows [1, 17].

Definition 1.1.2. Let (P, ω) be a symplectic manifold. A submanifold $N \subset P$ is called

- *isotropic* if $T_p N \subset (T_p N)^\perp$ for all $p \in N$,
- *coisotropic* if $(T_p N)^\perp \subset T_p N$ for all $p \in N$,
- *Lagrangian* if $(T_p N)^\perp = T_p N$ for all $p \in N$.

Equivalently, N is isotropic if the restriction of the symplectic form to N vanishes, namely $\omega|_N = 0$. A Lagrangian submanifold is therefore a maximal isotropic submanifold. In particular, if $\dim P = 2n$, then a submanifold $N \subset P$ is Lagrangian if and only if it is isotropic and $\dim N = n$.

Theorem 1.1.3 (Darboux theorem [1]). *Let (P, ω) be a symplectic manifold. Then, around every point $p \in P$, there exist local coordinates $\{q^i, p_i\}$, with $i = 1, \dots, n$, such that*

$$\omega = \sum_{i=1}^n dq^i \wedge dp_i.$$

The coordinates $\{q^1, \dots, q^n, p_1, \dots, p_n\}$ are called Darboux coordinates.

In Darboux coordinates $\{q^i, p_i\}_{i=1, \dots, n}$ the volume form is given by

$$\mu = dq^1 \wedge dp_1 \wedge \dots \wedge dq^n \wedge dp_n.$$

Darboux coordinates are far from unique. Darboux's theorem asserts only the local existence of coordinates in which the symplectic form assumes its canonical expression. These facts reflect the absence of local invariants in symplectic geometry. That is, locally, all symplectic manifolds of the same dimension are locally indistinguishable.

Cotangent bundles constitute one of the most important classes of symplectic manifolds. Indeed, given a manifold Q , its cotangent bundle T^*Q is naturally equipped with the canonical one-form, whose exterior derivative defines a symplectic form on T^*Q . Moreover, cotangent bundles provide the natural geometric framework for Hamiltonian mechanics, where points of T^*Q represent states of a mechanical system in terms of positions and momenta.

Definition 1.1.4. Let Q be a smooth manifold and let $\tau : T^*Q \rightarrow Q$ denote the natural projection of its cotangent bundle. The *canonical one-form*, also called the *Liouville form*, is the differential one-form $\theta_Q \in \Omega^1(T^*Q)$ defined by

$$(\theta_Q)_{\alpha_q}(v_{\alpha_q}) := \langle \alpha_q, T_{\alpha_q} \tau(v_{\alpha_q}) \rangle, \quad \forall q \in Q, \quad \forall \alpha_q \in T_q^*Q, \quad \forall v_{\alpha_q} \in T_{\alpha_q} T^*Q.$$

Here, $\langle \cdot, \cdot \rangle$ denotes the natural pairing between covectors and vectors. The *canonical two-form* (or *symplectic form*) on T^*Q is the differential two-form $\omega_Q \in \Omega^2(T^*Q)$ given by $\omega_Q := -d\theta_Q$.

In local coordinates $\{q^i, p_i\}_{i=1, \dots, n}$ adapted to T^*Q , the Liouville one-form and the canonical symplectic form [1] on T^*Q are given by

$$\theta_Q = \sum_{i=1}^n p_i dq^i, \quad \omega_Q = -d\theta_Q = \sum_{i=1}^n dq^i \wedge dp_i.$$

The symplectic manifold (T^*Q, ω_Q) plays a significant role in classical mechanics and has numerous physical applications [1, 89].

Definition 1.1.5. A vector field $X \in \mathfrak{X}(P)$ is *Hamiltonian* if there exists a function $f \in C^\infty(P)$ such that

$$\iota_X \omega = df.$$

The function f is called a *Hamiltonian function* associated with X .

Since the symplectic form ω is non-degenerate, for every $f \in C^\infty(P)$ there exists a unique vector field $X_f \in \mathfrak{X}(P)$ satisfying $\iota_{X_f} \omega = df$. We call X_f the *Hamiltonian vector field associated with f* . The set of all Hamiltonian vector fields on (P, ω) is denoted by $\text{Ham}(P, \omega)$. Hamiltonian vector fields preserve the symplectic structure. It is worth noting that the Hamiltonian function of a Hamiltonian vector field can be defined as $\iota_{X_f} \omega = -df$, but this approach is more appropriate for mathematical works [125].

The symplectic form also induces a natural algebraic structure on $C^\infty(P)$.

Definition 1.1.6. A *Poisson bracket* on a manifold P is a bilinear map $\{\cdot, \cdot\} : C^\infty(P) \times C^\infty(P) \rightarrow C^\infty(P)$ such that $(C^\infty(P), \{\cdot, \cdot\})$ is a Lie algebra and the *Leibniz rule* is satisfied, namely

$$\{f, gh\} = \{f, g\}h + g\{f, h\}, \quad \forall f, g, h \in C^\infty(P).$$

Given a symplectic manifold (P, ω) , the bracket

$$\{f, g\}_\omega := \omega(X_f, X_g), \quad f, g \in C^\infty(P),$$

defines a Poisson bracket on $C^\infty(P)$. Bilinearity and skew-symmetry follow directly from the corresponding properties of the symplectic form ω , while the Jacobi identity is a consequence of the closedness condition $d\omega = 0$ [1, 101]. Therefore, $\{\cdot, \cdot\}_\omega$ defines a *Lie bracket* on $C^\infty(P)$.

To show that $\{\cdot, \cdot\}_\omega$ is a Poisson bracket, it remains to verify the Leibniz rule. Let $f, g, h \in C^\infty(P)$. Since

$$\iota_{X_{gh}} \omega = d(gh) = g dh + h dg = g \iota_{X_h} \omega + h \iota_{X_g} \omega,$$

the non-degeneracy of ω implies that $X_{gh} = gX_h + hX_g$. Consequently,

$$\{f, gh\}_\omega = \omega(X_f, X_{gh}) = g\omega(X_f, X_h) + h\omega(X_f, X_g) = g\{f, h\}_\omega + h\{f, g\}_\omega.$$

Thus, $\{\cdot, \cdot\}_\omega$ satisfies the Leibniz rule and therefore defines a Poisson bracket on $C^\infty(P)$.

One has that $\iota_{[X_f, X_g]} = \mathcal{L}_{X_f} \iota_{X_g} - \iota_{X_g} \mathcal{L}_{X_f}$ for every $f, g \in C^\infty(P)$ (see [1, p. 121]). Combining this with Cartan's formula, it follows that

$$\iota_{[X_f, X_g]} \omega = \mathcal{L}_{X_f} \iota_{X_g} \omega - \iota_{X_g} \mathcal{L}_{X_f} \omega = \mathcal{L}_{X_f} \iota_{X_g} \omega = d(\iota_{X_f} dg) = d\{g, f\}_\omega$$

and thus $X_{\{g, f\}_\omega} = [X_f, X_g]$. In other words, the mapping $f \in C^\infty(P) \mapsto -X_f \in \text{Ham}(P, \omega)$ is a Lie algebra morphism relative to the Lie bracket $\{\cdot, \cdot\}_\omega$ in $C^\infty(P)$ and the commutator $[\cdot, \cdot]$ of vector fields in $\mathfrak{X}(P)$.

The construction above shows that a symplectic manifold induces a Poisson bracket on the algebra of functions on the manifold. This observation motivates the introduction of *Poisson manifolds*, for which a Poisson bracket is taken as the fundamental geometric object [125]. In this sense, Poisson geometry provides a natural generalisation of symplectic geometry, allowing for brackets that need not arise from a non-degenerate symplectic form.

Definition 1.1.7. A *Poisson manifold* is a pair $(P, \{\cdot, \cdot\})$ where P is a smooth manifold equipped with a Poisson bracket $\{\cdot, \cdot\} : C^\infty(P) \times C^\infty(P) \rightarrow C^\infty(P)$.

The Poisson bracket on a manifold can be equivalently described in terms of a bivector field. Indeed, since the bracket satisfies the Leibniz rule in each argument, the value of $\{f, g\}$ at a point depends only on the differentials df and dg . Consequently, there exists a unique bivector field $\Lambda \in \mathcal{V}^2(P)$ such that $\{f, g\}(p) = \Lambda_p((df)_p, (dg)_p)$, for all $f, g \in C^\infty(P)$ and every $p \in P$.

The bivector field Λ is called the *Poisson bivector* (or *Poisson tensor*) associated with the Poisson structure. Conversely, any bivector field $\Lambda \in \mathcal{V}^2(P)$ defines a bracket on $C^\infty(P)$ by

$$\{f, g\} := \Lambda(df, dg), \quad f, g \in C^\infty(P),$$

where the above expression is understood pointwise. This bracket is automatically bilinear, skew-symmetric, and satisfies the Leibniz rule. The Jacobi identity is equivalent to the condition $[\Lambda, \Lambda]_S = 0$, where $[\cdot, \cdot]_S$ denotes the Schouten–Nijenhuis bracket [125]. Thus, a Poisson structure may be equivalently described either by a Poisson bracket or by a bivector field Λ satisfying $[\Lambda, \Lambda]_S = 0$.

As shown above, every symplectic manifold (P, ω) defines a Poisson bracket $\{\cdot, \cdot\}_\omega$ on $C^\infty(P)$. Recall the vector bundle isomorphism $\omega^\flat : TP \rightarrow T^*P$, given by $\omega^\flat(v) = \iota_v \omega$, and define $\Lambda^\sharp := -(\omega^\flat)^{-1} : T^*P \rightarrow TP$. The corresponding bivector field $\Lambda \in \mathcal{V}^2(P)$ is given by

$$\Lambda(\alpha, \beta) := \beta(\Lambda^\sharp(\alpha)), \quad \alpha, \beta \in T^*P.$$

In particular, $\Lambda(df, dg) = \{f, g\}_\omega$, where $f, g \in C^\infty(P)$, so Λ is precisely the Poisson bivector associated with the symplectic form ω . With a slight abuse of notation, we also denote by $\omega^\flat$ the induced morphism of $C^\infty(P)$ -modules

$$\omega^\flat : \mathfrak{X}(P) \longrightarrow \Omega^1(P), \quad X \longmapsto \iota_X \omega.$$

In terms of these maps, the Hamiltonian vector field X_f associated with $f \in C^\infty(P)$ satisfies

$$\omega^\flat(X_f) = df, \quad \Lambda^\sharp(df) = -X_f.$$

Thus, the symplectic form and the corresponding Poisson bivector encode the same Hamiltonian dynamics, and their associated bundle maps are related by $\Lambda^\sharp = -(\omega^\flat)^{-1}$.

To conclude, every symplectic manifold is naturally a Poisson manifold, but the converse is not true in general, since a Poisson bivector may be degenerate and therefore need not arise from a symplectic form.

1.1.2 Contact geometry

Contact geometry can be regarded as the odd-dimensional counterpart of symplectic geometry and provides a natural geometric framework for systems with dissipation, thermodynamics, and non-conservative dynamics. Standard references include [15, 32].

Let M be a manifold of dimension $2n+1$, where $n \in \mathbb{N}$. A distribution $\mathcal{D} \subset TM$ of corank one is locally described as the kernel of a differential one-form $\eta \in \Omega^1(U)$, where $U \subset M$ is an open neighbourhood of each point $x \in M$. Then, a distribution $\mathcal{D}$ is said to be *maximally non-integrable* if, for every local defining form η , the differential form $\eta \wedge (d\eta)^n$ is nowhere vanishing on every neighbourhood $U \subset M$. Equivalently, $\eta \wedge (d\eta)^n$ defines a volume form on U . In this case, η is called a *contact form* on U .

Definition 1.1.8. A *contact manifold* is a pair $(M, \mathcal{D})$ where M is a smooth manifold of dimension $2n+1$ and $\mathcal{D}$ is a corank one maximally non-integrable distribution on M . We call $\mathcal{D}$ a *contact distribution* on M .

When the distribution is globally defined as the kernel of a one-form, one obtains the notion of a co-oriented contact manifold.

Definition 1.1.9. A *co-oriented contact manifold* is a pair (M, η) , where η is a differential one-form on M such that $(M, \ker \eta)$ is a contact manifold. We call η a *contact form* on M .

In much of the modern literature, the distinction between contact manifolds and co-oriented contact manifolds is often overlooked, and the latter are simply referred to as contact manifolds. In particular, every contact manifold can locally be regarded as a co-oriented contact manifold, which explains why this

distinction is not relevant when studying local aspects of dynamical systems on contact manifolds [80]. However, when global structures are considered, it is useful to keep the distinction explicit. Throughout this work, unless explicitly stated otherwise, a contact manifold will always mean a co-oriented contact manifold (M, η) .

It is important to note that a contact distribution does not determine a unique contact form. If η is a contact form and $f \in C^\infty(M)$ is nowhere vanishing, then $f\eta$ is also a contact form and defines the same contact distribution $\ker(f\eta) = \ker \eta$. Consequently, a contact structure is represented by an equivalence class of contact forms differing by multiplication by a non-vanishing function.

Moreover, the non-degeneracy condition of a contact form yields a distinguished splitting of the tangent bundle. More precisely, a differential one-form η on a $(2n + 1)$ -dimensional manifold M is a contact form if and only if $TM = \ker \eta \oplus \ker d\eta$.

Every co-oriented contact manifold (M, η) admits a canonical vector field that encodes a distinguished direction transverse to the contact distribution and plays a crucial role in contact dynamics.

Definition 1.1.10. Let (M, η) be a contact manifold. The unique vector field $R \in \mathfrak{X}(M)$ satisfying

$$\iota_R \eta = 1, \quad \iota_R d\eta = 0,$$

is called the *Reeb vector field* associated with η .

It follows from the above definition that the distribution $\ker d\eta$ is generated by the Reeb vector field. Using Cartan's formula we have the following proposition.

Proposition 1.1.11. If R denotes the Reeb vector field of a contact manifold (M, η) , then

$$\mathcal{L}_R \eta = 0, \quad \mathcal{L}_R (d\eta) = 0.$$

The Reeb vector field therefore generates a flow that preserves both the contact form η and its differential $d\eta$.

A fundamental result in contact geometry is the contact analogue of Darboux's theorem.

Theorem 1.1.12. (Darboux theorem for contact manifolds [33]) *Let (M, η) be a co-oriented contact manifold with $\dim M = 2n + 1$. Then, around every point $x \in M$, there exist local coordinates $\{q^i, p_i, z\}$, with $i = 1, \dots, n$, called Darboux coordinates, in which the contact form takes the canonical form*

$$\eta = dz - \sum_{i=1}^n p_i dq^i.$$

Furthermore, in these coordinates the Reeb vector field is $R = \frac{\partial}{\partial z}$.

The Darboux theorem shows that all contact manifolds of the same dimension are locally equivalent.

Let (M, η) be a co-oriented contact manifold. Let us define a vector bundle isomorphism $\flat_\eta : TM \rightarrow T^*M$ by

$$\flat_\eta(v) = \iota_v(d\eta) + (\iota_v \eta)\eta, \quad v \in T_x M.$$

Its inverse is denoted by $\sharp_\eta = \flat_\eta^{-1}$. In particular, the Reeb vector field can be expressed by $R = \sharp_\eta(\eta)$. A one-form η is a contact form precisely when the map $\flat_\eta$ is a vector bundle isomorphism.

Definition 1.1.13. A *contact Hamiltonian system* is a triple (M, η, h) , where (M, η) is a co-oriented contact manifold and $h \in C^\infty(M)$ is a function, called the *contact Hamiltonian* (or simply the *Hamiltonian* when the contact structure is understood from the context).

Every contact Hamiltonian system (M, η, h) admits a unique vector field X_h , called the *contact Hamiltonian vector field* associated with h , characterised by the conditions below.

Proposition 1.1.14. *For every contact Hamiltonian h , there exists a unique vector field $X_h \in \mathfrak{X}(M)$ determined by the conditions*

$$\iota_{X_h} d\eta = dh - (Rh)\eta, \quad \iota_{X_h} \eta = -h,$$

or equivalently $\mathcal{L}_{X_h} \eta = -(Rh)\eta$, $\iota_{X_h} \eta = -h$, or

$$\flat_\eta(X_h) = dh - (Rh + h)\eta.$$

In Darboux coordinates, the contact Hamiltonian vector field X_h takes the form

$$X_h = \sum_{i=1}^n \frac{\partial h}{\partial p_i} \frac{\partial}{\partial q^i} - \sum_{i=1}^n \left(\frac{\partial h}{\partial q^i} + p_i \frac{\partial h}{\partial z} \right) \frac{\partial}{\partial p_i} + \left(\sum_{i=1}^n p_i \frac{\partial h}{\partial p_i} - h \right) \frac{\partial}{\partial z}. \quad (1.1.1)$$

Therefore, an integral curve $\phi(t) = (q^i(t), p_i(t), z(t))$ of X_h satisfies the so-called *contact Hamilton equations*, that is, the system of ordinary differential equations

$$\frac{dq^i}{dt} = \frac{\partial h}{\partial p_i}, \quad \frac{dp_i}{dt} = - \left(\frac{\partial h}{\partial q^i} + p_i \frac{\partial h}{\partial z} \right), \quad \frac{dz}{dt} = \sum_{i=1}^n p_i \frac{\partial h}{\partial p_i} - h, \quad i = 1, \dots, n.$$

The equations for q^i and p_i reduce to the classical Hamilton equations whenever the Hamiltonian does not depend on the contact coordinate z . More precisely, unlike symplectic Hamiltonian mechanics, the contact Hamiltonian function is generally not conserved along its own flow. Indeed, $X_h(h) = -(Rh)h$. Consequently, whenever $Rh \neq 0$, the energy-like quantity h changes along trajectories. This property makes contact geometry particularly suitable for modelling dissipative systems. A function $f \in C^\infty(M)$ is called a *dissipated quantity* if $X_h(f) = -(Rh)f$ [80]. Such quantities decay (or grow) according to the same law as the Hamiltonian itself. Therefore, the extra z -dependence introduces dissipative terms, allowing contact Hamiltonian systems to describe friction, damping, thermodynamic processes, and irreversible phenomena within a unified geometric framework. Thus, contact geometry extends the reach of symplectic mechanics beyond conservative dynamics.

Definition 1.1.15. Using the notation introduced above, the vector field

$$E_h = hR + X_h \in \mathfrak{X}(M)$$

is called the *evolution contact Hamiltonian vector field* associated with the Hamiltonian $h : M \rightarrow \mathbb{R}$.

Indeed, if $\{q^i, p_i, z\}$ are Darboux coordinates on M , then by (1.1.1),

$$E_h := \sum_{i=1}^n \frac{\partial h}{\partial p_i} \frac{\partial}{\partial q^i} - \sum_{i=1}^n \left(\frac{\partial h}{\partial q^i} + p_i \frac{\partial h}{\partial z} \right) \frac{\partial}{\partial p_i} + \sum_{i=1}^n p_i \frac{\partial h}{\partial p_i} \frac{\partial}{\partial z}.$$

The integral curves of E_h satisfy

$$\frac{dq^i}{dt} = \frac{\partial h}{\partial p_i}, \quad \frac{dp_i}{dt} = - \left(\frac{\partial h}{\partial q^i} + p_i \frac{\partial h}{\partial z} \right), \quad \frac{dz}{dt} = \sum_{j=1}^n p_j \frac{\partial h}{\partial p_j}, \quad i = 1, \dots, n.$$

Evolution vector fields were introduced in [122].

1.2 Fundamentals on geometric field theory

The geometric description of classical field theories requires structures adapted to systems of partial differential equations. In this section, we recall the main geometric notions and formalisms that will be used later in the development of k -contact Hamiltonian field theory.

1.2.1 Distributions, k -vector fields and integral sections

The concept of k -vector fields provides one of the fundamental geometric tools for the study of systems of partial differential equations. In particular, they furnish an appropriate framework for describing the geometric structure underlying such systems and play a central role in the k -symplectic, k -cosymplectic, and most importantly, k -contact formalisms [72]. For completeness, we briefly recall the basic notions and results that will be used throughout this work.

A *distribution* on M is a subset $D \subset TM$ such that, for every $x \in M$, the fibre $D_x = D \cap T_x M$ is a vector subspace of $T_x M$. We call $\dim D_x$ the *rank* of D at $x \in M$. A distribution D is *smooth* if, for every $x \in M$, there exists a neighbourhood $U \subset M$ of x and vector fields $X_1, \dots, X_r \in \mathfrak{X}(U)$ such that $D_y = \langle X_1(y), \dots, X_r(y) \rangle$ for every $y \in U$. Note that r does not need to coincide with $\dim D_x$. A distribution D is said to be *regular* if it is smooth and of constant rank. Similarly, a *codistribution* on M is a subset $C \subset T^*M$ such that $C_x = C \cap T_x^*M$ is a vector subspace of T_x^*M for every $x \in M$. Meanwhile, the *annihilator* of a distribution D is the codistribution $D^\circ = \bigsqcup_{x \in M} D_x^\circ$, where $D_x^\circ = \{\alpha \in T_x^*M \mid \alpha(v) = 0 \text{ for all } v \in D_x\}$. The rank of the codistribution D° at $x \in M$ is called the *corank* of D at x .

A distribution D on M has a constant rank m if and only if D has constant corank $n - m$, where $\dim M = n$. We denote this by writing $\text{rk } D = m$, and $\text{cork } D = n - m$. It is important to note that if D is not regular, then D° may not be smooth.

Using the canonical identification $E^{**} \cong E$ for a finite-dimensional vector space E , it follows that $(D^\circ)^\circ = D$. We denote by $\Gamma(D)$ the space of vector fields on M taking values in the distribution D . Throughout this dissertation, we simply write ‘distribution’ rather than ‘generalised distribution’. Although the latter terminology is common in the literature for arbitrary distributions that are not assumed to be smooth or of constant rank, our terminology should cause no confusion.

Consider a differential one-form $\eta \in \Omega^1(M)$. Then, η spans a co-distribution $\mathcal{C} = \langle \eta \rangle = \{\langle \eta_x \rangle \mid x \in M\} \subset T^*M$. Then, $\mathcal{C}$ has rank one at every point where η does not vanish. The annihilator of $\mathcal{C}$ is the distribution $\ker \eta \subset TM$. The distribution $\mathcal{C}^\circ$ has corank one at every point where η does not vanish, and zero otherwise.

Consider the Whitney sum of k copies of the tangent bundle of M , namely $\bigoplus^k TM = TM \oplus_M \cdots \oplus_M^{(k)} TM$, where the fibre product is taken over M . Then, we have the natural projections

$$\text{pr}^\alpha: \bigoplus^k TM \rightarrow TM, \quad \text{pr}_M: \bigoplus^k TM \rightarrow M, \quad \alpha = 1, \dots, k,$$

where pr^α is the projection onto the α -th component of the Whitney sum.

Definition 1.2.1. A k -vector field on M is a section $\mathbf{Z}: M \rightarrow \bigoplus^k TM$ of the vector bundle $\text{pr}_M: \bigoplus^k TM \rightarrow M$.

The space of k -vector fields on M will be denoted by $\mathfrak{X}^k(M)$. In case of $k = 1$ we simply write $\mathfrak{X}^1(M) = \mathfrak{X}(M)$. A k -vector field $\mathbf{Z} \in \mathfrak{X}^k(M)$ defines a commutative diagram

$$\begin{array}{ccc} & \bigoplus^k TM & \\ \mathbf{Z} \nearrow & \downarrow \text{pr}^\alpha & \\ M & \xrightarrow{Z_\alpha} & TM \end{array} \quad (1.2.1)$$

where $Z_\alpha = \text{pr}^\alpha \circ \mathbf{Z}$ with $\alpha = 1, \dots, k$.

Thus, in view of diagram (1.2.1), a k -vector field $\mathbf{Z} \in \mathfrak{X}^k(M)$ is equivalently described by a family of vector fields $Z_1, \dots, Z_k \in \mathfrak{X}(M)$, where

$$Z_\alpha = \text{pr}^\alpha \circ \mathbf{Z}, \quad \alpha = 1, \dots, k.$$

Accordingly, we make an identification $\mathbf{Z} = (Z_1, \dots, Z_k)$.

Definition 1.2.2. Let $\phi: U \subset \mathbb{R}^k \rightarrow M$ be a smooth map. The *first prolongation* of ϕ is the map $\phi': U \subset \mathbb{R}^k \rightarrow \bigoplus^k TM$ defined by

$$\phi'(t) = \left(\phi(t); T_t \phi \left(\frac{\partial}{\partial t^1} \Big|_t \right), \dots, T_t \phi \left(\frac{\partial}{\partial t^k} \Big|_t \right) \right) =: (\phi(t); \phi'_\alpha(t)), \quad t \in \mathbb{R}^k,$$

where $t^1, \dots, t^k$ denote the standard Cartesian coordinates on $\mathbb{R}^k$.

The notion of an integral section provides the natural generalisation of an integral curve to the setting of k -vector fields.

Definition 1.2.3. An *integral section* of a k -vector field $\mathbf{Z} = (Z_1, \dots, Z_k) \in \mathfrak{X}^k(M)$ is a map $\phi: U \subset \mathbb{R}^k \rightarrow M$ such that $\phi' = \mathbf{Z} \circ \phi$, that is, $T\phi \left(\frac{\partial}{\partial t^\alpha} \right) = Z_\alpha \circ \phi$ for $\alpha = 1, \dots, k$. We say that k -vector field $\mathbf{Z} \in \mathfrak{X}^k(M)$ is integrable if and only if every point of M is in the image $\text{Im}\phi$ of an integral section of $\mathbf{Z}$.

Let $\{x_1, \dots, x_n\}$ be local coordinates on M . Consider a k -vector field $\mathbf{Z} = (Z_1, \dots, Z_k) \in \mathfrak{X}^k(M)$ with local expression $Z_\alpha = \sum_{i=1}^n Z_\alpha^i \frac{\partial}{\partial x^i}$ for $\alpha = 1, \dots, k$. Then, $\phi: U \subset \mathbb{R}^k \rightarrow M$ is an integral section of $\mathbf{Z}$ if and only if its coordinates form a solution of the system of partial differential equations

$$\frac{\partial \phi^i}{\partial t^\alpha} = Z_\alpha^i \circ \phi, \quad i = 1, \dots, n, \quad \alpha = 1, \dots, k. \quad (1.2.2)$$

The system of PDEs (1.2.2) is integrable if and only if $[Z_\alpha, Z_\beta] = 0$ for $\alpha, \beta = 1, \dots, k$ (see [58] for details). These conditions are equivalent to the integrability of the k -vector field $\mathbf{Z}$.

Let us consider the distribution generated by the components of $\mathbf{Z}$, denoted by $D^{\mathbf{Z}} := \text{span}\{Z_1, \dots, Z_k\}$. Assume that $D^{\mathbf{Z}}$ has constant rank. The notion of integrability for a k -vector field is more restrictive than that for its associated distribution. In fact, $D^{\mathbf{Z}}$ is integrable if and only if it is involutive, that is,

$$[Z_\alpha, Z_\beta] = \sum_{\gamma=1}^k f_{\alpha\beta}^\gamma Z_\gamma, \quad \alpha, \beta = 1, \dots, k,$$

for some smooth functions $f_{\alpha\beta}^\gamma \in C^\infty(M)$. By contrast, the k -vector field $\mathbf{Z}$ is integrable if and only if the vector fields $Z_1, \dots, Z_k$ commute pairwise. Whenever $Z_1, \dots, Z_k$ are linearly independent, this is equivalent to $f_{\alpha\beta}^\gamma = 0$ for $\alpha, \beta, \gamma = 1, \dots, k$. Consequently, every integrable k -vector field determines an integrable distribution, whereas the converse does not hold in general.

Throughout this dissertation, the symbol $\mathbf{Z}$ is reserved for an arbitrary k -vector field. Other boldface letters will be used to denote distinguished classes of k -vector fields, such as Hamiltonian and evolution k -vector fields, which will be introduced in the subsequent sections.

1.2.2 Geometric preliminaries for field theory

This section reviews the preliminaries needed for the geometric formulation of Hamiltonian field theories (see [9, 72, 110]). Throughout this thesis, we fix the canonical basis $\{e_1, \dots, e_k\}$ of $\mathbb{R}^k$, where k is a positive integer. The dual basis on $\mathbb{R}^{k*}$ will be denoted by $\{e^1, \dots, e^k\}$.

Let $\Omega^\ell(M, \mathbb{R}^k)$ denote the space of $\mathbb{R}^k$ -valued differential ℓ -forms on M and consider $\boldsymbol{\theta} \in \Omega^\ell(M, \mathbb{R}^k)$. This form can be represented uniquely as $\boldsymbol{\theta} = \sum_{\alpha=1}^k \theta^\alpha \otimes e_\alpha$ for some unique differential ℓ -forms $\theta^1, \dots, \theta^k \in \Omega^\ell(M)$. Unless otherwise stated, all algebraic operations on $\mathbb{R}^k$ -valued forms are understood componentwise. We say that $\boldsymbol{\theta} \in \Omega^\ell(M, \mathbb{R}^k)$ is *non-degenerate* if $\ker \boldsymbol{\theta} := \bigcap_{\alpha=1}^k \ker \theta^\alpha = \{0\}$. The contractions of a vector field $Z \in \mathfrak{X}(M)$ and a k -vector field $\mathbf{Z} \in \mathfrak{X}^k(M)$ on M with $\boldsymbol{\theta} \in \Omega^\ell(M, \mathbb{R}^k)$ are given, respectively, by

$$\iota_Z \boldsymbol{\theta} = \sum_{\alpha=1}^k \iota_Z \theta^\alpha \otimes e_\alpha \in \Omega^{\ell-1}(M, \mathbb{R}^k), \quad \iota_{\mathbf{Z}} \boldsymbol{\theta} = \sum_{\alpha=1}^k \iota_{Z_\alpha} \theta^\alpha \in \Omega^{\ell-1}(M).$$

Meanwhile, the exterior derivative of $\theta \in \Omega^\ell(M, \mathbb{R}^k)$ is given by

$$d\theta = \sum_{\alpha=1}^k d\theta^\alpha \otimes e_\alpha \in \Omega^{\ell+1}(M, \mathbb{R}^k).$$

Furthermore, given two $\mathbb{R}^k$ -valued differential forms $\vartheta = \sum_{\alpha=1}^k \vartheta^\alpha \otimes e_\alpha \in \Omega^{\ell_1}(M, \mathbb{R}^k)$ and $\nu = \sum_{\alpha=1}^k \nu^\alpha \otimes e_\alpha \in \Omega^{\ell_2}(M, \mathbb{R}^k)$, we define their componentwise exterior product by

$$\vartheta \bar{\wedge} \nu = \sum_{\alpha=1}^k (\vartheta^\alpha \wedge \nu^\alpha) \otimes e_\alpha \in \Omega^{\ell_1+\ell_2}(M, \mathbb{R}^k),$$

while the exterior derivative of the above reads

$$d(\vartheta \bar{\wedge} \nu) = d\vartheta \bar{\wedge} \nu + (-1)^{\ell_1} \vartheta \bar{\wedge} d\nu \in \Omega^{\ell_1+\ell_2+1}(M, \mathbb{R}^k).$$

Likewise, the Lie derivative of an $\mathbb{R}^k$ -valued differential form $\theta \in \Omega^\ell(M, \mathbb{R}^k)$ along a vector field $Z \in \mathfrak{X}(M)$ is defined by

$$\mathcal{L}_Z \theta = \sum_{\alpha=1}^k \mathcal{L}_Z \theta^\alpha \otimes e_\alpha \in \Omega^\ell(M, \mathbb{R}^k).$$

Consequently, Cartan's formula extends naturally to $\mathbb{R}^k$ -valued differential forms, namely $\mathcal{L}_Z \theta = \iota_Z d\theta + d(\iota_Z \theta)$.

Moreover, the Lie derivative satisfies the usual Leibniz rule with respect to the exterior product. More precisely, if $\vartheta \in \Omega^{\ell_1}(M, \mathbb{R}^k)$ and $\nu \in \Omega^{\ell_2}(M, \mathbb{R}^k)$, then

$$\mathcal{L}_Z (\vartheta \bar{\wedge} \nu) = (\mathcal{L}_Z \vartheta) \bar{\wedge} \nu + \vartheta \bar{\wedge} (\mathcal{L}_Z \nu) \in \Omega^{\ell_1+\ell_2}(M, \mathbb{R}^k).$$

The above definitions are useful for simplifying the description of our methods. A point-wise analogue can be defined similarly. In particular, these definitions apply to the contraction of elements of E or $E \otimes \mathbb{R}^k$ with $E^* \otimes \mathbb{R}^k$ for a vector space E . Note that we will freely switch between the intrinsic notation introduced above and the corresponding componentwise expressions whenever convenient. The intrinsic notation is preferred in statements of results, whereas componentwise expressions will be used when they simplify computations and applications.

1.2.3 k -Polysymplectic geometry

The geometric treatment of first-order classical field theories requires generalisations of the structures arising in symplectic geometry that are adapted to systems of partial differential equations. Among these approaches, *k-symplectic geometry*, also referred to as *k-polysymplectic geometry*, was introduced by A. Awane [9, 10] and later employed in the geometric description of first-order classical field theories by M. de León et al. [66–68] and L.K. Norris [93, 98] (see also [72]). The resulting geometric structure agrees with the *polysymplectic structures* considered by G.C. Günther [42], but should be distinguished from other approaches to polysymplectic geometry, such as those developed by G. Sardanashvily et al. [34, 120] and I.V. Kanatchikov [49]. Since closely related but non-equivalent geometric structures occur in the literature under similar names, the terminology adopted in this dissertation is fixed explicitly in Definition 1.2.4.

The k -polysymplectic framework provides a geometric formulation of several classes of partial differential equations arising in classical field theory. In particular, it has been used to describe the Euler–Lagrange and Hamilton–De Donder–Weyl equations and their associated dynamics, including those of first-order regular autonomous field theories [24, 64, 72, 74]. It also provides tools for the geometric study of symmetries, conservation laws and reduction in field theories [9, 42, 90, 114]. Beyond this field-theoretic setting, k -polysymplectic geometry has also been applied to systems of ordinary differential

equations and, in particular, to the study of their superposition rules [86]. This constitutes a different use of the framework from its usual application to partial differential equations in classical field theory.

The purpose of this subsection is to introduce the k -polysymplectic structures required throughout this dissertation and to relate them to the polysymplectic and k -symplectic terminology found in the literature. This distinction is necessary because the same or similar terminology has been used for different, non-equivalent geometric structures. In particular, the foundational works of Günther [42] and Awane [9] illustrate the different conventions that have appeared in this context.

Definition 1.2.4. A k -polysymplectic form on a manifold P is a closed, non-degenerate $\mathbb{R}^k$ -valued differential two-form

$$\boldsymbol{\omega} = \sum_{\alpha=1}^k \omega^\alpha \otimes e_\alpha \in \Omega^2(P, \mathbb{R}^k).$$

A manifold P endowed with a k -polysymplectic form $\boldsymbol{\omega}$ is called a k -polysymplectic manifold and is denoted by $(P, \boldsymbol{\omega})$. If, moreover, there exists $\boldsymbol{\theta} \in \Omega^1(P, \mathbb{R}^k)$ such that $\boldsymbol{\omega} = d\boldsymbol{\theta}$, the k -polysymplectic structure is said to be *exact*.

The term ‘ k -polysymplectic manifold’ is sometimes shortened to ‘polysymplectic manifold’ in the literature (see, for instance, [90]). Since ‘polysymplectic manifold’ is also used for a distinct geometric structure, which will be introduced below, we retain the prefix ‘ k -’ throughout this dissertation whenever the distinction is relevant.

Equivalently, a k -polysymplectic structure on P can be described by a family of closed two-forms $\omega^1, \dots, \omega^k \in \Omega^2(P)$ whose kernels have trivial intersection. Indeed, for $\boldsymbol{\omega} = \sum_{\alpha=1}^k \omega^\alpha \otimes e_\alpha$, the non-degeneracy condition is

$$\ker \boldsymbol{\omega} = \bigcap_{\alpha=1}^k \ker \omega^\alpha = 0.$$

Throughout this dissertation, $\mathbb{R}^k$ -valued differential forms will be denoted by bold symbols. We now recall the more restrictive notions of polysymplectic and k -symplectic structures that will be used in what follows.

Definition 1.2.5. Let P be an $n(k+1)$ -dimensional manifold.

- A *polysymplectic form* on P is an $\mathbb{R}^k$ -valued differential two-form

$$\boldsymbol{\omega} = \sum_{\alpha=1}^k \omega^\alpha \otimes e_\alpha \in \Omega^2(P, \mathbb{R}^k),$$

where $\omega^1, \dots, \omega^k \in \Omega^2(P)$ are closed and satisfy the non-degeneracy condition

$$\ker \boldsymbol{\omega} = \bigcap_{\alpha=1}^k \ker \omega^\alpha = 0.$$

A manifold endowed with such a form is called a *polysymplectic manifold* and is denoted by $(P, \boldsymbol{\omega})$.

- A *k -symplectic structure* on P consists of a polysymplectic form $\boldsymbol{\omega}$ together with an integrable distribution $V \subset TP$ of rank nk satisfying

$$\boldsymbol{\omega}|_{V \times V} = 0.$$

The triple $(P, \boldsymbol{\omega}, V)$ is then called a *k -symplectic manifold*, and V is referred to as its *polarisation*.

Moreover, a polysymplectic or k -symplectic structure is said to be *exact* if there exists $\boldsymbol{\theta} \in \Omega^1(P, \mathbb{R}^k)$ such that $\boldsymbol{\omega} = d\boldsymbol{\theta}$.

The notion of a k -symplectic manifold given above agrees with that introduced by A. Awane [9, 11]. It is also locally equivalent to the *standard polysymplectic structure* considered by C. Günther [42]. The two notions are globally equivalent when an atlas of compatible Darboux charts exists.¹ Moreover, the k -symplectic structure is globally equivalent to the *integrable p -almost cotangent structure* introduced by M. de León et al. [66, 67]. For $k = 1$, Awane's definition reduce to that of a *polarised symplectic manifold*, that is, a symplectic manifold endowed with a Lagrangian distribution [74].

In Günther's terminology, a polysymplectic manifold is obtained by considering the differential forms entering a k -symplectic structure without requiring the additional distribution V . A *standard polysymplectic manifold*, on the other hand, is a polysymplectic manifold admitting local Darboux coordinates and is therefore locally equivalent to a k -symplectic manifold. The role of the polarisation V is precisely to ensure the existence of an atlas of compatible k -symplectic Darboux coordinates, while the existence of such an atlas determines the corresponding polarisation (see [9, 38] and [110, p. 57]).

Theorem 1.2.6 (Darboux theorem for k -symplectic manifolds). *Let (P, ω, V) be a k -symplectic manifold. Then, around every point $p \in P$, there exist local coordinates $\{q^i, p_i^\alpha\}$, with $i = 1, \dots, n$ and $\alpha = 1, \dots, k$, such that*

$$\omega = \sum_{\alpha=1}^k \sum_{i=1}^n dq^i \wedge dp_i^\alpha \otimes e_\alpha, \quad V = \left\langle \frac{\partial}{\partial p_i^\alpha} \right\rangle.$$

Such coordinates are called k -symplectic Darboux coordinates.

The following canonical construction provides the fundamental example of a k -symplectic manifold and will be used repeatedly throughout this dissertation.

Example 1.2.7 (Canonical model for k -symplectic manifolds). Let Q be an n -dimensional manifold and consider the Whitney sum

$$\bigoplus^k T^*Q = T^*Q \oplus_Q \cdots \oplus_Q T^*Q.$$

Denote by $\pi^\alpha: \bigoplus^k T^*Q \rightarrow T^*Q$, for $\alpha = 1, \dots, k$, the projection onto the α -th factor, and by $\pi_Q: \bigoplus^k T^*Q \rightarrow Q$ the natural projection onto the base. Local coordinates $\{q^i\}$ on Q induce natural coordinates $\{q^i, p_i^\alpha\}$ on $\bigoplus^k T^*Q$, with $i = 1, \dots, n$ and $\alpha = 1, \dots, k$.

Let $\theta \in \Omega^1(T^*Q)$ be the Liouville one-form on the cotangent bundle and let $\omega = -d\theta$ be the corresponding canonical symplectic form. These forms induce the canonical $\mathbb{R}^k$ -valued forms on $\bigoplus^k T^*Q$ given by $\theta_k = \sum_{\alpha=1}^k (\pi^\alpha)^* \theta \otimes e_\alpha$, and $\omega_k = -d\theta_k$. In the induced natural coordinates, they take the form

$$\theta_k = \sum_{\alpha=1}^k \sum_{i=1}^n p_i^\alpha dq^i \otimes e_\alpha, \quad \omega_k = \sum_{\alpha=1}^k \sum_{i=1}^n dq^i \wedge dp_i^\alpha \otimes e_\alpha.$$

Therefore, setting $V_k = \ker T\pi_Q$, the triple $(\bigoplus^k T^*Q, \omega_k, V_k)$ defines a k -symplectic manifold. Moreover, the natural coordinates $\{q^i, p_i^\alpha\}$ are k -symplectic Darboux coordinates. $\triangle$

Given a k -polysymplectic manifold (P, ω) , the vector bundle morphism

$$\flat_\omega: (v_1, \dots, v_k) \in \bigoplus^k TP \mapsto \sum_{\alpha=1}^k \iota_{v_\alpha} \langle \omega, e^\alpha \rangle = \sum_{\alpha=1}^k \iota_{v_\alpha} \omega^\alpha \in T^*P,$$

induces a morphism of $C^\infty(P)$ -modules, namely $\flat_\omega: \mathfrak{X}^k(P) \rightarrow \Omega^1(P)$. The morphism $\flat_\omega$ is surjective, since the annihilator of its image is contained in $\bigcap_{\alpha=1}^k \ker \omega^\alpha = \ker \omega = 0$.

¹It is not clear, however, what Günther means precisely by an atlas of canonical charts, and, in particular, which compatibility conditions are imposed on different Darboux charts.

1.2.4 ω -Hamiltonian functions and vector fields

This subsection introduces the notions of Hamiltonian vector fields and Hamiltonian functions associated with a k -polysymplectic structure.

Definition 1.2.8. Let $(P, \omega = \sum_{\alpha=1}^k \omega^\alpha \otimes e_\alpha)$ be a k -polysymplectic manifold. A vector field $X \in \mathfrak{X}(P)$ is called ω -Hamiltonian if it is Hamiltonian with respect to each of the presymplectic forms $\omega^1, \dots, \omega^k$, that is, $\iota_X \omega^\alpha$ is closed for every $\alpha = 1, \dots, k$. The space of ω -Hamiltonian vector fields on (P, ω) is denoted by $\mathfrak{X}_\omega(P)$.

The closed one-forms $\iota_X \omega^\alpha$ need not admit globally defined potentials. They are, however, locally exact. Since this dissertation is primarily concerned with local aspects of the theory, this possible global obstruction will not play a role in what follows.

It is convenient to collect the local Hamiltonian functions associated with the different components of ω into a single $\mathbb{R}^k$ -valued function. This leads to the following notion (see [9, 86] for further details).

Definition 1.2.9. Let $(P, \omega = \sum_{\alpha=1}^k \omega^\alpha \otimes e_\alpha)$ be a k -polysymplectic manifold. An $\mathbb{R}^k$ -valued function $\mathbf{h} = \sum_{\alpha=1}^k h^\alpha \otimes e_\alpha \in C^\infty(P, \mathbb{R}^k)$ is called an ω -Hamiltonian function if there exists a vector field $X_{\mathbf{h}} \in \mathfrak{X}(P)$ satisfying $\iota_{X_{\mathbf{h}}} \omega = d\mathbf{h}$, or, equivalently, $\iota_{X_{\mathbf{h}}} \omega^\alpha = dh^\alpha$, for $\alpha = 1, \dots, k$. In this case, $\mathbf{h}$ is said to be associated with $X_{\mathbf{h}}$. The space of ω -Hamiltonian functions on (P, ω) is denoted by $C_\omega^\infty(P)$.

When the k -polysymplectic form ω is clear from the context, ω -Hamiltonian vector fields and functions are also referred to as k -Hamiltonian. In [94], a k -Hamiltonian system associated with an $\mathbb{R}^k$ -valued Hamiltonian function $\mathbf{h}$ is described by the corresponding vector field $X_{\mathbf{h}}$. A related terminology was introduced by A. Awane [9], who refers to $\mathbf{h}$ as a Hamiltonian map of a vector field X when X is, in addition, an infinitesimal automorphism of a distribution on which the presymplectic forms defining the k -symplectic structure vanish.

Notice that, whenever it exists, the vector field $X_{\mathbf{h}}$ associated with an ω -Hamiltonian function $\mathbf{h}$ is unique. Indeed, if $X_{\mathbf{h}}$ and $X'_{\mathbf{h}}$ satisfy the defining condition, then

$$X_{\mathbf{h}} - X'_{\mathbf{h}} \in \bigcap_{\alpha=1}^k \ker \omega^\alpha = \ker \omega = 0.$$

The following results summarise some basic properties of ω -Hamiltonian functions and vector fields. Further details can be found in [86].

Proposition 1.2.10. *Every ω -Hamiltonian vector field admits at least one associated ω -Hamiltonian function. Conversely, every ω -Hamiltonian function determines a unique ω -Hamiltonian vector field. Moreover, the space $C_\omega^\infty(P)$ associated with a k -polysymplectic manifold (P, ω) carries a Lie algebra structure. Its vector space operations are defined componentwise by*

$$\mathbf{h} + \mathbf{g} := \sum_{\alpha=1}^k (h^\alpha + g^\alpha) \otimes e_\alpha, \quad \lambda \cdot \mathbf{h} := \sum_{\alpha=1}^k \lambda h^\alpha \otimes e_\alpha,$$

where $\mathbf{h} = \sum_{\alpha=1}^k h^\alpha \otimes e_\alpha, \mathbf{g} = \sum_{\alpha=1}^k g^\alpha \otimes e_\alpha \in C_\omega^\infty(P)$, and $\lambda \in \mathbb{R}$. The corresponding Lie bracket $\{\cdot, \cdot\}_\omega : C_\omega^\infty(P) \times C_\omega^\infty(P) \rightarrow C_\omega^\infty(P)$ is given by

$$\{\mathbf{h}, \mathbf{g}\}_\omega = \sum_{\alpha=1}^k \{h^\alpha, g^\alpha\}_{\omega^\alpha} \otimes e_\alpha,$$

where $\{\cdot, \cdot\}_{\omega^\alpha}$ denotes the bracket associated with the presymplectic form ω^α , for $\alpha = 1, \dots, k$.

In contrast to the Lie bracket, the componentwise product

$$\mathbf{h} \star \mathbf{g} = \sum_{\alpha=1}^k (h^\alpha g^\alpha) \otimes e_\alpha$$

does not, in general, define an ω -Hamiltonian function [86, p. 2239]. Consequently, $(C_\omega^\infty(P), \star, \{\cdot, \cdot\}_\omega)$ does not generally form a Poisson algebra. Moreover, for a fixed $\mathbf{h} \in C_\omega^\infty(P)$, the map

$$\{\mathbf{h}, \cdot\}_\omega: C_\omega^\infty(P) \longrightarrow C_\omega^\infty(P), \quad \mathbf{g} \longmapsto \{\mathbf{g}, \mathbf{h}\}_\omega,$$

is not, in general, a derivation with respect to $\star$. These properties distinguish the k -polysymplectic setting from the usual Poisson framework. Nevertheless, $\{\mathbf{h}, \mathbf{g}\}_\omega = 0$ whenever $\mathbf{g}$ is locally constant. The following result records another property of this Lie algebra.

Proposition 1.2.11. *Let (P, ω) be a k -polysymplectic manifold. Every ω -Hamiltonian vector field $X_{\mathbf{h}}$ acts as a derivation of the Lie algebra $(C_\omega^\infty(P), \{\cdot, \cdot\}_\omega)$ according to*

$$X_{\mathbf{h}}\mathbf{f} = \{\mathbf{f}, \mathbf{h}\}_\omega, \quad \mathbf{f} \in C_\omega^\infty(P),$$

where $\mathbf{h}$ is an ω -Hamiltonian function associated with $X_{\mathbf{h}}$.

1.2.5 k -Polysymplectic geometry

The geometric description of non-autonomous field theories leads naturally to structures that extend the k -polysymplectic framework by incorporating the dependence on space-time variables. In this context, *k -polysymplectic geometry* generalises the cosymplectic formulation of non-autonomous mechanics [1] to field theories whose Lagrangian or Hamiltonian functions may depend explicitly on the space-time coordinates [69, 70]. Non-autonomous field theories can be formulated in terms of both k -polysymplectic [95] and k -cosymplectic structures [69]. Further details on the k -symplectic and k -cosymplectic formalisms can be found in [39, 72, 96], while their relationship with the multisymplectic formalism is discussed in [109].

We begin with the notion of a k -polysymplectic structure. Let $\{e_1, \dots, e_k\}$ be the canonical basis of $\mathbb{R}^k$ and write

$$\boldsymbol{\tau} = \sum_{\alpha=1}^k \tau^\alpha \otimes e_\alpha, \quad \boldsymbol{\omega} = \sum_{\alpha=1}^k \omega^\alpha \otimes e_\alpha,$$

where $\tau^\alpha \in \Omega^1(M)$ and $\omega^\alpha \in \Omega^2(M)$.

Definition 1.2.12. A *k -polysymplectic structure* on a manifold M is a pair $(\boldsymbol{\tau}, \boldsymbol{\omega})$, where $\boldsymbol{\tau} \in \Omega^1(M, \mathbb{R}^k)$, and $\boldsymbol{\omega} \in \Omega^2(M, \mathbb{R}^k)$ are closed $\mathbb{R}^k$ -valued differential forms satisfying

$$\text{rk ker } \boldsymbol{\omega} = \text{rk} \left(\bigcap_{\alpha=1}^k \text{ker } \omega^\alpha \right) = k, \quad \text{ker } \boldsymbol{\omega} \cap \text{ker } \boldsymbol{\tau} = 0.$$

A manifold endowed with such a structure is called a *k -polysymplectic manifold* and is denoted by $(M, \boldsymbol{\tau}, \boldsymbol{\omega})$. If, moreover, $\dim M = k + n(k+1)$ for some $n \in \mathbb{N}$, then $(\boldsymbol{\tau}, \boldsymbol{\omega})$ is called a *polysymplectic structure*, and $(M, \boldsymbol{\tau}, \boldsymbol{\omega})$ is called a *polysymplectic manifold*.

Definition 1.2.13. A *k -cosymplectic structure* on a manifold M is a triple $(\boldsymbol{\tau}, \boldsymbol{\omega}, V)$, where $(\boldsymbol{\tau}, \boldsymbol{\omega})$ is a polysymplectic structure on M and $V \subset TM$ is an integrable distribution of rank nk satisfying

$$\boldsymbol{\tau}|_V = 0, \quad \boldsymbol{\omega}|_{V \times V} = 0.$$

A manifold endowed with such a structure is called a *k -cosymplectic manifold* and is denoted by $(M, \boldsymbol{\tau}, \boldsymbol{\omega}, V)$.

If $\boldsymbol{\omega}$ is exact, that is, $\boldsymbol{\omega} = d\boldsymbol{\theta}$ for some $\boldsymbol{\theta} \in \Omega^1(M, \mathbb{R}^k)$, the corresponding k -polysymplectic, polysymplectic or k -cosymplectic structure is said to be *exact*.

The additional distribution appearing in a k -cosymplectic structure gives rise to a Darboux theorem analogous to that of k -symplectic geometry.

Theorem 1.2.14 (Darboux theorem for k -cosymplectic manifolds). *Let (M, τ, ω, V) be a k -cosymplectic manifold. Then, around every point $x \in M$, there exist local coordinates $\{t^\alpha, q^i, p_i^\alpha\}$, with $i = 1, \dots, n$ and $\alpha = 1, \dots, k$, such that*

$$\omega = \sum_{\alpha=1}^k \sum_{i=1}^n dq^i \wedge dp_i^\alpha \otimes e_\alpha, \quad \tau = \sum_{\alpha=1}^k dt^\alpha \otimes e_\alpha, \quad V = \left\langle \frac{\partial}{\partial p_i^\alpha} \right\rangle.$$

Such coordinates are called k -cosymplectic Darboux coordinates.

An important feature of a k -polycosymplectic structure is the existence of a distinguished family of vector fields determined by (τ, ω) . These vector fields provide the analogue of the Reeb vector field in cosymplectic geometry [18] and will also be useful in relating k -polycosymplectic and k -polysymplectic structures.

Proposition 1.2.15. *Let (M, τ, ω) be a k -polycosymplectic manifold. There exists a unique family of vector fields $R_1, \dots, R_k \in \mathfrak{X}(M)$, called the Reeb vector fields, satisfying*

$$\iota_{R_\alpha} \tau = e_\alpha, \quad \iota_{R_\alpha} \omega = 0, \quad \alpha = 1, \dots, k. \quad (1.2.3)$$

Proof. Set $D = \ker \omega = \bigcap_{\alpha=1}^k \ker \omega^\alpha$. By Definition 1.2.12, D has rank k and $D \cap \ker \tau = 0$. Hence the restriction $\tau|_D: D \rightarrow \mathbb{R}^k$ is injective. Since both D_x and $\mathbb{R}^k$ have dimension k , it is an isomorphism at every $x \in M$. Consequently, for each $\alpha = 1, \dots, k$, there exists a unique vector field R_α taking values in D such that $\tau(R_\alpha) = e_\alpha$. Since $R_\alpha \in D = \ker \omega$, one also has $\iota_{R_\alpha} \omega = 0$. Thus, the vector fields $R_1, \dots, R_k$ satisfy (1.2.3), and their uniqueness follows from the fact that $\tau|_D$ is an isomorphism. $\square$

1.2.6 k -Polycosymplectic Hamiltonian systems

This subsection introduces the Hamiltonian description of non-autonomous field theories within the k -polycosymplectic framework. The additional one-form τ accounts geometrically for the independent variables, while the Hamiltonian dynamics is described by a suitable k -vector field.

Definition 1.2.16. Let (M, τ, ω) be a k -polycosymplectic manifold and let $h \in C^\infty(M)$. The quadruple (M, τ, ω, h) is called a *k -polycosymplectic Hamiltonian system*. Meanwhile, a k -vector field $\mathbf{X} = (X_1, \dots, X_k) \in \mathfrak{X}^k(M)$ is called a *k -polycosymplectic Hamiltonian k -vector field* associated with h if it satisfies

$$\begin{cases} \iota_{\mathbf{X}} \omega = dh - \sum_{\alpha=1}^k (R_\alpha h) \tau^\alpha, \\ \iota_{X_\beta} \tau = e_\beta, \quad \beta = 1, \dots, k. \end{cases} \quad (1.2.4)$$

The space of k -polycosymplectic Hamiltonian k -vector fields associated with (M, τ, ω) is denoted by $\mathfrak{X}_{\text{Ham}}^k(M, \tau, \omega)$.

Unlike the Hamiltonian vector field associated with an ω -Hamiltonian function in the k -polysymplectic setting, a scalar Hamiltonian h does not, in general, determine a unique k -polycosymplectic Hamiltonian k -vector field. This non-uniqueness becomes transparent in local coordinates.

Suppose that, in a neighbourhood of a point $x \in M$, there exist k -cosymplectic Darboux coordinates $\{t^\alpha, q^i, p_i^\alpha\}$, with $i = 1, \dots, n$ and $\alpha = 1, \dots, k$. In these coordinates, a k -vector field $\mathbf{X} = (X_1, \dots, X_k)$ can be written as

$$X_\alpha = (X_\alpha)^\beta \frac{\partial}{\partial t^\beta} + (X_\alpha)^i \frac{\partial}{\partial q^i} + (X_\alpha)_i^\beta \frac{\partial}{\partial p_i^\beta}.$$

If $\mathbf{X}$ satisfies (1.2.4), then

$$(X_\alpha)^\beta = \delta_\alpha^\beta, \quad (X_\alpha)^i = \frac{\partial h}{\partial p_i^\alpha}, \quad \sum_{\alpha=1}^k (X_\alpha)_i^\alpha = -\frac{\partial h}{\partial q^i}. \quad (1.2.5)$$

Thus, the components $(X_\alpha)_i^\beta$ are constrained only through the sum of their diagonal terms, which accounts for the non-uniqueness of the Hamiltonian k -vector field associated with a given h .

Assume now that $\mathbf{X}$ is integrable and let $\psi: U \subset \mathbb{R}^k \rightarrow M$ be one of its integral sections. Locally,

$$\psi(s) = (t^\alpha(s), q^i(s), p_i^\alpha(s)), \quad s = (s^1, \dots, s^k) \in U.$$

The condition $T\psi(\partial/\partial s^\alpha) = X_\alpha \circ \psi$ together with (1.2.5) yields

$$\frac{\partial t^\beta}{\partial s^\alpha} = \delta_\alpha^\beta, \quad \frac{\partial q^i}{\partial s^\alpha} = \frac{\partial h}{\partial p_i^\alpha} \Big|_{\psi(s)}, \quad \sum_{\alpha=1}^k \frac{\partial p_i^\alpha}{\partial s^\alpha} = - \frac{\partial h}{\partial q^i} \Big|_{\psi(s)}. \quad (1.2.6)$$

These equations are the k -polysymplectic Hamilton–De Donder–Weyl equations associated with the Hamiltonian function h .

Example 1.2.17 (Vibrating membrane under an external force). Consider a vibrating membrane whose equilibrium configuration is described by Cartesian coordinates $\{x, y\}$ and which is subjected to a time-dependent external force $f(t, x, y)$. Let $M = \mathbb{R}^3 \times \bigoplus^3 T^*\mathbb{R}$ be the corresponding phase space, endowed with global coordinates $\{t, x, y, \zeta, p^t, p^x, p^y\}$. Here, ζ represents the transverse displacement of the membrane from its equilibrium configuration, while p^t, p^x, p^y denote the corresponding momentum variables.

Consider the Hamiltonian function $h \in C^\infty(M)$ given by

$$h(t, x, y, \zeta, p^t, p^x, p^y) = \frac{1}{2}(p^t)^2 - \frac{1}{2c^2}(p^x)^2 - \frac{1}{2c^2}(p^y)^2 - \zeta f(t, x, y),$$

where $c > 0$ is a constant determined by the physical properties of the membrane.

Let $\psi: \mathbb{R}^3 \rightarrow M$ be a section locally written as

$$\psi(t, x, y) = (t, x, y, \zeta(t, x, y), p^t(t, x, y), p^x(t, x, y), p^y(t, x, y)).$$

Then, the k -polysymplectic Hamilton–De Donder–Weyl equations (1.2.6) take the form

$$\begin{aligned} \frac{\partial p^t}{\partial t} + \frac{\partial p^x}{\partial x} + \frac{\partial p^y}{\partial y} &= f(t, x, y), \\ \frac{\partial \zeta}{\partial t} &= p^t, \quad \frac{\partial \zeta}{\partial x} = -\frac{1}{c^2}p^x, \quad \frac{\partial \zeta}{\partial y} = -\frac{1}{c^2}p^y. \end{aligned}$$

Substituting the momentum variables from the second set of equations into the first one gives

$$\frac{\partial^2 \zeta}{\partial t^2} - c^2 \left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \zeta}{\partial y^2} \right) = f(t, x, y),$$

which is the equation governing a forced vibrating membrane.

As is customary in the literature [110], we use the same symbols $\{t, x, y\}$ for the coordinates on the base and for the corresponding coordinates on the domain of the section ψ . This convention will be used throughout this dissertation.

The same system can also be expressed in polar coordinates $\{r, \theta\}$. In this case, consider the Hamiltonian

$$\tilde{h}(t, r, \theta, \zeta, p^t, p^r, p^\theta) = \frac{1}{2r} \left((p^t)^2 - \frac{1}{c^2}(p^r)^2 - \frac{r^2}{c^2}(p^\theta)^2 \right) - r \zeta f(t, r, \theta).$$

For a section $\psi: \mathbb{R}^3 \rightarrow M$ of the form

$$\psi(t, r, \theta) = (t, r, \theta, \zeta(t, r, \theta), p^t(t, r, \theta), p^r(t, r, \theta), p^\theta(t, r, \theta)),$$

the corresponding Hamilton–De Donder–Weyl equations are

$$\begin{aligned} \frac{\partial p^t}{\partial t} + \frac{\partial p^r}{\partial r} + \frac{\partial p^\theta}{\partial \theta} &= r f(t, r, \theta), \\ \frac{\partial \zeta}{\partial t} &= \frac{1}{r}p^t, \quad \frac{\partial \zeta}{\partial r} = -\frac{1}{rc^2}p^r, \quad \frac{\partial \zeta}{\partial \theta} = -\frac{r}{c^2}p^\theta. \end{aligned}$$

Eliminating the momentum variables yields

$$\frac{\partial^2 \zeta}{\partial t^2} - c^2 \left(\frac{\partial^2 \zeta}{\partial r^2} + \frac{1}{r} \frac{\partial \zeta}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \zeta}{\partial \theta^2} \right) = f(t, r, \theta),$$

which is the forced membrane equation expressed in polar coordinates. $\triangle$

1.2.7 k -Contact geometry

Let us present the basic notions of k -contact geometry [29, 30, 84]. Note that k -contact geometry is a generalisation of contact geometry developed to deal with dissipative first-order field theories [110]. The main idea was to generalise the notion of a contact form as follows.

Definition 1.2.18. A k -contact form on an open subset $U \subset M$ is a differential one-form on U taking values in $\mathbb{R}^k$, that is $\boldsymbol{\eta} \in \Omega^1(U, \mathbb{R}^k)$, such that $\ker \boldsymbol{\eta} \subset TU$ is a regular non-zero distribution of corank k and $\ker \boldsymbol{\eta} \oplus \ker d\boldsymbol{\eta} = TU$.

If $\boldsymbol{\eta}$ is a k -contact form defined on M , the pair $(M, \boldsymbol{\eta})$ is called a *co-oriented k -contact manifold*, and $\ker d\boldsymbol{\eta}$ is called the *Reeb distribution* of $(M, \boldsymbol{\eta})$. Moreover, if $\dim M = n + nk + k$ for some $n, k \in \mathbb{N}$ and there exists an integrable distribution $\mathcal{V} \subset \ker \boldsymbol{\eta}$ with $\text{rk } \mathcal{V} = nk$, then $(M, \boldsymbol{\eta}, \mathcal{V})$ is called a *polarised co-oriented k -contact manifold*. The distribution $\mathcal{V}$ is referred to as a *polarisation* of $(M, \boldsymbol{\eta})$.

The previous formalism recovers contact forms for $k = 1$. Nevertheless, Definition 1.2.18 requires $\ker \boldsymbol{\eta} \neq 0$ to avoid studying the case $k = 1$ on a one-dimensional manifold. In this instance, $\ker \boldsymbol{\eta} = 0$ is sometimes said to be maximally non-integrable [37], although it is an integrable distribution. Moreover, accepting $\ker \boldsymbol{\eta} = 0$ involves dealing with several technical nuances in k -contact geometry, which is undesirable (cf. [84]). Despite this, one should note that $\ker \boldsymbol{\eta} \neq 0$ is a mere technical condition and excludes an instance of no relevance in contact or k -contact geometry.

Theorem 1.2.19. Let $(M, \boldsymbol{\eta} = \sum_{\alpha=1}^k \eta^\alpha \otimes e_\alpha)$ be a co-oriented k -contact manifold. There exists a unique family of vector fields $R_1, \dots, R_k \in \mathfrak{X}(M)$ such that

$$\iota_{R_\alpha} \eta^\beta = \delta_\alpha^\beta, \quad \iota_{R_\alpha} d\eta^\beta = 0, \quad \alpha, \beta = 1, \dots, k.$$

The vector fields $R_1, \dots, R_k$ commute with each other, that is, $[R_\alpha, R_\beta] = 0$ for $\alpha, \beta = 1, \dots, k$. Moreover, these vector fields form a basis of the Reeb distribution, namely $\ker d\boldsymbol{\eta} = \langle R_1, \dots, R_k \rangle$.

Since the Reeb distribution $\ker d\boldsymbol{\eta}$ is the intersection of the kernels of the closed two-forms $d\eta^1, \dots, d\eta^k$ and has constant rank k , it is involutive and hence integrable by Frobenius' theorem.

Definition 1.2.20. Given a k -contact manifold $(M, \boldsymbol{\eta})$, the *Reeb k -vector field* of $(M, \boldsymbol{\eta})$ is the integrable k -vector field $\mathbf{R} = (R_1, \dots, R_k)$ on M whose components are described in Theorem 1.2.19. The vector fields $R_1, \dots, R_k$ are called the *Reeb vector fields* of the co-oriented k -contact manifold $(M, \boldsymbol{\eta})$.

The following example will be used repeatedly throughout this thesis, serving as the principal model of k -contact geometry considered hereafter. It also admits a natural interpretation in terms of first-order jet bundles and arises in numerous applications to classical field theory [84].

Example 1.2.21. The manifold

$$\mathcal{J}_{Q,k} = \bigoplus_{\alpha=1}^k T^*Q \times \mathbb{R}^k$$

carries a canonical k -contact manifold structure defined by the k -contact form

$$\boldsymbol{\eta}_{Q,k} = \sum_{\alpha=1}^k (dz^\alpha - \theta^\alpha) \otimes e_\alpha.$$

Here, $\{z^1, \dots, z^k\}$ are the pull-backs of linear coordinates on $\mathbb{R}^k$ to $\mathcal{J}_{Q,k}$ via its canonical projection onto $\mathbb{R}^k$. These coordinates are hereafter called *dissipative variables* due to their relation to dissipation in field theories [110]. Note that $\mathcal{J}_{Q,k}$ also admits a natural projection onto $\bigoplus^k T^*Q$. Moreover, each θ^α is the pull-back to $\mathcal{J}_{Q,k}$ of the Liouville one-form θ on the α -th copy of the cotangent bundle T^*Q via the natural projection $\mathcal{J}_{Q,k} \rightarrow \bigoplus^k T^*Q \xrightarrow{\text{Pr}^\alpha} T^*Q$. Additionally, $\mathcal{J}_{Q,k}$ admits a vertical distribution $\mathcal{V}_{Q,k}$ relative to the natural projection onto $Q \times \mathbb{R}^k$ of rank $k \cdot \dim Q$ contained in $\ker \eta_{Q,k}$. Therefore, $(\mathcal{J}_{Q,k}, \eta_{Q,k}, \mathcal{V}_{Q,k})$ is a polarised co-oriented k -contact manifold. From now on, we assume Q to be n -dimensional.

Any choice of local coordinates $\{q^1, \dots, q^n\}$ on Q , together with linear coordinates on $\mathbb{R}^k$, induces a natural coordinate system $\{q^i, p_i^\alpha, z^\alpha\}$ on $\mathcal{J}_{Q,k}$, with $\alpha = 1, \dots, k$. In these coordinates,

$$\eta_{Q,k} = \sum_{\alpha=1}^k \left(dz^\alpha - \sum_{i=1}^n p_i^\alpha dq^i \right) \otimes e_\alpha, \quad d\eta_{Q,k} = \sum_{\alpha=1}^k \left(\sum_{i=1}^n dq^i \wedge dp_i^\alpha \right) \otimes e_\alpha, \quad R_\beta = \frac{\partial}{\partial z^\beta},$$

for $\beta = 1, \dots, k$, and

$$\ker \eta_{Q,k} = \left\langle \frac{\partial}{\partial p_i^\alpha}, \frac{\partial}{\partial q^i} + \sum_{\beta=1}^k p_i^\beta \frac{\partial}{\partial z^\beta} \right\rangle_{\substack{\alpha=1, \dots, k \\ i=1, \dots, n}}, \quad \ker d\eta_{Q,k} = \langle R_1, \dots, R_k \rangle, \quad \mathcal{V}_{Q,k} = \left\langle \frac{\partial}{\partial p_i^\alpha} \right\rangle_{\substack{i=1, \dots, n \\ \alpha=1, \dots, k}}.$$

△

Example 1.2.22. Let us study the k -contactification of an exact k -symplectic manifold. More specifically, consider an exact k -symplectic manifold $(P, \omega = \sum_{\alpha=1}^k \omega^\alpha \otimes e_\alpha)$ with $\omega = -d\theta$, and the product manifold $M = P \times \mathbb{R}^k$. Let $\{z^1, \dots, z^k\}$ denote the pull-backs to M of linear coordinates on $\mathbb{R}^k$, and let θ_M^α be the pull-back of θ^α from P to M in the canonical manner.

Define the $\mathbb{R}^k$ -valued one-form

$$\eta_\theta = \sum_{\alpha=1}^k (dz^\alpha - \theta_M^\alpha) \otimes e_\alpha \in \Omega^1(M, \mathbb{R}^k).$$

Then, (M, η_θ) becomes a co-oriented k -contact manifold. Indeed, $\ker \eta_\theta$ has corank k and is non-zero, while $d\eta_\theta = -d\theta_M$ for $\theta_M = \sum_{\alpha=1}^k \theta_M^\alpha \otimes e_\alpha$. Then,

$$\ker d\eta_\theta = \left\langle \frac{\partial}{\partial z^1}, \dots, \frac{\partial}{\partial z^k} \right\rangle$$

has rank k , since (P, ω) is k -symplectic. Hence, η_θ is a k -contact form on M .

The canonical k -contact form $\eta_{Q,k}$ described in Example 1.2.21 is precisely the k -contactification of the exact k -symplectic manifold $(P = \bigoplus^k T^*Q, \omega_{Q,k} = \sum_{\alpha=1}^k \omega_Q^\alpha \otimes e_\alpha)$, where ω_Q^α denotes the pull-back to P of the canonical symplectic form from the α -th copy of T^*Q via the natural projection from $\bigoplus^k T^*Q$ onto its α -th copy of T^*Q . Note that one can construct a canonical one-form $\theta_Q = \sum_{\alpha=1}^k \theta_Q^\alpha \otimes e_\alpha \in \Omega^1(P, \mathbb{R}^k)$ from the pull-back θ_Q^α to P of the canonical Liouville form θ_Q on the α -th copy of T^*Q in P . Moreover, $\omega_{Q,k} = -d\theta_Q$.

△

Note that, for $k = 1$, the construction presented in Example 1.2.22 reduces to the usual contactification of an exact symplectic manifold. Indeed, let $(P, \omega = -d\theta)$ be exact and consider $M = P \times \mathbb{R}$ endowed with differential one-form $\eta = dz - \theta_M$, where θ_M denotes the pull-back of θ to M . Then

$$d\eta = -d\theta_M, \quad \ker d\eta = \left\langle \frac{\partial}{\partial z} \right\rangle,$$

since $-d\theta_M$ is the pull-back of a symplectic form ω to M . Thus, η is a contact form on M , and (M, η) is the usual (co-oriented) contactification of (P, ω) .

Theorem 1.2.23 (k -contact Darboux Theorem [110]). *Let $(M, \boldsymbol{\eta}, \mathcal{V})$ be a polarised co-oriented k -contact manifold. Around every point of M , there exist local coordinates $\{q^i, p_i^\alpha, z^\alpha\}$, with $1 \leq i \leq n$ and $1 \leq \alpha \leq k$, such that*

$$\boldsymbol{\eta} = \sum_{\alpha=1}^k \left(dz^\alpha - \sum_{i=1}^n p_i^\alpha dq^i \right) \otimes e_\alpha, \quad \ker d\boldsymbol{\eta} = \left\langle R_1 = \frac{\partial}{\partial z^1}, \dots, R_k = \frac{\partial}{\partial z^k} \right\rangle, \quad \mathcal{V} = \left\langle \frac{\partial}{\partial p_i^\alpha} \right\rangle_{\substack{i=1, \dots, n \\ \alpha=1, \dots, k}}.$$

These coordinates are called Darboux coordinates of $(M, \boldsymbol{\eta}, \mathcal{V})$.

Note that we can formulate a local converse part of the above theorem. If a co-oriented k -contact manifold $(M, \boldsymbol{\eta})$ admits on an open neighbourhood U of $p \in M$ coordinates $\{q^i, p_i^\alpha, z^\alpha\}$, with $1 \leq i \leq n$ and $1 \leq \alpha \leq k$, so that $\boldsymbol{\eta} = \sum_{\alpha=1}^k (dz^\alpha - \sum_{i=1}^n p_i^\alpha dq^i) \otimes e_\alpha$, then $\boldsymbol{\eta}|_U$ gives rise to a polarised co-oriented k -contact manifold $(U, \boldsymbol{\eta}|_U, \mathcal{V} = \langle \partial/\partial p_i^\alpha \rangle)$.

By Theorem 1.2.23, the manifold $\mathcal{J}_{Q,k}$ introduced in Example 1.2.21 provides the canonical local model for polarised co-oriented k -contact manifolds (see [84] for details). Moreover, the k -contactification of $P = \bigoplus^k T^*Q$ endowed with its canonical exact k -symplectic structure, as described in Example 1.2.22, realises precisely this canonical model.

We now introduce a vector bundle morphism that will be very useful in Chapter 3. Given a co-oriented k -contact manifold $(M, \boldsymbol{\eta})$, we can define a vector bundle morphism over M of the form $\chi: \bigoplus^k TM \rightarrow T^*M \times \mathbb{R}$ by

$$\chi(\mathbf{Z}) = (\iota_{\mathbf{Z}} d\boldsymbol{\eta}, \iota_{\mathbf{Z}} \boldsymbol{\eta}).$$

We say that a k -vector field $\mathbf{Y}$ on M is called an $\boldsymbol{\eta}$ -gauge k -vector field if it takes values in $\ker \chi$.

Note that the vector bundle morphism χ is injective if and only if $k = 1$. In this case, a co-oriented one-contact manifold $(M, \boldsymbol{\eta})$ is a contact manifold. The morphism χ will play a fundamental role in the study of k -contact Hamiltonian systems. In particular, the k -contact Hamilton–De Donder–Weyl equations for k -vector fields admit solutions for every Hamiltonian function, but these solutions are not unique when $k > 1$ (see [44]).

Proposition 1.2.24. *Let $\mathcal{J}_{Q,k} = \bigoplus^k T^*Q \times \mathbb{R}^k$ be endowed with Darboux coordinates $\{q^i, p_i^\alpha, z^\alpha\}$, and let $x \in \mathcal{J}_{Q,k}$. Consider $\mathbf{Z} = (Z_1, \dots, Z_k) \in \bigoplus^k T_x \mathcal{J}_{Q,k}$, with $Z_\alpha = \sum_{i=1}^n Z_\alpha^i \frac{\partial}{\partial q^i} + \sum_{i=1}^n \sum_{\beta=1}^k (Z_\alpha)_i^\beta \frac{\partial}{\partial p_i^\beta} + \sum_{\beta=1}^k Z_\alpha^\beta \frac{\partial}{\partial z^\beta}$, for $\alpha = 1, \dots, k$. Then*

$$\ker \chi_x = \left\{ \mathbf{Z} = (Z_1, \dots, Z_k) \in \bigoplus^k T_x \mathcal{J}_{Q,k} \left| \begin{array}{l} Z_\beta^i = 0, \sum_{\alpha=1}^k (Z_\alpha)_i^\alpha = 0, \sum_{\alpha=1}^k Z_\alpha^\alpha = 0, \\ i = 1, \dots, n, \beta = 1, \dots, k \end{array} \right. \right\}.$$

In particular, $\ker \chi_x \subset \bigoplus^k V_x(\pi_Q)$, and $\dim \ker \chi_x = (n+1)(k^2 - 1)$, where $\pi_Q: \mathcal{J}_{Q,k} \rightarrow Q$ is the canonical projection and $V(\pi_Q) = \ker T\pi_Q$ is its vertical bundle.

Proof. For the canonical k -contact form $\boldsymbol{\eta}_{Q,k}$, the two components of the morphism χ are

$$\iota_{\mathbf{Z}} d\boldsymbol{\eta}_{Q,k} = \sum_{i=1}^n \sum_{\alpha=1}^k (Z_\alpha^i dp_i^\alpha - (Z_\alpha)_i^\alpha dq^i), \quad \iota_{\mathbf{Z}} \boldsymbol{\eta}_{Q,k} = \sum_{\alpha=1}^k \left(Z_\alpha^\alpha - \sum_{i=1}^n p_i^\alpha Z_\alpha^i \right).$$

Hence, $\mathbf{Z} \in \ker \chi_x$ if and only if both expressions vanish. By the linear independence of the one-forms dq^i and dp_i^α , the first condition is equivalent to $Z_\beta^i = 0$, and $\sum_{\alpha=1}^k (Z_\alpha)_i^\alpha = 0$, for all $i = 1, \dots, n$ and $\beta = 1, \dots, k$. Once $Z_\alpha^i = 0$, the second condition reduces to $\sum_{\alpha=1}^k Z_\alpha^\alpha = 0$. Therefore, $\mathbf{Z} \in \ker \chi_x$ if and only if $Z_\beta^i = 0$, $\sum_{\alpha=1}^k (Z_\alpha)_i^\alpha = 0$, and $\sum_{\alpha=1}^k Z_\alpha^\alpha = 0$.

The conditions $Z_\alpha^i = 0$ are precisely the conditions for $T_x \pi_Q(Z_\alpha) = 0$. Thus, each Z_α belongs to $V_x(\pi_Q)$, and consequently $\ker \chi_x \subset \bigoplus^k V_x(\pi_Q)$. Within $\bigoplus^k V_x(\pi_Q)$, the coefficients $(Z_\alpha)_i^\beta$ and Z_α^β provide $nk^2 + k^2 = (n+1)k^2$ independent parameters. The n conditions $\sum_{\alpha=1}^k (Z_\alpha)_i^\alpha = 0$, for $i = 1, \dots, n$, together with $\sum_{\alpha=1}^k Z_\alpha^\alpha = 0$, give $n+1$ independent linear constraints. Hence, $\dim \ker \chi_x = (n+1)k^2 - (n+1) = (n+1)(k^2 - 1)$. $\square$

1.2.8 k -Contact Hamiltonian systems

This section reviews the Hamiltonian formulation of classical field theories with dissipation in the co-oriented k -contact framework. Further details can be found in [84, 85, 110].

Definition 1.2.25. A k -contact Hamiltonian system is a triple $(M, \boldsymbol{\eta}, h)$, where $(M, \boldsymbol{\eta})$ is a co-oriented k -contact manifold and $h \in C^\infty(M)$ is called the *Hamiltonian function*. Let $\psi : U \subset \mathbb{R}^k \rightarrow M$, where U is an open subset of $\mathbb{R}^k$. The associated k -contact Hamilton–De Donder–Weyl equations are

$$\sum_{\alpha=1}^k \iota_{\psi'_\alpha} d\eta^\alpha = \left(dh - \sum_{\alpha=1}^k (\mathcal{L}_{R_\alpha} h) \eta^\alpha \right) \circ \psi, \quad \sum_{\alpha=1}^k \iota_{\psi'_\alpha} \eta^\alpha = -h \circ \psi, \quad (1.2.7)$$

where $\psi'(t) = (\psi'_1(t), \dots, \psi'_k(t)) \in \bigoplus^k TM$ denotes the first prolongation of ψ (see Definition 1.2.2).

The corresponding Hamilton–De Donder–Weyl equations for a k -vector field $\mathbf{X} = (X_1, \dots, X_k) \in \mathfrak{X}^k(M)$ are

$$\iota_{\mathbf{X}} d\boldsymbol{\eta} = \sum_{\alpha=1}^k \iota_{X_\alpha} d\eta^\alpha = dh - \sum_{\alpha=1}^k (\mathcal{L}_{R_\alpha} h) \eta^\alpha, \quad \iota_{\mathbf{X}} \boldsymbol{\eta} = \sum_{\alpha=1}^k \iota_{X_\alpha} \eta^\alpha = -h. \quad (1.2.8)$$

A k -vector field $\mathbf{X}$ satisfying (1.2.8) is called a k -contact Hamiltonian k -vector field associated with $(M, \boldsymbol{\eta}, h)$, or simply $\boldsymbol{\eta}$ -Hamiltonian k -vector field if h is understood from the context.

In k -contact Darboux coordinates, equations (1.2.7) become

$$\begin{cases} \frac{\partial q^i}{\partial t^\beta} = \frac{\partial h}{\partial p_i^\beta} \circ \psi, \\ \sum_{\alpha=1}^k \frac{\partial p_i^\alpha}{\partial t^\alpha} = - \left(\frac{\partial h}{\partial q^i} + \sum_{\alpha=1}^k p_i^\alpha \frac{\partial h}{\partial z^\alpha} \right) \circ \psi, & i = 1, \dots, n, \quad \beta = 1, \dots, k. \\ \sum_{\alpha=1}^k \frac{\partial z^\alpha}{\partial t^\alpha} = \left(\sum_{\alpha=1}^k \sum_{i=1}^n p_i^\alpha \frac{\partial h}{\partial p_i^\alpha} - h \right) \circ \psi, \end{cases}$$

Likewise, if

$$X_\alpha = \sum_{i=1}^n (X_\alpha)^i \frac{\partial}{\partial q^i} + \sum_{\beta=1}^k \sum_{i=1}^n (X_\alpha)_i^\beta \frac{\partial}{\partial p_i^\beta} + \sum_{\beta=1}^k (X_\alpha)^\beta \frac{\partial}{\partial z^\beta},$$

then equations (1.2.8) are equivalent to

$$\begin{cases} (X_\beta)^i = \frac{\partial h}{\partial p_i^\beta}, \\ \sum_{\alpha=1}^k (X_\alpha)_i^\alpha = - \left(\frac{\partial h}{\partial q^i} + \sum_{\alpha=1}^k p_i^\alpha \frac{\partial h}{\partial z^\alpha} \right), & i = 1, \dots, n, \quad \beta = 1, \dots, k. \\ \sum_{\alpha=1}^k (X_\alpha)^\alpha = \sum_{\alpha=1}^k \sum_{i=1}^n p_i^\alpha \frac{\partial h}{\partial p_i^\alpha} - h, \end{cases} \quad (1.2.9)$$

The following result establishes the relationship between the Hamilton–De Donder–Weyl equations for sections and for k -vector fields.

Proposition 1.2.26. Let $\mathbf{X} \in \mathfrak{X}^k(M)$ be an integrable k -vector field. Then, $\mathbf{X}$ satisfies (1.2.8) if and only if every integral section of $\mathbf{X}$ satisfies (1.2.7).

It is important to emphasise that a k -vector field satisfying (1.2.8) need not be integrable and therefore need not admit integral sections through every point. Furthermore, equations (1.2.9) show that a Hamiltonian k -vector field is, in general, not uniquely determined, since they prescribe only certain combinations of components. Equivalently, if $\mathbf{X}$ is a k -contact Hamiltonian k -vector field associated with

$(M, \boldsymbol{\eta}, h)$, then $\mathbf{X} + \mathbf{Y}$ is also a k -contact Hamiltonian k -vector field for every $\boldsymbol{\eta}$ -gauge k -vector field $\mathbf{Y}$, that is, for every $\mathbf{Y} \in \ker \chi$.

Finally, we present an equivalent characterisation of the k -contact Hamilton–De Donder–Weyl equations, which follows directly from applying Cartan’s formula to equations (1.2.8).

Corollary 1.2.27. *The k -contact Hamilton–De Donder–Weyl equations (1.2.8) are equivalent to the following conditions*

$$\begin{cases} \mathcal{L}_{\mathbf{X}}\boldsymbol{\eta} = \sum_{\alpha=1}^k \mathcal{L}_{X_\alpha}\eta^\alpha = \iota_{\mathbf{X}}d\boldsymbol{\eta} + d(\iota_{\mathbf{X}}\boldsymbol{\eta}) = -\sum_{\alpha=1}^k (\mathcal{L}_{R_\alpha}h)\eta^\alpha, \\ \iota_{\mathbf{X}}\boldsymbol{\eta} = \sum_{\alpha=1}^k \iota_{X_\alpha}\eta^\alpha = -h, \end{cases}$$

where $\mathbf{X} = (X_1, \dots, X_k) \in \mathfrak{X}^k(M)$.

Definition 1.2.28. The k -contact orthogonal of a vector subspace $E_x \subset \ker \boldsymbol{\eta}_x$, for $x \in M$ and a k -contact manifold $(M, \boldsymbol{\eta})$, is the vector subspace

$$E_x^{\perp \boldsymbol{\eta}} := \{v_x \in \ker \boldsymbol{\eta}_x \mid d\boldsymbol{\eta}(v_x, w_x) = 0, \forall w_x \in E_x\}.$$

Then, E_x is *isotropic* if $E_x \subset E_x^{\perp \boldsymbol{\eta}}$. Meanwhile, E_x is *Legendrian* if E_x is isotropic and it admits an isotropic complement F_x such that $F_x \oplus E_x = \ker \boldsymbol{\eta}_x$. A distribution $E \subset TM$ is *isotropic* (resp. *Legendrian*) if every E_x , with $x \in M$, is isotropic (resp. Legendrian). A submanifold of M is called isotropic (resp. Legendrian) if its tangent space at each point is isotropic (resp. Legendrian). A subspace $W_x \subset T_x M$ is called *coisotropic* if $(W_x \cap \ker \boldsymbol{\eta}_x)^{\perp \boldsymbol{\eta}} \subset W_x \cap \ker \boldsymbol{\eta}_x$ and *maximally coisotropic* if $(W_x \cap \ker \boldsymbol{\eta}_x)^{\perp \boldsymbol{\eta}} = W_x \cap \ker \boldsymbol{\eta}_x$.

The following lemma is necessary so as to stress why our results retrieve natural Hamilton–Jacobi equations for contact Hamiltonian systems when $k = 1$.

Lemma 1.2.29. *Let (M, η) be a contact manifold with $\dim M = 2n + 1$ and let $N \subset M$ be a submanifold of dimension $n + 1$ such that $\eta|_{T_N}$ is nowhere vanishing. Then, N is coisotropic if and only if it is maximally coisotropic.*

Proof. Fix $x \in N$ and set $W_x := T_x N \cap \ker \eta_x$. Since $\eta_x|_{T_x N}$ is a nonzero linear form on the $(n + 1)$ -dimensional space $T_x N$, one has that $T_x N + \ker \eta_x$ has dimension $2n + 1$ and $\dim W_x = n$. On the other hand, $(\ker \eta_x, d\eta_x|_{\ker \eta_x})$ is a symplectic linear space of dimension $2n$, so $\dim W_x^{\perp \eta} = 2n - \dim W_x = n$. If N is coisotropic, then $W_x^{\perp \eta} \subset W_x$, and since both spaces have dimension n , they must be equal. The converse is immediate. $\square$

Example 1.2.30 (The damped wave equation). The dynamics of a vibrating string can be described within the k -contact Hamiltonian framework [29]. Let $\{t, x\}$ denote the standard coordinates on $\mathbb{R}^2$, and take the configuration space to be $Q = \mathbb{R}$. The corresponding phase space is $\bigoplus^2 T^*\mathbb{R} \times \mathbb{R}^2$, with coordinates $\{u, p^t, p^x, s^t, s^x\}$. Here, u represents the displacement of the string from its equilibrium position, while p^t and p^x are the momenta conjugate to u with respect to the independent variables t and x , respectively. As in Example 1.2.21, the canonical two-contact structure on $\bigoplus^2 T^*\mathbb{R} \times \mathbb{R}^2$ is given by

$$\eta^t = ds^t - p^t du, \quad \eta^x = ds^x - p^x du.$$

A Hamiltonian function $h \in C^\infty\left(\bigoplus^2 T^*\mathbb{R} \times \mathbb{R}^2\right)$ describing a damped vibrating string is

$$h(u, p^t, p^x, s^t, s^x) = \frac{1}{2\rho}(p^t)^2 - \frac{1}{2\tau}(p^x)^2 + ks^t,$$

where ρ denotes the linear mass density, τ the string tension, and $k > 0$ the constant damping coefficient. A corresponding two-contact Hamiltonian two-vector field $\mathbf{X} = (X^1, X^2)$ is

$$\begin{aligned} X^1 &= \frac{p^t}{\rho} \frac{\partial}{\partial u} + A_t^1 \frac{\partial}{\partial p^t} + A_x^1 \frac{\partial}{\partial p^x} + B_t^1 \frac{\partial}{\partial s^t} + B_x^1 \frac{\partial}{\partial s^x}, \\ X^2 &= -\frac{p^x}{\tau} \frac{\partial}{\partial u} + A_t^2 \frac{\partial}{\partial p^t} - (kp^t + A_t^1) \frac{\partial}{\partial p^x} + B_t^2 \frac{\partial}{\partial s^t} + \left(\frac{(p^t)^2}{2\rho} - \frac{(p^x)^2}{2\tau} - ks^t - B_t^1 \right) \frac{\partial}{\partial s^x}, \end{aligned}$$

where $A_t^1, A_x^1, B_t^1, B_x^1, A_t^2$, and B_t^2 are arbitrary smooth functions on M . For example, choosing

$$B_x^1 = \frac{\tau k (p^t)^2 u}{\rho p^x}, \quad B_t^2 = B_t^2(s^t, p^x),$$

and setting

$$A_t^2 = B_t^1 = A_x^1 = 0, \quad A_t^1 = -kp^t,$$

yields an integrable k -vector field $\mathbf{X}$. The associated Hamilton–De Donder–Weyl equations are therefore

$$\begin{cases} \frac{\partial u}{\partial t} = \frac{1}{\rho} p^t, \\ \frac{\partial u}{\partial x} = -\frac{1}{\tau} p^x, \\ \frac{\partial p^t}{\partial t} + \frac{\partial p^x}{\partial x} = -kp^t, \\ \frac{\partial s^t}{\partial t} + \frac{\partial s^x}{\partial x} = \frac{1}{2\rho} (p^t)^2 - \frac{1}{2\tau} (p^x)^2 - ks^t. \end{cases}$$

Substituting the first two equations into the third gives the damped wave equation

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} + k \frac{\partial u}{\partial t} = 0,$$

where $c^2 = \frac{\tau}{\rho}$. $\triangle$

Example 1.2.31. Let us illustrate our previous theory by studying a damped double sine–Gordon equation. Consider coordinates $\{t, x\}$ on $\mathbb{R}^2$ and let $Q = \mathbb{R}$. The phase space is $\mathcal{J}_{\mathbb{R},2} = (\mathbb{T}^*\mathbb{R} \oplus \mathbb{T}^*\mathbb{R}) \times \mathbb{R}^2$ with coordinates $\{u, p^t, p^x, z^t, z^x\}$. The variable u represents a scalar field, while p^t and p^x denote the conjugate momenta associated with the independent variables t and x , respectively. The double sine–Gordon equation appears in several physical contexts, for instance in long Josephson junctions and related non-integrable field-theoretic models [50, 103, 124]. Let us consider the Hamiltonian function $h_{\text{DSG}} \in C^\infty(\mathcal{J}_{\mathbb{R},2})$ given by

$$h_{\text{DSG}}(u, p^t, p^x, z^t, z^x) = \frac{1}{2} \left((p^t)^2 - \frac{1}{c^2} (p^x)^2 \right) + a(1 - \cos u) + b(1 - \cos 2u) + \frac{1}{2} (z^t)^2, \quad (1.2.10)$$

where $c > 0$ and $a, b \in \mathbb{R}$ are constants.

The Hamilton–De Donder–Weyl equations for the associated $\eta_{\mathbb{R},2}$ -Hamiltonian two-vector field read

$$\partial_t u = p^t, \quad \partial_x u = -\frac{1}{c^2} p^x, \quad \partial_t p^t + \partial_x p^x = -(a \sin u + 2b \sin 2u + z^t p^t),$$

together with

$$\partial_t z^t + \partial_x z^x = \frac{1}{2} \left((p^t)^2 - \frac{1}{c^2} (p^x)^2 \right) - a(1 - \cos u) - b(1 - \cos 2u) - \frac{1}{2} (z^t)^2.$$

Using the first two equations in the third one, we obtain

$$u_{tt} - c^2 u_{xx} + a \sin u + 2b \sin 2u + z^t u_t = 0. \quad (1.2.11)$$

Hence, h_{DSG} yields a damped double sine–Gordon equation with effective dissipation coefficient z^t . One may regard $z^t = \lambda(t, x)$ as a prescribed space-time dependent attenuation coefficient, while z^x is chosen so that the balance equation for the dissipative variables is satisfied. In this way, (1.2.11) becomes

$$u_{tt} - c^2 u_{xx} + a \sin u + 2b \sin 2u + \lambda(t, x) u_t = 0.$$

Therefore, the quadratic term $\frac{1}{2}(z^t)^2$ in the Hamiltonian allows one to encode variable dissipation directly within the two-contact formalism in a non-autonomous manner. $\triangle$

1.2.9 Evolution k -contact k -vector fields

We now generalise contact evolution vector fields [62] to the k -contact setting. This construction will be used in the development of the k -contact Hamilton–Jacobi theory in Chapter 3.

On a co-oriented one-contact manifold (M, η) , an evolution vector field is a vector field E on M that satisfies the equations

$$\iota_E d\eta = dh - (\mathcal{L}_R h)\eta, \quad \iota_E \eta = 0,$$

for a certain $h \in C^\infty(M)$. In fact, E is uniquely determined from h because $\ker \eta \oplus \ker d\eta = TM$. Meanwhile, for the k -contact setting with $k > 1$, an evolution k -contact k -vector field is defined as follows.

Definition 1.2.32. An *evolution k -vector field* associated with a k -contact Hamiltonian system $(M, \boldsymbol{\eta}, h)$ is a k -vector field $\mathbf{E}$ on M satisfying

$$\iota_{\mathbf{E}} d\boldsymbol{\eta} = \sum_{\alpha=1}^k \iota_{E_\alpha} d\eta^\alpha = dh - \sum_{\alpha=1}^k (\mathcal{L}_{R_\alpha} h) \eta^\alpha, \quad \iota_{\mathbf{E}} \boldsymbol{\eta} = \sum_{\alpha=1}^k \iota_{E_\alpha} \eta^\alpha = 0. \quad (1.2.12)$$

The above equations for a fixed $h \in C^\infty(M)$ are called the *evolution k -contact Hamilton–De Donder–Weyl equations* for k -contact vector fields. If h is understood from the context, we refer to $\mathbf{E}$ as an $\boldsymbol{\eta}$ -evolution k -vector field.

The main difference between the system (1.2.12) and the k -contact HdDW equations for $\boldsymbol{\eta}$ -Hamiltonian k -vector fields is that $\iota_{\mathbf{E}} \boldsymbol{\eta} = 0$ instead of $\iota_{\mathbf{E}} \boldsymbol{\eta} = -h$. The following result is a simple modification of [44, Proposition 3.5] to the evolution k -contact realm.

Proposition 1.2.33. *Evolution k -contact Hamilton–De Donder–Weyl equations (1.2.12) admit solutions. They are not unique if $k > 1$.*

Proof. Consider the mapping $\rho: \mathbf{E} \in \bigoplus^k TM \mapsto \iota_{\mathbf{E}} d\boldsymbol{\eta} \in T^*M$. One has that $\iota_{R_\alpha} \rho(\mathbf{E}) = 0$ for every $R_1, \dots, R_k$. Hence, $\text{Im } \rho \subset (\ker d\boldsymbol{\eta})^\circ$. Moreover, $\text{Im } \rho = (\ker d\boldsymbol{\eta})^\circ$, otherwise, $(\text{Im } \rho)^\circ$ would have dimension bigger than k and $(\text{Im } \rho)^\circ \cap \ker \boldsymbol{\eta} \neq 0$. But $(\text{Im } \rho)^\circ \subset \ker d\boldsymbol{\eta}$ and

$$(\text{Im } \rho)^\circ \cap \ker \boldsymbol{\eta} \subset \ker d\boldsymbol{\eta} \cap \ker \boldsymbol{\eta} = 0,$$

which is a contradiction. Therefore, for every h there exists some k -vector field $\mathbf{E}$ such that $\rho(\mathbf{E}) = dh - \sum_{\alpha=1}^k (\mathcal{L}_{R_\alpha} h) \eta^\alpha$. Then, $(E_1 - (\iota_{\mathbf{E}} \boldsymbol{\eta}) R_1, \dots, E_k)$ is an $\boldsymbol{\eta}$ -evolution k -vector field for h .

If $k > 1$ and $\mathbf{E}$ is a solution to the evolution k -contact HdDW equations, then any functions $f_1, \dots, f_k \in C^\infty(M)$ satisfying $\sum_{\alpha=1}^k f_\alpha = 0$ give another solution $\mathbf{E} + (f_1 R_1, \dots, f_k R_k)$. $\square$

Consider a k -vector field $\mathbf{E} = (E_1, \dots, E_k) \in \mathfrak{X}^k(M)$. Its local expression in *Darboux coordinates* in a co-oriented polarised k -contact manifold $(M, \boldsymbol{\eta}, \mathcal{V})$ reads

$$E_\alpha = \sum_{i=1}^n (E_\alpha)^i \frac{\partial}{\partial q^i} + \sum_{\beta=1}^k \sum_{i=1}^n (E_\alpha)^\beta_i \frac{\partial}{\partial p_i^\beta} + \sum_{\beta=1}^k (E_\alpha)^\beta \frac{\partial}{\partial z^\beta}, \quad \alpha = 1, \dots, k.$$

Moreover, equation (1.2.12) implies that

$$\begin{cases} (E_\beta)^i = \frac{\partial h}{\partial p_i^\beta}, \\ \sum_{\alpha=1}^k (E_\alpha)_i^\alpha = - \left(\frac{\partial h}{\partial q^i} + \sum_{\alpha=1}^k p_i^\alpha \frac{\partial h}{\partial z^\alpha} \right), & i = 1, \dots, n, \quad \beta = 1, \dots, k, \\ \sum_{\alpha=1}^k (E_\alpha)^\alpha = \sum_{j=1}^n \sum_{\alpha=1}^k p_j^\alpha \frac{\partial h}{\partial p_j^\alpha}, \end{cases}$$

Thus, the first two sets of equations coincide with those of the Hamiltonian formulation, whereas the last equation differs by the absence of the term $-h$.

The following observation is immediate from the local expressions above, so we only record the statement.

Lemma 1.2.34. *Let $\mathbf{E} = (E_1, \dots, E_k)$ be an η -evolution k -vector field. Then, $\mathbf{E}' = (E'_1, \dots, E'_k)$ with*

$$E'_\alpha = E_\alpha - \sum_{\beta \neq \alpha} (E_\alpha)^\beta R_\beta, \quad \alpha = 1, \dots, k,$$

satisfies the same k -contact Hamilton–De Donder–Weyl equations as $\mathbf{E}$. In particular, $\mathbf{E}'$ is again an η -evolution k -vector field for the same Hamiltonian function.

We now turn to the corresponding equations for integral sections, which provide the PDE formulation of the evolution dynamics.

Definition 1.2.35. Given a k -contact Hamiltonian system (M, η, h) , the *evolution k -contact Hamilton–De Donder–Weyl (HdDW) equations* are

$$\sum_{\alpha=1}^k \iota_{\psi'_\alpha} d\eta^\alpha = \left(dh - \sum_{\alpha=1}^k (\mathcal{L}_{R_\alpha} h) \eta^\alpha \right) \circ \psi, \quad \sum_{\alpha=1}^k \iota_{\psi'_\alpha} \eta^\alpha = 0, \quad (1.2.13)$$

where solutions are maps $\psi: D \subset \mathbb{R}^k \rightarrow M$, where D is an open subset of $\mathbb{R}^k$.

In *Darboux coordinates* for some $(M, \eta, \mathcal{V})$, one may write $\psi(t) = (q^i(t), p_i^\alpha(t), z^\alpha(t))$ and equations (1.2.13) read

$$\begin{cases} \frac{\partial q^i}{\partial t^\beta} = \frac{\partial h}{\partial p_i^\beta} \circ \psi, \\ \sum_{\alpha=1}^k \frac{\partial p_i^\alpha}{\partial t^\alpha} = - \left(\frac{\partial h}{\partial q^i} + \sum_{\alpha=1}^k p_i^\alpha \frac{\partial h}{\partial z^\alpha} \right) \circ \psi, \\ \sum_{\alpha=1}^k \frac{\partial z^\alpha}{\partial t^\alpha} = \sum_{\alpha=1}^k \sum_{j=1}^n p_j^\alpha \frac{\partial h}{\partial p_j^\alpha} \circ \psi, \end{cases} \quad (1.2.14)$$

for $i = 1, \dots, n$ and $\beta = 1, \dots, k$.

Lemma 1.2.36. *Let $(P, \omega = \sum_{\alpha=1}^k \omega^\alpha \otimes e_\alpha)$ be an exact k -symplectic manifold, with $\omega = -d\theta$, and let $(M = P \times \mathbb{R}^k, \eta = \sum_{\alpha=1}^k \eta^\alpha \otimes e_\alpha)$ be its induced k -contactification. Let $\rho: M \rightarrow P$ be the canonical projection and $H \in C^\infty(P)$, set $h := H \circ \rho$, and let $\mathbf{X} = (X_1, \dots, X_k)$ be a k -symplectic Hamiltonian k -vector field for H , i.e. $\sum_{\alpha=1}^k \iota_{X_\alpha} \omega^\alpha = dH$. Then, there exists a unique k -vector field $\mathbf{E} = (E_1, \dots, E_k)$ on M such that $\rho_* E_\alpha = X_\alpha$, $\iota_{E_\alpha} \eta^\alpha = 0$ and $\iota_{E_\alpha} dz^\beta = 0$ for every $\alpha, \beta = 1, \dots, k$ with $\alpha \neq \beta$. It is given by*

$$E_\alpha = \tilde{X}_\alpha + \rho^*(\theta^\alpha(X_\alpha)) \frac{\partial}{\partial z^\alpha}, \quad \alpha = 1, \dots, k, \quad (1.2.15)$$

where $\tilde{X}_\alpha$ denotes the lift of X_α to M tangent to the P -factor. Moreover, $\mathbf{E}$ is an η -evolution k -contact k -vector field for h .

Proof. Consider the vector fields E_α defined by (1.2.15). By construction, $\rho_* E_\alpha = X_\alpha$ and $\iota_{E_\alpha} dz^\beta = 0$ for every $\beta \neq \alpha$. Since $\eta^\alpha = dz^\alpha - \rho^* \theta^\alpha$, one has $d\eta^\alpha = \rho^* \omega^\alpha$. Moreover, $\partial/\partial z^\beta$ takes values in $\ker d\eta^\alpha$ for every α, β . Hence, we obtain

$$\sum_{\alpha=1}^k \iota_{E_\alpha} d\eta^\alpha = \sum_{\alpha=1}^k \iota_{\tilde{X}_\alpha} \rho^* \omega^\alpha = \rho^* \left(\sum_{\alpha=1}^k \iota_{X_\alpha} \omega^\alpha \right) = \rho^*(dH) = dh.$$

On the other hand,

$$\iota_{E_\alpha} \eta^\alpha = \iota_{\tilde{X}_\alpha} (dz^\alpha - \rho^* \theta^\alpha) + \rho^*(\theta^\alpha(X_\alpha)) \iota_{\partial/\partial z^\alpha} \eta^\alpha.$$

Since $\tilde{X}_\alpha$ is tangent to the P -factor, one has $\iota_{\tilde{X}_\alpha} dz^\alpha = 0$, while $\iota_{\tilde{X}_\alpha} \rho^* \theta^\alpha = \rho^*(\theta^\alpha(X_\alpha))$ and $\iota_{\partial/\partial z^\alpha} \eta^\alpha = 1$. Therefore, $\iota_{E_\alpha} \eta^\alpha = 0$ for every α , and thus $\sum_{\alpha=1}^k \iota_{E_\alpha} \eta^\alpha = 0$. Hence, $\mathbf{E}$ is an evolution k -contact k -vector field for h .

Finally, let $\mathbf{F} = (F_1, \dots, F_k)$ be another k -vector field on M such that $\rho_* F_\alpha = X_\alpha$, $\iota_{F_\alpha} \eta^\alpha = 0$ and $\iota_{F_\alpha} dz^\beta = 0$ for every $\beta \neq \alpha$. Then, $F_\alpha - \tilde{X}_\alpha$ is vertical with respect to ρ , so $F_\alpha = \tilde{X}_\alpha + \sum_{\beta=1}^k f_\alpha^\beta \partial/\partial z^\beta$ for certain functions f_α^β . Since $\eta^\alpha(\partial/\partial z^\beta) = \delta_\beta^\alpha$, the conditions $\iota_{F_\alpha} \eta^\alpha = 0$, for $\alpha = 1, \dots, k$, yield

$$0 = \eta^\alpha(F_\alpha) = -\rho^*(\theta^\alpha(X_\alpha)) + f_\alpha^\alpha, \quad \alpha = 1, \dots, k.$$

Thus, $f_\alpha^\alpha = \rho^*(\theta^\alpha(X_\alpha))$ for $\alpha = 1, \dots, k$. The conditions $\iota_{F_\alpha} dz^\beta = 0$ for $\beta \neq \alpha$ force all off-diagonal coefficients to vanish, that is, $f_\alpha^\beta = 0$ whenever $\beta \neq \alpha$. Then, the conditions $\iota_{F_\alpha} \eta^\alpha = 0$ for $\alpha = 1, \dots, k$, determine uniquely the remaining diagonal coefficients. Hence, $F_\alpha = E_\alpha$ for $\alpha = 1, \dots, k$. $\square$

1.2.10 Dissipation laws

In the k -contact setting, one is frequently interested in dissipative systems admitting quantities that are no longer conserved but still evolve according to a specific law [20]. As a motivation, recall that if X_h is a contact Hamiltonian vector field on P , then $\mathcal{L}_{X_h} h = -(Rh)h$ and more generally, a dissipative quantity is a function $F \in C^\infty(P)$ that satisfies $\mathcal{L}_{X_h} F = -(Rh)F$. This shows that, in the contact case, the Hamiltonian evolves according to a dissipation law governed by the Reeb dynamics.

Guided by this fact and the corresponding notion of conservation law in field theory, one may introduce the concept of dissipation law for a k -contact Hamiltonian system $(M, \boldsymbol{\eta}, h)$. For a map $\mathbf{F} = (F^1, \dots, F^k): M \rightarrow \mathbb{R}^k$, two related notions are defined. First, $\mathbf{F}$ satisfies the *dissipation law for sections* if, for every solution ψ of the k -contact HddW equations, the divergence of $\mathbf{F} \circ \psi$ satisfies

$$\operatorname{div}(\mathbf{F} \circ \psi) = - \sum_{\alpha=1}^k [(R_\alpha h) F^\alpha] \circ \psi.$$

Note that $\psi: U \subset \mathbb{R}^k \rightarrow M$ and $\mathbf{F} \circ \psi: t \in U \subset \mathbb{R}^k \mapsto \mathbf{F}(\psi(t)) \in \mathbb{R}^k \simeq T_t \mathbb{R}^k$ can be understood as a vector field, while $\mathbb{R}^k$ can be endowed with the standard volume form $\Omega = dt^1 \wedge \dots \wedge dt^k$ giving rise to the divergence of $\mathbf{F} \circ \psi$.

It is also worth analysing the meaning of the above expression. When the right-hand side is zero, this expression is what is called a *local conservation law for systems of PDEs*, where $\mathbf{F} \circ \psi$ is called the *conserved current* [42, 99, 100]. Meanwhile, the right-hand side measures how the left-hand side differs from a local conservation law, which is recovered, in particular, when h is invariant under all the Reeb vector fields, that is, when $\mathcal{L}_{R_\alpha} h = 0$ for every $\alpha = 1, \dots, k$.

Second, $\mathbf{F}$ satisfies the *dissipation law for k -vector fields* if, for every k -vector field $\mathbf{X} = (X_1, \dots, X_k)$ solving the geometric HddW equations for k -vector fields, one has

$$\sum_{\alpha=1}^k \mathcal{L}_{X_\alpha} F^\alpha = - \sum_{\alpha=1}^k (R_\alpha h) F^\alpha.$$

It can be proved that if $\mathbf{F}$ satisfies the dissipation law for sections, then the above identity holds for every integrable k -vector field $\mathbf{X}$ satisfying the geometric Hamilton–De Donder–Weyl equations. Conversely, if

the identity holds for a given integrable k -vector field, then its integral sections satisfy the corresponding divergence law.

The following results explain how infinitesimal dynamical symmetries generate dissipation laws (see [29, Proposition 6.6 & Proposition 6.10]).

Definition 1.2.37. An *infinitesimal dynamical symmetry* of a k -contact Hamiltonian system $(M, \boldsymbol{\eta}, h)$ is a vector field Y on M such that

$$\mathcal{L}_Y \boldsymbol{\eta} = 0, \quad \mathcal{L}_Y h = 0.$$

Theorem 1.2.38. *If Y is an infinitesimal dynamical symmetry, then $\mathbf{F} = -\iota_Y \boldsymbol{\eta}$ satisfies the dissipation law for k -vector fields.*

Hence, infinitesimal dynamical symmetries generate dissipation laws, in analogy with the role played by symmetries and conservation laws in conservative theories, but now adapted to the k -contact framework.

Chapter 2

Classical Hamilton–Jacobi theory

In its original form, Hamilton–Jacobi theory offers a method for solving Hamilton’s equations, i.e. the system of first-order ordinary differential equations of motion describing a mechanical system [1, 7, 36]. The classical approach is to find a canonical transformation leading to constants of motion, so that Hamilton equations become trivial. The idea underlying Hamilton–Jacobi theory is to replace the study of trajectories in phase space with the search for a function $S(q, P, t)$, called *Hamilton’s principal function*, which serves as a generating function for a canonical transformation and satisfies the *Hamilton–Jacobi equation*

$$\frac{\partial S}{\partial t} + h\left(q, \frac{\partial S}{\partial q}, t\right) = 0,$$

where $h(q, p, t)$ is the Hamiltonian function of the system. In this way, the problem of solving a system of ordinary differential equations is replaced by that of solving a first-order partial differential equation. Once a complete solution S is found, the dynamics of the system can be recovered by differentiation [7, 19, 113].

From a geometric point of view, Hamilton–Jacobi theory can be naturally formulated within the framework of symplectic geometry [19, 113]. In what follows, we shall primarily restrict our attention to autonomous systems, described by a time-independent Hamiltonian h . In this setting, the phase space is modelled by the cotangent bundle T^*M of the configuration space M , equipped with its canonical symplectic form ω . The Hamilton–Jacobi equation then takes the form

$$h\left(q, \frac{\partial S}{\partial q}\right) = E,$$

where E is a constant. In the geometric formulation, solutions of the Hamilton–Jacobi equation are associated with Lagrangian submanifolds of the phase space that are invariant under the Hamiltonian flow and transversal to the fibres of the cotangent bundle. More generally, a complete solution, depending on a suitable set of parameters, gives rise locally to a foliation of the phase space by such Lagrangian submanifolds. Each leaf of this foliation is obtained as the image of the differential of S for a fixed value of the parameters [19, 113]. This geometric formulation will constitute the main setting for the subsequent developments in this chapter. The non-autonomous case requires a suitable extension of this framework and will be briefly addressed separately.

The chapter is organised as follows. In Section 2.1, we first review the classical Hamilton–Jacobi theory on cotangent bundles, starting from its formulation in terms of canonical transformations and establishing its relation with the Hamilton–Jacobi equation. Some representative examples are presented in Section 2.2. In Section 2.3, we introduce the geometric formulation of Hamilton–Jacobi theory on symplectic manifolds [19]. We first consider the generalised Hamilton–Jacobi problem and then recover the classical Hamilton–Jacobi problem by imposing the appropriate conditions. This allows the classical theory on cotangent bundles to be understood as a particular instance of a more general geometric construction. The relation between the two formulations is further illustrated through examples. Finally,

in Section 2.4, we briefly discuss the modifications required for non-autonomous Hamiltonian systems and the corresponding time-dependent Hamilton–Jacobi theory.

2.1 The classical Hamilton–Jacobi method

Let M be a smooth manifold of dimension n . Consider the phase space T^*M equipped with adapted coordinates $\{q^i, p_i\}_{i=1,\dots,n}$, where $\{q^i\}_{i=1,\dots,n}$ stand for the configuration variables and $\{p_i\}_{i=1,\dots,n}$ denote their conjugate momenta. Consider a Hamiltonian function $h : T^*M \rightarrow \mathbb{R}$, which determines the dynamics of the mechanical system. The corresponding Hamilton’s equations are

$$\frac{dq^i}{dt} = \frac{\partial h}{\partial p_i}(q(t), p(t)), \quad \frac{dp_i}{dt} = -\frac{\partial h}{\partial q^i}(q(t), p(t)), \quad i = 1, \dots, n.$$

Now consider a time-dependent canonical transformation that maps the original Darboux coordinates $\{q^i, p_i\}$ into a new set of Darboux coordinates $\{Q^i, P_i\}$ (for more details on canonical transformations, see [36]). Locally, such a transformation can be generated by a function $S = S(q^i, P_i, t)$. If S is taken to be a generating function of this type, then

$$p_i = \frac{\partial S}{\partial q^i}, \quad Q^i = \frac{\partial S}{\partial P_i}, \quad i = 1, \dots, n.$$

Under this transformation, the Hamiltonian in the new coordinates Q^i, P_i takes the form

$$K = \frac{\partial S}{\partial t} + h\left(q^1, \dots, q^n, \frac{\partial S}{\partial q^1}, \dots, \frac{\partial S}{\partial q^n}\right).$$

The objective is to choose the generating function S so that the transformed Hamiltonian K vanishes identically, that is,

$$\frac{\partial S}{\partial t} + h\left(q^1, \dots, q^n, \frac{\partial S}{\partial q^1}, \dots, \frac{\partial S}{\partial q^n}\right) = 0.$$

This is the Hamilton–Jacobi equation for the autonomous Hamiltonian system, and a solution S is called *Hamilton’s principal function*. With this choice of S , the transformed Hamiltonian vanishes and, consequently, the new canonical variables Q^i, P_i are constant. The original trajectories are then recovered by inverting the canonical transformation,

$$q^i = q^i(Q, P, t), \quad p_i = p_i(Q, P, t),$$

and provide the solutions to Hamilton’s original equations. Therefore, instead of solving a coupled system of ordinary differential equations, one reduces the problem to solving a single partial differential equation for the function S . In many physically relevant situations, this reformulation proves considerably more effective.

A standard separation ansatz assumes that the principal function can be decomposed as

$$S(q^i, P_i, t) = W(q^i, P_i) - tE,$$

where E is a constant, interpreted as the total energy of the system, and W is called *Hamilton’s characteristic function* [36]. Substituting this expression into the transformed Hamiltonian equation yields

$$h\left(q^1, \dots, q^n, \frac{\partial W}{\partial q^1}, \dots, \frac{\partial W}{\partial q^n}\right) = E.$$

This equation is known as the *time-independent Hamilton–Jacobi equation*. Since it is the form most frequently encountered in applications, it is often referred to simply as the Hamilton–Jacobi equation.

Various constructive methods exist for determining the canonical variables $\{Q^i, P_i\}$. In particular, suppose that P_i are conserved quantities and that Hamilton’s characteristic function $W(q^i, P_i)$ satisfies the time-independent Hamilton–Jacobi equation. Then, using $S = W - tE$, the relations

$$p_i = \frac{\partial S}{\partial q^i}, \quad Q_i = \frac{\partial S}{\partial P_i},$$

determine the canonical transformation. Consequently, the functions

$$q(t) = q(Q, P, t), \quad p(t) = p(Q, P, t),$$

provide the general solution of the original Hamiltonian system.

2.2 Examples of the classical Hamilton–Jacobi method

To complement the preceding theoretical discussion, we now consider several classical examples of the Hamilton–Jacobi method. These examples demonstrate how the Hamilton–Jacobi equation can be employed to obtain solutions of classical dynamical systems and reveal the connection between the principal function and the system’s trajectories. Through these examples, we highlight the effectiveness of the method in reducing the complexity of the equations of motion. The following cases were selected to present different aspects of the Hamilton–Jacobi approach and its role in classical mechanics.

Example 2.2.1. Consider a particle of mass m moving in a constant gravitational field directed along the z -axis. The configuration space is $\mathbb{R}^3$ with coordinates $\{x, y, z\}$ and the phase space becomes $T^*\mathbb{R}^3$ with coordinates $\{x, y, z, p_x, p_y, p_z\}$. The potential energy is $V(x, y, z) = mgz$ and the Hamiltonian function reads

$$h(x, y, z, p_x, p_y, p_z) = \frac{1}{2m} (p_x^2 + p_y^2 + p_z^2) + mgz.$$

Since the Hamiltonian function does not depend explicitly on time, the system is conservative and the total energy E is constant. For such systems, the Hamilton–Jacobi equation can be written in terms of characteristic function W , namely

$$\frac{1}{2m} \left[\left(\frac{\partial W}{\partial x} \right)^2 + \left(\frac{\partial W}{\partial y} \right)^2 + \left(\frac{\partial W}{\partial z} \right)^2 \right] + mgz = E.$$

The goal is to find a function W that satisfies this equation. Once W is known, the particle’s trajectory can be obtained directly. Notice that the Hamiltonian is a sum of terms involving x , y , and z separately. This suggests trying a separable solution of the form

$$W = W_x(x) + W_y(y) + W_z(z).$$

Substituting this into the Hamilton–Jacobi equation gives

$$\frac{1}{2m} \left[\left(\frac{\partial W_x}{\partial x} \right)^2 + \left(\frac{\partial W_y}{\partial y} \right)^2 + \left(\frac{\partial W_z}{\partial z} \right)^2 \right] + mgz = E.$$

Since x and y do not appear explicitly in the potential energy, their corresponding momenta must be constants of motion. In that case we say x and y are cyclic coordinates. We therefore introduce the separation constants

$$k_x = \frac{dW_x}{dx}, \quad k_y = \frac{dW_y}{dy}.$$

Physically, those constants are simply conserved momenta in the x - and y -directions, that is, $p_x = k_x$ and $p_y = k_y$, which is also consistent with Hamilton equations. Substituting these into the energy equation yields

$$p_z = \frac{dW_z}{dz} = \sqrt{2m(E - mgz) - k_x^2 - k_y^2}, \quad (2.2.1)$$

where for definiteness, we restrict our consideration to the positive branch of p_z , corresponding to the particle’s upward motion. The above expression shows how the vertical momentum changes with height because gravity acts only in the z -direction. From the above relation, integration immediately gives

$$W_x(x) = \int_{x_0}^x k_x dx, \quad W_y(y) = \int_{y_0}^y k_y dy, \quad W_z(z) = \int_{z_0}^z \sqrt{2m(E - mgz) - k_x^2 - k_y^2} dz.$$

Therefore, the complete characteristic function reads

$$W = \int_{x_0}^x k_x dx + \int_{y_0}^y k_y dy + \int_{z_0}^z \sqrt{2m(E - mgz) - k_x^2 - k_y^2} dz.$$

The Hamilton–Jacobi formalism tells us that differentiating W with respect to the separation constants produces new canonical coordinates. For the energy parameter we obtain

$$C_E = \frac{\partial S}{\partial E} = \frac{\partial W}{\partial E} - t = \int_{z_0}^z \frac{m dz}{\sqrt{2m(E - mgz) - k_x^2 - k_y^2}} - t,$$

and similarly

$$C_x = x - x_0 - \int_{z_0}^z \frac{k_x dz}{\sqrt{2m(E - mgz) - k_x^2 - k_y^2}}, \quad C_y = y - y_0 - \int_{z_0}^z \frac{k_y dz}{\sqrt{2m(E - mgz) - k_x^2 - k_y^2}}$$

If the particle is located at (x_0, y_0, z_0) at $t = t_0$, then $C_x = C_y = 0$ and $C_E = -t_0$. Now, the new canonical coordinates are $Q_x = C_x, Q_y = C_y, Q_E = C_E, P_x = k_x, P_y = k_y, P_E = E$ with respect to which the transformed Hamiltonian function is identically zero. In terms of the original canonical coordinates, the above relations yield

$$x(t) = x_0 + \frac{k_x}{m}(t - t_0), \quad y(t) = y_0 + \frac{k_y}{m}(t - t_0), \quad z(t) = z_0 + \frac{p_{z_0}}{m}(t - t_0) - \frac{1}{2}g(t - t_0)^2,$$

where $p_{z_0} = \sqrt{2m(E - mgz_0) - k_x^2 - k_y^2}$. For completeness, we also give the corresponding evolution of the canonical momenta. Although the formula for p_z can be recovered from the Hamilton–Jacobi relation (2.2.1), in this example, p_z is most easily obtained directly from Hamilton’s equations. This gives

$$p_x(t) = k_x, \quad p_y(t) = k_y, \quad p_z(t) = p_{z_0} - mg(t - t_0).$$

Thus, we recover the familiar equations of motion for a particle in a uniform gravitational field. $\triangle$

Example 2.2.2. Let us consider a one-dimensional harmonic oscillator with energy E . In canonical coordinates $\{q, p\}$ on $T^*\mathbb{R}$, the Hamiltonian function reads

$$h(q, p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2,$$

where $\omega = \sqrt{\frac{k}{m}}$. Similarly to the previous example, due to the conservative nature of the system, we can write the Hamilton–Jacobi equation in terms of the characteristic function $W = W(q, E)$, which becomes

$$\frac{1}{2m} \left(\frac{\partial W}{\partial q} \right)^2 + \frac{1}{2}m\omega^2 q^2 = E.$$

Choosing the positive branch, we obtain

$$\frac{dW}{dq} = \sqrt{2mE - m^2\omega^2 q^2},$$

and differentiating with respect to E yields

$$C_E = \frac{\partial S}{\partial E} = \frac{\partial W}{\partial E} - t = \sqrt{\frac{m}{2E}} \int_{q_0}^q \frac{dq}{\sqrt{1 - \frac{m\omega^2 q^2}{2E}}} - t = \frac{1}{\omega} \left(\arcsin \sqrt{\frac{m\omega^2}{2E}} q - \arcsin \sqrt{\frac{m\omega^2}{2E}} q_0 \right) - t.$$

Choosing for simplicity the initial condition $q(t_0) = q_0 = 0$ yields $C_E = -t_0$ and the above relation can be rewritten as

$$q(t) = \sqrt{\frac{2E}{m\omega^2}} \sin(\omega(t - t_0)),$$

while the canonical momentum is given by

$$p(t) = \sqrt{2mE} \cos(\omega(t - t_0)).$$

$\triangle$

2.3 Geometric Hamilton–Jacobi theory on symplectic manifolds

Having introduced the Hamilton–Jacobi method in terms of canonical transformations and generating functions, we now reformulate it in geometric terms. This formulation reveals the underlying geometric structures and provides a natural setting for subsequent generalisations of the Hamilton–Jacobi problem. Indeed, the time-independent Hamilton–Jacobi equation

$$h\left(q^i, \frac{\partial W}{\partial q^i}\right) = E \quad (2.3.1)$$

admits a natural geometric interpretation on the cotangent bundle T^*M of the configuration manifold M . Let $\pi : T^*M \rightarrow M$ denote the canonical projection and let ω be the canonical symplectic form on T^*M , which in Darboux coordinates $\{q^i, p_i\}$ is given by

$$\omega = \sum_{i=1}^n dq^i \wedge dp_i.$$

Let us consider a time-independent Hamiltonian function $h : T^*M \rightarrow \mathbb{R}$. Let $\mathcal{P} \subseteq \mathbb{R}^n$ be an open set of parameters, with $P = (P_1, \dots, P_n) \in \mathcal{P}$, and consider a smooth function $W : M \times \mathcal{P} \rightarrow \mathbb{R}$. For each fixed $P \in \mathcal{P}$, define $W_P : M \rightarrow \mathbb{R}$ by $W_P(q) = W(q, P)$. Here, W is the characteristic function introduced in Section 2.1. For each fixed value of P , its differential defines a one-form, or equivalently a section of the cotangent bundle,

$$dW_P : M \longrightarrow T^*M.$$

The Hamilton–Jacobi equation (2.3.1) is then equivalent to

$$h \circ dW_P = E,$$

and hence to

$$d(h \circ dW_P) = 0.$$

This suggests replacing dW_P by a closed one-form $\gamma_P \in \Omega^1(M)$ and considering the condition

$$d(h \circ \gamma_P) = 0. \quad (2.3.2)$$

By Poincaré’s lemma, every closed one-form is locally exact. Thus, for every $q \in M$, there exists a neighbourhood $U \subset M$ and a function $W_P \in C^\infty(U)$ such that

$$\gamma_P|_U = dW_P.$$

Consequently, equation (2.3.2) provides a geometric formulation of the classical time-independent Hamilton–Jacobi equation.

To understand the dynamical meaning of this formulation, let $X_h \in \mathfrak{X}(T^*M)$ be the Hamiltonian vector field associated with h , defined by

$$\iota_{X_h} \omega = dh.$$

Given a one-form $\gamma \in \Omega^1(M)$, regarded as a section $\gamma : M \rightarrow T^*M$, the Hamiltonian vector field can be projected onto M along γ . This defines the vector field

$$X_h^\gamma := T\pi \circ X_h \circ \gamma \in \mathfrak{X}(M),$$

where $T\pi : TT^*M \rightarrow TM$ is the tangent map of the canonical projection. Thus, the following diagram commutes:

$$\begin{array}{ccc} T^*M & \xrightarrow{X_h} & TT^*M \\ \gamma \downarrow \pi & & \downarrow T\pi \\ M & \xrightarrow{X_h^\gamma} & TM \end{array}$$

The aim of the Hamilton–Jacobi method can then be expressed geometrically as follows. One seeks a suitable one-form γ such that the dynamics of X_h on T^*M can be recovered from the lower-dimensional dynamics of X_h^γ on M . More precisely, if $q(t)$ is an integral curve of X_h^γ , one requires its lift

$$\gamma \circ q(t) = (q(t), p(t))$$

to be an integral curve of X_h . In this case, once $q(t)$ has been obtained by integrating X_h^γ , the corresponding momentum is recovered directly from

$$(q(t), p(t)) = \gamma(q(t)).$$

For the classical Hamilton–Jacobi problem, the one-form γ is additionally required to be closed and is therefore locally of the form $\gamma = dW$. The function W then provides a local solution of the time-independent Hamilton–Jacobi equation.

2.3.1 The generalised Hamilton–Jacobi problem

The preceding geometric formulation of the Hamilton–Jacobi equation involves two distinct features: the reduction of the Hamiltonian dynamics from T^*M to M , and the closedness of the one-form defining this reduction. It is useful to separate these two requirements. We first ask whether there exist a one-form $\gamma \in \Omega^1(M)$ and a vector field $X \in \mathfrak{X}(M)$ such that every integral curve of X , when mapped to T^*M by γ , becomes an integral curve of the Hamiltonian vector field X_h . At this point no closedness condition is imposed on γ . This gives the generalised Hamilton–Jacobi problem. The usual Hamilton–Jacobi problem will subsequently be recovered by restricting to closed one-forms γ which locally have the form $\gamma = dS$, leading back to the classical time-independent Hamilton–Jacobi equation.

Definition 2.3.1. The *generalised Hamilton–Jacobi problem* for a Hamiltonian system (T^*M, ω, h) is to find a differential one-form $\gamma \in \Omega^1(M)$ and a vector field $X \in \mathfrak{X}(M)$, such that, if $\phi : I \subseteq \mathbb{R} \rightarrow M$ is an integral curve of X , then $\gamma \circ \phi : I \subseteq \mathbb{R} \rightarrow T^*M$ is an integral curve of X_h . In other words,

$$\phi'(t) = X \circ \phi(t) \implies (\gamma \circ \phi)'(t) = X_h \circ (\gamma \circ \phi)(t), \quad \forall t \in I.$$

Then, we say that the pair (X, γ) is a *solution to the generalised Hamilton–Jacobi problem*.

There are several equivalent ways of characterizing a solution to the generalised Hamilton–Jacobi problem. The following result (see [19, 113]) provides both dynamical and geometric interpretations of such a solution, relating the defining condition to the invariance of $\text{Im}\gamma$, the behaviour of integral curves, and a differential condition involving γ . We include a detailed proof of these equivalences, following the proof strategy outlined in [113].

Theorem 2.3.2. Let (T^*M, ω, h) be a Hamiltonian system, with Hamiltonian vector field X_h , and let $\pi : T^*M \rightarrow M$ be the canonical projection. Let $\gamma \in \Omega^1(M)$ and $X \in \mathfrak{X}(M)$. Then, the following statements are equivalent:

1. the pair (X, γ) is a solution to the generalised Hamilton–Jacobi problem;
2. the vector fields X and X_h are γ -related, that is

$$X_h \circ \gamma = T\gamma \circ X. \tag{2.3.3}$$

In consequence, $X = T\pi \circ X_h \circ \gamma$ and it is completely determined by X_h and γ ;

3. the submanifold $\text{Im}\gamma \subset T^*M$ is invariant under the flow of X_h , and $X = T\pi \circ X_h \circ \gamma$;
4. the integral curves of X_h with initial conditions in $\text{Im}\gamma$ project by π onto integral curves of X ;

5. the identity $\iota_X d\gamma = -d(\gamma^*h)$ holds for $X = T\pi \circ X_h \circ \gamma$.

Proof. First, let us prove the equivalence $1 \Leftrightarrow 2$. Let $I \subseteq \mathbb{R}$ be an interval, and let $\phi : I \rightarrow M$ be an integral curve of X . By the chain rule,

$$(\gamma \circ \phi)'(t) = T_{\phi(t)}\gamma(\phi'(t)) = T_{\phi(t)}\gamma(X(\phi(t))), \quad t \in I.$$

Hence, if $T\gamma \circ X = X_h \circ \gamma$, then $(\gamma \circ \phi)'(t) = X_h(\gamma(\phi(t)))$, so $\gamma \circ \phi$ is an integral curve of X_h . Conversely, suppose that, for every integral curve ϕ of X , the curve $\gamma \circ \phi$ is an integral curve of X_h . Given $q \in M$, choose ϕ such that $\phi(0) = q$. Then,

$$T_q\gamma(X(q)) = (\gamma \circ \phi)'(0) = X_h(\gamma(q)).$$

Since $q \in M$ was arbitrary, $T\gamma \circ X = X_h \circ \gamma$, meaning that X and X_h are γ -related, and conditions 1 and 2 are equivalent. Moreover, composing both sides of equation (2.3.3) with $T\pi$ gives

$$T\pi \circ T\gamma \circ X = T\pi \circ X_h \circ \gamma.$$

Since $\pi \circ \gamma = \text{id}_M$, it follows that $T\pi \circ T\gamma = \text{Id}_{TM}$. Therefore,

$$X = T\pi \circ X_h \circ \gamma,$$

and X is uniquely determined by γ and X_h .

Now, let us prove $2 \Rightarrow 3$. Assume that X_h and X are γ -related. Thus, $X_h \circ \gamma(q) = T_q\gamma \circ X(q)$ for all $q \in M$. Since γ is a smooth section of π , it is an embedding onto its image, and hence $T_{\gamma(q)}\text{Im}\gamma = \text{Im}T_q\gamma$. Therefore, $X_h(\gamma(q)) \in T_{\gamma(q)}\text{Im}\gamma$ and X_h is tangent to $\text{Im}\gamma$, or equivalently, $\text{Im}\gamma$ is invariant under the local flow of X_h . Let us prove the converse implication $3 \Rightarrow 2$. Assume that X_h is tangent to $\text{Im}\gamma$. Then, for every $q \in M$, $X_h(\gamma(q)) \in T_{\gamma(q)}\text{Im}\gamma = \text{Im}T_q\gamma$. Therefore, there exists $X(q) \in T_qM$ such that $X_h(\gamma(q)) = T_q\gamma(X(q))$. Moreover, since $T_q\gamma$ is injective, this vector is unique. Applying $T_{\gamma(q)}\pi$ to both sides, we obtain

$$X(q) = T_{\gamma(q)}\pi(X_h(\gamma(q))),$$

where we used $T_{\gamma(q)}\pi \circ T_q\gamma = \text{id}_{T_qM}$. Thus, $X = T\pi \circ X_h \circ \gamma$ and $X_h \circ \gamma = T\gamma \circ X$. Hence, conditions 2 and 3 are equivalent.

Next, we prove $2 \Rightarrow 4$. Assume that X and X_h are γ -related. Let $\Phi : I \subseteq \mathbb{R} \rightarrow T^*M$ be an integral curve of X_h satisfying $\Phi(0) = \gamma(q_0) \in \text{Im}\gamma$ for some $q_0 \in M$, and let $\phi : J \subseteq \mathbb{R} \rightarrow M$ be an integral curve of X such that $\phi(0) = q_0$. By condition 2, the curve $\gamma \circ \phi$ is an integral curve of X_h . Moreover, $(\gamma \circ \phi)(0) = \gamma(q_0) = \Phi(0)$. By the uniqueness of integral curves, $\Phi(t) = \gamma(\phi(t))$ for every $t \in I \cap J$. In particular, $\Phi(t) \in \text{Im}\gamma$, and applying π yields

$$\pi(\Phi(t)) = \pi(\gamma(\phi(t))) = \phi(t).$$

Therefore, every integral curve of X_h with an initial condition in $\text{Im}\gamma$ remains in $\text{Im}\gamma$ and projects onto an integral curve of X .

Conversely, let us prove $4 \Rightarrow 2$. Fix $q \in M$, and let $\Phi : I \subseteq \mathbb{R} \rightarrow T^*M$ be the integral curve of X_h satisfying $\Phi(0) = \gamma(q)$. By condition 4, the curve $\phi := \pi \circ \Phi$ is an integral curve of X , and Φ remains in $\text{Im}\gamma$. Since $\phi(0) = \pi(\gamma(q)) = q$, we have $\phi'(0) = X(q)$. Moreover, because $\Phi(t) \in \text{Im}\gamma$ and $\pi \circ \gamma = \text{id}_M$,

$$\Phi(t) = \gamma(\pi(\Phi(t))) = \gamma(\phi(t)).$$

Differentiating at $t = 0$, we obtain

$$X_h(\gamma(q)) = \Phi'(0) = T_q\gamma(\phi'(0)) = T_q\gamma(X(q)).$$

Since $q \in M$ was arbitrary, $X_h \circ \gamma = T\gamma \circ X$. This proves the equivalence of conditions 2 and 4.

Finally, we prove $2 \Leftrightarrow 5$. First, assume that X_h and X are γ -related. Recall that $\omega = -d\theta$, where θ is the canonical one-form on T^*M . Since γ is a differential one-form on M , it satisfies $\gamma^*\theta = \gamma$. Therefore,

$$\gamma^*\omega = \gamma^*(-d\theta) = -d(\gamma^*\theta) = -d\gamma.$$

Thus, for every $q \in M$ and $Y_q \in T_qM$ one has

$$\begin{aligned} -d\gamma_q(X_q, Y_q) &= (\gamma^*\omega)_q(X_q, Y_q) = \omega_{\gamma(q)}(T_q\gamma(X_q), T_q\gamma(Y_q)) \\ &= \omega_{\gamma(q)}(X_h(\gamma(q)), T_q\gamma(Y_q)) = dh_{\gamma(q)}(T_q\gamma(Y_q)) = d(h \circ \gamma)_q(Y_q), \end{aligned}$$

where in the second line we used equation (2.3.3). Hence, $\iota_X(d\gamma) = -d(h \circ \gamma) = -d(\gamma^*h)$, which is precisely condition 5.

It remains to prove $5 \Rightarrow 2$. For this purpose, define $D_q := X_h(\gamma(q)) - T_q\gamma(X_q) \in T_{\gamma(q)}T^*M$. By assumption, $X = T\pi \circ X_h \circ \gamma$, and since $T\pi \circ T\gamma = \text{Id}_{TM}$, we have

$$T_{\gamma(q)}\pi(D_q) = T_{\gamma(q)}\pi(X_h(\gamma(q))) - T_{\gamma(q)}\pi(T_q\gamma(X_q)) = X_q - X_q = 0.$$

In other words, $D_q \in \ker T_{\gamma(q)}\pi$, so D_q is vertical. Moreover, for every $q \in M$ and $Y_q \in T_qM$,

$$\begin{aligned} \omega_{\gamma(q)}(D_q, T_q\gamma(Y_q)) &= \omega_{\gamma(q)}(X_h(\gamma(q)), T_q\gamma(Y_q)) - \omega_{\gamma(q)}(T_q\gamma(X_q), T_q\gamma(Y_q)) \\ &= d(h \circ \gamma)_q(Y_q) - (\gamma^*\omega)_q(X_q, Y_q) \\ &= d(h \circ \gamma)_q(Y_q) + d\gamma_q(X_q, Y_q) \\ &= 0. \end{aligned}$$

Hence, $\omega_{\gamma(q)}(D_q, T_q\gamma(Y_q)) = 0$ for all $Y_q \in T_qM$, and therefore $D_q \in (\text{Im } T_q\gamma)^\perp$. Recall that $T_{\gamma(q)}\text{Im } \gamma = \text{Im } T_q\gamma$ and furthermore,

$$T_{\gamma(q)}T^*M = T_{\gamma(q)}\text{Im } \gamma \oplus \ker T_{\gamma(q)}\pi.$$

The vertical subspace $\ker T_{\gamma(q)}\pi$ is Lagrangian with respect to the canonical symplectic form, so $\ker T_{\gamma(q)}\pi = (\ker T_{\gamma(q)}\pi)^\perp$. Since D_q is vertical, it follows that $\omega_{\gamma(q)}(D_q, v) = 0$ for all $v \in \ker T_{\gamma(q)}\pi$. Together with $D_q \in (\text{Im } T_q\gamma)^\perp$, and the direct-sum decomposition, this gives $\omega_{\gamma(q)}(D_q, v) = 0$ for all $v \in T_{\gamma(q)}T^*M$. By the non-degeneracy of ω , it follows that $D_q = 0$. Since $q \in M$ is arbitrary, $X_h \circ \gamma = T\gamma \circ X$. Thus, conditions 2 and 5 are equivalent. $\square$

2.3.2 The classical Hamilton–Jacobi problem

We now impose the additional geometric structure characteristic of the classical theory on the generalised Hamilton–Jacobi problem. Recall that the geometric reformulation of the time-independent Hamilton–Jacobi equation (2.3.1) led naturally to a closed differential one-form γ , which locally can be written as $\gamma = dW$. In formulating the generalised Hamilton–Jacobi problem, the closedness assumption was dropped in order to isolate the underlying dynamical condition. The preceding characterisation further shows that, although the generalised problem was initially formulated in terms of a pair (X, γ) , the vector field X is in fact completely determined by γ , namely

$$X = T\pi \circ X_h \circ \gamma.$$

Thus, the essential object to be determined is the differential one-form γ . Restoring the closedness condition on γ brings us back to the setting from which the geometric formulation originated and leads to the classical Hamilton–Jacobi problem.

Definition 2.3.3. The *(classical) Hamilton–Jacobi problem* for a Hamiltonian system (T^*M, ω, h) consists in finding a differential one-form $\gamma \in \Omega^1(M)$ which solves the generalised Hamilton–Jacobi problem and, in addition, satisfies

$$d\gamma = 0.$$

Such a differential one-form γ is called a *solution to the (classical) Hamilton–Jacobi problem*.

The additional closedness condition imposed on γ has important geometric consequences.

Lemma 2.3.4. *Let $\gamma \in \Omega^1(M)$. Then, the submanifold $\text{Im}\gamma \subset T^*M$ is Lagrangian if and only if γ is closed.*

Proof. In the proof of Theorem 2.3.2, we have already seen that $\gamma^*\omega = -d\gamma$, where ω is the symplectic form on T^*M . Hence, $\gamma^*\omega = 0$ if and only if $d\gamma = 0$. Since $\gamma: M \rightarrow T^*M$ is an embedding onto its image, we have $\text{Im}T\gamma = T\text{Im}\gamma$. Therefore, $\gamma^*\omega = 0$ is equivalent to saying that $\omega|_{\text{Im}\gamma} = 0$, so γ is closed if and only if $\text{Im}\gamma$ is isotropic. Moreover, $\dim \text{Im}\gamma = \dim M = \frac{1}{2} \dim T^*M$, so that $\text{Im}\gamma$ is a maximally isotropic submanifold of T^*M . Consequently, $\text{Im}\gamma$ is isotropic if and only if it is Lagrangian. This concludes the proof. $\square$

Furthermore, by the Poincaré lemma, every point $q \in M$ has an open neighbourhood $U \subset M$ such that

$$\gamma|_U = dW$$

for some $W \in C^\infty(U)$. Thus, W is a local generating function for the Lagrangian submanifold $\text{Im}\gamma$, in the sense that locally

$$\text{Im}\gamma = \left\{ (q, p) \in T^*U : p_i = \frac{\partial W}{\partial q^i} \right\}.$$

This allows the geometric formulation to be related directly to the generating function W appearing in the time-independent Hamilton–Jacobi equation. Consequently, the equivalent formulations obtained for the generalised problem admit a natural specialisation to the classical case, as summarised in the following theorem.

Theorem 2.3.5. *Let $\gamma \in \Omega^1(M)$ be a closed differential one-form and let W be a local function such that $\gamma = dW$. Then the following statements are equivalent:*

1. *γ is a solution to the classical Hamilton–Jacobi problem;*
2. *the Lagrangian submanifold $\text{Im}\gamma \subset T^*M$ is invariant under the flow of the Hamiltonian vector field X_h ;*
3. *the identity $d(\gamma^*h) = 0$ holds for γ , or equivalently, $h \circ dW$ is locally constant.*

Proof. The result follows directly from Theorem 2.3.2 by imposing the closedness condition $d\gamma = 0$. Indeed, in this case the condition 5 of Theorem 2.3.2 reduces to $d(\gamma^*h) = 0$. Moreover, by Lemma 2.3.4, the condition $d\gamma = 0$ is equivalent to $\text{Im}\gamma$ being a Lagrangian submanifold of T^*M . Hence, condition 3 of Theorem 2.3.2 gives the invariance of the Lagrangian submanifold $\text{Im}\gamma$ under the flow of X_h . Finally, since locally $\gamma = dW$, one has

$$\gamma^*h = h \circ \gamma = h \circ dW,$$

and therefore $d(\gamma^*h) = 0$ is equivalent to $h \circ dW$ being locally constant. Thus, all three statements are equivalent. $\square$

2.3.3 Examples

We shall now revisit the examples from Section 2.2 and formulate them in the framework of symplectic geometry.

Example 2.3.6. Now, let us consider the problem given in Example 2.2.1 in a geometric manner. We start with the configuration manifold $M = \mathbb{R}^3$ with coordinates $\{x, y, z\}$, and the phase space is the cotangent bundle $T^*\mathbb{R}^3$ equipped with canonical coordinates $\{x, y, z, p_x, p_y, p_z\}$ and symplectic form

$$\omega = dx \wedge dp_x + dy \wedge dp_y + dz \wedge dp_z.$$

The associated Hamiltonian vector field X_h is defined by $\iota_{X_h}\omega = dh$. For the Hamiltonian function introduced at the beginning of this example, we obtain

$$X_h = \frac{p_x}{m} \frac{\partial}{\partial x} + \frac{p_y}{m} \frac{\partial}{\partial y} + \frac{p_z}{m} \frac{\partial}{\partial z} - mg \frac{\partial}{\partial p_z},$$

which is simply Hamilton's equations written as a vector field on phase space. The problem of solving Hamilton–Jacobi equation reduces to finding a closed one-form $\gamma \in \Omega^1(\mathbb{R}^3)$, whose image $\text{Im}\gamma \subset T^*\mathbb{R}^3$ is invariant under X_h . Writing

$$\gamma = \gamma_x dx + \gamma_y dy + \gamma_z dz,$$

the condition $d\gamma = 0$ implies locally $\gamma = dW$ for some generating function W . The Hamilton–Jacobi equation is then $h \circ \gamma = E$, that is,

$$\frac{1}{2m} (\gamma_x^2 + \gamma_y^2 + \gamma_z^2) + mgz = E.$$

Because the Hamiltonian h is invariant under translations in x and y , it is natural to choose constant horizontal momenta, that is, $\gamma_x = P_x$ and $\gamma_y = P_y$. Then, γ_z is obtained from the above equation and by selecting the upward branch we obtain

$$\gamma = P_x dx + P_y dy + \sqrt{2m(E - mgz) - P_x^2 - P_y^2} dz.$$

It follows directly that $d\gamma = 0$, since the coefficient of dz depends only on z . Now consider a submanifold $L = \text{Im}\gamma \subset T^*\mathbb{R}^3$, which geometrically is a common level set given by $p_x = P_x$, $p_y = P_y$, and $h = E$, and consequently $\dim L = 3$. By straightforward calculation, it also follows that $\omega|_L = 0$, and L is a Lagrangian submanifold of T^*M . Now, projected vector field $X_h^\gamma = T\pi \circ X_h \circ \gamma$ becomes

$$X_h^\gamma = \frac{P_x}{m} \frac{\partial}{\partial x} + \frac{P_y}{m} \frac{\partial}{\partial y} + \frac{\sqrt{2m(E - mgz) - P_x^2 - P_y^2}}{m} \frac{\partial}{\partial z},$$

and simple integration yields the integral curves of X_h^γ , and hence the corresponding solution already obtained in Example 2.2.1. $\triangle$

Example 2.3.7. Now, let us consider the one-dimensional harmonic oscillator introduced in Example 2.2.2 from a geometric point of view. The configuration manifold is $M = \mathbb{R}$ with coordinate $\{q\}$, and the phase space is the cotangent bundle $T^*\mathbb{R}$ equipped with canonical coordinates $\{q, p\}$ and a symplectic form

$$\omega = dq \wedge dp.$$

The Hamiltonian function is

$$h(q, p) = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 q^2,$$

where $\omega = \sqrt{k/m}$. The associated Hamiltonian vector field X_h , defined by $\iota_{X_h}\omega = dh$, is given by

$$X_h = \frac{p}{m} \frac{\partial}{\partial q} - m \omega^2 q \frac{\partial}{\partial p}.$$

Thus, the Hamilton's equations for the harmonic oscillator read

$$\frac{dq}{dt} = \frac{p}{m}, \quad \frac{dp}{dt} = -m \omega^2 q.$$

The geometric Hamilton–Jacobi problem consists in finding a closed one-form $\gamma \in \Omega^1(\mathbb{R})$ whose image $\text{Im}\gamma \subset T^*\mathbb{R}$ is invariant under the Hamiltonian vector field X_h . Writing

$$\gamma = \gamma_q(q) dq,$$

where $\gamma_q \in C^\infty(\mathbb{R})$, we observe that $d\gamma = 0$ automatically, since every one-form on the one-dimensional configuration manifold is closed. Hence, locally, $\gamma = dW$ for some generating function W . The Hamilton–Jacobi equation is then expressed geometrically as

$$h \circ \gamma = E,$$

which gives

$$\frac{\gamma_q^2}{2m} + \frac{1}{2}m\omega^2 q^2 = E.$$

Consequently,

$$\gamma_q(q) = \pm \sqrt{2mE - m^2\omega^2 q^2}.$$

Choosing, for definiteness, the positive branch, corresponding locally to motion with positive momentum, we obtain

$$\gamma = \sqrt{2mE - m^2\omega^2 q^2} dq.$$

This one-form is defined in the region

$$|q| < \sqrt{\frac{2E}{m\omega^2}},$$

with the endpoints corresponding to the turning points of the oscillator.

Consider now the submanifold $L = \text{Im}\gamma \subset T^*\mathbb{R}$. It is given locally by

$$p = \sqrt{2mE - m^2\omega^2 q^2},$$

and therefore lies in the energy level set

$$h^{-1}(E) = \left\{ (q, p) \in T^*\mathbb{R} : \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2 = E \right\}.$$

Since $\dim L = 1 = \frac{1}{2} \dim T^*\mathbb{R}$ and $\omega|_L = 0$, the submanifold L is Lagrangian. Geometrically, it corresponds to one local branch of the constant-energy ellipse in phase space. In contrast to Example 2.3.6, a single graph $\text{Im}\gamma$ cannot describe the entire periodic orbit globally, since the momentum changes sign at the turning points. The positive and negative branches of γ therefore provide local Hamilton–Jacobi descriptions of the two portions of the motion.

The projected Hamiltonian vector field $X_h^\gamma = T\pi \circ X_h \circ \gamma$ on the configuration manifold becomes

$$X_h^\gamma = \frac{\sqrt{2mE - m^2\omega^2 q^2}}{m} \frac{\partial}{\partial q}.$$

Its integral curves satisfy

$$\frac{dq}{dt} = \frac{1}{m} \sqrt{2mE - m^2\omega^2 q^2},$$

and by direct integration, we recover $q(t)$, and the corresponding phase-space trajectory is obtained from $(q(t), p(t)) = \gamma(q(t))$, as in Example 2.2.2. Thus, the geometric Hamilton–Jacobi formulation reproduces the familiar harmonic-oscillator trajectory. In this description, the characteristic function appearing in Example 2.2.2 is replaced by the closed one-form γ , whose image gives a local Lagrangian submanifold contained in the constant-energy level set, while the dynamics on configuration space is generated by the projected vector field X_h^γ .

△

2.4 Time-dependent classical Hamilton–Jacobi

So far, we have considered Hamiltonian systems determined by a time-independent Hamiltonian function $h : T^*M \rightarrow \mathbb{R}$. We shall now briefly extend the geometric formulation of Hamilton–Jacobi theory to

explicitly time-dependent Hamiltonian systems (see [107]). In this case, the Hamiltonian is a smooth function $h : T^*M \times \mathbb{R} \rightarrow \mathbb{R}$, and, for every fixed $t \in \mathbb{R}$, we denote by

$$h_t : T^*M \longrightarrow \mathbb{R}, \quad h_t(q, p) = h(q, p, t),$$

the corresponding Hamiltonian at time t .

Let (T^*M, ω) be the canonical symplectic manifold. For every $t \in \mathbb{R}$, the function h_t determines a Hamiltonian vector field $X_{h_t} \in \mathfrak{X}(T^*M)$ defined by $\iota_{X_{h_t}}\omega = dh_t$. Consequently, the dynamics is described by a time-dependent vector field $X_h : \mathbb{R} \times T^*M \rightarrow TT^*M$. In canonical coordinates $\{q^i, p_i\}$, its integral curves satisfy the time-dependent Hamilton’s equations

$$\frac{dq^i}{dt} = \frac{\partial h}{\partial p_i}(q(t), p(t), t), \quad \frac{dp_i}{dt} = -\frac{\partial h}{\partial q^i}(q(t), p(t), t), \quad i = 1, \dots, n.$$

In analogy with the time-independent case, consider now a time-dependent family of differential one-forms $\gamma_t \in \Omega^1(M)$, or, equivalently, a smooth map

$$\gamma : M \times \mathbb{R} \longrightarrow T^*M, \quad (q, t) \longmapsto \gamma_t(q),$$

such that $\pi \circ \gamma_t = \text{id}_M$ for every t , where $\pi : T^*M \rightarrow M$ denotes the canonical projection. The corresponding projected time-dependent vector field on the configuration manifold is defined by

$$X_{h_t}^{\gamma_t} := T\pi \circ X_{h_t} \circ \gamma_t.$$

Contrary to the autonomous case, the submanifold $\text{Im } \gamma_t \subset T^*M$ is now allowed to evolve with time. Moreover, the Hamiltonian vector field X_{h_t} need not be tangent to $\text{Im } \gamma_t$ at a fixed value of t . Instead, the motion of the family $\text{Im } \gamma_t$ itself has to be taken into account.

Let $\phi : I \subset \mathbb{R} \rightarrow M$ be an integral curve of the projected time-dependent vector field, that is,

$$\phi'(t) = X_{h_t}^{\gamma_t}(\phi(t)).$$

We require that its time-dependent lift $t \mapsto \gamma_t(\phi(t))$ be an integral curve of the original Hamiltonian system. Differentiating this curve gives

$$\frac{d}{dt}(\gamma_t(\phi(t))) = \frac{\partial \gamma_t}{\partial t}(\phi(t)) + T_{\phi(t)}\gamma_t(\phi'(t)),$$

where we denote by

$$\frac{\partial \gamma_t}{\partial t}(q) := \frac{d}{dt}(\gamma_t(q)) \in T_{\gamma_t(q)}T^*M, \quad q \in M$$

the tangent vector to the curve $t \mapsto \gamma_t(q)$. Hence, a time-dependent analogue of the γ -relatedness condition appearing in the autonomous Hamilton–Jacobi problem is

$$X_{h_t} \circ \gamma_t = T\gamma_t \circ X_{h_t}^{\gamma_t} + \frac{\partial \gamma_t}{\partial t}. \quad (2.4.1)$$

This leads naturally to the following formulation.

Definition 2.4.1. The *generalised time-dependent Hamilton–Jacobi problem for a Hamiltonian system* (T^*M, ω, h) consists in finding a time-dependent family of differential one-forms $\gamma_t \in \Omega^1(M)$ such that

$$X_{h_t} \circ \gamma_t = T\gamma_t \circ X_{h_t}^{\gamma_t} + \frac{\partial \gamma_t}{\partial t},$$

where $X_{h_t}^{\gamma_t} = T\pi \circ X_{h_t} \circ \gamma_t$. In this case, we say that γ is a *solution to the generalised time-dependent Hamilton–Jacobi problem*.

The geometric meaning of this condition is analogous to that of the time-independent problem. If ϕ is an integral curve of $X_{h_t}^{\gamma_t}$, then $\gamma_t(\phi(t))$ determines an integral curve of the Hamiltonian dynamics on T^*M . Thus, the problem of integrating the Hamiltonian system can again be reduced to the integration of a vector field on the lower-dimensional configuration manifold M .

To recover the classical Hamilton–Jacobi equation, we impose, as in the time-independent case, the additional condition

$$d\gamma_t = 0$$

for every t . By Poincaré’s lemma, the one-form γ_t is then locally exact, and hence there exists a time-dependent function $S : M \times \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\gamma_t = dS_t, \quad S_t(q) := S(q, t).$$

Since each γ_t is closed, its image $\text{Im } \gamma_t \subset T^*M$ is a Lagrangian submanifold. Consequently, a time-dependent solution may be regarded as a one-parameter family of Lagrangian submanifolds whose evolution is compatible with the Hamiltonian dynamics.

For a closed family γ_t , condition (2.4.1) is equivalent locally to

$$d\left(\frac{\partial S}{\partial t} + h_t \circ dS_t\right) = 0.$$

Therefore, on each connected neighbourhood, there exists a function $c : \mathbb{R} \rightarrow \mathbb{R}$ depending only on time such that

$$\frac{\partial S}{\partial t} + h(q, dS_t(q), t) = c(t).$$

Since adding a function depending only on time to S does not change the one-form dS_t , the function $c(t)$ can be absorbed into the definition of S . Thus, one obtains the standard time-dependent Hamilton–Jacobi equation

$$\frac{\partial S}{\partial t} + h\left(q, \frac{\partial S}{\partial q}, t\right) = 0. \quad (2.4.2)$$

In local canonical coordinates this equation reads

$$\frac{\partial S}{\partial t} + h\left(q^1, \dots, q^n, \frac{\partial S}{\partial q^1}, \dots, \frac{\partial S}{\partial q^n}, t\right) = 0.$$

The relation with the time-independent theory discussed in the preceding sections follows immediately. Suppose that the Hamiltonian has no explicit time dependence, $h = h(q, p)$, and consider a solution of the separated form $S(q, t) = W(q) - Et$, where $E \in \mathbb{R}$ is constant. Then,

$$\frac{\partial S}{\partial t} = -E, \quad dS_t = dW,$$

and equation (2.4.2) reduces to

$$h \circ dW = E,$$

which is precisely the time-independent Hamilton–Jacobi equation considered in Section 2.3. Hence, the autonomous Hamilton–Jacobi theory can be regarded as a particular case of the time-dependent formulation in which the evolution of the family of Lagrangian submanifolds can be encoded by a separated principal function $S(q, t) = W(q) - Et$.

The geometric interpretation therefore remains essentially the same in both settings. In the autonomous case, a solution is represented by a fixed Lagrangian submanifold $\text{Im } dW \subset T^*M$ invariant under the Hamiltonian flow. In the explicitly time-dependent case, this fixed submanifold is replaced by a time-dependent family

$$L_t = \text{Im } dS_t \subset T^*M,$$

whose evolution is compatible with the time-dependent Hamiltonian dynamics. This provides the natural extension of the geometric Hamilton–Jacobi formulation to non-autonomous Hamiltonian systems.

Chapter 3

Hamilton–Jacobi equations in the k -contact setting

The primary objective in this chapter is to extend Hamilton–Jacobi theory to k -contact manifolds. This framework provides a natural geometric description of systems with multiple dissipation mechanisms and, more generally, of non-conservative dynamics depending on several contact variables. From this perspective, the proposed Hamilton–Jacobi formalism extends its contact counterpart (see [37, 61]), in analogy with the extension of the symplectic Hamilton–Jacobi formalism to the k -symplectic setting for field theories (cf. [64]).

The geometric structures underlying k -contact Hamiltonian systems have been introduced in Chapter 1 and will not be reviewed here. Instead, we focus on the development of Hamilton–Jacobi theory in this setting, considering both $\eta_{Q,k}$ -Hamiltonian and evolution k -vector fields. In addition, we distinguish between the z -independent setting, in which the projected dynamics is defined on Q , and the z -dependent setting, in which the dissipative variables are retained and the projected dynamics is defined on $Q \times \mathbb{R}^k$. We study these formulations and analyse their relation with the k -contact Hamilton–De Donder–Weyl equations.

Throughout this chapter, the term Hamilton–Jacobi problem, without further qualification, refers to the problem associated with $\eta_{Q,k}$ -Hamiltonian k -vector fields. When useful for emphasis or comparison with the evolution formulation, we also refer to it as the Hamiltonian Hamilton–Jacobi problem. The problem associated with $\eta_{Q,k}$ -evolution k -vector fields is referred to as the evolution Hamilton–Jacobi problem. The same convention applies to the corresponding equations, solutions, and formulations, as well as to related terminology used throughout the chapter. We use the notation introduced in Section 1.2.

3.1 Regular k -contact Hamiltonian systems and higher-order PDEs

In this section, we study a class of k -contact Hamiltonian systems and the corresponding HdDW equations, with particular emphasis on their relation to higher-order systems of PDEs and their applications [110].

We begin by introducing the notion of regularity for canonical k -contact Hamiltonian systems. This extends the usual regularity condition for Hamiltonian functions in symplectic and contact geometry and, at the same time, provides the Hamiltonian counterpart of the notion of regularity for k -contact Lagrangian systems considered in [110].

Definition 3.1.1. We say that $(\mathcal{J}_{Q,k}, \eta_{Q,k}, h)$ is *regular* if the fibre derivative $\mathbb{F}h : \bigoplus^k T^*Q \times \mathbb{R}^k \rightarrow \bigoplus^k TQ \times \mathbb{R}^k$, given in canonical coordinates by $\mathbb{F}h(q^i, p_i^\alpha, z^\alpha) = (q^i, \partial h / \partial p_i^\alpha, z^\alpha)$, is a local diffeomorphism. If $\mathbb{F}h$ is a global diffeomorphism, then $(\mathcal{J}_{Q,k}, \eta_{Q,k}, h)$ is said to be *hyperregular*.

Whenever the underlying k -contact structure $(\mathcal{J}_{Q,k}, \eta_{Q,k})$ is clear from the context, we will simply

say that the Hamiltonian function h is regular or hyperregular. As in the classical Hamiltonian setting, regularity is characterised by the non-degeneracy of the Hessian with respect to the momentum variables. Equivalently, regularity ensures that the relations between the momenta and the corresponding velocity-type variables can be locally inverted. The analogous statement for the Lagrangian setting can be found in [110, Proposition 3.2].

Proposition 3.1.2. *Given $(\mathcal{J}_{Q,k}, \eta_{Q,k}, h)$, the following conditions are equivalent:*

1. h is regular;
2. The Hessian matrix $(\partial^2 h / \partial p_i^\alpha \partial p_j^\beta)$ is non-singular at every point;
3. The relations $v_\alpha^i = \partial h / \partial p_i^\alpha$ can be solved locally for the variables p_i^α as smooth functions of $(q^i, v_\alpha^i, z^\alpha)$.

We now consider an application of k -contact geometry to the study of higher-order systems of PDEs. We restrict our attention to Hamiltonian functions that depend affinely on the dissipative variables, a class that arises naturally in a variety of models and applications (cf. [40, 110–112]). For Hamiltonians of this form, the following result shows that integral sections of both $\eta_{Q,k}$ -Hamiltonian and $\eta_{Q,k}$ -evolution k -vector fields associated with the same Hamiltonian function project onto maps whose components satisfy the same second-order system of PDEs. In particular, the resulting system depends only on the Hamiltonian function and not on the choice of a particular k -vector field. This property will be useful in the subsequent study of PDEs within the k -contact framework.

Theorem 3.1.3. *Let $h \in C^\infty(\mathcal{J}_{Q,k})$ be a regular Hamiltonian of the form*

$$h(q^i, p_i^\alpha, z^\alpha) = g(q^i, p_i^\alpha) + \sum_{\alpha=1}^k A_\alpha z^\alpha, \quad (3.1.1)$$

where $A_1, \dots, A_k \in \mathbb{R}$ are constants. Then the projection onto Q of any integral section of an $\eta_{Q,k}$ -Hamiltonian k -vector field $\mathbf{X}_h$ satisfies a second-order system of partial differential equations. More precisely, let $p_i^\alpha = P_i^\alpha(q, v_\beta)$ denote the local inverse of the fibre derivative, defined by $v_\beta^i = \partial h / \partial p_i^\beta = \partial g / \partial p_i^\beta$. If $\psi(t) = (q^i(t), p_i^\alpha(t), z^\alpha(t))$ is an integral section of $\mathbf{X}_h$, then its projection onto Q , namely $q(t) = (q^1(t), \dots, q^n(t))$ satisfies

$$\sum_{\alpha=1}^k \frac{\partial}{\partial t^\alpha} (P_i^\alpha(q, v_\beta)) + \frac{\partial g}{\partial q^i} (q, P_j^\beta(q, v_\gamma)) + \sum_{\alpha=1}^k A_\alpha P_i^\alpha(q, v_\beta) = 0, \quad i = 1, \dots, n, \quad (3.1.2)$$

where $v_\beta = \partial q / \partial t^\beta$. In particular, this system does not involve the dissipative variables (z^α) . Moreover, the projection of every integral section of an $\eta_{Q,k}$ -evolution k -vector field associated with the same Hamiltonian function h satisfies the same system (3.1.2).

Proof. We first consider an integral section $\psi(t) = (q^i(t), p_i^\alpha(t), z^\alpha(t))$ of an $\eta_{Q,k}$ -Hamiltonian k -vector field $\mathbf{X}_h$. The local k -contact HdDW equations give

$$\frac{\partial q^i}{\partial t^\beta} = \frac{\partial h}{\partial p_i^\beta} \circ \psi, \quad \sum_{\alpha=1}^k \frac{\partial p_i^\alpha}{\partial t^\alpha} = - \left(\frac{\partial h}{\partial q^i} + \sum_{\alpha=1}^k p_i^\alpha \frac{\partial h}{\partial z^\alpha} \right) \circ \psi.$$

For the Hamiltonian (3.1.1), one has

$$\frac{\partial h}{\partial p_i^\beta} = \frac{\partial g}{\partial p_i^\beta}, \quad \frac{\partial h}{\partial q^i} = \frac{\partial g}{\partial q^i}, \quad \frac{\partial h}{\partial z^\alpha} = A_\alpha.$$

Thus, the HdDW equations reduce to

$$\frac{\partial q^i}{\partial t^\beta} = \frac{\partial g}{\partial p_i^\beta} \circ \psi, \quad \sum_{\alpha=1}^k \frac{\partial p_i^\alpha}{\partial t^\alpha} = - \left(\frac{\partial g}{\partial q^i} + \sum_{\alpha=1}^k A_\alpha p_i^\alpha \right) \circ \psi, \quad i = 1, \dots, n. \quad (3.1.3)$$

By regularity of h , Proposition 3.1.2 implies that the relations $v_\beta^i = \partial g / \partial p_i^\beta$ can be inverted locally with respect to the momenta. Hence, $p_i^\alpha = P_i^\alpha(q, v_\beta)$. Since g does not depend on z^α , neither does this inverse fibre map. Replacing p_i^α with $P_i^\alpha(q, v_\beta)$ in the second equation of (3.1.3) yields precisely (3.1.2). This proves, in particular, that the projected second-order equations are independent of the variables z^α . It remains to compare this result with the evolution formulation. Let ψ now be an integral section of an $\eta_{Q,k}$ -evolution k -vector field associated with h . For a Hamiltonian that is affine in the dissipative variables with constant coefficients, the local evolution Hamilton–De Donder–Weyl equations for q^i and p_i^α reduce to the same equations as in (3.1.3). Thus, both formulations induce the same first-order subsystem for (q^i, p_i^α) . Applying once more the local inverse $p_i^\alpha = P_i^\alpha(q, v_\beta)$ gives the same second-order system of PDEs (3.1.2) for the projected variables q^i . $\square$

Note that the resulting second-order system of PDEs (3.1.2) is determined solely by the Hamiltonian h . In particular, it is independent both of the choice between the Hamiltonian and evolution formulations and of the particular k -vector field representing either formulation.

As an illustration, consider the damped vibrating membrane on an elastic foundation studied in [83], with Hamiltonian

$$h(u, p^t, p^x, p^y, z^t, z^x, z^y) = \frac{1}{2} ((p^t)^2 - c^{-2}(p^x)^2 - c^{-2}(p^y)^2) + \frac{\kappa}{2} u^2 + \lambda z^t,$$

the corresponding PDE

$$u_{tt} - c^2(u_{xx} + u_{yy}) + \lambda u_t + \kappa u = 0$$

can be recovered as a particular instance of the second-order system (3.1.2). Indeed, the Hamiltonian has precisely the affine dependence on the dissipative variables required in Theorem 3.1.3. It follows that the projection of any integral section of either an $\eta_{Q,k}$ -Hamiltonian or an $\eta_{Q,k}$ -evolution k -vector field associated with this Hamiltonian satisfies the damped membrane equation above.

Theorem 3.1.4. *Let $h \in C^\infty(\mathcal{J}_{Q,k})$, and consider two $\eta_{Q,k}$ -Hamiltonian k -vector fields $\mathbf{X} = (X_1, \dots, X_k)$ and $\mathbf{Y} = (Y_1, \dots, Y_k)$ associated with h . Then, $\mathbf{X} - \mathbf{Y} \in \ker \chi$. Consequently, $\mathbf{X} - \mathbf{Y}$ is π_Q -vertical. The analogous statement holds for any pair of $\eta_{Q,k}$ -evolution k -vector fields corresponding to the same Hamiltonian h .*

Proof. Since $\mathbf{X}$ and $\mathbf{Y}$ are associated with the same Hamiltonian h , the defining equations for $\eta_{Q,k}$ -Hamiltonian k -vector fields imply that their images under χ coincide, that is $\chi(\mathbf{X}) = \chi(\mathbf{Y})$. Using the fibrewise linearity of χ , we therefore obtain $\chi(\mathbf{X} - \mathbf{Y}) = \chi(\mathbf{X}) - \chi(\mathbf{Y}) = 0$. Hence, $\mathbf{X} - \mathbf{Y}$ takes values in $\ker \chi$. By Proposition 1.2.24, $\ker \chi \subset \bigoplus^k V(\pi_Q)$, where $V(\pi_Q) = \ker T\pi_Q$. Consequently, $\left(\bigoplus^k T\pi_Q\right)(\mathbf{X} - \mathbf{Y}) = 0$, so $\mathbf{X} - \mathbf{Y}$ is π_Q -vertical. The same argument applies to two $\eta_{Q,k}$ -evolution k -vector fields associated with h , since their defining equations likewise yield the same image under χ . $\square$

Proposition 1.2.33 and [44, Proposition 3.5] guarantee the existence of $\eta_{Q,k}$ -Hamiltonian and $\eta_{Q,k}$ -evolution k -vector fields, respectively, for every $(\mathcal{J}_{Q,k}, \eta_{Q,k}, h)$. Together with the preceding result, this immediately yields the following corollary.

Corollary 3.1.5. *Let $h \in C^\infty(\mathcal{J}_{Q,k})$. The set of $\eta_{Q,k}$ -Hamiltonian k -vector fields associated with h is a non-empty affine space modelled on $\Gamma(\ker \chi)$. The same is true for the set of $\eta_{Q,k}$ -evolution k -vector fields associated with h .*

Although the discussion above concerns second-order systems of PDEs, the framework can also be applied to higher-order equations after an appropriate order-reduction procedure. By introducing suitable derivatives of the dependent variables as additional variables, a higher-order system may be rewritten as a system of lower-order equations, together with the corresponding compatibility conditions. Whenever the resulting system admits a regular k -contact Hamiltonian description of the type considered above, the preceding construction can then be applied.

3.2 The k -contact Hamilton–Jacobi equations in the z -independent framework

We now introduce two Hamilton–Jacobi theories for k -contact Hamiltonian systems. The underlying idea is to study the Hamilton–De Donder–Weyl equations associated with

$$(\mathcal{J}_{Q,k}, \boldsymbol{\eta}_{Q,k}, h), \quad \mathcal{J}_{Q,k} = \bigoplus^k T^*Q \times \mathbb{R}^k,$$

by restricting an $\boldsymbol{\eta}_{Q,k}$ -Hamiltonian k -vector field to a suitable submanifold of $\mathcal{J}_{Q,k}$ projecting onto Q . An important feature of the case $k > 1$ is that the Hamiltonian k -vector field need not be tangent to this submanifold. This leads naturally to a weaker notion of relatedness between k -vector fields, which will play a crucial role in the Hamilton–Jacobi theory developed below.

Let $\pi_Q: \mathcal{J}_{Q,k} = \bigoplus^k T^*Q \times \mathbb{R}^k \rightarrow Q$ be the canonical projection, and consider a section γ of π_Q . In adapted coordinates, it is written as

$$\begin{aligned} \gamma: \quad Q &\longrightarrow \mathcal{J}_{Q,k} \\ q = (q^i) &\longmapsto (q^i, \gamma_i^\alpha(q), \gamma^\alpha(q)). \end{aligned}$$

For a smooth map $f: M \rightarrow N$, we use the notation

$$\bigoplus^k Tf: \bigoplus^k TM \longrightarrow \bigoplus^k TN, \quad (v_1, \dots, v_k) \longmapsto (Tf(v_1), \dots, Tf(v_k)).$$

Let $\mathbf{Z} = (Z_1, \dots, Z_k)$ be a k -vector field on $\mathcal{J}_{Q,k}$. Since γ is a section of π_Q , the restriction of $\mathbf{Z}$ along γ determines a k -vector field on Q , defined by

$$\mathbf{Z}^\gamma = \left(\bigoplus^k T\pi_Q \right) \circ \mathbf{Z} \circ \gamma.$$

We shall refer to $\mathbf{Z}^\gamma$ as the k -vector field on Q induced by $\mathbf{Z}$ along γ .

In general, the k -vector field $\mathbf{Z} \circ \gamma$ along γ need not coincide with the image of $\mathbf{Z}^\gamma$ under $\bigoplus^k T\gamma$. This motivates the following weaker notion of relatedness.

Definition 3.2.1. Let $\gamma: Q \rightarrow \mathcal{J}_{Q,k}$ be a section of π_Q , and let $\mathbf{Z}$ be a k -vector field on $\mathcal{J}_{Q,k}$. The k -vector fields $\mathbf{Z}$ and $\mathbf{Z}^\gamma$ are said to be *almost γ -related* if

$$\mathbf{Z} \circ \gamma - \left(\bigoplus^k T\gamma \right) \circ \mathbf{Z}^\gamma \in \ker \chi \quad (3.2.1)$$

along $\gamma(Q)$. Moreover, if

$$\mathbf{Z} \circ \gamma = \left(\bigoplus^k T\gamma \right) \circ \mathbf{Z}^\gamma,$$

then $\mathbf{Z}$ and $\mathbf{Z}^\gamma$ are said to be *γ -related*.

The almost γ -relation admits an equivalent description in terms of the quotient by $\ker \chi$. To see this, let $[\cdot]_\chi: \bigoplus^k T\mathcal{J}_{Q,k} \rightarrow \bigoplus^k T\mathcal{J}_{Q,k}/\ker \chi$ be the canonical quotient projection. By Proposition 1.2.24, every element of $\ker \chi$ is π_Q -vertical. Consequently, $\ker \chi \subset \ker \left(\bigoplus^k T\pi_Q \right)$. It follows that $\bigoplus^k T\pi_Q$ is constant on the equivalence classes determined by $\ker \chi$, and therefore induces a well-defined map

$$\widetilde{\bigoplus^k T\pi_Q}: \bigoplus^k T\mathcal{J}_{Q,k}/\ker \chi \longrightarrow \bigoplus^k TQ,$$

given by

$$\widetilde{\bigoplus^k T\pi_Q}([Z]_\chi) = \left(\bigoplus^k T\pi_Q \right)(Z).$$

Likewise, we denote by $\left[\bigoplus^k T\gamma \right]_\chi = [\cdot]_\chi \circ \left(\bigoplus^k T\gamma \right)$ the map from $\bigoplus^k TQ$ to $\bigoplus^k T\mathcal{J}_{Q,k}/\ker \chi$.

With this notation, condition (3.2.1) is equivalent to the commutativity of the diagram

$$\begin{array}{ccc}
 \mathcal{J}_{Q,k} & \xrightarrow{[\mathbf{Z}]_\chi} & \bigoplus^k \mathrm{T}\mathcal{J}_{Q,k} / \ker \chi \\
 \uparrow \gamma & & \downarrow \left(\bigoplus^k \mathrm{T}\pi_Q \right) \left(\bigoplus^k \mathrm{T}\gamma \right)_\chi \\
 Q & \xrightarrow{\mathbf{Z}^\gamma} & \bigoplus^k \mathrm{T}Q
 \end{array}$$

where $[\mathbf{Z}]_\chi = [\cdot]_\chi \circ \mathbf{Z}$.

For $k > 1$, almost γ -relatedness is, in general, weaker than γ -relatedness. Indeed, condition (3.2.1) allows $\mathbf{Z} \circ \gamma - \left(\bigoplus^k \mathrm{T}\gamma \right) \circ \mathbf{Z}^\gamma$ to be a non-zero element of $\ker \chi$. Thus, the restriction of $\mathbf{Z}$ to $\mathrm{Im} \gamma$ need not coincide with the lift of $\mathbf{Z}^\gamma$ by $\bigoplus^k \mathrm{T}\gamma$. On the other hand, $\ker \chi = 0$ when $k = 1$. Hence, the almost γ -relation reduces to the usual γ -relation in the contact case.

Since γ is a section of π_Q , its image $\mathrm{Im} \gamma$ is an embedded submanifold of $\mathcal{J}_{Q,k}$. In particular, if $\mathbf{X}_h$ is an $\eta_{Q,k}$ -Hamiltonian k -vector field associated with h given locally by (1.2.9), then its projection along γ is locally given by

$$(X_h^\gamma)_\alpha(q) = \sum_{i=1}^n \frac{\partial h}{\partial p_i^\alpha} \circ \gamma(q) \frac{\partial}{\partial q^i}, \quad \alpha = 1, \dots, k, \quad q \in Q.$$

The same expression holds for the projection of an $\eta_{Q,k}$ -evolution k -vector field $\mathbf{E}_h$ associated with h .

We now turn our attention to those sections $\gamma: Q \rightarrow \mathcal{J}_{Q,k}$ whose images are Legendrian submanifolds of the k -contact manifold $(\mathcal{J}_{Q,k}, \eta_{Q,k})$. This condition may be regarded as the k -contact counterpart of the Lagrangian condition appearing in the symplectic formulation of Hamilton–Jacobi theory, discussed in Chapter 2. Indeed, in the classical symplectic setting, sections of $\mathrm{T}^*Q \rightarrow Q$ whose images are Lagrangian are locally given by exact one-forms. As we shall see, the analogous role in the present setting is played by holonomic sections of $\mathcal{J}_{Q,k}$.

To make this relation more transparent, recall the interpretation of $\mathcal{J}_{Q,k}$ as a first-order jet bundle $J^1(Q, Q \times \mathbb{R}^k)$ of sections of the bundle $\pi: Q \times \mathbb{R}^k \rightarrow Q$ (see [99, 121]). A local section of this bundle is determined by a smooth map $\sigma = (\sigma^1, \dots, \sigma^k): Q \rightarrow \mathbb{R}^k$. Two local sections σ and $\tilde{\sigma}$ are said to have the same first jet at $x \in Q$ if they have the same value and the same first derivative at x , that is, $\sigma(x) = \tilde{\sigma}(x)$, and $d\sigma(x) = d\tilde{\sigma}(x)$. The corresponding equivalence class is denoted by $j_x^1 \sigma$. Thus, a point of the *first jet bundle* $J^1(Q, Q \times \mathbb{R}^k)$ records the value of a section together with its first-order derivatives at a given point of Q . Since $d\sigma(x) = (d\sigma^1(x), \dots, d\sigma^k(x)) \in \bigoplus^k \mathrm{T}_x^*Q$, there is a canonical diffeomorphism

$$J^1(Q, Q \times \mathbb{R}^k) \longrightarrow \bigoplus^k \mathrm{T}^*Q \times \mathbb{R}^k, \quad j_x^1 \sigma \longmapsto (x, d\sigma^\alpha(x), \sigma^\alpha(x)).$$

Accordingly, the coordinates $(q^i, p_i^\alpha, z^\alpha)$ on $\mathcal{J}_{Q,k}$ may be interpreted as first-jet coordinates: the variables (z^α) represent the values of the components of a section, whereas (p_i^α) represent their first derivatives with respect to (q^i) . This jet-bundle interpretation motivates the notation $\mathcal{J}_{Q,k} = \bigoplus^k \mathrm{T}^*Q \times \mathbb{R}^k$. It also provides a natural notion of holonicity.

Definition 3.2.2. A section $\gamma: Q \rightarrow \mathcal{J}_{Q,k}$ of π_Q is said to be *holonomic* if there exist functions $\gamma^\alpha \in C^\infty(Q)$, $\alpha = 1, \dots, k$, such that

$$\gamma(q) = (q, d\gamma^\alpha(q), \gamma^\alpha(q)).$$

Note that in terms of the local representation $\gamma(q) = (q^i, \gamma_i^\alpha(q), \gamma^\alpha(q))$, the condition of holonicity is equivalently expressed as

$$\gamma_i^\alpha(q) = \frac{\partial \gamma^\alpha}{\partial q^i}(q), \quad i = 1, \dots, n, \quad \alpha = 1, \dots, k.$$

The following result characterises holonomic sections in terms of the k -contact geometry of $\mathcal{J}_{Q,k}$, and will be used later in the development of the Hamilton–Jacobi theory.

Theorem 3.2.3. *Let $\gamma: Q \rightarrow \mathcal{J}_{Q,k}$ be a section of π_Q , locally represented in adapted coordinates by $\gamma(q) = (q^i, \gamma_i^\alpha(q), \gamma^\alpha(q))$. Then the tangent bundle of its image is locally spanned by*

$$\mathrm{TIm} \gamma = \left\langle \frac{\partial}{\partial q^i} + \sum_{\alpha=1}^k \sum_{j=1}^n \frac{\partial \gamma_j^\alpha}{\partial q^i} \frac{\partial}{\partial p_j^\alpha} + \sum_{\alpha=1}^k \frac{\partial \gamma^\alpha}{\partial q^i} \frac{\partial}{\partial z^\alpha} \right\rangle_{i=1, \dots, n}.$$

Moreover, the following conditions are equivalent:

- (i) γ is holonomic;
- (ii) $\mathrm{Im} \mathrm{T}\gamma \subset \ker \boldsymbol{\eta}_{Q,k}$;
- (iii) $\mathrm{Im} \gamma$ is a Legendrian submanifold of $(\mathcal{J}_{Q,k}, \boldsymbol{\eta}_{Q,k})$.

Proof. Since γ is a section of π_Q , its tangent map is locally given by

$$\mathrm{T}\gamma \left(\frac{\partial}{\partial q^i} \right) = \frac{\partial}{\partial q^i} + \sum_{\alpha=1}^k \sum_{j=1}^n \frac{\partial \gamma_j^\alpha}{\partial q^i} \frac{\partial}{\partial p_j^\alpha} + \sum_{\alpha=1}^k \frac{\partial \gamma^\alpha}{\partial q^i} \frac{\partial}{\partial z^\alpha},$$

which yields the standard local expression for $\mathrm{TIm} \gamma$.

Recall that the canonical k -contact form on $\mathcal{J}_{Q,k}$ reads $\boldsymbol{\eta}_{Q,k} = \sum_{\alpha=1}^k (dz^\alpha - \sum_{i=1}^n p_i^\alpha dq^i) \otimes e_\alpha$. Hence,

$$\gamma^* \boldsymbol{\eta}_{Q,k} = \sum_{\alpha=1}^k \left(d\gamma^\alpha - \sum_{i=1}^n \gamma_i^\alpha dq^i \right) \otimes e_\alpha.$$

Suppose first that (ii) holds. Since $\mathrm{Im} \mathrm{T}\gamma \subset \ker \boldsymbol{\eta}_{Q,k}$ is equivalent to $\gamma^* \boldsymbol{\eta}_{Q,k} = 0$, we obtain

$$d\gamma^\mu = \sum_{i=1}^n \gamma_i^\mu dq^i, \quad \mu = 1, \dots, k.$$

Alternatively, in local coordinates, the above condition may be written as

$$\gamma_i^\mu = \frac{\partial \gamma^\mu}{\partial q^i}, \quad \mu = 1, \dots, k, \quad i = 1, \dots, n. \quad (3.2.2)$$

By definition, these relations are the holonomicity conditions for γ . Thus, (ii) implies (i).

Conversely, if γ is holonomic, then (3.2.2) holds, and substitution into the expression for $\gamma^* \boldsymbol{\eta}_{Q,k}$ gives $\gamma^* \boldsymbol{\eta}_{Q,k} = 0$. Therefore, $\mathrm{Im} \mathrm{T}\gamma \subset \ker \boldsymbol{\eta}_{Q,k}$, and hence (i) implies (ii). Thus, (i) and (ii) are equivalent.

We now prove that a holonomic section has Legendrian image. By (3.2.2) and the symmetry of mixed partial derivatives,

$$\frac{\partial \gamma_i^\mu}{\partial q^j} = \frac{\partial \gamma_j^\mu}{\partial q^i}, \quad \mu = 1, \dots, k, \quad i, j = 1, \dots, n. \quad (3.2.3)$$

Since $d\boldsymbol{\eta}_{Q,k} = -\sum_{\alpha=1}^k \sum_{i=1}^n dp_i^\alpha \wedge dq^i \otimes e_\alpha$, its pull-back by γ is

$$\gamma^* d\boldsymbol{\eta}_{Q,k} = -\sum_{\mu=1}^k \sum_{i,j=1}^n \frac{\partial \gamma_i^\mu}{\partial q^j} dq^j \wedge dq^i \otimes e_\mu = -\frac{1}{2} \sum_{\mu=1}^k \sum_{i,j=1}^n \left(\frac{\partial \gamma_i^\mu}{\partial q^j} - \frac{\partial \gamma_j^\mu}{\partial q^i} \right) dq^j \wedge dq^i \otimes e_\mu.$$

It follows from (3.2.3) that $\gamma^* d\boldsymbol{\eta}_{Q,k} = 0$. Together with $\mathrm{TIm} \gamma \subset \ker \boldsymbol{\eta}_{Q,k}$, this shows that $\mathrm{TIm} \gamma$ is isotropic, namely,

$$\mathrm{TIm} \gamma \subset (\mathrm{TIm} \gamma)^\perp_{\boldsymbol{\eta}_{Q,k}}.$$

Furthermore, the vertical distribution $\mathcal{V}_{Q,k} = \langle \partial / \partial p_i^\alpha \rangle$ is isotropic and provides a complement to $\mathrm{TIm} \gamma$ in $\ker \boldsymbol{\eta}_{Q,k}$, so that $\ker \boldsymbol{\eta}_{Q,k} = \mathrm{TIm} \gamma \oplus \mathcal{V}_{Q,k}$. It follows from Definition 1.2.28 that $\mathrm{Im} \gamma$ is Legendrian. This proves (i) $\Rightarrow$ (iii).

Finally, suppose that $\mathrm{Im} \gamma$ is Legendrian. Then, condition $\mathrm{TIm} \gamma \subset \ker \boldsymbol{\eta}_{Q,k}$ follows immediately from Definition 1.2.28. Thus, (iii) implies (ii), which proves the equivalence of the three conditions. $\square$

3.2.1 The Hamiltonian approach

Let us now formulate a z -independent Hamilton–Jacobi theorem for k -contact Hamiltonian k -vector fields. We then discuss some of its consequences and applications.

Theorem 3.2.4 (z -independent k -contact Hamilton–Jacobi theorem). *Let $\mathbf{X}_h$ be an $\eta_{Q,k}$ -Hamiltonian k -vector field associated with $h \in C^\infty(\mathcal{J}_{Q,k})$, and let $\gamma: Q \rightarrow \mathcal{J}_{Q,k}$ be a holonomic section. Assume that the projected k -vector field $\mathbf{X}_h^\gamma$ is integrable. Then the following statements are equivalent:*

- (a) *Every integral section σ of $\mathbf{X}_h^\gamma$ lifts via γ to a solution $\gamma \circ \sigma$ of the z -independent $\eta_{Q,k}$ -Hamiltonian HdDW equations for h ;*
- (b) *$h \circ \gamma = 0$.*

Proof. We first prove that (a) implies (b). Let σ be an integral section of $\mathbf{X}_h^\gamma$. By assumption,

$$(\gamma \circ \sigma)(t) = (\sigma^i(t), \gamma_i^\alpha(\sigma(t)), \gamma^\alpha(\sigma(t)))$$

is a solution of the z -independent $\eta_{Q,k}$ -Hamiltonian HdDW equations. Hence,

$$\begin{cases} \left. \frac{\partial \sigma^i}{\partial t^\beta} \right|_t = \left. \frac{\partial h}{\partial p_i^\beta} \right|_{\gamma \circ \sigma(t)}, & i = 1, \dots, n, \quad \beta = 1, \dots, k, \\ \sum_{\alpha=1}^k \left. \frac{\partial(\gamma_i^\alpha \circ \sigma)}{\partial t^\alpha} \right|_t = - \left(\left. \frac{\partial h}{\partial q^i} \right|_{\gamma \circ \sigma(t)} + \sum_{\alpha=1}^k (\gamma_i^\alpha \circ \sigma)(t) \left. \frac{\partial h}{\partial z^\alpha} \right|_{\gamma \circ \sigma(t)} \right), & i = 1, \dots, n, \\ \sum_{\alpha=1}^k \left. \frac{\partial(\gamma^\alpha \circ \sigma)}{\partial t^\alpha} \right|_t = \sum_{\alpha=1}^k \sum_{j=1}^n (\gamma_j^\alpha \circ \sigma)(t) \left. \frac{\partial h}{\partial p_j^\alpha} \right|_{\gamma \circ \sigma(t)} - h \circ \gamma \circ \sigma(t). \end{cases}$$

Since γ is holonomic, one has $\frac{\partial \gamma_i^\alpha}{\partial q^i} = \gamma_i^\alpha$ for $i = 1, \dots, n$ and $\alpha = 1, \dots, k$. Using the first family of HdDW equations and the chain rule, we obtain

$$\sum_{\alpha=1}^k \frac{\partial(\gamma^\alpha \circ \sigma)}{\partial t^\alpha} = \sum_{\alpha=1}^k \sum_{i=1}^n \frac{\partial \gamma_i^\alpha}{\partial q^i}(\sigma(t)) \frac{\partial \sigma^i}{\partial t^\alpha} = \sum_{\alpha=1}^k \sum_{i=1}^n (\gamma_i^\alpha \circ \sigma)(t) \frac{\partial h}{\partial p_i^\alpha} \circ \gamma \circ \sigma(t).$$

Comparison with the third HdDW equation gives $h \circ \gamma \circ \sigma = 0$. Since $\mathbf{X}_h^\gamma$ is integrable, through every $q \in Q$ there exists a local integral section σ of $\mathbf{X}_h^\gamma$ such that $\sigma(t_0) = q$ for some t_0 . Consequently, $h \circ \gamma(q) = h \circ \gamma \circ \sigma(t_0) = 0$. As q is arbitrary, $h \circ \gamma = 0$, which proves (b).

Conversely, suppose that $h \circ \gamma = 0$, and let σ be an integral section of $\mathbf{X}_h^\gamma$. We show that $\gamma \circ \sigma$ satisfies the z -independent $\eta_{Q,k}$ -Hamiltonian HdDW equations. Since γ is holonomic, we have $\frac{\partial \gamma_i^\alpha}{\partial q^i} = \gamma_i^\alpha$ and the symmetry of mixed partial derivatives yields

$$\frac{\partial \gamma_i^\alpha}{\partial q^j} = \frac{\partial \gamma_j^\alpha}{\partial q^i}. \quad (3.2.4)$$

Moreover, $h \circ \gamma = 0$ implies $d(h \circ \gamma) = 0$, which in local coordinates reads

$$0 = \sum_{i=1}^n \left(\frac{\partial h}{\partial q^i} \circ \gamma + \sum_{\alpha=1}^k \sum_{j=1}^n \left(\frac{\partial h}{\partial p_j^\alpha} \circ \gamma \right) \frac{\partial \gamma_j^\alpha}{\partial q^i} + \sum_{\alpha=1}^k \left(\frac{\partial h}{\partial z^\alpha} \circ \gamma \right) \frac{\partial \gamma^\alpha}{\partial q^i} \right) dq^i. \quad (3.2.5)$$

Recall that the projection of an $\eta_{Q,k}$ -Hamiltonian k -vector field $\mathbf{X}_h$ along γ is locally given by

$$(X_h^\gamma)_\alpha = \sum_{i=1}^n \left(\frac{\partial h}{\partial p_i^\alpha} \circ \gamma \right) \frac{\partial}{\partial q^i}, \quad \alpha = 1, \dots, k.$$

Since σ is an integral section of $\mathbf{X}_h^\gamma$, it follows that

$$\left. \frac{\partial \sigma^i}{\partial t^\alpha} \right|_t = \left. \frac{\partial h}{\partial p_i^\alpha} \right|_{\sigma(t)} \circ \gamma \circ \sigma(t), \quad \alpha = 1, \dots, k, \quad i = 1, \dots, n. \quad (3.2.6)$$

This is precisely the first family of HdDW equations for $\gamma \circ \sigma$.

To verify the second family, using (3.2.4), (3.2.5), and (3.2.6), we obtain

$$\begin{aligned} \sum_{\alpha=1}^k \left. \frac{\partial(\gamma_i^\alpha \circ \sigma)}{\partial t^\alpha} \right|_t &= \sum_{\alpha=1}^k \sum_{j=1}^n \left. \frac{\partial \gamma_i^\alpha}{\partial q^j} \right|_{\sigma(t)} \left. \frac{\partial \sigma^j}{\partial t^\alpha} \right|_t = \sum_{\alpha=1}^k \sum_{j=1}^n \left. \frac{\partial \gamma_j^\alpha}{\partial q^i} \right|_{\sigma(t)} \left. \frac{\partial h}{\partial p_j^\alpha} \right|_{\gamma \circ \sigma(t)} \\ &= - \left. \frac{\partial h}{\partial q^i} \right|_{\gamma \circ \sigma(t)} - \sum_{\alpha=1}^k \left. \frac{\partial \gamma^\alpha}{\partial q^i} \right|_{\sigma(t)} \left. \frac{\partial h}{\partial z^\alpha} \right|_{\gamma \circ \sigma(t)} \\ &= - \left. \frac{\partial h}{\partial q^i} \right|_{\gamma \circ \sigma(t)} - \sum_{\alpha=1}^k (\gamma_i^\alpha \circ \sigma)(t) \left. \frac{\partial h}{\partial z^\alpha} \right|_{\gamma \circ \sigma(t)}, \end{aligned}$$

for $i = 1, \dots, n$. Thus, the second family of HdDW equations is satisfied.

Finally, holonomicity of γ and (3.2.6) give

$$\sum_{\alpha=1}^k \left. \frac{\partial(\gamma^\alpha \circ \sigma)}{\partial t^\alpha} \right|_t = \sum_{\alpha=1}^k \sum_{i=1}^n \left. \frac{\partial \gamma^\alpha}{\partial q^i} \right|_{\sigma(t)} \left. \frac{\partial \sigma^i}{\partial t^\alpha} \right|_t = \sum_{\alpha=1}^k \sum_{i=1}^n (\gamma_i^\alpha \circ \sigma)(t) \left. \frac{\partial h}{\partial p_i^\alpha} \right|_{\gamma \circ \sigma(t)}.$$

Since $h \circ \gamma = 0$, this is exactly the third HdDW equation. Therefore, $\gamma \circ \sigma$ is a solution of the z -independent $\eta_{Q,k}$ -Hamiltonian HdDW equations. This shows that (b) implies (a) and concludes the proof. $\square$

The preceding theorem also provides a geometric interpretation of the Hamilton–Jacobi equation $h \circ \gamma = 0$. Indeed, if this condition is satisfied and σ is an integral section of $\mathbf{X}_h^\gamma$, then $\gamma \circ \sigma$ is a solution of the z -independent $\eta_{Q,k}$ -Hamiltonian HdDW equations for h , whose first prolongation takes values in $T \operatorname{Im} \gamma$. Thus, $\mathbf{X}_h^\gamma$ determines, via γ , dynamics along the Legendrian submanifold $\operatorname{Im} \gamma$.

It is important, however, to distinguish this lifted dynamics from the restriction of the original k -vector field $\mathbf{X}_h$ to $\operatorname{Im} \gamma$. In the contact case $k = 1$, one has $\ker \chi = 0$, and hence the almost γ -relation reduces to the usual γ -relation. Consequently, $\mathbf{X}_h \circ \gamma = T\gamma \circ \mathbf{X}_h^\gamma$, so the lift of the projected dynamics coincides with the restriction of $\mathbf{X}_h$ to $\operatorname{Im} \gamma$.

For $k > 1$, however, this equality need not hold. In this case, $\mathbf{X}_h \circ \gamma$ and $(\bigoplus^k T\gamma) \circ \mathbf{X}_h^\gamma$ may differ by a k -vector field taking values in $\ker \chi$. This reflects the non-uniqueness of the k -vector fields representing the k -contact Hamiltonian dynamics. For instance, if $\mathbf{X}_h$ is an $\eta_{Q,k}$ -Hamiltonian k -vector field, then so is $\mathbf{X}'_h = \mathbf{X}_h + \mathbf{Z}$, where $\mathbf{Z} = (Z_1, \dots, Z_k)$ is a k -vector field whose components are given in Darboux coordinates by

$$Z_\alpha = \sum_{\beta \neq \alpha} Z_\alpha^\beta \frac{\partial}{\partial z^\beta}, \quad \alpha = 1, \dots, k,$$

for arbitrary functions $Z_\alpha^\beta \in C^\infty(\mathcal{J}_{Q,k})$. Since $\mathbf{Z}$ takes values in $\ker \chi$, the k -vector fields $\mathbf{X}_h$ and $\mathbf{X}'_h$ are associated with the same Hamiltonian function h and induce the same projected k -vector field on Q . Nevertheless, their restrictions to $\operatorname{Im} \gamma$ do not coincide in general.

Finally, let us emphasise that Theorem 3.2.4 requires only the integrability of the projected k -vector field $\mathbf{X}_h^\gamma$, rather than that of the original k -vector field $\mathbf{X}_h$. This distinction is particularly relevant in applications, where the Hamilton–Jacobi approach aims to construct solutions from integral sections of the reduced dynamics on Q . In this respect, requiring the integrability of $\mathbf{X}_h^\gamma$ is a weaker and, in general, more manageable condition than imposing integrability on $\mathbf{X}_h$ itself.

3.2.2 The evolution approach

We now establish the counterpart of the preceding Hamilton–Jacobi theorem for $\eta_{Q,k}$ -evolution k -vector fields in the z -independent setting.

Theorem 3.2.5 (z -independent k -contact evolution Hamilton–Jacobi theorem). *Let $\mathbf{E}_h$ be an $\eta_{Q,k}$ -evolution k -vector field associated with $h \in C^\infty(\mathcal{J}_{Q,k})$, and let $\gamma: Q \rightarrow \mathcal{J}_{Q,k}$ be a holonomic section. Assume that the projected k -vector field $\mathbf{E}_h^\gamma$ is integrable. Then the following statements are equivalent:*

- (a) *Every integral section σ of $\mathbf{E}_h^\gamma$ lifts via γ to a solution $\gamma \circ \sigma$ of the z -independent $\eta_{Q,k}$ -evolution HdDW equations for h ;*
- (b) $d(h \circ \gamma) = 0$.

Proof. We first prove that (a) implies (b). Let σ be an integral section of $\mathbf{E}_h^\gamma$ and suppose that

$$(\gamma \circ \sigma)(t) = (\sigma^i(t), \gamma_i^\alpha(\sigma(t)), \gamma^\alpha(\sigma(t)))$$

is a solution of the z -independent $\eta_{Q,k}$ -evolution HdDW equations. Then,

$$\begin{cases} \left. \frac{\partial \sigma^i}{\partial t^\beta} \right|_t = \left. \frac{\partial h}{\partial p_i^\beta} \right|_{\gamma \circ \sigma(t)}, & i = 1, \dots, n, \quad \beta = 1, \dots, k, \\ \sum_{\alpha=1}^k \left. \frac{\partial(\gamma_i^\alpha \circ \sigma)}{\partial t^\alpha} \right|_t = - \left(\left. \frac{\partial h}{\partial q^i} \right|_{\gamma \circ \sigma(t)} + \sum_{\alpha=1}^k (\gamma_i^\alpha \circ \sigma)(t) \left. \frac{\partial h}{\partial z^\alpha} \right|_{\gamma \circ \sigma(t)} \right), & i = 1, \dots, n, \\ \sum_{\alpha=1}^k \left. \frac{\partial(\gamma^\alpha \circ \sigma)}{\partial t^\alpha} \right|_t = \sum_{\alpha=1}^k \sum_{i=1}^n (\gamma_i^\alpha \circ \sigma)(t) \left. \frac{\partial h}{\partial p_i^\alpha} \right|_{\gamma \circ \sigma(t)}. \end{cases}$$

On the other hand, the differential of $h \circ \gamma$ is given locally by

$$d(h \circ \gamma) = \sum_{i=1}^n \left(\frac{\partial h}{\partial q^i} \circ \gamma + \sum_{\alpha=1}^k \sum_{j=1}^n \left(\frac{\partial h}{\partial p_j^\alpha} \circ \gamma \right) \frac{\partial \gamma_j^\alpha}{\partial q^i} + \sum_{\alpha=1}^k \left(\frac{\partial h}{\partial z^\alpha} \circ \gamma \right) \frac{\partial \gamma^\alpha}{\partial q^i} \right) dq^i.$$

Since γ is holonomic,

$$\frac{\partial \gamma^\alpha}{\partial q^i} = \gamma_i^\alpha, \quad \frac{\partial \gamma_j^\alpha}{\partial q^j} = \frac{\partial \gamma_j^\alpha}{\partial q^i}, \quad i, j = 1, \dots, n, \quad \alpha = 1, \dots, k.$$

Using these identities together with the first two families of the evolution HdDW equations, we obtain

$$\begin{aligned} d(h \circ \gamma)|_{\sigma(t)} &= \sum_{i=1}^n \left(- \sum_{\alpha=1}^k (\gamma_i^\alpha \circ \sigma)(t) \left. \frac{\partial h}{\partial z^\alpha} \right|_{\gamma \circ \sigma(t)} - \sum_{\alpha=1}^k \left. \frac{\partial(\gamma_i^\alpha \circ \sigma)}{\partial t^\alpha} \right|_t \right. \\ &\quad \left. + \sum_{\alpha=1}^k \sum_{j=1}^n \left. \frac{\partial \sigma^j}{\partial t^\alpha} \right|_t \left. \frac{\partial \gamma_j^\alpha}{\partial q^j} \right|_{\sigma(t)} + \sum_{\alpha=1}^k \left. \frac{\partial h}{\partial z^\alpha} \right|_{\gamma \circ \sigma(t)} \left. \frac{\partial \gamma^\alpha}{\partial q^i} \right|_{\sigma(t)} \right) dq^i(\sigma(t)) \\ &= \sum_{i=1}^n \sum_{\alpha=1}^k \left(-(\gamma_i^\alpha \circ \sigma)(t) + \left. \frac{\partial \gamma^\alpha}{\partial q^i} \right|_{\sigma(t)} \right) \left. \frac{\partial h}{\partial z^\alpha} \right|_{\gamma \circ \sigma(t)} dq^i(\sigma(t)) = 0. \end{aligned}$$

Thus, $d(h \circ \gamma)(\sigma(t)) = 0$. Since $\mathbf{E}_h^\gamma$ is integrable, for every $q \in Q$ there exists a local integral section σ such that $\sigma(t_0) = q$ for some t_0 . Hence $d(h \circ \gamma)(q) = 0$ for every $q \in Q$, and therefore $d(h \circ \gamma) = 0$. This proves (b).

We now prove the converse. Assume that $d(h \circ \gamma) = 0$, and let σ be an integral section of $\mathbf{E}_h^\gamma$. We show that $\gamma \circ \sigma$ satisfies the z -independent $\eta_{Q,k}$ -evolution HdDW equations.

The condition $d(h \circ \gamma) = 0$ reads locally

$$0 = \sum_{i=1}^n \left(\frac{\partial h}{\partial q^i} \circ \gamma + \sum_{\alpha=1}^k \sum_{j=1}^n \left(\frac{\partial h}{\partial p_j^\alpha} \circ \gamma \right) \frac{\partial \gamma_j^\alpha}{\partial q^i} + \sum_{\alpha=1}^k \left(\frac{\partial h}{\partial z^\alpha} \circ \gamma \right) \frac{\partial \gamma^\alpha}{\partial q^i} \right) dq^i. \quad (3.2.7)$$

By the local expression of an $\eta_{Q,k}$ -evolution k -vector field,

$$(\mathbf{E}_h^\gamma)_\alpha = \sum_{i=1}^n \left(\frac{\partial h}{\partial p_i^\alpha} \circ \gamma \right) \frac{\partial}{\partial q^i}, \quad \alpha = 1, \dots, k.$$

Since σ is an integral section of $\mathbf{E}_h^\gamma$, it follows that

$$\left. \frac{\partial \sigma^i}{\partial t^\alpha} \right|_t = \left. \frac{\partial h}{\partial p_i^\alpha} \circ \gamma \circ \sigma(t) \right|_t, \quad i = 1, \dots, n, \quad \alpha = 1, \dots, k. \quad (3.2.8)$$

Thus, the first family of the evolution HdDW equations is satisfied.

Using holonomicity of γ , (3.2.7), and (3.2.8), we obtain

$$\begin{aligned} \sum_{\alpha=1}^k \left. \frac{\partial(\gamma_i^\alpha \circ \sigma)}{\partial t^\alpha} \right|_t &= \sum_{\alpha=1}^k \sum_{j=1}^n \left. \frac{\partial \gamma_i^\alpha}{\partial q^j} \right|_{\sigma(t)} \left. \frac{\partial \sigma^j}{\partial t^\alpha} \right|_t = \sum_{\alpha=1}^k \sum_{j=1}^n \left. \frac{\partial \gamma_j^\alpha}{\partial q^i} \right|_{\sigma(t)} \left. \frac{\partial h}{\partial p_j^\alpha} \right|_{\gamma \circ \sigma(t)} \\ &= - \left. \frac{\partial h}{\partial q^i} \right|_{\gamma \circ \sigma(t)} - \sum_{\alpha=1}^k \left. \frac{\partial \gamma^\alpha}{\partial q^i} \right|_{\sigma(t)} \left. \frac{\partial h}{\partial z^\alpha} \right|_{\gamma \circ \sigma(t)} \\ &= - \left. \frac{\partial h}{\partial q^i} \right|_{\gamma \circ \sigma(t)} - \sum_{\alpha=1}^k (\gamma_i^\alpha \circ \sigma)(t) \left. \frac{\partial h}{\partial z^\alpha} \right|_{\gamma \circ \sigma(t)}. \end{aligned}$$

This proves the second family of the evolution HdDW equations.

Finally, by holonomicity and (3.2.8),

$$\begin{aligned} \sum_{\alpha=1}^k \left. \frac{\partial(\gamma^\alpha \circ \sigma)}{\partial t^\alpha} \right|_t &= \sum_{\alpha=1}^k \sum_{i=1}^n \left. \frac{\partial \gamma^\alpha}{\partial q^i} \right|_{\sigma(t)} \left. \frac{\partial \sigma^i}{\partial t^\alpha} \right|_t \\ &= \sum_{\alpha=1}^k \sum_{i=1}^n (\gamma_i^\alpha \circ \sigma)(t) \left. \frac{\partial h}{\partial p_i^\alpha} \right|_{\gamma \circ \sigma(t)}, \end{aligned}$$

which is precisely the third family. Hence $\gamma \circ \sigma$ is a solution of the z -independent $\eta_{Q,k}$ -evolution HdDW equations. This proves (a) and completes the proof. $\square$

Remark 3.2.6. For $k = 1$, one has $\ker \chi = 0$, and therefore the almost γ -relation reduces to the usual γ -relation. In particular, along $\text{Im } \gamma$, the lift of $\mathbf{E}_h^\gamma$ coincides with $\mathbf{E}_h$.

The evolution Hamilton–Jacobi equation $d(h \circ \gamma) = 0$ differs from the equation $h \circ \gamma = 0$ obtained in the Hamiltonian formulation of the preceding section. This distinction admits a natural geometric interpretation through the relation with the classical k -symplectic Hamilton–Jacobi theory.

Indeed, let $(P, \omega = \sum_{\alpha=1}^k \omega^\alpha \otimes e_\alpha)$ be an exact k -symplectic manifold, with $\omega = -d\theta$. Its k -contactification is $P \times \mathbb{R}^k$, equipped with the k -contact form $\eta_\theta = \sum_{\alpha=1}^k (dz^\alpha - \rho^* \theta^\alpha) \otimes e_\alpha$, where $\rho: P \times \mathbb{R}^k \rightarrow P$ denotes the canonical projection. Given $H \in C^\infty(P)$ and a k -symplectic Hamiltonian k -vector field $\mathbf{X}_H$ associated with H , the function $h := H \circ \rho$ admits a canonically associated η_θ -evolution k -vector field on $P \times \mathbb{R}^k$ whose projection onto P is $\mathbf{X}_H$.

In the canonical case $P = \bigoplus^k T^*Q$, let $\gamma_0: Q \rightarrow \bigoplus^k T^*Q$ be a section whose components γ_0^α are closed one-forms. Locally, there exist functions $S^\alpha \in C^\infty(Q)$ such that

$$\gamma_0^\alpha = dS^\alpha, \quad \alpha = 1, \dots, k.$$

These functions determine a holonomic section

$$\hat{\gamma}: Q \longrightarrow \mathcal{J}_{Q,k}, \quad \hat{\gamma}(q) = (q, \gamma_0^\alpha(q), S^\alpha(q)).$$

Since $h = H \circ \rho$ and $\rho \circ \hat{\gamma} = \gamma_0$, it follows that $h \circ \hat{\gamma} = H \circ \gamma_0$. Consequently,

$$d(h \circ \hat{\gamma}) = 0 \quad \Longleftrightarrow \quad d(H \circ \gamma_0) = 0,$$

which is precisely the classical k -symplectic Hamilton–Jacobi equation.

Thus, under k -contactification, the z -independent evolution Hamilton–Jacobi equation recovers the classical k -symplectic Hamilton–Jacobi equation. Moreover, the corresponding projected k -vector fields on Q coincide, and hence their integrability requirements agree as well. This provides a geometric explanation for the difference between the Hamilton–Jacobi equations in Theorems 3.2.5 and 3.2.4, and shows why $d(h \circ \gamma) = 0$ is the natural Hamilton–Jacobi condition for $\eta_{Q,k}$ -evolution k -vector fields.

3.3 The k -contact Hamilton–Jacobi equations in the z -dependent framework

We now turn to a z -dependent version of the k -contact Hamilton–Jacobi theory. The main difference with the previous setting is that the sections under consideration are defined over $Q \times \mathbb{R}^k$, rather than over Q . This requires a corresponding modification of the notion of almost γ -relatedness.

Let $(\mathcal{J}_{Q,k} = \bigoplus^k T^*Q \times \mathbb{R}^k, \eta_{Q,k}, h)$ be a k -contact Hamiltonian system, and consider a section of the canonical projection $\pi_{Q \times \mathbb{R}^k} : \mathcal{J}_{Q,k} \longrightarrow Q \times \mathbb{R}^k$ of the form

$$\gamma : Q \times \mathbb{R}^k \longrightarrow \mathcal{J}_{Q,k}, \quad (q^i, z^\alpha) \longmapsto (q^i, \gamma_i^\alpha(q, z), z^\alpha).$$

Let $\mathbf{Z} = (Z_1, \dots, Z_k)$ be a k -vector field on $\mathcal{J}_{Q,k}$, locally written as

$$Z_\alpha = \sum_{i=1}^n (Z_\alpha)^i \frac{\partial}{\partial q^i} + \sum_{\beta=1}^k \sum_{i=1}^n (Z_\alpha)^\beta_i \frac{\partial}{\partial p_i^\beta} + \sum_{\beta=1}^k (Z_\alpha)^\beta \frac{\partial}{\partial z^\beta}, \quad \alpha = 1, \dots, k.$$

Its projection along γ is the k -vector field on $Q \times \mathbb{R}^k$ defined by

$$\mathbf{Z}^\gamma := \left(\bigoplus^k T\pi_{Q \times \mathbb{R}^k} \right) \circ (\mathbf{Z} \circ \gamma).$$

In adapted coordinates, its components are

$$Z_\alpha^\gamma = \sum_{i=1}^n (Z_\alpha)^i \circ \gamma \frac{\partial}{\partial q^i} + \sum_{\beta=1}^k (Z_\alpha)^\beta \circ \gamma \frac{\partial}{\partial z^\beta}, \quad \alpha = 1, \dots, k.$$

Recall that $\mathbf{Z}$ and $\mathbf{Z}^\gamma$ are said to be γ -related whenever

$$\left(\bigoplus^k T\gamma \right) \circ \mathbf{Z}^\gamma = \mathbf{Z} \circ \gamma.$$

In the z -independent theory, the ambiguity in the choice of a Hamiltonian k -vector field disappears after projection onto Q , since $\ker \chi$ is contained in the kernel of the corresponding tangent projection. This is no longer true in the z -dependent setting. Indeed, in general, $\ker \chi \not\subset \ker \left(\bigoplus^k T\pi_{Q \times \mathbb{R}^k} \right)$. Consequently, two representatives of the same Hamiltonian class may induce different k -vector fields on $Q \times \mathbb{R}^k$. Nevertheless, their projections differ by the image of $\ker \chi$. This leads naturally to a quotient description of the projected dynamics. Let

$$[\cdot]_\chi : \bigoplus^k T\mathcal{J}_{Q,k} \longrightarrow \bigoplus^k T\mathcal{J}_{Q,k} / \ker \chi$$

denote the canonical quotient projection, and set

$$\mathcal{K} := \left(\bigoplus^k T\pi_{Q \times \mathbb{R}^k} \right) (\ker \chi) \subset \bigoplus^k T(Q \times \mathbb{R}^k).$$

We denote the corresponding quotient projection by

$$[\cdot]_\mathcal{K} : \bigoplus^k T(Q \times \mathbb{R}^k) \longrightarrow \bigoplus^k T(Q \times \mathbb{R}^k) / \mathcal{K}.$$

The projection $\bigoplus^k T\pi_{Q \times \mathbb{R}^k}$ induces a well-defined map between the corresponding quotient bundles,

$$\bigoplus^k \widetilde{T\pi_{Q \times \mathbb{R}^k}} : \bigoplus^k T\mathcal{J}_{Q,k} / \ker \chi \longrightarrow \bigoplus^k T(Q \times \mathbb{R}^k) / \mathcal{K},$$

defined by

$$\bigoplus^k \widetilde{T\pi_{Q \times \mathbb{R}^k}}([Z]_\chi) = \left[\left(\bigoplus^k T\pi_{Q \times \mathbb{R}^k} \right) (Z) \right]_{\mathcal{K}}.$$

Indeed, replacing Z by another representative differing from it by a section of $\ker \chi$ changes its projection by a section of $\mathcal{K}$, and therefore does not change its class in the quotient. Accordingly, the projected class of Z along γ is defined by

$$[Z^\gamma]_{\mathcal{K}} := \bigoplus^k \widetilde{T\pi_{Q \times \mathbb{R}^k}} \circ [Z]_\chi \circ \gamma.$$

Definition 3.3.1. Let $[Z]_\chi$ be a class of k -vector fields on $\mathcal{J}_{Q,k}$, and let $[Z^\gamma]_{\mathcal{K}}$ be its projected class along γ . We say that these classes are *almost γ -related in the z -dependent sense* if there exist representatives $\bar{Z} \in [Z]_\chi$ and $\bar{Z}^\gamma \in [Z^\gamma]_{\mathcal{K}}$, such that

$$[\bar{Z} \circ \gamma]_\chi = \left[\left(\bigoplus^k T\gamma \right) \circ \bar{Z}^\gamma \right]_\chi.$$

Thus, almost γ -relatedness requires the lift of a suitable representative of the projected class to agree with a representative of the original dynamics along γ , up to a section taking values in $\ker \chi$. This extends the corresponding notion from the z -independent setting while taking into account that $\ker \chi$ is no longer contained in the kernel of $\bigoplus^k T\pi_{Q \times \mathbb{R}^k}$.

Equivalently, $[Z]_\chi$ and $[Z^\gamma]_{\mathcal{K}}$ are almost γ -related in the z -dependent sense if there exist representatives $\bar{Z} \in [Z]_\chi$ and $\bar{Z}^\gamma \in [Z^\gamma]_{\mathcal{K}}$ for which the following diagram commutes:

$$\begin{array}{ccccc} \mathcal{J}_{Q,k} & \xrightarrow{\bar{Z}} & \bigoplus^k T\mathcal{J}_{Q,k} & \xrightarrow{[\cdot]_\chi} & \bigoplus^k T\mathcal{J}_{Q,k} / \ker \chi \\ \uparrow \gamma & & \downarrow \bigoplus^k T\pi_{Q \times \mathbb{R}^k} & \nearrow [\bigoplus^k T\gamma]_\chi & \downarrow \bigoplus^k \widetilde{T\pi_{Q \times \mathbb{R}^k}} \\ Q \times \mathbb{R}^k & \xrightarrow{\bar{Z}^\gamma} & \bigoplus^k T(Q \times \mathbb{R}^k) & \xrightarrow{[\cdot]_{\mathcal{K}}} & \bigoplus^k T(Q \times \mathbb{R}^k) / \mathcal{K} \end{array}$$

To formulate a z -dependent Hamilton–Jacobi theorem, it remains to identify suitable geometric conditions on the section γ . These conditions play the role of holonomicity in the z -independent theory and will allow us to relate integral sections of the projected dynamics on $Q \times \mathbb{R}^k$ to solutions of the k -contact HddW equations.

Lemma 3.3.2. Let $(\mathcal{J}_{Q,k} = \bigoplus^k T^*Q \times \mathbb{R}^k, \eta_{Q,k})$ be the canonical k -contact manifold, and let $\gamma: Q \times \mathbb{R}^k \rightarrow \mathcal{J}_{Q,k}$, $\gamma(q, z) = (q, \gamma_i^\alpha(q, z), z)$, be a section of the projection $\pi_{Q \times \mathbb{R}^k}: \mathcal{J}_{Q,k} \rightarrow Q \times \mathbb{R}^k$. Then $\text{Im } \gamma$ is maximally coisotropic if and only if

$$\frac{\partial \gamma_j^\alpha}{\partial q^i} + \sum_{\beta=1}^k \gamma_j^\beta \frac{\partial \gamma_j^\alpha}{\partial z^\beta} = \frac{\partial \gamma_i^\alpha}{\partial q^j} + \sum_{\beta=1}^k \gamma_j^\beta \frac{\partial \gamma_i^\alpha}{\partial z^\beta}, \quad i, j = 1, \dots, n, \quad \alpha = 1, \dots, k. \quad (3.3.1)$$

Moreover, for every fixed $z \in \mathbb{R}^k$, consider the section $\gamma_z: Q \rightarrow \bigoplus^k T^*Q$, $\gamma_z(q) = (q, \gamma_i^\alpha(q, z))$. Its image is isotropic with respect to the canonical k -symplectic structure $\omega_{Q,k}$ on $\bigoplus^k T^*Q$ if and only if

$$\frac{\partial \gamma_j^\alpha}{\partial q^i} = \frac{\partial \gamma_i^\alpha}{\partial q^j}, \quad i, j = 1, \dots, n, \quad \alpha = 1, \dots, k. \quad (3.3.2)$$

Proof. We begin by determining the subspace of $T \text{Im } \gamma$ contained in the contact distribution. In adapted Darboux coordinates $\mathcal{J}_{Q,k}$, one has

$$T \text{Im } \gamma = \left\langle \frac{\partial}{\partial q^i} + \sum_{\alpha=1}^k \sum_{j=1}^n \frac{\partial \gamma_j^\alpha}{\partial q^i} \frac{\partial}{\partial p_j^\alpha}, \sum_{\alpha=1}^k \sum_{j=1}^n \frac{\partial \gamma_j^\alpha}{\partial z^\beta} \frac{\partial}{\partial p_j^\alpha} + \frac{\partial}{\partial z^\beta} \right\rangle_{\substack{i=1, \dots, n \\ \beta=1, \dots, k}}.$$

It follows that $\text{T Im } \gamma \cap \ker \eta_{Q,k}$ is locally spanned by the linearly independent vector fields

$$X_i = \frac{\partial}{\partial q^i} + \sum_{\alpha=1}^k \gamma_i^\alpha \frac{\partial}{\partial z^\alpha} + \sum_{\alpha=1}^k \sum_{j=1}^n \left(\frac{\partial \gamma_j^\alpha}{\partial q^i} + \sum_{\beta=1}^k \gamma_i^\beta \frac{\partial \gamma_j^\alpha}{\partial z^\beta} \right) \frac{\partial}{\partial p_j^\alpha}, \quad i = 1, \dots, n.$$

For convenience, set

$$A_{ij}^\alpha := \frac{\partial \gamma_j^\alpha}{\partial q^i} + \sum_{\beta=1}^k \gamma_i^\beta \frac{\partial \gamma_j^\alpha}{\partial z^\beta}.$$

Fix $x \in \text{Im } \gamma$ and let $v \in \ker(\eta_{Q,k})_x$. In what follows, vector fields, differential forms, and functions appearing in pointwise expressions are understood to be evaluated at x , unless otherwise indicated. Since the vectors $X_1, \dots, X_n$ and $\frac{\partial}{\partial p_j^\alpha}$ for $\alpha = 1, \dots, k$ and $j = 1, \dots, n$ form a basis of $\ker(\eta_{Q,k})_x$, there is a unique decomposition $v = \sum_{m=1}^n a^m X_m + \sum_{\alpha=1}^k \sum_{j=1}^n u_j^\alpha \frac{\partial}{\partial p_j^\alpha}$ for suitable coefficients a^m and u_j^α . Using $d\eta_{Q,k}^\alpha = \sum_{r=1}^n dq^r \wedge dp_r^\alpha$, we obtain

$$d\eta_{Q,k}^\alpha(X_m, X_i) = A_{im}^\alpha - A_{mi}^\alpha, \quad d\eta_{Q,k}^\alpha\left(\frac{\partial}{\partial p_j^\beta}, X_i\right) = -\delta_\beta^\alpha \delta_{ij}.$$

Consequently, the conditions $d\eta_{Q,k}^\alpha(v, X_i) = 0$ for $\alpha = 1, \dots, k$ and $i = 1, \dots, n$ are equivalent to

$$u_i^\alpha = \sum_{m=1}^n a^m (A_{im}^\alpha - A_{mi}^\alpha).$$

By Definition 1.2.28, these equations characterise $(\text{T Im } \gamma \cap \ker \eta_{Q,k})_x^{\perp \eta_{Q,k}}$. This orthogonal space is contained in $(\text{T Im } \gamma \cap \ker \eta_{Q,k})_x$ precisely when $u_i^\alpha = 0$ for every choice of the coefficients a^m . Equivalently,

$$A_{ij}^\alpha = A_{ji}^\alpha, \quad \alpha = 1, \dots, k, \quad i, j = 1, \dots, n.$$

Since both $(\text{T Im } \gamma \cap \ker \eta_{Q,k})_x^{\perp \eta_{Q,k}}$ and $(\text{T Im } \gamma \cap \ker \eta_{Q,k})_x$ have dimension n , this inclusion is then necessarily an equality. Thus, $\text{Im } \gamma$ is maximally coisotropic if and only if (3.3.1) holds.

For the second assertion, fix $z \in \mathbb{R}^k$. The pull-back of the canonical k -symplectic forms by γ_z is

$$\gamma_z^* \omega_Q^\alpha = \sum_{1 \leq i < j \leq n} \left(\frac{\partial \gamma_i^\alpha}{\partial q^j}(q, z) - \frac{\partial \gamma_j^\alpha}{\partial q^i}(q, z) \right) dq^i \wedge dq^j, \quad \alpha = 1, \dots, k.$$

Hence, $\gamma_z^* \omega_Q^\alpha = 0$ for $\alpha = 1, \dots, k$ if and only if (3.3.2) is satisfied. This is precisely the condition for $\text{Im } \gamma_z$ to be isotropic with respect to $\omega_{Q,k}$, which completes the proof $\square$

In the contact case $k = 1$, the coisotropy condition used in the z -dependent contact Hamilton–Jacobi theory turns out to be equivalent to the maximal coisotropy condition adopted here. The following lemma makes this correspondence precise.

Lemma 3.3.3. *Let $\gamma: Q \times \mathbb{R} \rightarrow \mathcal{J}_{Q,1} = \text{T}^*Q \times \mathbb{R}$, $\gamma(q, z) = (q, \gamma_i(q, z), z)$, be a section of $\pi_{Q \times \mathbb{R}}: \mathcal{J}_{Q,1} \rightarrow Q \times \mathbb{R}$. Then, the restriction of the contact form $\eta_{Q,1}$ to $\text{T Im } \gamma$ is nowhere vanishing. Consequently, $\text{Im } \gamma$ is coisotropic if and only if it is maximally coisotropic.*

Proof. Let (q^i, p_i, z) be adapted Darboux coordinates on $\mathcal{J}_{Q,1}$, so that $\eta_{Q,1} = dz - \theta_Q = dz - p_i dq^i$. Since $\gamma(q, z) = (q, \gamma_i(q, z), z)$, we have

$$\text{T}\gamma\left(\frac{\partial}{\partial z}\right) = \frac{\partial}{\partial z} + \sum_{i=1}^n \frac{\partial \gamma_i}{\partial z} \frac{\partial}{\partial p_i}.$$

Therefore,

$$\eta_{Q,1}\left(\text{T}\gamma\left(\frac{\partial}{\partial z}\right)\right) = 1.$$

Thus, $\eta_{Q,1}|_{\text{T Im } \gamma}$ is nowhere vanishing, and Lemma 1.2.29 implies that $\text{Im } \gamma$ is coisotropic if and only if it is maximally coisotropic. $\square$

Condition (3.3.2) also admits a useful local interpretation. For each fixed $z \in \mathbb{R}^k$, it states that the one-forms

$$\gamma_z^\alpha = \sum_{i=1}^n \gamma_i^\alpha(q, z) dq^i, \quad \alpha = 1, \dots, k,$$

are closed. Hence, by the Poincaré lemma, locally there exist functions $W^\alpha(q, z)$ such that

$$\gamma_i^\alpha(q, z) = \frac{\partial W^\alpha}{\partial q^i}(q, z), \quad \alpha = 1, \dots, k, \quad i = 1, \dots, n.$$

In the z -dependent setting, it will be convenient to distinguish differentiation with respect to the variables on Q from differentiation with respect to the dissipative variables z^α . We therefore introduce an exterior derivative along the Q -variables, treating $z \in \mathbb{R}^k$ as a fixed parameter. This construction extends the corresponding definition given in [61] in the contact case and will be used to formulate the z -dependent k -contact Hamilton–Jacobi equations.

Definition 3.3.4. Let $\alpha: Q \times \mathbb{R}^k \rightarrow \bigwedge^\ell T^*Q \times \mathbb{R}^k$ be a section. For each fixed $z \in \mathbb{R}^k$, define the ℓ -form

$$\alpha_z: Q \rightarrow \bigwedge^\ell T^*Q, \quad q \mapsto \text{pr}_{\bigwedge^\ell T^*Q}(\alpha(q, z)),$$

where $\text{pr}_{\bigwedge^\ell T^*Q}$ denotes the canonical projection. The *exterior derivative of α at fixed z* is the section $d_Q \alpha: Q \times \mathbb{R}^k \rightarrow \bigwedge^{\ell+1} T^*Q \times \mathbb{R}^k$ defined by

$$d_Q \alpha(q, z) = (d\alpha_z(q), z).$$

3.3.1 The Hamiltonian approach

We now develop the z -dependent Hamilton–Jacobi theory for $\eta_{Q,k}$ -Hamiltonian k -vector fields. Unlike in the z -independent setting, the freedom associated with $\ker \chi$ is not eliminated by projection onto $Q \times \mathbb{R}^k$. Hence, different representatives associated with the same Hamiltonian h may give rise to different projected k -vector fields on $Q \times \mathbb{R}^k$, although they determine the same z -dependent HdDW equations. It is therefore natural to formulate the Hamilton–Jacobi problem in terms of the corresponding equivalence classes, rather than by requiring a particular projected k -vector field to lift exactly to $\mathbf{X}_h$ along $\text{Im } \gamma$. The following theorem gives this formulation for maximally coisotropic sections and identifies the resulting z -dependent Hamilton–Jacobi equation.

Theorem 3.3.5 (z -dependent k -contact Hamilton–Jacobi theorem). *Let $(\mathcal{J}_{Q,k} = \bigoplus^k T^*Q \times \mathbb{R}^k, \eta_{Q,k}, h)$ be a k -contact Hamiltonian system, and let $\gamma: Q \times \mathbb{R}^k \rightarrow \mathcal{J}_{Q,k}$ be a section of the canonical projection $\pi_{Q \times \mathbb{R}^k}: \mathcal{J}_{Q,k} \rightarrow Q \times \mathbb{R}^k$ such that $\text{Im } \gamma$ is maximally coisotropic. Denote by $[\mathbf{X}_h]_\chi$ the equivalence class, modulo $\ker \chi$, of $\eta_{Q,k}$ -Hamiltonian k -vector fields associated with h . Moreover, set $\mathcal{K} := \left(\bigoplus^k T\pi_{Q \times \mathbb{R}^k} \right) (\ker \chi)$ and denote by $[\mathbf{X}_h^\gamma]_{\mathcal{K}}$ the corresponding projected class along γ . Then the following statements are equivalent:*

- (a) *There exist representatives $\mathbf{X} \in [\mathbf{X}_h]_\chi$, $\overline{\mathbf{X}}^\gamma \in [\mathbf{X}_h^\gamma]_{\mathcal{K}}$, such that $\left(\bigoplus^k T\gamma \right) (\overline{\mathbf{X}}^\gamma)$ and $\mathbf{X} \circ \gamma$ determine the same z -dependent HdDW equations associated with h ;*
- (b) *There exists a smooth matrix-valued function $C = (C_\alpha^\beta) \in C^\infty(Q \times \mathbb{R}^k, \text{Mat}_{k \times k}(\mathbb{R}))$ satisfying $\text{tr } C = \sum_{\alpha=1}^k C_\alpha^\alpha = -h \circ \gamma$ and such that*

$$d_Q(h \circ \gamma) + \sum_{\beta=1}^k \Gamma_\beta \gamma^* \theta^\beta + \sum_{\alpha, \beta=1}^k C_\alpha^\beta \iota_{\partial/\partial z^\beta} d(\gamma^* \theta^\alpha) = 0, \quad (3.3.3)$$

where $\Gamma_\beta := \frac{\partial(h \circ \gamma)}{\partial z^\beta} = dh|_\gamma \left(T\gamma \left(\frac{\partial}{\partial z^\beta} \right) \right)$ for $\beta = 1, \dots, k$.

Proof. Choose adapted Darboux coordinates $(q^i, p_i^\alpha, z^\alpha)$ on $\mathcal{J}_{Q,k}$ and write $\gamma(q, z) = (q, \gamma_i^\alpha(q, z), z)$. For convenience, set $U_\alpha^i := (\partial h / \partial p_i^\alpha) \circ \gamma$. Moreover, by the chain rule,

$$\Gamma_\beta = \frac{\partial(h \circ \gamma)}{\partial z^\beta} = \left(\frac{\partial h}{\partial z^\beta} \right) \circ \gamma + \sum_{\alpha=1}^k \sum_{i=1}^n U_\alpha^i \frac{\partial \gamma_i^\alpha}{\partial z^\beta}, \quad \beta = 1, \dots, k.$$

Taking into account the gauge freedom induced by $\ker \chi$ and its projection onto $Q \times \mathbb{R}^k$, a representative $\bar{\mathbf{X}}^\gamma \in [\mathbf{X}_h^\gamma]_{\mathcal{K}}$ can be written locally as

$$\bar{X}_\alpha^\gamma = \sum_{i=1}^n U_\alpha^i \frac{\partial}{\partial q^i} + \sum_{\beta=1}^k \left(\sum_{j=1}^n \gamma_j^\beta U_\alpha^j + C_\alpha^\beta \right) \frac{\partial}{\partial z^\beta}, \quad \alpha = 1, \dots, k,$$

where $C = (C_\alpha^\beta)$ satisfies $\text{tr } C = \sum_{\alpha=1}^k C_\alpha^\alpha = -(h \circ \gamma)$. This trace condition corresponds to the z -equation $\sum_{\alpha=1}^k \eta_{Q,k}^\alpha(X_\alpha) = -h$.

Assume first that (a) holds. Applying $T\gamma$ to the components of $\bar{\mathbf{X}}^\gamma$, we obtain

$$T\gamma(\bar{X}_\alpha^\gamma) = \sum_{i=1}^n U_\alpha^i T\gamma \left(\frac{\partial}{\partial q^i} \right) + \sum_{\beta=1}^k \left(\sum_{j=1}^n \gamma_j^\beta U_\alpha^j + C_\alpha^\beta \right) T\gamma \left(\frac{\partial}{\partial z^\beta} \right).$$

Consequently, the p_j^α -components satisfy

$$\sum_{\alpha=1}^k (T\gamma(\bar{X}_\alpha^\gamma))_{p_j^\alpha} = \sum_{\alpha=1}^k \sum_{i=1}^n U_\alpha^i \frac{\partial \gamma_j^\alpha}{\partial q^i} + \sum_{\alpha, \beta=1}^k \left(\sum_{i=1}^n \gamma_i^\beta U_\alpha^i + C_\alpha^\beta \right) \frac{\partial \gamma_j^\alpha}{\partial z^\beta}. \quad (3.3.4)$$

By (a), there exists $\mathbf{X} \in [\mathbf{X}_h]_{\mathcal{X}}$ such that $(\bigoplus^k T\gamma)(\bar{\mathbf{X}}^\gamma)$ and $\mathbf{X} \circ \gamma$ determine the same z -dependent HddW equations. In particular,

$$\sum_{\alpha=1}^k (T\gamma(\bar{X}_\alpha^\gamma))_{p_j^\alpha} = -\frac{\partial h}{\partial q^j} \circ \gamma - \sum_{\alpha=1}^k \gamma_j^\alpha \frac{\partial h}{\partial z^\alpha} \circ \gamma, \quad j = 1, \dots, n. \quad (3.3.5)$$

Since $\text{Im } \gamma$ is maximally coisotropic, Lemma 3.3.2 gives (3.3.1), namely

$$\frac{\partial \gamma_j^\alpha}{\partial q^i} + \sum_{\beta=1}^k \gamma_i^\beta \frac{\partial \gamma_j^\alpha}{\partial z^\beta} = \frac{\partial \gamma_i^\alpha}{\partial q^j} + \sum_{\beta=1}^k \gamma_j^\beta \frac{\partial \gamma_i^\alpha}{\partial z^\beta}, \quad \alpha = 1, \dots, k, \quad i, j = 1, \dots, n.$$

Using these identities, the terms in (3.3.4) involving U_α^i can be rewritten as

$$\sum_{\alpha=1}^k \sum_{i=1}^n U_\alpha^i \frac{\partial \gamma_i^\alpha}{\partial q^j} + \sum_{\beta=1}^k \gamma_j^\beta \sum_{\alpha=1}^k \sum_{i=1}^n U_\alpha^i \frac{\partial \gamma_i^\alpha}{\partial z^\beta}.$$

Combining this expression with (3.3.5) and using the definition of Γ_β , we obtain

$$\frac{\partial(h \circ \gamma)}{\partial q^j} + \sum_{\beta=1}^k \Gamma_\beta \gamma_j^\beta + \sum_{\alpha, \beta=1}^k C_\alpha^\beta \frac{\partial \gamma_j^\alpha}{\partial z^\beta} = 0, \quad j = 1, \dots, n.$$

On the other hand, $\gamma^* \theta^\alpha = \sum_{i=1}^n \gamma_i^\alpha dq^i$ for $\alpha = 1, \dots, k$, and hence

$$\iota_{\partial/\partial z^\beta} d(\gamma^* \theta^\alpha) = \sum_{i=1}^n \frac{\partial \gamma_i^\alpha}{\partial z^\beta} dq^i.$$

Therefore, the preceding system is precisely (3.3.3). This proves (b).

Conversely, suppose that (b) holds, and choose a matrix $C = (C_\alpha^\beta)$ satisfying $\text{tr } C = -(h \circ \gamma)$ and (3.3.3). Define a representative of the projected class $[\mathbf{X}_h^\gamma]_{\mathcal{K}}$ by

$$\overline{X}_\alpha^\gamma = \sum_{i=1}^n U_\alpha^i \frac{\partial}{\partial q^i} + \sum_{\beta=1}^k \left(\sum_{j=1}^n \gamma_j^\beta U_\alpha^j + C_\alpha^\beta \right) \frac{\partial}{\partial z^\beta}, \quad \alpha = 1, \dots, k.$$

Its q -components are, by construction, $U_\alpha^i = (\partial h / \partial p_i^\alpha) \circ \gamma$, and therefore satisfy the corresponding q -equations. Furthermore,

$$\sum_{\alpha=1}^k \eta_{Q,k}^\alpha \left(T\gamma(\overline{X}_\alpha^\gamma) \right) = \sum_{\alpha=1}^k C_\alpha^\alpha = -(h \circ \gamma),$$

so the z -equation is satisfied as well.

Finally, using (3.3.3) together with the maximal coisotropy condition (3.3.1) and reversing the computation above, we obtain

$$\sum_{\alpha=1}^k (T\gamma(\overline{X}_\alpha^\gamma))_{p_j^\alpha} = -\frac{\partial h}{\partial q^j} \circ \gamma - \sum_{\alpha=1}^k \gamma_j^\alpha \frac{\partial h}{\partial z^\alpha} \circ \gamma, \quad j = 1, \dots, n.$$

Thus, $(\bigoplus^k T\gamma)(\overline{\mathbf{X}}^\gamma)$ satisfies the same z -dependent HddW equations along $\text{Im } \gamma$ as a representative $\mathbf{X} \in [\mathbf{X}_h]_{\mathcal{X}}$. Hence, (a) follows. $\square$

The preceding Hamilton–Jacobi condition admits a geometric reformulation that makes the role of the projected gauge freedom more transparent. Before presenting this reformulation, it is useful to clarify the role of some of the assumptions entering the z -dependent Hamilton–Jacobi theory.

First, the isotropy of the sections $\gamma_z: Q \rightarrow \bigoplus^k T^*Q$, $q \mapsto (q^i, \gamma_i^\alpha(q, z))$, for fixed $z \in \mathbb{R}^k$, with respect to the canonical k -symplectic structure is not required for the proof of the z -dependent Hamilton–Jacobi equation itself. Indeed, the argument only uses the maximal coisotropy of $\text{Im } \gamma$, through condition (3.3.1), in order to rewrite the momentum equations. The isotropy of the sections γ_z becomes relevant only when one wishes to introduce local generating functions, since the condition $\partial \gamma_i^\alpha / \partial q^j = \partial \gamma_j^\alpha / \partial q^i$ ensures the local existence of functions $W^\alpha(q, z)$ such that $\gamma_i^\alpha(q, z) = \partial W^\alpha / \partial q^i$.

Second, no integrability assumption on the k -vector fields involved is used in the proof of Theorem 3.3.5. Integrability becomes relevant only when the theorem is interpreted in terms of integral sections, and hence in terms of actual solutions of the corresponding Hamilton–De Donder–Weyl equations. In this case, one seeks integral sections of the projected system whose lifts through γ provide solutions of these equations. For this purpose, it is sufficient to require the integrability of the projected representative $\overline{\mathbf{X}}^\gamma$.

We now give an intrinsic reformulation of condition (b) of Theorem 3.3.5, which makes the role of the projected gauge freedom more transparent.

Remark 3.3.6. Let $\{e_1, \dots, e_k\}$ be the canonical basis of $\mathbb{R}^k$, and let $\{e^1, \dots, e^k\}$ be its dual basis. Define the $\mathbb{R}^k$ -valued one-form and the $(\mathbb{R}^k)^*$ -valued function

$$\gamma^* \boldsymbol{\theta} := \sum_{\alpha=1}^k \gamma^* \theta^\alpha e_\alpha \in \Omega^1(Q \times \mathbb{R}^k, \mathbb{R}^k), \quad \Gamma := \sum_{\alpha=1}^k \Gamma_\alpha e^\alpha \in C^\infty(Q \times \mathbb{R}^k, (\mathbb{R}^k)^*).$$

Furthermore, define the $\text{End}(\mathbb{R}^k)$ -valued one-form $\mathcal{Z}_\gamma \in \Omega^1(Q \times \mathbb{R}^k, \text{End}(\mathbb{R}^k))$ by

$$(\mathcal{Z}_\gamma(Y))(v) := (\iota_v \vee d(\gamma^* \boldsymbol{\theta}))(Y), \quad Y \in \mathfrak{X}(Q \times \mathbb{R}^k), \quad v \in \mathbb{R}^k,$$

where v^V denotes the constant vertical vector field on $Q \times \mathbb{R}^k$ associated with $v \in \mathbb{R}^k$. With respect to the canonical basis, one has

$$(\mathcal{Z}_\gamma)^\alpha_\beta = \iota_{\partial/\partial z^\beta} d(\gamma^* \theta^\alpha), \quad \alpha, \beta = 1, \dots, k.$$

Since $\gamma^*\theta^\alpha = \sum_{i=1}^n \gamma_i^\alpha(q, z) dq^i$, it follows that

$$(\mathcal{Z}_\gamma)^\alpha{}_\beta = \sum_{i=1}^n \frac{\partial \gamma_i^\alpha}{\partial z^\beta}(q, z) dq^i, \quad \alpha, \beta = 1, \dots, k.$$

Thus, $\mathcal{Z}_\gamma$ encodes the vertical z -dependence of the pulled-back canonical form $\gamma^*\theta$. Set

$$\Xi_\gamma := d_Q(h \circ \gamma) + \langle \Gamma, \gamma^*\theta \rangle \in \Omega^1(Q \times \mathbb{R}^k).$$

Then condition (b) of Theorem 3.3.5 is equivalent to the requirement $\Xi_\gamma \in \mathcal{T}_\gamma(h)$, where

$$\mathcal{T}_\gamma(h) := \left\{ -\text{tr}(\mathcal{A} \circ \mathcal{Z}_\gamma) \mid \mathcal{A} \in C^\infty(Q \times \mathbb{R}^k, \text{End}(\mathbb{R}^k)), \quad \text{tr}(\mathcal{A}) = -(h \circ \gamma) \right\}.$$

Equivalently, there exists $\mathcal{A} \in C^\infty(Q \times \mathbb{R}^k, \text{End}(\mathbb{R}^k))$, satisfying $\text{tr}(\mathcal{A}) = -(h \circ \gamma)$, and such that

$$d_Q(h \circ \gamma) + \langle \Gamma, \gamma^*\theta \rangle + \text{tr}(\mathcal{A} \circ \mathcal{Z}_\gamma) = 0.$$

Thus, the deviation from the equation $d_Q(h \circ \gamma) + \langle \Gamma, \gamma^*\theta \rangle = 0$ is measured precisely by the affine family $\mathcal{T}_\gamma(h)$, determined by $\mathcal{Z}_\gamma$ subject to the prescribed trace condition. The freedom in choosing $\mathcal{A}$ with fixed trace corresponds to the freedom in the matrix $C = (C_\alpha^\beta)$ appearing in Theorem 3.3.5. In particular, any two such choices differ by a trace-free endomorphism. This provides a geometric manifestation of the projected gauge freedom induced by $\ker \chi$.

Note that for $k = 1$, one has $\text{End}(\mathbb{R}) \cong \mathbb{R}$, so the trace condition determines $\mathcal{A}$ uniquely, namely $\mathcal{A} = -(h \circ \gamma)$. Consequently, $\mathcal{T}_\gamma(h)$ consists of a single one-form, and there is no remaining trace-free freedom. Equation (3.3.3) therefore reduces to

$$d_Q(h \circ \gamma) + \Gamma \gamma^*\theta - (h \circ \gamma) \iota_{\partial/\partial z} d(\gamma^*\theta) = 0, \quad (3.3.6)$$

which is precisely the z -dependent contact Hamilton–Jacobi equation presented in [61]. Moreover, by Lemma 3.3.3, when $k = 1$, the condition that $\text{Im } \gamma$ be maximally coisotropic is equivalent to the usual coisotropy condition for the image of a section $\gamma: Q \times \mathbb{R} \rightarrow T^*Q \times \mathbb{R}$. As said before, the isotropy condition for γ_z is not relevant for the proof of the Hamilton–Jacobi equation itself as used in [61], but it is relevant if one wishes to introduce local generating functions, as performed in the cited work.

3.3.2 The evolution approach

The next result provides the z -dependent Hamilton–Jacobi theorem for evolution $\eta_{Q,k}$ -Hamiltonian systems. In contrast with the Hamiltonian case, the projected gauge term is trace-free, and the resulting Hamilton–Jacobi equation therefore gives the corresponding evolution counterpart of equation (3.3.3).

Theorem 3.3.7 (z -dependent k -contact evolution Hamilton–Jacobi theorem). *Consider the k -contact Hamiltonian system $(\mathcal{J}_{Q,k} = \bigoplus^k T^*Q \times \mathbb{R}^k, \eta_{Q,k}, h)$. Let $[\mathbf{E}_h]_{\chi_{\text{ev}}}$ denote the class of $\eta_{Q,k}$ -evolution k -vector fields associated with h , modulo gauge terms taking values in $\ker \chi_{\text{ev}} := \{\mathbf{Y} \in \ker \chi \mid \iota_{\mathbf{Y}} \eta_{Q,k} = 0\}$. Let $\gamma: Q \times \mathbb{R}^k \rightarrow \mathcal{J}_{Q,k}$ be a section of $\pi_{Q \times \mathbb{R}^k}: \mathcal{J}_{Q,k} \rightarrow Q \times \mathbb{R}^k$, and assume that $\text{Im } \gamma$ is maximally coisotropic. For each $\mathbf{E} \in [\mathbf{E}_h]_{\chi_{\text{ev}}}$, define $\mathbf{E}^\gamma := \left(\bigoplus^k T\pi_{Q \times \mathbb{R}^k} \right) (\mathbf{E} \circ \gamma)$. Moreover, set $\mathcal{K}_{\text{ev}} := \left(\bigoplus^k T\pi_{Q \times \mathbb{R}^k} \right) (\ker \chi_{\text{ev}})$, and denote by $[(\mathbf{E}_h)^\gamma]_{\mathcal{K}_{\text{ev}}}$ the corresponding projected class. Then the following statements are equivalent:*

- (a) *There exist representatives $\mathbf{E} \in [\mathbf{E}_h]_{\chi_{\text{ev}}}$ and $\bar{\mathbf{E}}^\gamma \in [(\mathbf{E}_h)^\gamma]_{\mathcal{K}_{\text{ev}}}$ such that $\left(\bigoplus^k T\gamma \right) (\bar{\mathbf{E}}^\gamma)$ and $\mathbf{E} \circ \gamma$ determine the same z -dependent evolution Hamilton–De Donder–Weyl equations for h ;*
- (b) *There exists a smooth matrix-valued function $C = (C_\alpha^\beta) \in C^\infty(Q \times \mathbb{R}^k, \text{Mat}_{k \times k}(\mathbb{R}))$ satisfying $\text{tr } C = \sum_{\alpha=1}^k C_\alpha^\alpha = 0$ and such that*

$$d_Q(h \circ \gamma) + \sum_{\beta=1}^k \Gamma_\beta \gamma^*\theta^\beta + \sum_{\alpha, \beta=1}^k C_\alpha^\beta \iota_{\partial/\partial z^\beta} d(\gamma^*\theta^\alpha) = 0, \quad (3.3.7)$$

where $\Gamma_\beta := \frac{\partial(h \circ \gamma)}{\partial z^\beta} = dh|_\gamma \left(T\gamma \left(\frac{\partial}{\partial z^\beta} \right) \right)$ for $\beta = 1, \dots, k$.

Proof. Take adapted Darboux coordinates $(q^i, p_i^\alpha, z^\alpha)$ on $\mathcal{J}_{Q,k}$, so that the section γ is locally given by $\gamma(q, z) = (q^i, \gamma_i^\alpha(q, z), z^\alpha)$. Set $U_\alpha^i := \frac{\partial h}{\partial p_i^\alpha} \circ \gamma$, for $\alpha = 1, \dots, k$ and $i = 1, \dots, n$. Using the chain rule yields $\Gamma_\beta = \frac{\partial h}{\partial z^\beta} \circ \gamma + \sum_{\alpha=1}^k \sum_{i=1}^n U_\alpha^i \frac{\partial \gamma_i^\alpha}{\partial z^\beta}$ for $\beta = 1, \dots, k$. Every representative $\bar{\mathbf{E}}^\gamma = (\bar{E}_1^\gamma, \dots, \bar{E}_k^\gamma)$ of the projected class $[(\mathbf{E}_h)^\gamma]_{\mathcal{K}_{\text{ev}}}$ can be written locally as

$$\bar{E}_\alpha^\gamma = \sum_{i=1}^n U_\alpha^i \frac{\partial}{\partial q^i} + \sum_{\beta=1}^k \left(\sum_{j=1}^n \gamma_j^\beta U_\alpha^j + C_\alpha^\beta \right) \frac{\partial}{\partial z^\beta}, \quad \alpha = 1, \dots, k,$$

for some smooth matrix-valued function $C = (C_\alpha^\beta)$. Since $\gamma^* \eta_{Q,k}^\beta = dz^\beta - \sum_{i=1}^n \gamma_i^\beta dq^i$, we have $\gamma^* \eta_{Q,k}^\beta(\bar{E}_\alpha^\gamma) = C_\alpha^\beta$. As the projected gauge freedom is induced by $\ker \chi_{\text{ev}}$, the evolution condition is preserved, and therefore $0 = \sum_{\alpha=1}^k \gamma^* \eta_{Q,k}^\alpha(\bar{E}_\alpha^\gamma) = \sum_{\alpha=1}^k C_\alpha^\alpha$. Hence, $\text{tr } C = 0$.

Assume first that (a) holds. Then there exist representatives $\mathbf{E} \in [\mathbf{E}_h]_{\mathcal{K}_{\text{ev}}}$ and $\bar{\mathbf{E}}^\gamma \in [(\mathbf{E}_h)^\gamma]_{\mathcal{K}_{\text{ev}}}$ such that $\left(\bigoplus^k \text{T}\gamma \right) (\bar{\mathbf{E}}^\gamma)$ and $\mathbf{E} \circ \gamma$ determine the same z -dependent evolution Hamilton–De Donder–Weyl equations for h . In particular, their momentum equations coincide. Proceeding as in the proof of Theorem 3.3.5, and using the maximal coisotropy condition (3.3.1) we obtain

$$\frac{\partial(h \circ \gamma)}{\partial q^j} + \sum_{\beta=1}^k \Gamma_\beta \gamma_j^\beta + \sum_{\alpha,\beta=1}^k C_\alpha^\beta \frac{\partial \gamma_j^\alpha}{\partial z^\beta} = 0, \quad j = 1, \dots, n.$$

Using $\gamma^* \theta^\alpha = \sum_{j=1}^n \gamma_j^\alpha dq^j$, the above system is exactly (3.3.7). Thus, (b) follows.

Conversely, assume (b). Choose a representative $\bar{\mathbf{E}}^\gamma$ of $[(\mathbf{E}_h)^\gamma]_{\mathcal{K}_{\text{ev}}}$ having the local form above. By the definition of the projected class, there exists a representative $\mathbf{E} \in [\mathbf{E}_h]_{\mathcal{K}_{\text{ev}}}$ whose projection along γ represents the same projected class. The q -components of $\left(\bigoplus^k \text{T}\gamma \right) (\bar{\mathbf{E}}^\gamma)$ are $U_\alpha^i = (\partial h / \partial p_i^\alpha) \circ \gamma$, and hence the q -equations of the evolution Hamilton–De Donder–Weyl system are satisfied. Moreover,

$$\sum_{\alpha=1}^k \eta_{Q,k}^\alpha \left(\text{T}\gamma \left(\bar{E}_\alpha^\gamma \right) \right) = \sum_{\alpha=1}^k \gamma^* \eta_{Q,k}^\alpha \left(\bar{E}_\alpha^\gamma \right) = \sum_{\alpha=1}^k C_\alpha^\alpha = 0,$$

so the evolution condition, and therefore the corresponding z -equation, is also satisfied.

Finally, reversing the preceding computation and using again the maximal coisotropy condition, equation (3.3.7) yields

$$\sum_{\alpha=1}^k \left(\text{T}\gamma \left(\bar{E}_\alpha^\gamma \right) \right)_{p_j^\alpha} = -\frac{\partial h}{\partial q^j} \circ \gamma - \sum_{\alpha=1}^k \gamma_j^\alpha \frac{\partial h}{\partial z^\alpha} \circ \gamma, \quad j = 1, \dots, n,$$

where $\left(\text{T}\gamma \left(\bar{E}_\alpha^\gamma \right) \right)_{p_j^\alpha}$ denotes the coefficient of $\partial / \partial p_j^\alpha$. Thus the momentum equations also coincide. It follows that $\left(\bigoplus^k \text{T}\gamma \right) (\bar{\mathbf{E}}^\gamma)$ and $\mathbf{E} \circ \gamma$ determine the same z -dependent evolution Hamilton–De Donder–Weyl equations for h , proving (a). $\square$

Remark 3.3.8. As in the Hamiltonian case, no isotropy assumption on the slices γ_z is required for the Hamilton–Jacobi equation itself. Such an assumption becomes relevant only when one wishes to introduce local generating functions.

Note also, that for $k = 1$, the trace-free condition implies $C_1^1 = 0$, and equation (3.3.7) reduces to

$$d_Q(h \circ \gamma) + \Gamma \gamma^* \theta = 0, \quad \Gamma = \frac{\partial(h \circ \gamma)}{\partial z}.$$

Moreover, by Lemma 3.3.3, the maximal coisotropy of $\text{Im } \gamma$ is equivalent to its coisotropy. Thus, in the contact case, the z -dependent evolution Hamilton–Jacobi equation takes the particularly simple form above, with no residual gauge term.

3.4 Integrable contact Hamiltonian systems and perspectives for the k -contact case

As discussed in Chapter 2, complete solutions of the classical Hamilton–Jacobi problem give rise locally to foliations of the symplectic phase space by Lagrangian submanifolds. Moreover, the image of each Hamilton–Jacobi solution is invariant under the Hamiltonian flow. This geometric picture provides a natural point of departure for considering complete solutions and integrability in the k -contact setting.

The results obtained in the previous sections reveal two different geometric situations. In the z -independent theory, the image of a Hamilton–Jacobi solution is Legendrian, whereas in the z -dependent theory the relevant geometric condition is maximal coisotropy. It is therefore natural to expect complete solutions to give rise, respectively, to Legendrian and maximally coisotropic foliations.

We first consider the z -independent case.

Definition 3.4.1. A *complete solution of the z -independent k -contact Hamilton–Jacobi (resp. evolution Hamilton–Jacobi) problem* for the k -contact Hamiltonian system $(\mathcal{J}_{Q,k}, \eta_{Q,k}, h)$ is a local bundle diffeomorphism $\Phi: Q \times \mathbb{R}^{k(n+1)} \rightarrow \mathcal{J}_{Q,k}$ such that, for every $\lambda \in \mathbb{R}^{k(n+1)}$, the map $\Phi_\lambda: Q \rightarrow \mathcal{J}_{Q,k}$, given by $q \mapsto \Phi(q, \lambda)$, is a solution of the z -independent k -contact Hamilton–Jacobi (resp. evolution Hamilton–Jacobi) problem.

Since each Φ_λ is a z -independent Hamilton–Jacobi solution, $\text{Im } \Phi_\lambda$ is an n -dimensional Legendrian submanifold of $\mathcal{J}_{Q,k}$. Moreover, $\dim \mathcal{J}_{Q,k} = n + k(n+1)$, so the $k(n+1)$ parameters of a complete solution provide precisely the codimension required for the family $\mathcal{F} = \{\mathcal{F}_\lambda := \text{Im } \Phi_\lambda \mid \lambda \in \mathbb{R}^{k(n+1)}\}$ to define, locally, a foliation of $\mathcal{J}_{Q,k}$ by Legendrian submanifolds. The corresponding construction in the z -dependent setting reflects the coisotropic geometry underlying the Hamilton–Jacobi problem.

Definition 3.4.2. A *complete solution of the z -dependent k -contact Hamilton–Jacobi (resp. evolution Hamilton–Jacobi) problem* for the k -contact Hamiltonian system $(\mathcal{J}_{Q,k}, \eta_{Q,k}, h)$ is a local bundle diffeomorphism $\Phi: Q \times \mathbb{R}^{kn} \times \mathbb{R}^k \rightarrow \mathcal{J}_{Q,k}$ such that, for every $\lambda \in \mathbb{R}^{kn}$, the map $\Phi_\lambda: Q \times \mathbb{R}^k \rightarrow \mathcal{J}_{Q,k}$, given by $(q, z) \mapsto \Phi(q, \lambda, z)$, is a solution of the z -dependent k -contact Hamilton–Jacobi (resp. evolution Hamilton–Jacobi) problem.

In this case, each $\text{Im } \Phi_\lambda$ is an $(n+k)$ -dimensional maximally coisotropic submanifold. Since $\dim \mathcal{J}_{Q,k} - (n+k) = kn$, the number of parameters in Definition 3.4.2 again coincides with the codimension of the corresponding leaves. Consequently, a complete solution determines locally a foliation $\mathcal{F} = \{\mathcal{F}_\lambda := \text{Im } \Phi_\lambda \mid \lambda \in \mathbb{R}^{kn}\}$ of $\mathcal{J}_{Q,k}$ by maximally coisotropic submanifolds. Thus, while complete solutions of the z -independent problem give rise to Legendrian foliations, their z -dependent counterparts naturally lead to maximally coisotropic foliations.

This picture is consistent with the action-dependent Hamilton–Jacobi theory in contact geometry. Consider a contact Hamiltonian system $(T^*Q \times \mathbb{R}, \eta, h)$. For a section $\gamma: Q \times \mathbb{R} \rightarrow T^*Q \times \mathbb{R}$, the action-dependent Hamilton–Jacobi equation can be written as

$$d_Q(h \circ \gamma) + \frac{\partial(h \circ \gamma)}{\partial z} \gamma^* \theta = (h \circ \gamma) \mathcal{L}_{\partial/\partial z}(\gamma^* \theta),$$

which is equivalent to (3.3.6). Accordingly, a *complete solution of the action-dependent Hamilton–Jacobi problem* is a local diffeomorphism $\Phi: Q \times \mathbb{R}^n \times \mathbb{R} \rightarrow T^*Q \times \mathbb{R}$ such that, for every $\lambda \in \mathbb{R}^n$, the map $\Phi_\lambda: Q \times \mathbb{R} \rightarrow T^*Q \times \mathbb{R}$, given by $(q, z) \mapsto \Phi(q, \lambda, z)$, is a solution of the action-dependent Hamilton–Jacobi problem. The corresponding family $\mathcal{F} = \{\mathcal{F}_\lambda := \text{Im } \Phi_\lambda \mid \lambda \in \mathbb{R}^n\}$ defines locally a foliation of $T^*Q \times \mathbb{R}$ by $(n+1)$ -dimensional coisotropic submanifolds. This is precisely the $k=1$ instance of the geometric picture described in Definition 3.4.2. Indeed, in this case maximal coisotropy of the image of a section $\gamma: Q \times \mathbb{R} \rightarrow T^*Q \times \mathbb{R}$ is equivalent to coisotropy. Hence, in the z -dependent contact setting, complete Hamilton–Jacobi solutions are naturally associated with coisotropic, rather than Legendrian, foliations.

This observation also provides a link with the notion of integrability in contact Hamiltonian dynamics. In the z -dependent setting, the relevant notion can be formulated in terms of an invariant coisotropic foliation.

Definition 3.4.3. Let (M, η, h) be a contact Hamiltonian system with $\dim M = 2n + 1$. We say that (M, η, h) is an *integrable contact Hamiltonian system in the z -dependent sense* if there exists a foliation $\mathcal{F}$ of M by $(n + 1)$ -dimensional coisotropic submanifolds that is invariant under the flow of the Hamiltonian vector field X_h .

Motivated by the preceding discussion, we propose the following k -contact counterpart. Since a k -vector field does not determine a single Hamiltonian flow, invariance is naturally expressed in terms of the distribution generated by its components.

Definition 3.4.4. Let $(\mathcal{J}_{Q,k}, \boldsymbol{\eta}_{Q,k}, h)$ be a k -contact Hamiltonian system. We say that it is an *integrable k -contact Hamiltonian system in the z -dependent sense* if there exist a foliation $\mathcal{F}$ of $\mathcal{J}_{Q,k}$ by $(n + k)$ -dimensional maximally coisotropic submanifolds and an integrable $\boldsymbol{\eta}_{Q,k}$ -Hamiltonian k -vector field $\mathbf{X}_h$ associated with h such that $D^{\mathbf{X}_h} \subset T\mathcal{F}$.

The condition $D^{\mathbf{X}_h} \subset T\mathcal{F}$ expresses the invariance of the foliation under the k -contact Hamiltonian dynamics: the integral manifolds of $D^{\mathbf{X}_h}$ are contained in the leaves of $\mathcal{F}$. Thus, this definition provides the natural k -contact analogue of invariance under the Hamiltonian flow in contact geometry. Whenever a complete z -dependent Hamilton–Jacobi solution generates such a foliation, it provides the geometric structure underlying this notion of integrability.

3.5 Applications

This section illustrates the Hamilton–De Donder–Weyl and Hamilton–Jacobi formalisms developed in the previous sections through several representative field-theoretic examples. Throughout the section, we consider a scalar field $u = u(t, x)$, with $(t, x) \in \mathbb{R}^2$, and take the configuration space to be $Q = \mathbb{R}$, with coordinate u . The corresponding two-contact phase space is $\mathcal{J}_{\mathbb{R},2} = \bigoplus^2 T^*\mathbb{R} \times \mathbb{R}^2$, with canonical coordinates (u, p^t, p^x, z^t, z^x) . The canonical two-contact form is

$$\boldsymbol{\eta}_{\mathbb{R},2} = (dz^t - p^t du) \otimes e_1 + (dz^x - p^x du) \otimes e_2,$$

and the corresponding Reeb vector fields are $R_t = \partial_{z^t}$, $R_x = \partial_{z^x}$.

Consider a two-vector field $\mathbf{Z} = (Z_t, Z_x)$ on $\mathcal{J}_{\mathbb{R},2}$ given locally by

$$Z_\alpha = (Z_\alpha)^u \frac{\partial}{\partial u} + \sum_{\beta=t,x} (Z_\alpha)^\beta_u \frac{\partial}{\partial p^\beta} + \sum_{\beta=t,x} (Z_\alpha)^\beta \frac{\partial}{\partial z^\beta}, \quad \alpha = t, x.$$

For later use, we briefly recall the projections associated with a section γ of a two-vector field $\mathbf{Z} = (Z_t, Z_x)$. In the z -independent setting, for a section $\gamma: Q \rightarrow \mathcal{J}_{\mathbb{R},2}$, the projected two-vector field on Q is $\mathbf{Z}^\gamma := \left(\bigoplus^2 T\pi_Q \right) \circ (\mathbf{Z} \circ \gamma)$. For a Hamiltonian two-vector field associated with h , its components take the local form

$$Z_\alpha^\gamma = \left((Z_\alpha)^u \circ \gamma \right) \frac{\partial}{\partial u}, \quad \alpha = t, x.$$

Thus, the projection of an $\boldsymbol{\eta}_{\mathbb{R},2}$ -Hamiltonian two-vector field $\mathbf{X}_h$ associated with Hamiltonian h reads

$$X_{h,\alpha}^\gamma = \left(\frac{\partial h}{\partial p^\alpha} \circ \gamma \right) \frac{\partial}{\partial u}, \quad \alpha = t, x.$$

In the z -dependent setting, a section is instead of the form $\gamma: Q \times \mathbb{R}^2 \rightarrow \mathcal{J}_{\mathbb{R},2}$, and the relevant projected two-vector field on $Q \times \mathbb{R}^2$ is $\mathbf{Z}^\gamma := \left(\bigoplus^2 T\pi_{Q \times \mathbb{R}^2} \right) \circ (\mathbf{Z} \circ \gamma)$. Its components are locally given by

$$Z_\alpha^\gamma = \left((Z_\alpha)^u \circ \gamma \right) \frac{\partial}{\partial u} + \sum_{\beta=t,x} \left((Z_\alpha)^\beta \circ \gamma \right) \frac{\partial}{\partial z^\beta}, \quad \alpha = t, x.$$

Meanwhile, the projection of an $\eta_{\mathbb{R},2}$ -Hamiltonian two-vector field $\mathbf{X}_h$ associated with Hamiltonian h is

$$X_{h,\alpha}^\gamma = \left(\frac{\partial h}{\partial p^\alpha} \circ \gamma \right) \frac{\partial}{\partial u} + \sum_{\beta=t,x} ((X_{h,\alpha})^\beta \circ \gamma) \frac{\partial}{\partial z^\beta}, \quad \alpha = t, x,$$

where the z -components satisfy

$$\sum_{\alpha=t,x} (X_{h,\alpha}^\gamma)^\alpha = \sum_{\alpha=t,x} (X_{h,\alpha})^\alpha \circ \gamma = \sum_{\alpha=t,x} \gamma^\alpha \left(\frac{\partial h}{\partial p^\alpha} \circ \gamma \right) - h \circ \gamma.$$

In the following, we analyse the following representative examples:

- the damped telegrapher/Klein–Gordon equation;
- a dissipative Hunter–Saxton equation;
- a dissipative wave equation;
- an EIT-type thermodynamic model.

Before proceeding, we note that the maximal coisotropy condition simplifies considerably in the present setting. Since $\dim Q = 1$, condition (3.3.1) is automatically satisfied. Therefore, every section $\gamma: \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathcal{J}_{\mathbb{R},2}$ considered in the z -dependent Hamilton–Jacobi problem has maximally coisotropic image.

3.5.1 The damped telegrapher/Klein–Gordon equation

We first consider the damped telegrapher/Klein–Gordon equation

$$u_{tt} - \kappa u_{xx} + \lambda u_t + \epsilon u = 0, \quad \kappa > 0, \quad \lambda \geq 0, \quad \epsilon \in \mathbb{R}. \quad (3.5.1)$$

This equation includes, as particular cases, certain damped Klein–Gordon equations [88] and the classical telegrapher equation describing the propagation of electric signals along a transmission line [108]. More precisely, if $u(x, t)$ denotes the voltage at position x and time t , then an infinitesimal segment of the transmission line of length δx is modelled by a series resistance $R \delta x$, a series inductance $L \delta x$, a shunt conductance $G \delta x$, and a shunt capacitance $C \delta x$, where R, L, G, C denote the corresponding physical parameters per unit length.

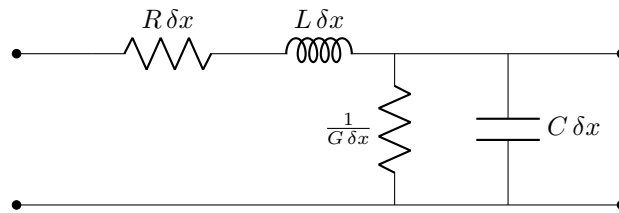


Figure 3.1: Infinitesimal element of a transmission line.

The corresponding second-order telegrapher equation is

$$LC u_{tt} + (RC + LG) u_t + RG u - u_{xx} = 0.$$

Assuming $L, C > 0$ and dividing by LC , we obtain

$$u_{tt} - \frac{1}{LC} u_{xx} + \frac{RC + LG}{LC} u_t + \frac{RG}{LC} u = 0.$$

Hence, equation (3.5.1) reproduces the physical telegrapher equation under the identifications

$$\kappa = \frac{1}{LC}, \quad \lambda = \frac{RC + LG}{LC} = \frac{R}{L} + \frac{G}{C}, \quad \epsilon = \frac{RG}{LC}.$$

Thus, κ is determined by the product LC , while λ encodes the damping contribution arising from resistance and leakage, and ϵ represents the zeroth-order contribution due to the simultaneous presence of resistance and conductance. In particular, in the lossless case $R = G = 0$, one has $\lambda = \epsilon = 0$, and equation (3.5.1) reduces to the wave equation $u_{tt} - \kappa u_{xx} = 0$, with propagation speed $\sqrt{\kappa} = 1/\sqrt{LC}$.

Conversely, for given values of $\kappa, \lambda, \epsilon$, the physical parameters satisfy $LC = 1/\kappa$, $RC + LG = \lambda/\kappa$, and $RG = \epsilon/\kappa$. Therefore, the triple $(\kappa, \lambda, \epsilon)$ determines the combinations LC , $RC + LG$, and RG , but does not determine the four parameters R, L, G, C separately. We now describe equation (3.5.1) within the two-contact Hamiltonian formalism. Consider the Hamiltonian

$$h_{\text{tel}}(u, p^t, p^x, z^t, z^x) = \frac{1}{2} ((p^t)^2 - \kappa^{-1}(p^x)^2) + \frac{\epsilon}{2} u^2 + \lambda z^t. \quad (3.5.2)$$

The Hamiltonian h_{tel} is regular and affine in the dissipative variables. Moreover, its dependence on these variables is entirely through z^t . As discussed in Section 3.1, this has relevant consequences for the partial differential equations satisfied by the dissipative variables and $u(t, x)$.

Let $\psi(t, x) = (u(t, x), p^t(t, x), p^x(t, x), z^t(t, x), z^x(t, x))$ be a solution section of the Hamiltonian Hamilton–De Donder–Weyl equations associated with h_{tel} . The first set of equations gives

$$u_t = \frac{\partial h_{\text{tel}}}{\partial p^t} = p^t, \quad u_x = \frac{\partial h_{\text{tel}}}{\partial p^x} = -\kappa^{-1} p^x, \quad (3.5.3)$$

whereas the momentum balance equation becomes

$$\partial_t p^t + \partial_x p^x = - \left(\frac{\partial h_{\text{tel}}}{\partial u} + p^t \frac{\partial h_{\text{tel}}}{\partial z^t} + p^x \frac{\partial h_{\text{tel}}}{\partial z^x} \right) = -(\epsilon u + \lambda p^t).$$

Using $p^t = u_t$ and $p^x = -\kappa u_x$, we immediately recover

$$u_{tt} - \kappa u_{xx} + \lambda u_t + \epsilon u = 0,$$

that is, equation (3.5.1). This is precisely the particular case of equation (3.1.2) determined by h_{tel} . The variables z^t and z^x complete the two-contact description, but they do not enter the resulting closed second-order equation for u . In fact, since h_{tel} is affine in the dissipative variables, the Hamiltonian and evolution Hamilton–De Donder–Weyl formulations associated with the same Hamiltonian h_{tel} produce the same equations for u, p^t, p^x , and hence the same second-order equation (3.5.1). The distinction between the two formulations appears in the balance equations satisfied by $z^t(t, x)$ and $z^x(t, x)$.

The affine dependence on the dissipative variables also illustrates the class of models considered in Section 3.1. A different behaviour arises if one allows a nonlinear dependence on these variables. For instance, consider

$$h'_{\text{tel}}(u, p^t, p^x, z^t, z^x) = \frac{1}{2} ((p^t)^2 - \kappa^{-1}(p^x)^2) + \frac{\epsilon}{2} u^2 + \frac{\lambda}{2} (z^t)^2.$$

The corresponding Hamilton–De Donder–Weyl equations yield

$$u_{tt} - \kappa u_{xx} + \lambda z^t(t, x) u_t + \epsilon u = 0, \quad \kappa > 0, \quad \lambda \geq 0, \quad \epsilon \in \mathbb{R},$$

where z^t is itself governed by the balance equations for the dissipative variables. Thus, the two-contact formalism provides a geometric mechanism for obtaining damped Klein–Gordon-type equations with an effective damping coefficient $\lambda z^t(t, x)$ [88]. In this case, the behaviour of z^t is constrained by the balance equation for the dissipative variables and is coupled, in particular, to z^x . This provides a simple mechanism for obtaining systems of PDEs whose coefficients depend on the dissipative variables and, consequently, on the independent variables through their solutions. In what follows, however, we restrict our analysis to the Hamiltonian h_{tel} in (3.5.2).

z -independent holonomic sections and projected integrable fields

A z -independent holonomic section $\gamma: \mathbb{R} \rightarrow \mathcal{J}_{\mathbb{R},2}$ is of the form

$$\gamma(u) = (u, \gamma_u^t(u), \gamma_u^x(u), \gamma^t(u), \gamma^x(u)), \quad \gamma_u^t(u) = \frac{d\gamma^t}{du}(u), \quad \gamma_u^x(u) = \frac{d\gamma^x}{du}(u).$$

We first consider the projected dynamics associated with the Hamiltonian h_{tel} . For both the $\boldsymbol{\eta}_{\mathbb{R},2}$ -Hamiltonian and $\boldsymbol{\eta}_{\mathbb{R},2}$ -evolution formulations, the first Hamilton–De Donder–Weyl equations coincide. By (3.5.3), the corresponding projected two-vector field $\mathbf{Z}^\gamma$ on $\mathbb{R}$ has components

$$Z_t^\gamma = \gamma_u^t(u) \frac{\partial}{\partial u}, \quad Z_x^\gamma = -\kappa^{-1} \gamma_u^x(u) \frac{\partial}{\partial u}.$$

We restrict our attention to the case in which $\mathbf{Z}^\gamma$ is integrable, so that its integral sections can be used to construct solutions of the damped telegrapher equation through Theorems 3.2.4 and 3.2.5. In particular, it is sufficient to require integrability of the projected two-vector field $\mathbf{Z}^\gamma$, while integrability of the corresponding two-vector field $\mathbf{Z}$ on $\mathcal{J}_{\mathbb{R},2}$ is not required.

Since both components of $\mathbf{Z}^\gamma$ are multiples of ∂_u , its integrability is equivalent to $[(Z^\gamma)_t, (Z^\gamma)_x] = 0$. Explicitly, this condition gives

$$\gamma_u^t(u) \frac{d\gamma_u^x}{du}(u) - \gamma_u^x(u) \frac{d\gamma_u^t}{du}(u) = 0.$$

Hence, on any domain where $\gamma_u^x \neq 0$,

$$\frac{d}{du} \left(\frac{\gamma_u^t(u)}{\gamma_u^x(u)} \right) = 0,$$

and therefore

$$\frac{\gamma_u^t(u)}{\gamma_u^x(u)} = c,$$

for some constant $c \in \mathbb{R}$. Thus, $\gamma_u^t = c \gamma_u^x$. In what follows, we restrict ourselves to this case, excluding $\gamma_u^x = 0$ from consideration. Let $\sigma: \mathbb{R}^2 \rightarrow Q = \mathbb{R}$ be an integral section of $\mathbf{Z}^\gamma$, and write

$$u(t, x) := \sigma(t, x).$$

Then

$$u_t = \gamma_u^t(u), \quad u_x = -\kappa^{-1} \gamma_u^x(u).$$

Using $\gamma_u^t = c \gamma_u^x$, define locally

$$F(u) := \int^u \frac{ds}{\gamma_u^x(s)},$$

on a domain where $\gamma_u^x \neq 0$. It follows that

$$\partial_t(F(u(t, x))) = \frac{\gamma_u^t(u(t, x))}{\gamma_u^x(u(t, x))} = c, \quad \partial_x(F(u(t, x))) = -\kappa^{-1}.$$

Consequently,

$$F(u(t, x)) = ct - \kappa^{-1}x + C, \quad u(t, x) = F^{-1}(ct - \kappa^{-1}x + C),$$

for some $C \in \mathbb{R}$, on any domain where F is invertible.

Whenever the corresponding z -independent Hamilton–Jacobi condition $h_{\text{tel}} \circ \gamma = 0$ is satisfied, an integral section σ of the projected dynamics lifts to a solution $\psi = \gamma \circ \sigma$. Its components are

$$p^t(t, x) = \gamma_u^t(u(t, x)) = c \gamma_u^x(u(t, x)), \quad p^x(t, x) = \gamma_u^x(u(t, x)),$$

and

$$z^t(t, x) = \gamma^t(u(t, x)), \quad z^x(t, x) = \gamma^x(u(t, x)), \quad \frac{d\gamma^t}{du} = \gamma_u^t, \quad \frac{d\gamma^x}{du} = \gamma_u^x.$$

Moreover, the relation $\gamma_u^t = c \gamma_u^x$ implies

$$\gamma^t(u) = c \gamma^x(u) + C_0, \tag{3.5.4}$$

for some $C_0 \in \mathbb{R}$.

We now distinguish the Hamiltonian and evolution z -independent Hamilton–Jacobi equations.

Hamiltonian z -independent HJ equation and a solvable family

The Hamiltonian z -independent Hamilton–Jacobi condition is $h_{\text{tel}} \circ \gamma = 0$, which explicitly reads

$$\frac{1}{2} ((\gamma_u^t(u))^2 - \kappa^{-1} (\gamma_u^x(u))^2) + \frac{\epsilon}{2} u^2 + \lambda \gamma^t(u) = 0. \quad (3.5.5)$$

As shown above, the projected two-vector field is integrable, on any domain where $\gamma_u^x \neq 0$, if and only if $\gamma_u^t = c \gamma_u^x$ for some constant $c \in \mathbb{R}$. Using this relation together with $(\gamma^t)' = \gamma_u^t$, differentiation of (3.5.5) with respect to u gives

$$(c^2 - \kappa^{-1}) \gamma_u^x(u) (\gamma_u^x)'(u) + \lambda c \gamma_u^x(u) + \epsilon u = 0. \quad (3.5.6)$$

Equation (3.5.6) is a necessary compatibility condition for the Hamiltonian z -independent Hamilton–Jacobi equation. Since it is obtained by differentiating (3.5.5), the original Hamilton–Jacobi equation must still be imposed.

To obtain an explicit family of solutions, assume $c^2 \neq \kappa^{-1}$ and consider the Ansatz $\gamma_u^x(u) = a u$ and $\gamma_u^t(u) = c a u$, where $a \neq 0$ is constant. Substitution into (3.5.6) gives

$$(c^2 - \kappa^{-1}) a^2 + \lambda c a + \epsilon = 0. \quad (3.5.7)$$

The corresponding roots are

$$a = \frac{-\lambda c \pm \sqrt{\lambda^2 c^2 - 4(c^2 - \kappa^{-1})\epsilon}}{2(c^2 - \kappa^{-1})}.$$

Thus, real solutions of this form exist whenever the discriminant is nonnegative and at least one of the resulting roots is nonzero. Integrating $(\gamma^x)' = \gamma_u^x$, $(\gamma^t)' = \gamma_u^t$, and using (3.5.4), we obtain

$$\gamma^x(u) = \frac{a}{2} u^2 + C_1, \quad \gamma^t(u) = \frac{c a}{2} u^2 + c C_1 + C_0,$$

where $C_0, C_1 \in \mathbb{R}$. Substituting these expressions into (3.5.5) and using (3.5.7), the terms proportional to u^2 vanish and the remaining condition is $\lambda(c C_1 + C_0) = 0$. Hence, when $\lambda \neq 0$, the constants must satisfy $C_0 + c C_1 = 0$, whereas for $\lambda = 0$, the combination $C_0 + c C_1$ is arbitrary. Therefore, provided that $\lambda(C_0 + c C_1) = 0$, the above Ansatz defines a family of solutions of the Hamiltonian z -independent $\eta_{\mathbb{R},2}$ -Hamilton–Jacobi equation.

The corresponding projected integral sections are obtained from

$$F(u) = \int^u \frac{ds}{\gamma_u^x(s)} = \frac{1}{a} \ln |u|,$$

up to an irrelevant additive constant. Hence,

$$F(u(t, x)) = c t - \kappa^{-1} x + C,$$

and therefore

$$u(t, x) = u_0 \exp(a(c t - \kappa^{-1} x)),$$

where $u_0 \neq 0$ is a constant. The induced lifted solutions $\psi = \gamma \circ \sigma$ are

$$p^t(t, x) = c a u(t, x), \quad p^x(t, x) = a u(t, x),$$

together with

$$z^t(t, x) = \frac{c a}{2} u(t, x)^2 + c C_1 + C_0, \quad z^x(t, x) = \frac{a}{2} u(t, x)^2 + C_1,$$

where

$$\lambda(c C_1 + C_0) = 0.$$

Evolution z -independent HJ equation and a solvable family

We now consider the evolution formulation associated with the same Hamiltonian h_{tel} . Since the first two Hamilton–De Donder–Weyl equations coincide with those of the Hamiltonian formulation, they yield the same damped telegrapher/Klein–Gordon equation for $u(t, x)$. The difference between the two formulations appears in the equation governing the dissipative variables. In the evolution formulation, the Hamilton–De Donder–Weyl equations (1.2.14) read

$$\begin{cases} u_t = \frac{\partial h_{\text{tel}}}{\partial p^t} \circ \psi = p^t, & u_x = \frac{\partial h_{\text{tel}}}{\partial p^x} \circ \psi = -\kappa^{-1} p^x, \\ \partial_t p^t + \partial_x p^x = - \left(\frac{\partial h_{\text{tel}}}{\partial u} + p^t \frac{\partial h_{\text{tel}}}{\partial z^t} + p^x \frac{\partial h_{\text{tel}}}{\partial z^x} \right) \circ \psi = -(\epsilon u + \lambda p^t), \\ \partial_t z^t + \partial_x z^x = \left(p^t \frac{\partial h_{\text{tel}}}{\partial p^t} + p^x \frac{\partial h_{\text{tel}}}{\partial p^x} \right) \circ \psi = (p^t)^2 - \kappa^{-1} (p^x)^2. \end{cases}$$

Thus, the equations for u, p^t, p^x are the same as in the Hamiltonian formulation, while the balance equation for the dissipative variables z^t, z^x differs by the absence of the term $-h_{\text{tel}} \circ \psi$.

Since the first Hamilton–De Donder–Weyl equations are unchanged, the projected two-vector field on $Q = \mathbb{R}$ is also the same as in the Hamiltonian formulation, and hence so is its integrability condition. Hence, $\gamma_u^t = c \gamma_u^x$ provided $\gamma_u^x \neq 0$. Consequently, the projected integral sections are again determined by

$$F(u(t, x)) = ct - \kappa^{-1}x + C, \quad u(t, x) = F^{-1}(ct - \kappa^{-1}x + C),$$

on any domain where F is invertible. Whenever the evolution z -independent Hamilton–Jacobi condition $d(h_{\text{tel}} \circ \gamma) = 0$ is satisfied, the integral sections σ lift via γ to solutions $\psi = \gamma \circ \sigma$ of the evolution HddW equations, whose components are

$$p^t(t, x) = \gamma_u^t(u(t, x)) = c \gamma_u^x(u(t, x)), \quad p^x(t, x) = \gamma_u^x(u(t, x)),$$

and

$$z^t(t, x) = \gamma^t(u(t, x)), \quad z^x(t, x) = \gamma^x(u(t, x)), \quad \frac{d\gamma^t}{du} = \gamma_u^t, \quad \frac{d\gamma^x}{du} = \gamma_u^x.$$

The essential difference from the Hamiltonian formulation lies in the Hamilton–Jacobi equation. In the evolution setting, the z -independent Hamilton–Jacobi condition is $d(h_{\text{tel}} \circ \gamma) = 0$. Since $Q = \mathbb{R}$, this is equivalent to

$$\frac{d}{du} \left[\frac{1}{2} ((\gamma_u^t(u))^2 - \kappa^{-1} (\gamma_u^x(u))^2) + \frac{\epsilon}{2} u^2 + \lambda \gamma^t(u) \right] = 0.$$

Using the holonomicity relations, this can be written as

$$\frac{d}{du} \left[\frac{1}{2} \left(\left(\frac{d\gamma^t}{du} \right)^2 - \kappa^{-1} \left(\frac{d\gamma^x}{du} \right)^2 \right) + \frac{\epsilon}{2} u^2 + \lambda \gamma^t(u) \right] = 0.$$

After substituting $\gamma_u^t = c \gamma_u^x$ and $(\gamma^t)' = \gamma_u^t$, this equation becomes precisely (3.5.6). In contrast to the Hamiltonian case, however, (3.5.6) is now the Hamilton–Jacobi equation itself rather than merely a necessary condition obtained by differentiating it.

In particular, consider again the Ansatz $\gamma_u^x(u) = a u$, $\gamma_u^t(u) = c a u$, where $a \neq 0$ satisfies (3.5.7). Integrating the holonomicity relations gives

$$\gamma^x(u) = \frac{a}{2} u^2 + C_1, \quad \gamma^t(u) = \frac{c a}{2} u^2 + c C_1 + C_0,$$

where $C_0, C_1 \in \mathbb{R}$. The corresponding projected integral sections are

$$u(t, x) = u_0 \exp(a(ct - \kappa^{-1}x)),$$

with $u_0 \neq 0$, and the induced lifted solutions are

$$p^t(t, x) = c a u(t, x), \quad p^x(t, x) = a u(t, x),$$

together with

$$z^t(t, x) = \frac{c a}{2} u(t, x)^2 + c C_1 + C_0, \quad z^x(t, x) = \frac{a}{2} u(t, x)^2 + C_1.$$

Indeed, substitution into the Hamiltonian gives

$$h_{\text{tel}} \circ \gamma = \frac{1}{2} [(c^2 - \kappa^{-1})a^2 + \epsilon + \lambda c a] u^2 + \lambda(C_0 + c C_1).$$

By (3.5.7), the coefficient of u^2 vanishes, and therefore $h_{\text{tel}} \circ \gamma = \lambda(C_0 + c C_1)$, which is constant, as required by the evolution Hamilton–Jacobi equation. Hence, unlike in the Hamiltonian formulation, no additional condition on the integration constants C_0 and C_1 is required.

The preceding z -independent constructions provide explicit families of solutions of both the Hamiltonian and evolution Hamilton–Jacobi problems, together with their corresponding lifted solutions. However, the families obtained from the above Ansatz do not define complete solutions in the sense of Definition 3.4.1. Indeed, for $Q = \mathbb{R}$ and $k = 2$, a complete z -independent solution would require a local bundle diffeomorphism $\Phi: \mathbb{R} \times \mathbb{R}^4 \rightarrow \mathcal{J}_{\mathbb{R}, 2}$. In the family considered above, the parameters a and c are constrained by (3.5.7), so that a is determined by c , up to the choice of a root, while C_0 and C_1 provide two additional continuous parameters. Thus, the resulting family does not contain the four independent parameters required for Φ to be a local diffeomorphism and therefore does not constitute a complete z -independent solution.

Hamiltonian z -dependent HJ equation and a complete solution

Let $\gamma: \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathcal{J}_{\mathbb{R}, 2}$ be a section of $\pi_{Q \times \mathbb{R}^2}: \mathcal{J}_{\mathbb{R}, 2} \rightarrow \mathbb{R} \times \mathbb{R}^2$, written as

$$\gamma(u, z^t, z^x) = (u, \gamma_u^t(u, z), \gamma_u^x(u, z), z^t, z^x).$$

Since $\dim Q = 1$, the maximal coisotropy condition for $\text{Im } \gamma$ is automatically satisfied. Moreover,

$$\gamma^* \theta^t = \gamma_u^t du, \quad \gamma^* \theta^x = \gamma_u^x du.$$

Hence, Theorem 3.3.5 yields the following Hamiltonian z -dependent Hamilton–Jacobi condition for h_{tel} : there exists a matrix-valued function $C = (C_\alpha^\beta)_{\alpha, \beta \in \{t, x\}}$ satisfying $C_t^t + C_x^x = -(h_{\text{tel}} \circ \gamma)$ and

$$\partial_u(h_{\text{tel}} \circ \gamma) + \Gamma_t \gamma_u^t + \Gamma_x \gamma_u^x + \sum_{\alpha, \beta \in \{t, x\}} C_\alpha^\beta \partial_{z^\beta} \gamma_u^\alpha = 0, \quad (3.5.8)$$

where

$$\Gamma_t = \lambda + \gamma_u^t \partial_{z^t} \gamma_u^t - \kappa^{-1} \gamma_u^x \partial_{z^t} \gamma_u^x, \quad \Gamma_x = \gamma_u^t \partial_{z^x} \gamma_u^t - \kappa^{-1} \gamma_u^x \partial_{z^x} \gamma_u^x.$$

We now construct an explicit family satisfying this condition. Fix $a \neq 0$ and let $(\mu, \nu) \in \mathbb{R}^2$. Consider

$$\gamma_{(\mu, \nu)}(u, z^t, z^x) = (u, a z^t - \lambda u + \mu, -a z^x + \nu, z^t, z^x). \quad (3.5.9)$$

For this family, $\partial_{z^t} \gamma_u^t = a$, $\partial_{z^x} \gamma_u^x = -a$, and $\partial_{z^t} \gamma_u^x = \partial_{z^x} \gamma_u^t = 0$. Therefore, $\Gamma_t = a \gamma_u^t + \lambda$, and $\Gamma_x = a \kappa^{-1} \gamma_u^x$. A direct computation gives

$$\partial_u(h_{\text{tel}} \circ \gamma_{(\mu, \nu)}) + \Gamma_t \gamma_u^t + \Gamma_x \gamma_u^x = \epsilon u + a ((\gamma_u^t)^2 + \kappa^{-1} (\gamma_u^x)^2).$$

Furthermore, for a diagonal choice

$$C^{\text{ham}} = \begin{pmatrix} A_{\text{ham}} & 0 \\ 0 & B_{\text{ham}} \end{pmatrix},$$

the last term in (3.5.8) becomes

$$\sum_{\alpha, \beta \in \{t, x\}} C_{\alpha}^{\beta} \partial_{z^{\beta}} \gamma_u^{\alpha} = aA_{\text{ham}} - aB_{\text{ham}}.$$

Thus, it is enough to choose A_{ham} and B_{ham} so that both the Hamilton–Jacobi equation and the trace condition are satisfied. One such choice is

$$A_{\text{ham}} = -\frac{1}{2} \left(h_{\text{tel}} \circ \gamma_{(\mu, \nu)} + \frac{\epsilon}{a} u + (\gamma_u^t)^2 + \kappa^{-1} (\gamma_u^x)^2 \right),$$

and

$$B_{\text{ham}} = -\frac{1}{2} \left(h_{\text{tel}} \circ \gamma_{(\mu, \nu)} - \frac{\epsilon}{a} u - (\gamma_u^t)^2 - \kappa^{-1} (\gamma_u^x)^2 \right).$$

Indeed, $A_{\text{ham}} + B_{\text{ham}} = -(h_{\text{tel}} \circ \gamma_{(\mu, \nu)})$, while $aA_{\text{ham}} - aB_{\text{ham}} = -\epsilon u - a((\gamma_u^t)^2 + \kappa^{-1}(\gamma_u^x)^2)$, so that (3.5.8) is satisfied.

Finally, define $\Phi: \mathbb{R} \times \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathcal{J}_{\mathbb{R}, 2}$ by

$$\Phi(u, \mu, \nu, z^t, z^x) = (u, az^t - \lambda u + \mu, -az^x + \nu, z^t, z^x).$$

Its inverse is given by

$$\Phi^{-1}(u, p^t, p^x, z^t, z^x) = (u, p^t - az^t + \lambda u, p^x + az^x, z^t, z^x).$$

Hence, Φ is a global diffeomorphism. Since, for every $(\mu, \nu) \in \mathbb{R}^2$, the corresponding section $\gamma_{(\mu, \nu)}$ satisfies the Hamiltonian z -dependent Hamilton–Jacobi equation, Φ defines a complete solution of the Hamiltonian z -dependent two-contact Hamilton–Jacobi problem for h_{tel} .

Evolution z -dependent HJ equation and a complete solution

We now consider the evolution z -dependent formulation. By Theorem 3.3.7, the corresponding Hamilton–Jacobi equation has the same form as (3.5.8), namely

$$\partial_u(h_{\text{tel}} \circ \gamma) + \Gamma_t \gamma_u^t + \Gamma_x \gamma_u^x + \sum_{\alpha, \beta \in \{t, x\}} C_{\alpha}^{\beta} \partial_{z^{\beta}} \gamma_u^{\alpha} = 0,$$

but the matrix-valued function $C = (C_{\alpha}^{\beta})_{\alpha, \beta \in \{t, x\}}$ now satisfies the traceless condition $C_t^t + C_x^x = 0$.

For the same family (3.5.9), the computation carried out above gives

$$\partial_u(h_{\text{tel}} \circ \gamma_{(\mu, \nu)}) + \Gamma_t \gamma_u^t + \Gamma_x \gamma_u^x = \epsilon u + a((\gamma_u^t)^2 + \kappa^{-1}(\gamma_u^x)^2).$$

Hence, the evolution z -dependent Hamilton–Jacobi equation is satisfied by choosing

$$C^{\text{ev}} = \begin{pmatrix} A_{\text{ev}} & 0 \\ 0 & -A_{\text{ev}} \end{pmatrix}, \quad A_{\text{ev}} = -\frac{1}{2} \left(\frac{\epsilon}{a} u + (\gamma_u^t)^2 + \kappa^{-1} (\gamma_u^x)^2 \right).$$

Indeed, $\text{tr } C^{\text{ev}} = 0$, and, since $\partial_{z^t} \gamma_u^t = a$, $\partial_{z^x} \gamma_u^x = -a$, while the mixed derivatives vanish, we have

$$\sum_{\alpha, \beta \in \{t, x\}} (C^{\text{ev}})_{\alpha}^{\beta} \partial_{z^{\beta}} \gamma_u^{\alpha} = aA_{\text{ev}} + (-A_{\text{ev}})(-a) = 2aA_{\text{ev}}.$$

Therefore,

$$2aA_{\text{ev}} = -\epsilon u - a((\gamma_u^t)^2 + \kappa^{-1}(\gamma_u^x)^2),$$

and the evolution z -dependent Hamilton–Jacobi equation (3.5.8) is satisfied.

Consequently, the same map Φ as in the Hamiltonian z -dependent case defines a complete solution of the evolution z -dependent two-contact Hamilton–Jacobi problem for h_{tel} .

3.5.2 A dissipative Hunter–Saxton equation as a k -contact Hamiltonian system

We consider the dissipative Hunter–Saxton equation

$$u_{tx} + u u_{xx} + \frac{1}{2}u_x^2 + \mu u_x = 0, \quad (3.5.10)$$

where $\mu \geq 0$. For $\mu = 0$, equation (3.5.10) reduces to the classical Hunter–Saxton equation, which arises, for instance, in the description of orientation waves in nematic liquid crystals. Differentiating (3.5.10) with respect to x gives

$$u_{txx} + 2u_x u_{xx} + u u_{xxx} + \mu u_{xx} = 0,$$

a form that is also commonly considered in the literature [46]. For further information on Hunter–Saxton and dissipative Hunter–Saxton-type equations, see [45, 46, 59, 77, 78, 131, 132].

Consider the Hamiltonian

$$h_{\text{HS}}(u, p^t, p^x, z^t, z^x) = -2p^t p^x + 2u(p^t)^2 + \mu p^t + 2\mu z^t. \quad (3.5.11)$$

The term $2\mu z^t$ gives rise to the dissipative contribution in (3.5.10), while the term μp^t modifies the first Hamilton–De Donder–Weyl equation relating u_t to the polymomenta. The Hamiltonian h_{HS} is affine in the dissipative variables, with its dependence on these variables occurring entirely through z^t , and it is regular. Indeed, its Hessian with respect to the polymomenta is

$$\begin{pmatrix} 4u & -2 \\ -2 & 0 \end{pmatrix},$$

whose determinant is -4 . Thus, the general considerations developed above for the Hamiltonian and evolution formulations apply.

The Hamiltonian Hamilton–De Donder–Weyl equations associated with (3.5.11) give

$$u_t = -2p^x + 4u p^t + \mu, \quad u_x = -2p^t, \quad (3.5.12)$$

together with

$$\partial_t p^t + \partial_x p^x = -2(p^t)^2 - 2\mu p^t. \quad (3.5.13)$$

Eliminating the polymomenta from (3.5.12)–(3.5.13), we have

$$p^t = -\frac{1}{2}u_x, \quad p^x = -\frac{1}{2}u_t - u u_x + \frac{\mu}{2},$$

and therefore

$$u_{tx} + u u_{xx} + \frac{1}{2}u_x^2 + \mu u_x = 0.$$

Hence, the dissipative Hunter–Saxton equation (3.5.10) is recovered.

z -independent holonomic sections and projected integrable fields

A z -independent holonomic section $\gamma: \mathbb{R} \rightarrow \mathcal{J}_{\mathbb{R},2}$ is locally written as

$$\gamma(u) = (u, \gamma_u^t(u), \gamma_u^x(u), \gamma^t(u), \gamma^x(u)), \quad \gamma_u^t(u) = \frac{d\gamma^t}{du}(u), \quad \gamma_u^x(u) = \frac{d\gamma^x}{du}(u).$$

By the first Hamilton–De Donder–Weyl equations (3.5.12), the corresponding projected two-vector field on $Q = \mathbb{R}$ has components

$$(Z^\gamma)_t = (-2\gamma_u^x(u) + 4u \gamma_u^t(u) + \mu) \frac{\partial}{\partial u}, \quad (Z^\gamma)_x = -2\gamma_u^t(u) \frac{\partial}{\partial u}. \quad (3.5.14)$$

We restrict our attention to the case in which $\mathbf{Z}^\gamma$ is integrable and $\gamma_u^t \neq 0$ on the domain under consideration. Since both components of $\mathbf{Z}^\gamma$ are multiples of ∂_u , the integrability condition is

$[(Z^\gamma)_t, (Z^\gamma)_x] = 0$. Equivalently, the ratio of their coefficients is constant, and hence there exists $c \in \mathbb{R}$ such that $-2\gamma_u^x + 4u\gamma_u^t + \mu = -2c\gamma_u^t$. Therefore,

$$\gamma_u^x(u) = (2u + c)\gamma_u^t(u) + \frac{\mu}{2}. \quad (3.5.15)$$

Let $\sigma: \mathbb{R}^2 \rightarrow Q = \mathbb{R}$ be an integral section of $\mathbf{Z}^\gamma$, and write $u(t, x) := \sigma(t, x)$. Then,

$$u_t = -2\gamma_u^x(u) + 4u\gamma_u^t(u) + \mu, \quad u_x = -2\gamma_u^t(u).$$

Using (3.5.15), these equations reduce to $u_t = -2c\gamma_u^t(u)$, $u_x = -2\gamma_u^t(u)$, and consequently $u_t = cu_x$. Define locally

$$F(u) := \int^u \frac{ds}{-2\gamma_u^t(s)}.$$

It follows that

$$\partial_t(F(u(t, x))) = c, \quad \partial_x(F(u(t, x))) = 1,$$

and hence

$$F(u(t, x)) = ct + x + C, \quad u(t, x) = F^{-1}(ct + x + C), \quad (3.5.16)$$

on any domain where F is invertible.

Whenever the Hamiltonian z -independent Hamilton–Jacobi condition is satisfied, that is, $h_{\text{HS}} \circ \gamma = 0$, an integral section σ lifts via γ to a solution $\psi = \gamma \circ \sigma$ of the Hamiltonian Hamilton–De Donder–Weyl equations, whose components are

$$p^t(t, x) = \gamma_u^t(u(t, x)), \quad p^x(t, x) = \gamma_u^x(u(t, x)),$$

and

$$z^t(t, x) = \gamma^t(u(t, x)), \quad z^x(t, x) = \gamma^x(u(t, x)).$$

Moreover, (3.5.15), together with the holonomicity relations, gives

$$\frac{d\gamma^x}{du}(u) = (2u + c)\frac{d\gamma^t}{du}(u) + \frac{\mu}{2}.$$

Integrating by parts, we obtain

$$\gamma^x(u) = (2u + c)\gamma^t(u) - 2 \int^u \gamma^t(s) ds + \frac{\mu}{2}u + C_0,$$

for some $C_0 \in \mathbb{R}$. The corresponding z -independent Hamilton–Jacobi condition remains to be imposed and may therefore introduce additional restrictions on the sections γ obtained above.

Hamiltonian z -independent HJ equation and a solvable family

The Hamiltonian z -independent Hamilton–Jacobi condition is $h_{\text{HS}} \circ \gamma = 0$, that is,

$$-2\gamma_u^t(u)\gamma_u^x(u) + 2u(\gamma_u^t(u))^2 + \mu\gamma_u^t(u) + 2\mu\gamma^t(u) = 0.$$

Using the integrability condition (3.5.15), this reduces to

$$-(u + c)(\gamma_u^t(u))^2 + \mu\gamma^t(u) = 0. \quad (3.5.17)$$

Thus, (3.5.17) is the Hamiltonian Hamilton–Jacobi equation expressed in terms of γ_u^t and γ^t . Differentiating it and using $(\gamma^t)'(u) = \gamma_u^t(u)$, we obtain the necessary compatibility condition

$$-(\gamma_u^t(u))^2 - 2(u + c)\gamma_u^t(u)(\gamma_u^t(u))' + \mu\gamma_u^t(u) = 0. \quad (3.5.18)$$

On any domain where $\gamma_u^t \neq 0$, this is equivalent to

$$2(u + c)(\gamma_u^t(u))' + \gamma_u^t(u) - \mu = 0.$$

Hence, on a connected domain where $u + c \neq 0$, we have

$$\gamma_u^t(u) = \mu + \frac{A}{\sqrt{|u+c|}}, \quad A \in \mathbb{R}.$$

Using (3.5.15), we then obtain

$$\gamma_u^x(u) = (2u+c) \left(\mu + \frac{A}{\sqrt{|u+c|}} \right) + \frac{\mu}{2}.$$

Let $\delta := \operatorname{sgn}(u+c)$ on the chosen connected domain. For $\mu > 0$, the choice

$$\gamma^t(u) = \mu(u+c) + 2A\delta\sqrt{|u+c|} + \frac{\delta A^2}{\mu}$$

satisfies both $(\gamma^t)'(u) = \gamma_u^t(u)$ and the original Hamilton–Jacobi equation (3.5.17).

A particularly simple explicit subfamily is obtained, for $\mu > 0$, by taking $A = 0$. Then

$$\gamma_u^t(u) = \mu, \quad \gamma_u^x(u) = \mu(2u+c) + \frac{\mu}{2},$$

and we may choose

$$\gamma^t(u) = \mu(u+c), \quad \gamma^x(u) = \mu u^2 + \frac{(2c+1)\mu}{2}u + C_1.$$

In this case, $F(u) = -u/(2\mu)$, up to an additive constant, and (3.5.16) therefore gives

$$u(t, x) = u_0 - 2\mu(x + ct),$$

for some constant $u_0 \in \mathbb{R}$. Since the Hamiltonian z -independent Hamilton–Jacobi condition is now satisfied, this projected integral section lifts via γ to a solution $\psi = \gamma \circ \sigma$ of the Hamiltonian Hamilton–De Donder–Weyl equations, given by

$$p^t(t, x) = \mu, \quad p^x(t, x) = \mu(2u(t, x) + c) + \frac{\mu}{2},$$

and

$$z^t(t, x) = \mu(u(t, x) + c), \quad z^x(t, x) = \mu u(t, x)^2 + \frac{(2c+1)\mu}{2}u(t, x) + C_1.$$

Evolution z -independent HJ equation and a solvable family

In the evolution formulation, the equations for u, p^t, p^x coincide with those of the Hamiltonian formulation, whereas the balance equation for the dissipative variables z^t, z^x differs by the absence of the term $-h_{\text{HS}}$. Therefore, the projected two-vector field is still given by (3.5.14), and the integrability condition (3.5.15) and the projected integral sections (3.5.16) remain unchanged. The evolution Hamilton–De Donder–Weyl equations associated with h_{HS} are

$$\begin{cases} u_t = -2p^x + 4u p^t + \mu, & u_x = -2p^t, \\ \partial_t p^t + \partial_x p^x = -2(p^t)^2 - 2\mu p^t, \\ \partial_t z^t + \partial_x z^x = -4p^t p^x + 4u(p^t)^2 + \mu p^t. \end{cases}$$

For a z -independent holonomic section γ , the evolution Hamilton–Jacobi condition is $d(h_{\text{HS}} \circ \gamma) = 0$, that is,

$$\frac{d}{du} [-2\gamma_u^t(u)\gamma_u^x(u) + 2u(\gamma_u^t(u))^2 + \mu\gamma_u^t(u) + 2\mu\gamma^t(u)] = 0.$$

Using the integrability condition (3.5.15), this is equivalent to

$$-(u+c)(\gamma_u^t(u))^2 + \mu\gamma^t(u) = K, \quad K \in \mathbb{R}. \quad (3.5.19)$$

Differentiating (3.5.19) with respect to u , we recover exactly (3.5.18). Hence, the previous computation applies directly to the evolution formulation. In particular, for $\mu > 0$ and on a connected domain where $u + c \neq 0$, we may take

$$\gamma_u^t(u) = \mu + \frac{A}{\sqrt{|u+c|}}, \quad \gamma_u^x(u) = (2u+c) \left(\mu + \frac{A}{\sqrt{|u+c|}} \right) + \frac{\mu}{2}.$$

Let $\delta := \operatorname{sgn}(u+c)$ on the chosen connected domain. Then

$$\gamma^t(u) = \mu(u+c) + 2A\delta\sqrt{|u+c|} + \frac{\delta A^2 + K}{\mu}$$

satisfies both $(\gamma^t)'(u) = \gamma_u^t(u)$ and (3.5.19). In contrast with the Hamiltonian z -independent case, the constant K is arbitrary, reflecting the fact that the evolution Hamilton–Jacobi condition requires $h_{\text{HS}} \circ \gamma$ to be constant rather than zero.

A particularly simple explicit subfamily is obtained by taking $A = 0$. Then

$$\gamma_u^t(u) = \mu, \quad \gamma_u^x(u) = \mu(2u+c) + \frac{\mu}{2},$$

and we may choose

$$\gamma^t(u) = \mu(u+c) + \frac{K}{\mu}, \quad \gamma^x(u) = \mu u^2 + \frac{(2c+1)\mu}{2}u + C_1.$$

The projected integral section is again $u(t, x) = u_0 - 2\mu(x + ct)$, while its lift through γ is given by

$$p^t(t, x) = \mu, \quad p^x(t, x) = \mu(2u(t, x) + c) + \frac{\mu}{2},$$

and

$$z^t(t, x) = \mu(u(t, x) + c) + \frac{K}{\mu}, \quad z^x(t, x) = \mu u(t, x)^2 + \frac{(2c+1)\mu}{2}u(t, x) + C_1.$$

A more general explicit subfamily is obtained by taking $A = 1$ and, for simplicity, $K = 0$. Then

$$\gamma_u^t(u) = \mu + \frac{1}{\sqrt{|u+c|}}, \quad \gamma_u^x(u) = (2u+c) \left(\mu + \frac{1}{\sqrt{|u+c|}} \right) + \frac{\mu}{2}.$$

On a connected domain where $u+c \neq 0$, with $\delta = \operatorname{sgn}(u+c)$, we may choose

$$\gamma^t(u) = \mu(u+c) + 2\delta\sqrt{|u+c|} + \frac{\delta}{\mu}.$$

Moreover, integrating $\gamma_u^x(u)$ gives

$$\gamma^x(u) = \mu u^2 + \frac{(2c+1)\mu}{2}u + \frac{4}{3}|u+c|^{3/2} - 2c\delta\sqrt{|u+c|} + C_1.$$

In this case,

$$F(u) = \int^u \frac{ds}{-2 \left(\mu + \frac{1}{\sqrt{|s+c|}} \right)},$$

and a direct computation gives

$$F(u) = -\frac{u+c}{2\mu} + \frac{\delta}{\mu^2}\sqrt{|u+c|} - \frac{\delta}{\mu^3} \ln \left(1 + \mu\sqrt{|u+c|} \right),$$

up to an additive constant. Hence, $F(u(t, x)) = x + ct + C$, or equivalently,

$$-\frac{u(t, x) + c}{2\mu} + \frac{\delta}{\mu^2}\sqrt{|u(t, x) + c|} - \frac{\delta}{\mu^3} \ln \left(1 + \mu\sqrt{|u(t, x) + c|} \right) = x + ct + C.$$

Define

$$G_\delta(r) := -\frac{\delta}{2\mu}r^2 + \frac{\delta}{\mu^2}r - \frac{\delta}{\mu^3}\ln(1 + \mu r), \quad r > 0.$$

On any domain where G_δ is invertible, we obtain

$$\sqrt{|u(t, x) + c|} = G_\delta^{-1}(x + ct + C),$$

and therefore

$$u(t, x) = \delta \left(G_\delta^{-1}(x + ct + C) \right)^2 - c.$$

Since the evolution z -independent Hamilton–Jacobi condition is satisfied, this projected integral section lifts via γ to a solution of the evolution Hamilton–De Donder–Weyl equations, whose components are

$$\begin{aligned} p^t(t, x) &= \mu + \frac{1}{\sqrt{|u(t, x) + c|}}, & p^x(t, x) &= (2u(t, x) + c) \left(\mu + \frac{1}{\sqrt{|u(t, x) + c|}} \right) + \frac{\mu}{2}, \\ z^t(t, x) &= \mu(u(t, x) + c) + 2\delta\sqrt{|u(t, x) + c|} + \frac{\delta}{\mu}, \\ z^x(t, x) &= \mu u(t, x)^2 + \frac{(2c + 1)\mu}{2} u(t, x) + \frac{4}{3}|u(t, x) + c|^{3/2} - 2c\delta\sqrt{|u(t, x) + c|} + C_1. \end{aligned}$$

Hamiltonian and evolution z -dependent HJ equations and a complete solution

Let $\gamma: \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathcal{J}_{\mathbb{R}, 2}$ be a section of $\pi_{Q \times \mathbb{R}^2}: \mathcal{J}_{\mathbb{R}, 2} \rightarrow \mathbb{R} \times \mathbb{R}^2$, written as

$$\gamma(u, z^t, z^x) = (u, \gamma_u^t(u, z), \gamma_u^x(u, z), z^t, z^x), \quad z = (z^t, z^x).$$

Since $Q = \mathbb{R}$ is one-dimensional, the maximal coisotropy condition for $\text{Im } \gamma$ is automatically satisfied. Moreover,

$$\gamma^* \theta^t = \gamma_u^t du, \quad \gamma^* \theta^x = \gamma_u^x du,$$

and

$$\iota_{\partial/\partial z^\beta} d(\gamma^* \theta^\alpha) = \frac{\partial \gamma_u^\alpha}{\partial z^\beta} du.$$

For the Hamiltonian (3.5.11), we have

$$h_{\text{HS}} \circ \gamma = -2\gamma_u^t \gamma_u^x + 2u(\gamma_u^t)^2 + \mu\gamma_u^t + 2\mu z^t.$$

Furthermore,

$$\Gamma_t = 2\mu + (-2\gamma_u^x + 4u\gamma_u^t + \mu) \frac{\partial \gamma_u^t}{\partial z^t} - 2\gamma_u^t \frac{\partial \gamma_u^x}{\partial z^t},$$

and

$$\Gamma_x = (-2\gamma_u^x + 4u\gamma_u^t + \mu) \frac{\partial \gamma_u^t}{\partial z^x} - 2\gamma_u^t \frac{\partial \gamma_u^x}{\partial z^x}.$$

Hence, the z -dependent Hamilton–Jacobi equation has the same form in the Hamiltonian and evolution formulations,

$$\partial_u(h_{\text{HS}} \circ \gamma) + \Gamma_t \gamma_u^t + \Gamma_x \gamma_u^x + \sum_{\alpha, \beta \in \{t, x\}} C_{\alpha}{}^{\beta} \frac{\partial \gamma_u^\alpha}{\partial z^\beta} = 0, \quad (3.5.20)$$

for a suitable matrix-valued function $C = (C_{\alpha}{}^{\beta})_{\alpha, \beta \in \{t, x\}}$. The difference between the two formulations lies in the corresponding trace condition. In the Hamiltonian case,

$$C_t{}^t + C_x{}^x = -(h_{\text{HS}} \circ \gamma), \quad (3.5.21)$$

whereas in the evolution case,

$$C_t{}^t + C_x{}^x = 0. \quad (3.5.22)$$

We now construct a genuinely z -dependent complete solution for both formulations. Fix $a \neq 0$ and let $(\rho, \sigma) \in \mathbb{R}^2$. Consider the family

$$\gamma_{(\rho, \sigma)}(u, z^t, z^x) = (u, az^t + \rho, -az^x + \sigma, z^t, z^x).$$

For convenience, set

$$T := az^t + \rho, \quad X := -az^x + \sigma.$$

Then $\gamma_u^t = T$, $\gamma_u^x = X$, and $\partial_{z^t}\gamma_u^t = a$, $\partial_{z^x}\gamma_u^x = -a$, while $\partial_{z^t}\gamma_u^x = \partial_{z^x}\gamma_u^t = 0$. Moreover,

$$h_{\text{HS}} \circ \gamma_{(\rho, \sigma)} = -2TX + 2uT^2 + \mu T + 2\mu z^t,$$

and

$$\Gamma_t = 2\mu + a(-2X + 4uT + \mu), \quad \Gamma_x = 2aT.$$

Therefore,

$$\Xi_\gamma := \partial_u(h_{\text{HS}} \circ \gamma_{(\rho, \sigma)}) + \Gamma_t T + \Gamma_x X = (2 + 4au)T^2 + \mu(2 + a)T.$$

In the Hamiltonian case, (3.5.20) is satisfied by choosing

$$C^{\text{ham}} = \begin{pmatrix} A_{\text{ham}} & 0 \\ 0 & B_{\text{ham}} \end{pmatrix},$$

where

$$A_{\text{ham}} = -\frac{1}{2} \left(h_{\text{HS}} \circ \gamma_{(\rho, \sigma)} + \frac{\Xi_\gamma}{a} \right), \quad B_{\text{ham}} = -\frac{1}{2} \left(h_{\text{HS}} \circ \gamma_{(\rho, \sigma)} - \frac{\Xi_\gamma}{a} \right).$$

Indeed, $A_{\text{ham}} + B_{\text{ham}} = -(h_{\text{HS}} \circ \gamma_{(\rho, \sigma)})$, and $aA_{\text{ham}} - aB_{\text{ham}} = -\Xi_\gamma$. Thus, both (3.5.20) and (3.5.21) are satisfied.

In the evolution case, we may instead choose

$$C^{\text{ev}} = \begin{pmatrix} A_{\text{ev}} & 0 \\ 0 & -A_{\text{ev}} \end{pmatrix}, \quad A_{\text{ev}} = -\frac{\Xi_\gamma}{2a}.$$

Then, $\text{tr } C^{\text{ev}} = 0$, and $aA_{\text{ev}} + (-a)(-A_{\text{ev}}) = 2aA_{\text{ev}} = -\Xi_\gamma$. Hence, (3.5.20) and (3.5.22) are satisfied.

Finally, define $\Phi: \mathbb{R} \times \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathcal{J}_{\mathbb{R}, 2}$ by

$$\Phi(u, \rho, \sigma, z^t, z^x) = (u, az^t + \rho, -az^x + \sigma, z^t, z^x).$$

Its inverse is

$$\Phi^{-1}(u, p^t, p^x, z^t, z^x) = (u, p^t - az^t, p^x + az^x, z^t, z^x).$$

Hence, Φ is a global diffeomorphism. Since each section $\gamma_{(\rho, \sigma)}$ satisfies the corresponding z -dependent Hamilton–Jacobi equation with the appropriate trace condition, Φ defines a complete solution of both the Hamiltonian and evolution z -dependent two-contact Hamilton–Jacobi problems for h_{HS} .

Remark 3.5.1. The distinction between the Hamiltonian and evolution z -dependent formulations is encoded in the trace condition imposed on the matrix C and, at the level of the Hamilton–De Donder–Weyl equations, in the balance equation for the dissipative variables z^t, z^x .

A quadratic solution from the z -dependent evolution Hamilton–Jacobi equation

The z -dependent Hamilton–Jacobi equation is more flexible than its z -independent counterpart. In particular, it allows us to construct sections satisfying the Hamilton–Jacobi condition whose projected dynamics yields explicit non-affine solutions of the dissipative Hunter–Saxton equation.

Consider the section $\tilde{\gamma}: \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathcal{J}_{\mathbb{R}, 2}$ defined by

$$\tilde{\gamma}(u, z^t, z^x) = \left(u, 0, \frac{\mu}{2} - z^t, z^t, z^x \right).$$

Then, $\tilde{\gamma}_u^t = 0$, $\tilde{\gamma}_u^x = \mu/2 - z^t$, and $\partial_{z^t}\tilde{\gamma}_u^x = -1$, whereas all the remaining derivatives of $\tilde{\gamma}_u^t, \tilde{\gamma}_u^x$ with respect to z^t, z^x vanish. Moreover,

$$h_{\text{HS}} \circ \tilde{\gamma} = 2\mu z^t, \quad \Gamma_t = 2\mu, \quad \Gamma_x = 0.$$

Therefore, the left-hand side of the z -dependent evolution Hamilton–Jacobi equation becomes

$$\partial_u(h_{\text{HS}} \circ \tilde{\gamma}) + \Gamma_t \tilde{\gamma}_u^t + \Gamma_x \tilde{\gamma}_u^x + \sum_{\alpha, \beta \in \{t, x\}} C_\alpha^\beta \frac{\partial \tilde{\gamma}_u^\alpha}{\partial z^\beta} = -C_x^t.$$

Hence, the equation is satisfied by any traceless matrix $C = (C_\alpha^\beta)$ such that $C_x^t = 0$. For instance, choose

$$C^{\text{ev}} = \begin{pmatrix} 1 & -2z^t \left(\frac{\mu}{2} - z^t\right) \\ 0 & -1 \end{pmatrix}.$$

This matrix is traceless and satisfies $C_x^t = 0$, and therefore solves the corresponding z -dependent evolution Hamilton–Jacobi equation.

Let $\tilde{\mathbf{E}}^\gamma = (\tilde{E}_t^\gamma, \tilde{E}_x^\gamma)$ be the associated projected representative. Its u -components are

$$(\tilde{E}_t^\gamma)^u = \left(\frac{\partial h_{\text{HS}}}{\partial p^t} \circ \tilde{\gamma} \right) = 2z^t, \quad (\tilde{E}_x^\gamma)^u = \left(\frac{\partial h_{\text{HS}}}{\partial p^x} \circ \tilde{\gamma} \right) = 0.$$

Using

$$(\tilde{E}_\alpha^\gamma)^\beta = \tilde{\gamma}_u^\beta (\tilde{E}_\alpha^\gamma)^u + C_\alpha^\beta,$$

we obtain

$$(\tilde{E}_t^\gamma)^t = 1, \quad (\tilde{E}_t^\gamma)^x = 2z^t \left(\frac{\mu}{2} - z^t\right) - 2z^t \left(\frac{\mu}{2} - z^t\right) = 0,$$

and

$$(\tilde{E}_x^\gamma)^t = 0, \quad (\tilde{E}_x^\gamma)^x = -1.$$

Thus,

$$\tilde{E}_t^\gamma = 2z^t \frac{\partial}{\partial u} + \frac{\partial}{\partial z^t}, \quad \tilde{E}_x^\gamma = -\frac{\partial}{\partial z^x}.$$

These vector fields commute, and hence $\tilde{\mathbf{E}}^\gamma$ is integrable. Let $\sigma(t, x) = (u(t, x), z^t(t, x), z^x(t, x))$ be an integral section. Then,

$$u_t = 2z^t, \quad u_x = 0, \quad (z^t)_t = 1, \quad (z^t)_x = 0, \quad (z^x)_t = 0, \quad (z^x)_x = -1.$$

Therefore,

$$z^t(t, x) = t + c_1, \quad z^x(t, x) = -x + c_2,$$

and consequently

$$u_t(t, x) = 2(t + c_1), \quad u_x(t, x) = 0,$$

where $c_0, c_1, c_2 \in \mathbb{R}$. Hence, $u(t, x) = (t + c_1)^2 + c_0$ and the corresponding lifted section $\psi = \tilde{\gamma} \circ \sigma$ is

$$u(t, x) = (t + c_1)^2 + c_0, \quad p^t(t, x) = 0, \quad p^x(t, x) = \frac{\mu}{2} - t - c_1,$$

together with

$$z^t(t, x) = t + c_1, \quad z^x(t, x) = -x + c_2.$$

A direct substitution shows that this section satisfies the evolution z -dependent HdDW equations. In particular, its projected component $u(t, x) = (t + c_1)^2 + c_0$ satisfies the dissipative Hunter–Saxton equation

$$u_{tx} + u u_{xx} + \frac{1}{2} u_x^2 + \mu u_x = 0.$$

This example illustrates how the z -dependent evolution Hamilton–Jacobi equation admits sections whose projected dynamics produces explicit solutions that do not arise from the simple affine families considered in the z -independent setting.

Further remarks on the z -dependent Hamilton–Jacobi equation

It is worth stressing that the z -dependent Hamilton–Jacobi problem differs from its z -independent counterpart. Indeed, it is formulated for sections $\gamma: Q \times \mathbb{R}^2 \rightarrow \mathcal{J}_{Q,2}$ and describes classes of projected two-vector fields on $Q \times \mathbb{R}^2$. This formulation provides additional freedom, both through the dependence of the sections on the dissipative variables and through the choice of representatives of the projected class. At the same time, this additional freedom requires some care when reconstructing solutions of the original field equation. In particular, satisfying the z -dependent Hamilton–Jacobi equation does not by itself determine a solution of the dissipative Hunter–Saxton equation. One must additionally choose a suitable representative of the projected class and determine an integral section of the corresponding projected two-vector field.

To further illustrate the z -dependent Hamilton–Jacobi formulation, fix $c \in \mathbb{R}$ and consider the section

$$\tilde{\gamma}(u, z^t, z^x) = \left(u, \mu, \mu(2u + c) + \frac{\mu}{2}, z^t, z^x \right).$$

Although this section belongs to the z -dependent setting, its momentum components are independent of z^t, z^x . Hence, $\partial_{z^\beta} \tilde{\gamma}_u^\alpha = 0$, for $\alpha, \beta = t, x$, and therefore $\Gamma_t = 2\mu$, $\Gamma_x = 0$. Moreover,

$$h_{\text{HS}} \circ \tilde{\gamma} = -2\mu^2(u + c) + 2\mu z^t, \quad \partial_u(h_{\text{HS}} \circ \tilde{\gamma}) = -2\mu^2.$$

It follows that

$$\partial_u(h_{\text{HS}} \circ \tilde{\gamma}) + \Gamma_t \tilde{\gamma}_u^t + \Gamma_x \tilde{\gamma}_u^x = -2\mu^2 + 2\mu^2 = 0.$$

Since all the z -derivatives of the momentum components vanish, the z -dependent Hamiltonian Hamilton–Jacobi equation is therefore satisfied for any matrix $C = (C_\alpha^\beta)$ whose trace satisfies

$$C_t^t + C_x^x = -(h_{\text{HS}} \circ \tilde{\gamma}) = 2\mu^2(u + c) - 2\mu z^t.$$

For instance, one may choose

$$C^{\text{ham}} = \begin{pmatrix} 0 & 0 \\ 0 & 2\mu^2(u + c) - 2\mu z^t \end{pmatrix}.$$

The u -components of the associated projected representative are

$$(\bar{X}_{h,t}^{\tilde{\gamma}})^u = \left(\frac{\partial h_{\text{HS}}}{\partial p^t} \circ \tilde{\gamma} \right) = -2c\mu, \quad (\bar{X}_{h,x}^{\tilde{\gamma}})^u = \left(\frac{\partial h_{\text{HS}}}{\partial p^x} \circ \tilde{\gamma} \right) = -2\mu.$$

Thus, for a compatible integrable representative, the u -component of an integral section satisfies $u_t = -2c\mu$ and $u_x = -2\mu$, and hence

$$u(t, x) = u_0 - 2\mu(x + ct), \quad u_0 \in \mathbb{R}.$$

The same affine solution appeared in the z -independent analysis. This coincidence follows from the fact that the momentum components

$$\tilde{\gamma}_u^t = \mu, \quad \tilde{\gamma}_u^x = \mu(2u + c) + \frac{\mu}{2}$$

coincide with those of the corresponding $A = 0$ z -independent family. The resulting function satisfies the dissipative Hunter–Saxton equation (3.5.10), since

$$u_{tx} = u_{xx} = 0, \quad \frac{1}{2}u_x^2 + \mu u_x = 2\mu^2 - 2\mu^2 = 0.$$

This observation shows that the same projected solution may arise from both the z -independent and z -dependent Hamilton–Jacobi formulations, despite the different geometric settings in which the two problems are posed. More generally, the z -dependent theory provides additional freedom through the dependence of the sections on the dissipative variables and through the choice of representatives of the projected class. As emphasised above, however, the Hamilton–Jacobi condition alone does not guarantee the existence of an integral section since the integrability of the chosen projected representative must also be taken into account.

3.5.3 A dissipative wave equation as a k -contact Hamiltonian system

Let us consider the Hamiltonian function

$$h(u, p^t, p^x, z^t, z^x) = \frac{1}{2} (u^2 + (p^x)^2) + \lambda z^t, \quad (3.5.23)$$

where $\lambda \geq 0$ is a dissipation parameter. Since $R_t(h) = \lambda$ and $R_x(h) = 0$, the Hamiltonian is not invariant under the Reeb flow in the z^t -direction, reflecting the non-conservative character of the dynamics. Moreover, the Hamiltonian is non-regular, since its Hessian with respect to the momentum variables is

$$\left(\frac{\partial^2 h}{\partial p^\alpha \partial p^\beta} \right)_{\alpha, \beta \in \{t, x\}} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

This example will therefore allow us to illustrate some features of non-regular Hamiltonians. We will focus on the z -dependent Hamilton–Jacobi equations.

The Hamiltonian HdDW equations associated with (3.5.23) are

$$u_t = \frac{\partial h}{\partial p^t} = 0, \quad u_x = \frac{\partial h}{\partial p^x} = p^x,$$

together with

$$\partial_t p^t + \partial_x p^x = - \left(\frac{\partial h}{\partial u} + p^t \frac{\partial h}{\partial z^t} + p^x \frac{\partial h}{\partial z^x} \right) = -(u + \lambda p^t),$$

and

$$\partial_t z^t + \partial_x z^x = p^t \frac{\partial h}{\partial p^t} + p^x \frac{\partial h}{\partial p^x} - h = \frac{1}{2} ((p^x)^2 - u^2) - \lambda z^t.$$

The evolution HdDW equations associated with (3.5.23) read

$$\begin{cases} u_t = \frac{\partial h}{\partial p^t} = 0, & u_x = \frac{\partial h}{\partial p^x} = p^x, \\ \partial_t p^t + \partial_x p^x = - \frac{\partial h}{\partial u} - p^t \frac{\partial h}{\partial z^t} - p^x \frac{\partial h}{\partial z^x} = -u - \lambda p^t, \\ \partial_t z^t + \partial_x z^x = p^t \frac{\partial h}{\partial p^t} + p^x \frac{\partial h}{\partial p^x} = (p^x)^2. \end{cases}$$

Thus, the relations involving u and the momentum balance are the same in the Hamiltonian and evolution formulations, whereas the evolution law for the dissipative variables differs. In particular, the parameter λ does not appear explicitly in the latter balance equation. The non-regularity of the Hamiltonian is also reflected in the first HdDW equations: while $p^x = u_x$, there is no corresponding relation determining p^t in terms of derivatives of u , since $u_t = 0$. Instead, p^t remains an additional variable subject to the momentum balance equation. Consequently, unlike in the regular examples considered above, the momentum variables cannot both be eliminated in favour of u and its derivatives, and the HdDW system does not directly reduce to a second-order equation for $u(t, x)$.

z -dependent Hamiltonian and evolution two-contact Hamilton–Jacobi equations

Let $\gamma: \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathcal{J}_{\mathbb{R}, 2}$ be a section of $\pi_{\mathbb{R} \times \mathbb{R}^2}: \mathcal{J}_{\mathbb{R}, 2} \rightarrow \mathbb{R} \times \mathbb{R}^2$, written as

$$\gamma(u, z^t, z^x) = (u, \gamma_u^t(u, z), \gamma_u^x(u, z), z^t, z^x), \quad z = (z^t, z^x).$$

Since $Q = \mathbb{R}$ is one-dimensional, the maximal coisotropy condition is automatically satisfied. Moreover,

$$\gamma^* \theta^t = \gamma_u^t du, \quad \gamma^* \theta^x = \gamma_u^x du, \quad \iota_{\partial/\partial z^\beta} d(\gamma^* \theta^\alpha) = \frac{\partial \gamma_u^\alpha}{\partial z^\beta} du.$$

For the Hamiltonian (3.5.23), one has

$$h \circ \gamma = \frac{1}{2} (u^2 + (\gamma_u^x)^2) + \lambda z^t.$$

Furthermore,

$$\Gamma_t = \lambda + \gamma_u^x \frac{\partial \gamma_u^x}{\partial z^t}, \quad \Gamma_x = \gamma_u^x \frac{\partial \gamma_u^x}{\partial z^x}.$$

Hence, the z -dependent Hamilton–Jacobi equation takes the form

$$\partial_u(h \circ \gamma) + \Gamma_t \gamma_u^t + \Gamma_x \gamma_u^x + \sum_{\alpha, \beta \in \{t, x\}} C_{\alpha}^{\beta} \frac{\partial \gamma_u^{\alpha}}{\partial z^{\beta}} = 0, \quad (3.5.24)$$

for a suitable matrix $C = (C_{\alpha}^{\beta})_{\alpha, \beta \in \{t, x\}}$. As in the previous examples, the difference between the Hamiltonian and evolution formulations is encoded in the trace condition. In the Hamiltonian case,

$$C_t^t + C_x^x = -(h \circ \gamma), \quad (3.5.25)$$

whereas, in the evolution case,

$$C_t^t + C_x^x = 0. \quad (3.5.26)$$

Let us now exhibit a z -dependent complete solution. Fix $a \neq 0$ and parameters $(\rho, \sigma) \in \mathbb{R}^2$, and consider the family of sections

$$\gamma_{(\rho, \sigma)}(u, z^t, z^x) = (u, az^t + \rho, -az^x + \sigma, z^t, z^x).$$

Set $T := az^t + \rho$, and $X := -az^x + \sigma$. Then,

$$\gamma_u^t = T, \quad \gamma_u^x = X, \quad \partial_{z^t} \gamma_u^t = a, \quad \partial_{z^x} \gamma_u^x = -a,$$

while the mixed z -derivatives vanish. Moreover,

$$h \circ \gamma_{(\rho, \sigma)} = \frac{1}{2}(u^2 + X^2) + \lambda z^t, \quad \Gamma_t = \lambda, \quad \Gamma_x = -aX.$$

Therefore, setting

$$\Xi_{\gamma} := \partial_u(h \circ \gamma_{(\rho, \sigma)}) + \Gamma_t T + \Gamma_x X,$$

one obtains

$$\Xi_{\gamma} = u + \lambda T - aX^2.$$

In the Hamiltonian case, (3.5.24) is satisfied by choosing

$$C^{\text{ham}} = \begin{pmatrix} A_{\text{ham}} & 0 \\ 0 & B_{\text{ham}} \end{pmatrix}, \quad A_{\text{ham}} = -\frac{1}{2} \left((h \circ \gamma_{(\rho, \sigma)}) + \frac{\Xi_{\gamma}}{a} \right), \quad B_{\text{ham}} = -\frac{1}{2} \left((h \circ \gamma_{(\rho, \sigma)}) - \frac{\Xi_{\gamma}}{a} \right).$$

Indeed,

$$A_{\text{ham}} + B_{\text{ham}} = -(h \circ \gamma_{(\rho, \sigma)}), \quad aA_{\text{ham}} - aB_{\text{ham}} = -\Xi_{\gamma},$$

so both (3.5.24) and (3.5.25) are satisfied.

In the evolution case, one may instead take

$$C^{\text{ev}} = \begin{pmatrix} A_{\text{ev}} & 0 \\ 0 & -A_{\text{ev}} \end{pmatrix}, \quad A_{\text{ev}} = -\frac{\Xi_{\gamma}}{2a}.$$

Then C^{ev} is traceless and

$$aA_{\text{ev}} - a(-A_{\text{ev}}) = -\Xi_{\gamma},$$

so both (3.5.24) and (3.5.26) are satisfied.

Finally, define

$$\Phi: \mathbb{R} \times \mathbb{R}^2 \times \mathbb{R}^2 \longrightarrow \mathcal{J}_{\mathbb{R}, 2}, \quad \Phi(u, \rho, \sigma, z^t, z^x) = (u, az^t + \rho, -az^x + \sigma, z^t, z^x).$$

Its inverse is

$$\Phi^{-1}(u, p^t, p^x, z^t, z^x) = (u, p^t - az^t, p^x + az^x, z^t, z^x).$$

Hence, Φ is a global diffeomorphism and therefore defines a complete solution of both the Hamiltonian and evolution z -dependent two-contact Hamilton–Jacobi problems associated with (3.5.23).

Remark 3.5.2. The main difference between the z -dependent Hamiltonian and evolution formulations in this example lies not in the family of sections itself, but in the trace condition imposed on the matrix C in the corresponding Hamilton–Jacobi equation. Thus, both formulations admit the same complete family of sections, while the distinction between them is reflected in the corresponding choices of the projected representatives.

3.5.4 k -Contact Hamilton–Jacobi equations for a covariant EIT-type thermodynamic model

This example illustrates how the k -contact formalism can also be applied in a thermodynamic field-theory setting. In this case, the geometric structure is not intended to describe time evolution in the usual mechanical sense, but rather a covariant system involving constitutive relations, balance laws, and entropy production. The construction is motivated in part by the k -contact formulation of relativistic thermodynamic systems developed in [44]. More specifically, the model considered here has features of extended irreversible thermodynamics, in which dissipative fluxes are treated as independent non-equilibrium variables, and can also be viewed as a reduced divergence-type model, with the balance laws determined by a generating Hamiltonian. The purpose of the present example is primarily to illustrate, at a mathematical level, how the Hamilton–Jacobi formalism developed above may be applied to this type of thermodynamic system.

Consider the manifold $M = \mathbb{R}^{k^2+4k+2}$ with coordinates $(\tilde{S}^\mu, N^\mu, T^{\lambda\mu}, P^\mu, \xi, \beta_\lambda, V)$, where $\mu, \lambda = 1, \dots, k$. We introduce the variables $\tilde{S}^\mu = S^\mu + \xi N^\mu - \sum_{\lambda=1}^k \beta_\lambda T^{\lambda\mu}$, where S^μ denote the entropy fluxes, N^μ the particle-number fluxes, $T^{\lambda\mu}$ generalised stress-energy variables, not assumed to be symmetric, and P^μ pressure-type or dissipative fluxes. The variables (ξ, β_λ, V) play the role of thermodynamic configuration variables.

On M , consider the $\mathbb{R}^k$ -valued one-form

$$\boldsymbol{\eta}_{\text{Ther}} = \sum_{\mu=1}^k \left(d\tilde{S}^\mu - N^\mu d\xi + \sum_{\lambda=1}^k T^{\lambda\mu} d\beta_\lambda - P^\mu dV \right) \otimes e_\mu.$$

This form is in Darboux k -contact form under the identifications

$$z^\mu = \tilde{S}^\mu, \quad q^i = (\xi, \beta_\lambda, V), \quad p_i^\mu = (N^\mu, -T^{\lambda\mu}, P^\mu), \quad i = 1, \dots, n, \quad \mu, \lambda = 1, \dots, k,$$

where $n = k + 2$. Hence, $(M, \boldsymbol{\eta}_{\text{Ther}})$ is a co-oriented k -contact manifold, with Reeb vector fields

$$R_\mu = \frac{\partial}{\partial \tilde{S}^\mu}, \quad \mu = 1, \dots, k,$$

and polarisation

$$\mathcal{V} = \left\langle \frac{\partial}{\partial N^\mu}, \frac{\partial}{\partial T^{\lambda\mu}}, \frac{\partial}{\partial P^\mu} \right\rangle_{\lambda, \mu=1, \dots, k}.$$

Therefore, $(M, \boldsymbol{\eta}_{\text{Ther}}, \mathcal{V})$ is a polarised co-oriented k -contact manifold.

This description differs from the thermodynamic coordinates used in [44], but the two formulations are related by the change of variables from S^μ to $\tilde{S}^\mu$, with the remaining coordinates unchanged. The present choice has the advantage of putting $\boldsymbol{\eta}_{\text{Ther}}$ directly into Darboux k -contact form, which makes the application of the Hamilton–De Donder–Weyl formalism more transparent. With the index convention adopted here, the resulting field equations involve the tensor $T^{\mu\nu}$, which corresponds to the transpose of the convention commonly used in physics, $T_{\text{Phys}}^{\nu\mu} = T^{\mu\nu}$. Of course, if the stress-energy tensor is symmetric, the two conventions coincide.

Let $h = h(\xi, \beta_\lambda, V, N^\mu, T^{\lambda\mu}, P^\mu)$ be a Hamiltonian satisfying $\partial h / \partial \tilde{S}^\mu = 0$, for every $\mu = 1, \dots, k$. This assumption means that the Hamiltonian does not depend on the entropy-flux variables, while the constitutive content is encoded in its dependence on the intensive variables and on the fluxes.

Hamiltonian and evolution thermodynamic HddW equations

Let $\mathbf{X} = (X_1, \dots, X_k)$ be an $\boldsymbol{\eta}_{\text{Ther}}$ -Hamiltonian k -vector field for $(M, \boldsymbol{\eta}_{\text{Ther}}, h)$. Using the Darboux identification introduced above, the Hamilton–De Donder–Weyl equations take the form

$$\frac{\partial \xi}{\partial x^\mu} = \frac{\partial h}{\partial N^\mu}, \quad \frac{\partial \beta_\lambda}{\partial x^\mu} = -\frac{\partial h}{\partial T^{\lambda\mu}}, \quad \frac{\partial V}{\partial x^\mu} = \frac{\partial h}{\partial P^\mu}, \quad \mu = 1, \dots, k,$$

together with

$$\sum_{\mu=1}^k \frac{\partial N^\mu}{\partial x^\mu} = -\frac{\partial h}{\partial \xi}, \quad \sum_{\mu=1}^k \frac{\partial T^{\lambda\mu}}{\partial x^\mu} = \frac{\partial h}{\partial \beta_\lambda}, \quad \sum_{\mu=1}^k \frac{\partial P^\mu}{\partial x^\mu} = -\frac{\partial h}{\partial V}, \quad \lambda = 1, \dots, k,$$

and

$$\sum_{\mu=1}^k \frac{\partial \tilde{S}^\mu}{\partial x^\mu} = \sum_{\mu=1}^k \left(N^\mu \frac{\partial h}{\partial N^\mu} + \sum_{\lambda=1}^k T^{\lambda\mu} \frac{\partial h}{\partial T^{\lambda\mu}} + P^\mu \frac{\partial h}{\partial P^\mu} \right) - h.$$

Thus, the k -contact Hamiltonian formalism gives rise to a first-order thermodynamic field theory of balance-law type, with the constitutive relations determined by the Hamiltonian h .

A particularly useful class of Hamiltonians is

$$h = U(\xi, \beta_\lambda, V) + \Phi(N^\mu, T^{\lambda\mu}, P^\mu).$$

For this class, the constitutive equations become

$$\frac{\partial \xi}{\partial x^\mu} = \frac{\partial \Phi}{\partial N^\mu}, \quad \frac{\partial \beta_\lambda}{\partial x^\mu} = -\frac{\partial \Phi}{\partial T^{\lambda\mu}}, \quad \frac{\partial V}{\partial x^\mu} = \frac{\partial \Phi}{\partial P^\mu}, \quad \lambda, \mu = 1, \dots, k,$$

whereas the corresponding balance equations are

$$\sum_{\mu=1}^k \frac{\partial N^\mu}{\partial x^\mu} = -\frac{\partial U}{\partial \xi}, \quad \sum_{\mu=1}^k \frac{\partial T^{\lambda\mu}}{\partial x^\mu} = \frac{\partial U}{\partial \beta_\lambda}, \quad \sum_{\mu=1}^k \frac{\partial P^\mu}{\partial x^\mu} = -\frac{\partial U}{\partial V}, \quad \lambda = 1, \dots, k.$$

In particular, when $U = 0$, these reduce to the conservation laws

$$\sum_{\mu=1}^k \partial_\mu N^\mu = 0, \quad \sum_{\mu=1}^k \partial_\mu T^{\lambda\mu} = 0, \quad \sum_{\mu=1}^k \partial_\mu P^\mu = 0, \quad \lambda = 1, \dots, k.$$

This provides a simple covariant EIT-type instance within the present framework. Conservation and balance equations for particle-number and stress-energy fluxes commonly arise in relativistic thermodynamic theories (see, for instance, [44, 47]).

Let $\mathbf{E} = (E_1, \dots, E_k)$ now be an $\boldsymbol{\eta}_{\text{Ther}}$ -evolution k -vector field for the same Hamiltonian h . The equations for ξ , β_λ , V , N^μ , $T^{\lambda\mu}$, and P^μ remain unchanged, whereas the equation for the entropy-flux variables becomes

$$\sum_{\mu=1}^k \frac{\partial \tilde{S}^\mu}{\partial x^\mu} = \sum_{\mu=1}^k \left(N^\mu \frac{\partial h}{\partial N^\mu} + \sum_{\lambda=1}^k T^{\lambda\mu} \frac{\partial h}{\partial T^{\lambda\mu}} + P^\mu \frac{\partial h}{\partial P^\mu} \right).$$

Thus, the two formulations differ only in the balance equation for $\tilde{S}^\mu$, through the additional term $-h$ in the Hamiltonian case.

The z -independent Hamilton–Jacobi problem

Let $Q_{\text{th}} = \mathbb{R}^{k+2}$ with global coordinates (ξ, β_λ, V) , and let $\gamma: Q_{\text{th}} \rightarrow M$ be a section of the projection $(\xi, \beta_\lambda, V, \tilde{S}^\mu, N^\mu, T^{\lambda\mu}, P^\mu) \mapsto (\xi, \beta_\lambda, V)$. Writing

$$\gamma(\xi, \beta_\lambda, V) = (\tilde{S}^\mu(\xi, \beta_\lambda, V), N^\mu(\xi, \beta_\lambda, V), T^{\lambda\mu}(\xi, \beta_\lambda, V), P^\mu(\xi, \beta_\lambda, V), \xi, \beta_\lambda, V),$$

the condition $\gamma^* \eta_{\text{Ther}} = 0$ is equivalent to

$$d\tilde{S}^\mu - N^\mu d\xi + \sum_{\lambda=1}^k T^{\lambda\mu} d\beta_\lambda - P^\mu dV = 0, \quad \mu = 1, \dots, k.$$

Equivalently,

$$N^\mu = \frac{\partial \tilde{S}^\mu}{\partial \xi}, \quad T^{\lambda\mu} = -\frac{\partial \tilde{S}^\mu}{\partial \beta_\lambda}, \quad P^\mu = \frac{\partial \tilde{S}^\mu}{\partial V}, \quad \mu, \lambda = 1, \dots, k.$$

Hence, the functions $\tilde{S}^\mu = \tilde{S}^\mu(\xi, \beta_\lambda, V)$ form a generating family for the thermodynamic section.

The z -independent Hamiltonian Hamilton–Jacobi equation is $h \circ \gamma = 0$, that is,

$$h\left(\xi, \beta_\lambda, V, \frac{\partial \tilde{S}^\mu}{\partial \xi}, -\frac{\partial \tilde{S}^\mu}{\partial \beta_\lambda}, \frac{\partial \tilde{S}^\mu}{\partial V}\right) = 0.$$

In the thermodynamic setting considered here, it is physically meaningful to consider hydrodynamic solutions contained in Legendrian submanifolds, which leads to the zero-Hamiltonian condition [44]. The z -independent Hamiltonian Hamilton–Jacobi formalism is therefore naturally suited to this class of thermodynamic solutions.

For the evolution formulation, the corresponding Hamilton–Jacobi equation is $d(h \circ \gamma) = 0$, or, equivalently,

$$h\left(\xi, \beta_\lambda, V, \frac{\partial \tilde{S}^\mu}{\partial \xi}, -\frac{\partial \tilde{S}^\mu}{\partial \beta_\lambda}, \frac{\partial \tilde{S}^\mu}{\partial V}\right) = \text{const.}$$

Thus, the evolution formulation allows the pullback of the Hamiltonian to take an arbitrary constant value, while the physically motivated zero-Hamiltonian solutions are recovered as the particular case in which this constant vanishes.

The z -dependent Hamilton–Jacobi problem

Let $\gamma: Q_{\text{th}} \times \mathbb{R}^k \rightarrow M$ be a section of the projection $\pi_{Q_{\text{th}} \times \mathbb{R}^k}: M \rightarrow Q_{\text{th}} \times \mathbb{R}^k$, given by

$$(\tilde{S}^\mu, N^\mu, T^{\lambda\mu}, P^\mu, \xi, \beta_\lambda, V) \mapsto (\xi, \beta_\lambda, V, \tilde{S}^\mu).$$

For simplicity, we use the same symbols $\tilde{S}^\mu$ for the coordinates on M and for the corresponding induced coordinates on the $\mathbb{R}^k$ factor of the base $Q_{\text{th}} \times \mathbb{R}^k$. Thus, the section γ can be written locally as

$$\gamma(\xi, \beta_\lambda, V, \tilde{S}^\mu) = (\tilde{S}^\mu, N^\mu, T^{\lambda\mu}, P^\mu, \xi, \beta_\lambda, V),$$

where N^μ , $T^{\lambda\mu}$, and P^μ are functions of $(\xi, \beta_\lambda, V, \tilde{S}^\mu)$. The pull-backs of the canonical one-forms read

$$\gamma^* \theta^\alpha = N^\alpha d\xi - \sum_{\lambda=1}^k T^{\lambda\alpha} d\beta_\lambda + P^\alpha dV, \quad \alpha = 1, \dots, k.$$

Assume that $\text{Im } \gamma$ is maximally coisotropic. Then, the general z -dependent Hamilton–Jacobi equation reads

$$d_Q(h \circ \gamma) + \sum_{\beta=1}^k \Gamma_\beta \gamma^* \theta^\beta + \sum_{\alpha, \beta=1}^k C_\alpha{}^\beta \iota_{\partial/\partial \tilde{S}^\beta} d(\gamma^* \theta^\alpha) = 0,$$

where $\Gamma_\beta = dh|_\gamma(T\gamma(\partial/\partial \tilde{S}^\beta))$ for $\beta = 1, \dots, k$. In the Hamiltonian case, the matrix $C = (C_\alpha{}^\beta)$ satisfies $\sum_{\alpha=1}^k C_\alpha{}^\alpha = -(h \circ \gamma)$, whereas in the evolution case $\sum_{\alpha=1}^k C_\alpha{}^\alpha = 0$.

Expanding the Hamilton–Jacobi equation in the coordinates (ξ, β_λ, V) , one obtains

$$\frac{\partial(h \circ \gamma)}{\partial \xi} + \sum_{\beta=1}^k \Gamma_\beta N^\beta + \sum_{\alpha, \beta=1}^k C_\alpha{}^\beta \frac{\partial N^\alpha}{\partial \tilde{S}^\beta} = 0,$$

$$\frac{\partial(h \circ \gamma)}{\partial \beta_\lambda} - \sum_{\beta=1}^k \Gamma_\beta T^{\lambda\beta} - \sum_{\alpha,\beta=1}^k C_\alpha^\beta \frac{\partial T^{\lambda\alpha}}{\partial \tilde{S}^\beta} = 0, \quad \lambda = 1, \dots, k,$$

and

$$\frac{\partial(h \circ \gamma)}{\partial V} + \sum_{\beta=1}^k \Gamma_\beta P^\beta + \sum_{\alpha,\beta=1}^k C_\alpha^\beta \frac{\partial P^\alpha}{\partial \tilde{S}^\beta} = 0.$$

These are the z -dependent Hamilton–Jacobi equations in the thermodynamic variables.

The construction above provides a k -contact realisation of a covariant EIT-type model in which the flux variables $(N^\mu, T^{\lambda\mu}, P^\mu)$ are treated as independent non-equilibrium fields, while the entropy fluxes are incorporated through the k -contact variables. Since the corresponding field equations are balance laws determined by the Hamiltonian h , the model can also be regarded as a reduced divergence-type system. In particular, the choice $h = \Phi(N^\mu, T^{\lambda\mu}, P^\mu)$ gives the conservation laws discussed above, whereas $h = U(\xi, \beta_\lambda, V) + \Phi(N^\mu, T^{\lambda\mu}, P^\mu)$ introduces source terms in the balance equations without changing the underlying k -contact geometric structure.

The Maxwell–Cattaneo model

We now illustrate the thermodynamic Hamilton–Jacobi formalism with a simple model from extended irreversible thermodynamics. In particular, we consider the one-dimensional Maxwell–Cattaneo system

$$\partial_t e + \partial_x J = 0, \quad \tau \partial_t J + J = \ell \partial_x \beta,$$

where e is the internal energy density, J is the heat flux, $\tau > 0$ is the relaxation time, $\ell > 0$ is a mobility coefficient, and $\beta(e) = s'_{\text{eq}}(e)$ is the inverse temperature associated with the local equilibrium entropy s_{eq} . We assume throughout that τ and ℓ are constant. The Maxwell–Cattaneo system is a basic model of extended irreversible thermodynamics, in which the heat flux is treated as an independent thermodynamic variable rather than as an algebraic function of the temperature gradient (see, for instance, [31, 105, 116]). The construction below can also be regarded as a field-theoretic generalisation of previous contact-geometric approaches to thermodynamics [14].

Let Q_{th} be the thermodynamic state space with coordinates (e, J) , and consider the two-contact phase space $\mathcal{J}_{Q_{\text{th}},2} = \bigoplus_{\mu=0}^1 T^*Q_{\text{th}} \times \mathbb{R}^2$, with coordinates $(e, J, p_e^\mu, p_J^\mu, z^\mu)$, where $\mu = 0, 1$, and canonical two-contact form

$$\eta_{Q_{\text{th}},2} = \sum_{\mu=0}^1 (dz^\mu - p_e^\mu de - p_J^\mu dJ) \otimes e_\mu.$$

Consider the entropy density and entropy flux [48, 105]

$$S^0(e, J) = s_{\text{eq}}(e) - \frac{\tau}{2\ell} J^2, \quad S^1(e, J) = \beta(e)J.$$

They define a z -independent holonomic section $\gamma: Q_{\text{th}} \rightarrow \mathcal{J}_{Q_{\text{th}},2}$ by

$$z^\mu = S^\mu, \quad p_e^\mu = \frac{\partial S^\mu}{\partial e}, \quad p_J^\mu = \frac{\partial S^\mu}{\partial J}, \quad \mu = 0, 1.$$

Explicitly,

$$p_e^0 = \beta(e), \quad p_J^0 = -\frac{\tau}{\ell} J, \quad p_e^1 = \beta'(e)J, \quad p_J^1 = \beta(e).$$

Hence, $\gamma^* \eta_{Q_{\text{th}},2} = 0$, and therefore $\text{Im } \gamma$ is Legendrian.

We next construct a two-contact Hamiltonian whose Hamilton–De Donder–Weyl equations recover the Maxwell–Cattaneo equations upon projection. Let $F, G \in C^\infty(Q_{\text{th}})$, and define

$$V_0^e = F, \quad V_1^e = G, \quad V_0^J = \frac{\ell \beta'(e)G - J}{\tau}, \quad V_1^J = -F.$$

Thus, any integral section satisfying

$$e_t = V_0^e, \quad e_x = V_1^e, \quad J_t = V_0^J, \quad J_x = V_1^J$$

also satisfies

$$e_t + J_x = 0, \quad \tau J_t + J = \ell \beta'(e) e_x = \ell \partial_x \beta.$$

Accordingly, consider the Hamiltonian

$$h_{\text{MC}}^{F,G} = p_e^0 F + p_e^1 G + p_J^0 \frac{\ell \beta'(e) G - J}{\tau} - p_J^1 F - \frac{J^2}{\ell}.$$

Since $h_{\text{MC}}^{F,G}$ is independent of the contact variables z^μ , its Hamilton–De Donder–Weyl equations are

$$\frac{\partial q^i}{\partial x^\mu} = \frac{\partial h_{\text{MC}}^{F,G}}{\partial p_i^\mu}, \quad \sum_{\mu=0}^1 \frac{\partial p_i^\mu}{\partial x^\mu} = -\frac{\partial h_{\text{MC}}^{F,G}}{\partial q^i},$$

together with

$$\partial_t z^0 + \partial_x z^1 = \sum_{\mu=0}^1 \left(p_e^\mu \frac{\partial h_{\text{MC}}^{F,G}}{\partial p_e^\mu} + p_J^\mu \frac{\partial h_{\text{MC}}^{F,G}}{\partial p_J^\mu} \right) - h_{\text{MC}}^{F,G}.$$

The first group gives

$$e_t = F, \quad e_x = G, \quad J_t = \frac{\ell \beta'(e) G - J}{\tau}, \quad J_x = -F.$$

Therefore, whenever these equations admit an integral section, its projection onto Q_{th} satisfies the Maxwell–Cattaneo system. Conversely, along a solution of the Maxwell–Cattaneo equations, the functions F and G represent respectively the derivatives e_t and e_x .

The contact equation reduces to

$$\partial_t z^0 + \partial_x z^1 = \frac{J^2}{\ell}.$$

We now show that the entropy pair defines a Hamilton–Jacobi solution for this Hamiltonian. Substitution of the section γ gives

$$h_{\text{MC}}^{F,G} \circ \gamma = \beta F + \beta'(e) J G - \frac{\tau}{\ell} J \frac{\ell \beta'(e) G - J}{\tau} - \beta F - \frac{J^2}{\ell} = 0.$$

Hence, γ satisfies the z -independent Hamiltonian Hamilton–Jacobi equation $h_{\text{MC}}^{F,G} \circ \gamma = 0$.

The projected Hamilton–Jacobi two-vector field on Q_{th} is given by

$$X_0^\gamma = F \frac{\partial}{\partial e} + \frac{\ell \beta'(e) G - J}{\tau} \frac{\partial}{\partial J}, \quad X_1^\gamma = G \frac{\partial}{\partial e} - F \frac{\partial}{\partial J}.$$

Therefore, if $\psi: \mathbb{R}^2 \rightarrow Q_{\text{th}}$, $\psi(t, x) = (e(t, x), J(t, x))$, is an integral section of (X_0^γ, X_1^γ) , then ψ satisfies the Maxwell–Cattaneo equations and, by the Hamilton–Jacobi theorem, $\gamma \circ \psi$ is a solution of the corresponding two-contact Hamilton–De Donder–Weyl equations.

Finally, the contact equation recovers the entropy-production law. Along the Hamilton–Jacobi lift one has $z^\mu = S^\mu \circ \psi$, and hence

$$\partial_t z^0 + \partial_x z^1 = \frac{J^2}{\ell}$$

becomes

$$\partial_t S^0 + \partial_x S^1 = \frac{J^2}{\ell} \geq 0.$$

The same identity follows directly from the Maxwell–Cattaneo equations:

$$\partial_t S^0 + \partial_x S^1 = \beta \partial_t e - \frac{\tau}{\ell} J \partial_t J + \partial_x (\beta J) = J \partial_x \beta - \frac{\tau}{\ell} J \partial_t J = \frac{J^2}{\ell} \geq 0.$$

Thus, particular solutions of the Maxwell–Cattaneo system are recovered as projections of solutions of the two-contact Hamilton–De Donder–Weyl equations, while the entropy density and entropy flux determine the contact variables through the Hamilton–Jacobi section. In this sense, the two-contact Hamilton–Jacobi formalism does not merely encode the entropy pair; it also provides a geometric lift of the physical dynamics together with its entropy production. In particular, it is remarkable that, along the Hamilton–Jacobi section, the contact variables z^μ are identified with the components S^μ of the entropy current.

Chapter 4

New k -contact physical models

The aim of this chapter is to develop k -contact Hamiltonian descriptions of several classes of non-conservative partial differential equations. The additional variables z^α of the k -contact formalism allow dissipative, attenuating, and more general action-dependent terms to be incorporated directly into the geometric description of the dynamics [29, 30, 111]. This provides a natural framework for the study of non-conservative field equations that do not fit directly into the usual conservative k -symplectic or multisymplectic setting.

Two complementary constructions will be considered. First, we study canonical k -contact Hamiltonian systems on $\mathcal{J}_{Q,k}$ whose Hamiltonians depend polynomially on the dissipative variables. Under a regularity assumption, their Hamilton–De Donder–Weyl equations can be used to reconstruct second-order systems of partial differential equations. The familiar case of Hamiltonians affine in the dissipative variables appears as a particularly important specialisation of this family, since the resulting equations for the configuration variables are then independent of the z^α . Second, we consider k -contactifications of exact k -symplectic manifolds. In this setting, the Hamilton–De Donder–Weyl equations admit a natural splitting into equations on the underlying k -symplectic manifold and a balance equation for the dissipative variables. For two-contact systems, we introduce a class of adapted reduced phase spaces of cotangent type and a corresponding notion of partial regularity. This makes it possible to eliminate only the spatial momenta, rather than requiring regularity with respect to a full collection of momentum variables.

These constructions provide the geometric framework for the applications developed in Section 4.3. In particular, they broaden the class of partial differential equations that can be described within k -contact geometry and include examples in which the Hamiltonian has nonlinear polynomial dependence on the dissipative variables, extending the predominantly linear dependence considered in previous applications.

4.1 HdDW equations for Hamiltonians with polynomial dependence on the dissipative variables

In this section, we consider canonical k -contact Hamiltonian systems on $\mathcal{J}_{Q,k}$ whose Hamiltonians depend polynomially on the dissipative variables. This extends the class of Hamiltonians considered in Section 3.1 and provides a useful mechanism for constructing higher-order partial differential equations from k -contact Hamiltonian systems (cf. [40, 110–112]). The results of this section are presented in [81].

The following result states that, for such Hamiltonians, every integral section of an $\eta_{Q,k}$ -Hamiltonian k -vector field associated with the same Hamiltonian function projects onto a map whose coordinates satisfy the same second-order system of PDEs. It also explains certain additional properties of this case, which will be important for applications of k -contact geometry to PDEs, both in the literature and in the remainder of this dissertation. Note that its proof is quite straightforward and it is left to the reader. The essential assumption is regularity of the Hamiltonian with respect to the momentum variables, which allows the momentum equations to be inverted locally.

Theorem 4.1.1. *Let $h \in C^\infty(\mathcal{J}_{Q,k})$ take the form*

$$h(q^i, p_i^\alpha, z^\alpha) = g(q^i, p_i^\alpha) + \sum_{\alpha=1}^k A_\alpha(z^\alpha)^{\lambda_\alpha},$$

where $A_1, \dots, A_k$ are real constants, $\lambda_1, \dots, \lambda_k$ are non-negative integers, and h is regular. Let us assume that $\psi(t) = (q^i(t), p_i^\alpha(t), z^\alpha(t))$ is an integral section of an $\eta_{Q,k}$ -Hamiltonian k -vector field $\mathbf{X}_h$. Then, $\psi(t)$ projects onto a map $q(t) = (q^1(t), \dots, q^n(t))$ satisfying a second-order system of PDEs. More precisely, if $p_i^\alpha = P_i^\alpha(q, v_\beta)$ denotes the local inverse of the fibre derivative, determined by $v_\beta^i = \partial h / \partial p_i^\beta = \partial g / \partial p_i^\beta$, then the functions $q^i(t)$ satisfy the second-order system of PDEs of the form

$$\sum_{\alpha=1}^k \frac{\partial}{\partial t^\alpha} \left(P_i^\alpha(q, v_\beta) \right) + \frac{\partial g}{\partial q^i} \left(q, P_j^\gamma(q, v_\beta) \right) + \sum_{\alpha=1}^k R(z^\alpha, \lambda_\alpha) A_\alpha P_i^\alpha(q, v_\beta) = 0, \quad i = 1, \dots, n, \quad (4.1.1)$$

where $q = q(t)$ and $v_\beta = \partial q / \partial t^\beta$ and $R(z, \mu) = z^{\mu-1} \mu$ for $\mu \in \mathbb{N}_+$ and $R(z, 0) = 0$ for every $z \in \mathbb{R}$.

The system of second-order PDEs (1.2.11) is the particular case of the second-order system (4.1.1) for the Hamiltonian function (1.2.10), which is polynomial in the dissipative variables in the manner given in Theorem 4.1.1. Hence, Theorem 4.1.1 ensures that every integral section of an $\eta_{Q,k}$ -Hamiltonian k -vector field associated with the Hamiltonian (1.2.10) projects onto a solution of the damped double sine-Gordon equation (1.2.11).

If some $\lambda_\alpha > 1$, then (4.1.1) still involves the variables $z^\alpha(t)$. The HdDW equations for h also impose balance equations for $z^1(t), \dots, z^k(t)$, in addition to (4.1.1); these equations may depend on $q^i(t)$ and $p_i^\beta(t)$. Hence, in general, (4.1.1) has to be considered together with the corresponding z -balance equations. In the examples below, however, either the resulting second-order equations do not involve the variables z^α , or one restricts to compatible solutions for which the $z^\alpha(t)$ take prescribed values, often constants.

It is worth noting that higher-order systems of PDEs can be written as second-order systems of PDEs by considering higher-order partial derivatives of the coordinates of solutions as new variables. In this manner, the resulting system can be described by our k -contact formalism.

Theorem 4.1.1 can also be extended to Hamiltonian functions of the form

$$h(q^i, p_i^\alpha, z^\alpha) = g(q^i, p_i^\alpha) + \sum_{\alpha=1}^k A_\alpha(q) (z^\alpha)^{\lambda_\alpha},$$

where $A_1, \dots, A_k \in C^\infty(Q)$. In this case, the q -dependence of the functions A_α gives rise to additional terms in the corresponding second-order system of PDEs. More precisely, equation (4.1.1) becomes

$$\sum_{\alpha=1}^k \frac{\partial}{\partial t^\alpha} \left(P_i^\alpha(q, v_\beta) \right) + \frac{\partial g}{\partial q^i} \left(q, P_j^\gamma(q, v_\beta) \right) + \sum_{\alpha=1}^k \frac{\partial A_\alpha}{\partial q^i}(q) (z^\alpha)^{\lambda_\alpha} + \sum_{\alpha=1}^k R(z^\alpha, \lambda_\alpha) A_\alpha(q) P_i^\alpha(q, v_\beta) = 0.$$

On the other hand, Theorem 3.1.3 is recovered as the particular case of Theorem 4.1.1 in which $\lambda_\alpha = 1$ for every $\alpha = 1, \dots, k$, namely,

$$h(q^i, p_i^\alpha, z^\alpha) = g(q^i, p_i^\alpha) + \sum_{\alpha=1}^k A_\alpha z^\alpha, \quad A_1, \dots, A_k \in \mathbb{R}.$$

Indeed, since $R(z^\alpha, 1) = 1$, equation (4.1.1) reduces to (3.1.2). In particular, as already observed in Theorem 3.1.3, the resulting second-order system is independent of the dissipative variables $z^1, \dots, z^k$.

Proposition 4.1.2. *Let $(\mathcal{J}_{Q,k}, \eta_{Q,k}, h)$ be a canonical k -contact Hamiltonian system with $\dim Q = 1$, and assume the associated second-order system of PDEs as in Theorem 4.1.1. If $A = \left(\frac{\partial^2 h}{\partial p^\alpha \partial p^\beta} \right)$ is definite, the resulting second-order equation is elliptic. If A has signature $(k-1, 1)$ or $(1, k-1)$, one obtains a hyperbolic equation. If A is indefinite with at least two positive and two negative eigenvalues, the equation is of ultrahyperbolic type.*

Proof. Since $\dim Q = 1$, the reconstructed second-order system of PDEs (4.1.1) consists of a single scalar equation. Its analytic type is therefore determined by the quadratic form associated with its principal symbol. By Theorem 4.1.1, setting $v_\beta = \partial q / \partial t^\beta$, the reconstructed equation is

$$\sum_{\alpha=1}^k \frac{\partial}{\partial t^\alpha} \left(P^\alpha(q, v_\beta) \right) + \frac{\partial g}{\partial q} \left(q, P^\gamma(q, v_\beta) \right) + \sum_{\alpha=1}^k R(z^\alpha, \lambda_\alpha) A_\alpha P^\alpha(q, v_\beta) = 0.$$

The only terms containing second-order derivatives of q arise from the first summand. Hence, the principal part is

$$\sum_{\alpha, \beta=1}^k \frac{\partial P^\alpha}{\partial v_\beta} (q, v_\gamma) \frac{\partial^2 q}{\partial t^\alpha \partial t^\beta},$$

and the corresponding principal symbol is

$$\sigma(\xi) = \sum_{\alpha, \beta=1}^k \frac{\partial P^\alpha}{\partial v_\beta} (q, v_\gamma) \xi_\alpha \xi_\beta.$$

By the regularity of h , the momentum Hessian $A = \left(\frac{\partial^2 h}{\partial p^\alpha \partial p^\beta} \right)$ is non-singular. Moreover, P^α is obtained locally by inverting the fibre derivative $v_\alpha = \partial h / \partial p^\alpha$. Differentiating these inverse relations gives $\left(\frac{\partial P^\alpha}{\partial v_\beta} \right) = A^{-1}$. Therefore, the principal symbol is represented by the quadratic form associated with A^{-1} . Since A is symmetric and non-singular, A and A^{-1} have the same inertia.

It follows that if A is definite, then the principal symbol σ is definite and the equation is elliptic. If A has signature $(k-1, 1)$ or $(1, k-1)$, then σ has the same signature and the equation is hyperbolic. Finally, if A has at least two positive and two negative eigenvalues, then σ has the same property and the equation is of ultrahyperbolic type. $\square$

Remark 4.1.3. The assumption $\dim Q = 1$ is essential in the previous proposition, since only in the scalar case is the analytic type determined by a single quadratic form. When $\dim Q > 1$, the principal symbol of the reconstructed system is matrix-valued, and its analytic type must be determined from the full principal symbol rather than from the signature of a single quadratic form.

4.2 HdDW equations for k -contactifications

We now describe the general form of the HdDW equations on k -contactifications of exact k -symplectic manifolds. In particular, we show that these equations naturally split into a system on the underlying k -symplectic manifold and a balance equation for the dissipative variables. This formulation will be useful for studying particular classes of Hamiltonian functions and the associated systems of PDEs appearing in the applications of Section 4.3.

Proposition 4.2.1. *Let $(P, \omega = -d\theta)$ be an exact k -symplectic manifold and let $(M = P \times \mathbb{R}^k, \eta_\theta)$ be its k -contactification and $\rho : M \rightarrow P$ the natural projection. Then, a section $\psi = (\phi, z^1, \dots, z^k) : D \subset \mathbb{R}^k \rightarrow M$ satisfies the k -contact Hamilton–De Donder–Weyl equations for $h \in C^\infty(M)$ if and only if*

$$\sum_{\alpha=1}^k \iota_{\phi'_\alpha} \omega^\alpha = \left(d_P h + \sum_{\alpha=1}^k \frac{\partial h}{\partial z^\alpha} \rho^* \theta^\alpha \right) \circ \psi, \quad \sum_{\alpha=1}^k \frac{\partial z^\alpha}{\partial t^\alpha} = \left(\sum_{\alpha=1}^k \rho^* \theta^\alpha (\phi'_\alpha) - h \right) \circ \psi. \quad (4.2.1)$$

In particular, the equations on the k -symplectic factor P and the equation for the dissipative variables split naturally.

Proof. Since $R_\alpha = \partial / \partial z^\alpha$ for $\alpha = 1, \dots, k$, the k -contact Hamilton–De Donder–Weyl equations read

$$\sum_{\alpha=1}^k \iota_{\psi'_\alpha} d\eta_\theta^\alpha = \left(dh - \sum_{\alpha=1}^k \frac{\partial h}{\partial z^\alpha} \eta_\theta^\alpha \right) \circ \psi, \quad \sum_{\alpha=1}^k \iota_{\psi'_\alpha} \eta_\theta^\alpha = -h \circ \psi.$$

Since $d\eta_\theta = \rho^*\omega$, one has that $\iota_{\psi'_\alpha} d\eta_\theta^\alpha = \iota_{\phi'_\alpha} \rho^*\omega^\alpha$. Moreover, since $\eta_\theta^\alpha = dz^\alpha - \rho^*\theta^\alpha$, one has

$$dh - \sum_{\alpha=1}^k \frac{\partial h}{\partial z^\alpha} \eta_\theta^\alpha = d_P h + \sum_{\alpha=1}^k \frac{\partial h}{\partial z^\alpha} \rho^*\theta^\alpha.$$

This yields the first equation in (4.2.1). On the other hand,

$$\sum_{\alpha=1}^k \iota_{\psi'_\alpha} \eta_\theta^\alpha = \sum_{\alpha=1}^k \left(\frac{\partial z^\alpha}{\partial t^\alpha} - \rho^*\theta^\alpha(\phi'_\alpha) \right),$$

which gives the second equation in (4.2.1). $\square$

The main point for applications to second-order systems of PDEs is that, for suitable Hamiltonian functions, the first equation in (4.2.1) can be used to obtain a second-order system of PDEs for the relevant configuration variables, whereas the second equation provides a balance equation for the dissipative variables. In some cases, the resulting equations for the configuration variables are independent of the dissipative variables. In other cases, the dissipative variables can be chosen in a particular form compatible with their balance equation. The following result describes the equations arising from Hamiltonian functions that are affine in the dissipative variables.

Corollary 4.2.2. *In the setting of Proposition 4.2.1, define $h(x, z) = g(x) + \sum_{\alpha=1}^k a_\alpha(x) z^\alpha$, where $x \in P$. Then, equations (4.2.1) become*

$$\sum_{\alpha=1}^k \iota_{\phi'_\alpha} \omega^\alpha = \left(dg + \sum_{\alpha=1}^k z^\alpha da_\alpha + \sum_{\alpha=1}^k a_\alpha \rho^*\theta^\alpha \right) \circ \psi$$

and

$$\sum_{\alpha=1}^k \frac{\partial z^\alpha}{\partial t^\alpha} = \left(\sum_{\alpha=1}^k \rho^*\theta^\alpha(\phi'_\alpha) - g - \sum_{\alpha=1}^k a_\alpha z^\alpha \right) \circ \psi.$$

Proof. Indeed, $\frac{\partial h}{\partial z^\alpha} = a_\alpha(x)$ and $d_P h = dg + \sum_{\alpha=1}^k z^\alpha da_\alpha$. Substituting these identities into Proposition 4.2.1 yields the result. $\square$

4.2.1 Adapted reduced two-contact phase spaces and partial regularity

We now introduce a class of exact two-symplectic models that will appear in several applications in Section 4.3. In these examples, the reduced exact two-symplectic phase space is of cotangent type: one component of the primitive is the Liouville one-form of a cotangent bundle, whereas the other is the pull-back of a one-form on its base manifold. This structure provides a natural distinction between the field variables and the spatial momenta and, under an appropriate regularity condition on the Hamiltonian, allows the latter to be eliminated in order to obtain a second-order system of PDEs.

Definition 4.2.3. Let $\tau : T^*N \rightarrow N$ be the canonical projection of the cotangent bundle of an n -dimensional manifold N , let θ_N be the Liouville one-form on T^*N , and let $\theta_e \in \Omega^1(T^*N)$ be a basic one-form, namely, the pull-back by τ of a differential one-form on N . Then, T^*N is endowed with an adapted exact two-symplectic structure given by the $\mathbb{R}^2$ -valued two-form $\omega = -d\theta_e \otimes e_1 - d\theta_N \otimes e_2$. The associated two-contactification is the manifold $M = T^*N \times \mathbb{R}^2$ endowed with the two-contact form $\eta_\theta = (dz^t - \rho^*\theta_e) \otimes e_1 + (dz^x - \rho^*\theta_N) \otimes e_2$, where $\rho : M \rightarrow T^*N$ is the canonical projection and (z^t, z^x) denote the canonical coordinates on $\mathbb{R}^2$, as well as their pull-backs to M by the projection onto the second factor.

The assumption that θ_e is basic restricts the class of adapted exact two-symplectic structures considered here. It is motivated by the applications developed below: θ_e involves only the variables on the base

manifold N , whereas the Liouville form θ_N reflects the cotangent-bundle structure and hence the distinction between the base variables and the spatial momenta. Indeed, let $(y^1, \dots, y^n)$ be local coordinates on N , with induced fibre coordinates $(\pi_1, \dots, \pi_n)$ on T^*N . Then

$$\theta_e = \sum_{i=1}^n (\theta_e)_i(y) dy^i, \quad \theta_N = \sum_{i=1}^n \pi_i dy^i,$$

and consequently

$$\omega = \frac{1}{2} \sum_{i,j=1}^n \Omega_{ij}(y) dy^i \wedge dy^j \otimes e_1 + \sum_{i=1}^n dy^i \wedge d\pi_i \otimes e_2,$$

where $\Omega_{ij} := \partial_j(\theta_e)_i - \partial_i(\theta_e)_j$. In what follows, we write

$$\theta = \theta_1 \otimes e_1 + \theta_2 \otimes e_2 = \theta_e \otimes e_1 + \theta_N \otimes e_2 \in \Omega^1(T^*N, \mathbb{R}^2),$$

so that $\omega = -d\theta$. We continue to denote by $\rho : T^*N \times \mathbb{R}^2 \rightarrow T^*N$ the canonical projection. Thus, the two components of θ play different geometric roles. The component θ_N is the Liouville form associated with the cotangent variables (y^i, π_i) , whereas the basic component θ_e defines a pairing involving only the variables on the base manifold N . This geometric mechanism underlies the examples on adapted two-contact phase spaces of cotangent type considered later on and plays a key role in describing their HddW equations.

We next introduce the corresponding notion of regularity for this reduced setting.

Definition 4.2.4. Let $M = T^*N \times \mathbb{R}^2$ be the two-contactification of an adapted exact two-symplectic manifold on T^*N , and let $h \in C^\infty(M)$. We say that h is π -regular if the partial fibre derivative $\mathbb{F}_\pi h : T^*N \times \mathbb{R}^2 \rightarrow TN \times \mathbb{R}^2$, given by $(y^1, \dots, y^n, \pi_1, \dots, \pi_n, z^t, z^x) \mapsto (y^1, \dots, y^n, \frac{\partial h}{\partial \pi_1}, \dots, \frac{\partial h}{\partial \pi_n}, z^t, z^x)$, is a local diffeomorphism. If $\mathbb{F}_\pi h$ is a global diffeomorphism, then h is said to be π -hyperregular.

The following is a standard consequence of the inverse function theorem applied to the partial fibre derivative $\mathbb{F}_\pi h$.

Proposition 4.2.5. *The following conditions are equivalent:*

1. h is π -regular;
2. the Hessian matrix $(\partial^2 h / \partial \pi_i \partial \pi_j)_{i,j=1,\dots,n}$ is non-singular at every point;
3. the relations $v^i = \partial h / \partial \pi_i$, with $i = 1, \dots, n$, can be solved locally for the variables $\pi_1, \dots, \pi_n$ as smooth functions of $(y^1, \dots, y^n, v^1, \dots, v^n, z^t, z^x)$.

The relevance of π -regularity is that it allows the momentum variables π_i to be locally eliminated from the two-contact Hamilton–De Donder–Weyl equations on M , leading to a second-order system of PDEs for the configuration variables.

Proposition 4.2.6. *Let $M = T^*N \times \mathbb{R}^2$ be the two-contactification of an adapted exact two-symplectic manifold, and let $h \in C^\infty(M)$ be π -regular. Every solution, let us say $\psi = (y^1, \dots, y^n, \pi_1, \dots, \pi_n, z^t, z^x) : D \subset \mathbb{R}^2 \rightarrow M$, of the two-contact Hamilton–De Donder–Weyl equations satisfies locally a second-order system of PDEs for the variables $y^1, \dots, y^n$. More precisely, if the local inverse of the relations $(y^i)_x = \frac{\partial h}{\partial \pi_i}$ with $i = 1, \dots, n$ read $\pi = \Pi(y, y_x, z^t, z^x)$, then*

$$\sum_{j=1}^n \Omega_{ji}(y) (y^j)_t - \partial_x(\Pi_i(y, y_x, z^t, z^x)) = \left(\frac{\partial h}{\partial y^i} + \frac{\partial h}{\partial z^t} (\theta_e)_i(y) + \frac{\partial h}{\partial z^x} \pi_i \right) \Big|_{\pi=\Pi(y, y_x, z^t, z^x)}, \quad (4.2.2)$$

for $\Omega_{ki} := \partial_i(\theta_e)_k - \partial_k(\theta_e)_i$ and $i, k = 1, \dots, n$. Moreover, the dissipative variables satisfy

$$\partial_t z^t + \partial_x z^x = \sum_{i=1}^n (\theta_e)_i(y) (y^i)_t + \sum_{i=1}^n \Pi_i(y, y_x, z^t, z^x) (y^i)_x - h(y, \Pi(y, y_x, z^t, z^x), z^t, z^x). \quad (4.2.3)$$

Proof. By Proposition 4.2.1, the HdDW equations split into an equation on T^*N and an equation for the dissipative variables. In the above coordinates, the first equation in (4.2.1) reads

$$\iota_{\phi'_t}\omega_1 + \iota_{\phi'_x}\omega_2 = d_{T^*N}h + \frac{\partial h}{\partial z^t}\rho^*\theta_e + \frac{\partial h}{\partial z^x}\rho^*\theta_N.$$

Since

$$\omega_1 = \frac{1}{2} \sum_{i,j=1}^n \Omega_{ij}(y) dy^i \wedge dy^j, \quad \omega_2 = \sum_{i=1}^n dy^i \wedge d\pi_i,$$

one obtains

$$(y^i)_x = \frac{\partial h}{\partial \pi_i}, \quad i = 1, \dots, n, \quad (4.2.4)$$

and

$$\sum_{j=1}^n \Omega_{ji}(y) (y^j)_t - \partial_x \pi_i = \frac{\partial h}{\partial y^i} + \frac{\partial h}{\partial z^t} (\theta_e)_i(y) + \frac{\partial h}{\partial z^x} \pi_i, \quad i = 1, \dots, n. \quad (4.2.5)$$

By π -regularity, the first relations (4.2.4) can be solved locally as $\pi_i = \Pi_i(y, y_x, z^t, z^x)$, for $i = 1, \dots, n$. Substituting them into (4.2.5) yields (4.2.2). Equation (4.2.3) follows from Proposition 4.2.1, since

$$\rho^*\theta_1(\phi'_t) + \rho^*\theta_2(\phi'_x) = \sum_{i=1}^n (\theta_e)_i(y) (y^i)_t + \sum_{i=1}^n \pi_i (y^i)_x.$$

□

The previous proposition explains the structure of the reduced examples studied later on. The component associated with θ_N gives rise to the constitutive relations that identify the spatial momenta in terms of spatial derivatives of the solutions, while the component associated with θ_e determines the evolution pairing on the base manifold. The target system of second-order PDEs is then obtained by eliminating the momenta from the balance equations. In this way, adapted two-contactifications of exact two-symplectic manifolds provide a natural geometric mechanism for constructing second-order dissipative partial differential equations from two-contact Hamiltonian systems. In particular, the terms $\frac{\partial h}{\partial z^x} \pi_i$, for $i = 1, \dots, n$, in (4.2.2) account for the additional reduced models in which the Hamiltonian depends on the spatial dissipative variable z^x .

Remark 4.2.7. If, in addition, h is affine in the dissipative variables, say

$$h(y, \pi, z^t, z^x) = g(y, \pi) + a(y, \pi)z^t + b(y, \pi)z^x,$$

then (4.2.2) becomes

$$\sum_{j=1}^n \Omega_{ji}(y) (y^j)_t - \partial_x (\Pi_i(y, y_x, z^t, z^x)) = \left(\frac{\partial g}{\partial y^i} + z^t \frac{\partial a}{\partial y^i} + z^x \frac{\partial b}{\partial y^i} + a (\theta_e)_i(y) + b \pi_i \right) \Big|_{\pi = \Pi(y, y_x, z^t, z^x)},$$

for $i = 1, \dots, n$, where Π is locally determined by

$$(y^i)_x = \frac{\partial g}{\partial \pi_i} + z^t \frac{\partial a}{\partial \pi_i} + z^x \frac{\partial b}{\partial \pi_i}, \quad i = 1, \dots, n.$$

Moreover, (4.2.3) becomes

$$\begin{aligned} \partial_t z^t + \partial_x z^x &= \sum_{i=1}^n (\theta_e)_i(y) (y^i)_t + \sum_{i=1}^n \Pi_i(y, y_x, z^t, z^x) (y^i)_x - g(y, \Pi(y, y_x, z^t, z^x)) \\ &\quad - a(y, \Pi(y, y_x, z^t, z^x)) z^t - b(y, \Pi(y, y_x, z^t, z^x)) z^x. \end{aligned}$$

This framework covers both the examples with Hamiltonians depending on z^t and those with Hamiltonians depending on z^x considered in Section 4.3.

4.3 Applications

The constructions developed in the previous sections will now be applied to several dissipative systems of PDEs. The examples illustrate different ways in which the dependence of the Hamiltonian on the dissipative variables modifies the Hamilton–De Donder–Weyl equations and gives rise, after elimination of the momentum variables, to familiar or new second-order field equations. We consider both Hamiltonians on canonical two-contact phase spaces and models formulated on the adapted two-contactifications introduced in the previous section. Particular attention will be paid to the role of the dissipative variables and to the balance equations governing their evolution.

All Hamiltonians considered below are regular on the domains specified in the corresponding examples. For the reduced models introduced in Subsection 4.2.1, the corresponding notion is π -regularity. Therefore, the relevant momentum variables can be locally eliminated in favour of the derivatives of the fields. We shall use this fact throughout without explicitly computing the corresponding Hessian matrices in each example.

4.3.1 A damped Klein–Gordon model

Consider the Hamiltonian function $h \in C^\infty(\mathcal{J}_{\mathbb{R},2})$ given by

$$h(u, p^t, p^x, z^t, z^x) = \frac{1}{2} \left((p^t)^2 - \frac{1}{c^2} (p^x)^2 \right) + \frac{1}{2} (z^t)^2 + \frac{\epsilon}{2} u^2, \quad c > 0, \epsilon \geq 0,$$

where (u, p^t, p^x, z^t, z^x) are canonical Darboux coordinates on $(\mathcal{J}_{\mathbb{R},2}, \boldsymbol{\eta}_{\mathbb{R},2}, \mathcal{V}_{\mathbb{R},2})$.

Hence, the two-contact Hamilton–De Donder–Weyl equations for a section $\psi = \psi(t, x)$ read

$$u_t = \frac{\partial h}{\partial p^t}, \quad u_x = \frac{\partial h}{\partial p^x}, \quad \partial_t p^t + \partial_x p^x = - \left(\frac{\partial h}{\partial u} + p^t \frac{\partial h}{\partial z^t} + p^x \frac{\partial h}{\partial z^x} \right),$$

together with

$$\partial_t z^t + \partial_x z^x = p^t \frac{\partial h}{\partial p^t} + p^x \frac{\partial h}{\partial p^x} - h.$$

Since h is regular, one can write the momenta in terms of u_t, u_x . Indeed, $\partial h / \partial p^t = p^t$ and $\partial h / \partial p^x = -p^x / c^2$, and the first two equations give $p^t = u_t$ and $p^x = -c^2 u_x$. On the other hand, $\partial h / \partial u = \epsilon u$, $\partial h / \partial z^t = z^t$, and $\partial h / \partial z^x = 0$, so the momentum balance equation becomes

$$\partial_t p^t + \partial_x p^x = -(\epsilon u + z^t p^t).$$

Substituting the above expressions for the momenta, one obtains

$$u_{tt} - c^2 u_{xx} + z^t u_t + \epsilon u = 0.$$

Hence, h yields a damped Klein–Gordon-type equation with effective damping coefficient z^t and restoring term ϵu . In the limiting case $\epsilon = 0$, the equation reduces to a damped wave, or telegrapher-type, equation. In one spatial dimension, the latter is classically interpreted as a telegrapher-type equation describing signal propagation in transmission lines with resistance and inductance [118].

The evolution of the dissipative variables is governed by

$$\partial_t z^t + \partial_x z^x = p^t \frac{\partial h}{\partial p^t} + p^x \frac{\partial h}{\partial p^x} - h = \frac{1}{2} (p^t)^2 - \frac{1}{2c^2} (p^x)^2 - \frac{1}{2} (z^t)^2 - \frac{\epsilon}{2} u^2,$$

that is,

$$\partial_t z^t + \partial_x z^x = \frac{1}{2} u_t^2 - \frac{1}{2} c^2 u_x^2 - \frac{1}{2} (z^t)^2 - \frac{\epsilon}{2} u^2.$$

Therefore, the dissipative variable z^t , which acts as the damping coefficient in the reconstructed Klein–Gordon equation, is not prescribed externally but evolves according to a balance law coupled to the kinetic,

elastic, and potential contributions of the field. Alternatively, one may prescribe $z^t(t, x) = \lambda(t, x)$. Then, for every solution $u(t, x)$ of

$$u_{tt} - c^2 u_{xx} + \lambda(t, x)u_t + \epsilon u = 0,$$

the remaining dissipative variable z^x is locally determined, up to the corresponding integration data, by

$$\partial_x z^x = \frac{1}{2}u_t^2 - \frac{1}{2}c^2 u_x^2 - \frac{1}{2}\lambda^2 - \frac{\epsilon}{2}u^2 - \partial_t \lambda.$$

This example may be regarded as a natural two-contact candidate for damped Klein–Gordon or telegrapher equations [4, 128].

Let us now interpret dissipation in the sense of k -contact geometry. By the general notion of dissipation law for k -contact Hamiltonian systems [29], a map $\mathbf{F} = (F^t, F^x): M \rightarrow \mathbb{R}^2$ is said to be dissipated along solutions if

$$\partial_t(F^t \circ \psi) + \partial_x(F^x \circ \psi) = -[(\mathcal{L}_{R_t} h)F^t + (\mathcal{L}_{R_x} h)F^x] \circ \psi.$$

In the present case, $R_t = \partial/\partial z^t$ and $R_x = \partial/\partial z^x$, so $\mathcal{L}_{R_t} h = z^t$, $\mathcal{L}_{R_x} h = 0$, and consequently

$$\partial_t(F^t \circ \psi) + \partial_x(F^x \circ \psi) = -(z^t F^t) \circ \psi.$$

Thus, z^t plays the role of an instantaneous dissipation rate.

When $\epsilon = 0$, the Hamiltonian is independent of u , and therefore $Y = \partial/\partial u$ is an infinitesimal dynamical symmetry. The associated dissipated quantity is $\mathbf{F} = (-\iota_Y \eta^t, -\iota_Y \eta^x) = (p^t, p^x)$. Hence, the general dissipation theorem yields

$$\partial_t p^t + \partial_x p^x = -z^t p^t.$$

Using $p^t = u_t$ and $p^x = -c^2 u_x$, one obtains

$$\partial_t(u_t) + \partial_x(-c^2 u_x) = -z^t u_t.$$

Thus, for $z^t = \lambda(t, x)$, one recovers the standard local momentum-type balance law for the damped wave equation $u_{tt} - c^2 u_{xx} + \lambda(t, x)u_t = 0$. In the particular case of constant damping $z^t = \lambda$, and under suitable boundary conditions, this yields

$$\frac{d}{dt} \int p^t dx = -\lambda \int p^t dx, \quad e^{\lambda t} \int p^t dx = \text{const.}$$

Therefore, the k -contact formalism geometrises a classical exponentially weighted balance law for damped wave equations. This is consistent with the generalized momentum-type conservation laws described in the symmetry analysis of one-dimensional damped wave equations, and also with the conformal balance laws arising in damped acoustic wave models written in multisymplectic form [16].

4.3.2 Reaction–diffusion / Allen–Cahn type equation

Consider the equation

$$u_t - \nu u_{xx} + V'(u) + \lambda u = 0, \quad \nu > 0, \lambda \geq 0, \quad (4.3.1)$$

where $V \in C^\infty(\mathbb{R})$, for instance $V(u) = \frac{1}{4}(u^2 - 1)^2$. This class includes, in particular, Allen–Cahn type reaction–diffusion equations [3, 26, 97].

Let us study (4.3.1) via the two-contact manifold obtained as the contactification of a reduced exact two-symplectic manifold. Let $P_0 = \mathbb{R}^4 \simeq T^*\mathbb{R}^2$ with coordinates (u, v, p^x, q^x) , endowed with the one-form

$$\theta = \frac{1}{2}(-v du + u dv) \otimes e_1 + (p^x du + q^x dv) \otimes e_2$$

Hence, one obtains an exact two-symplectic manifold $(P_0, \omega = -d\theta)$. Let us now consider the contactification $M = P_0 \times \mathbb{R}^2$ with coordinates $(u, v, p^x, q^x, z^t, z^x)$ and the two-contact form $\eta = (dz^t - \theta_e) \otimes e_1 + (dz^x - \theta_{P_0}) \otimes e_2$. Then, consider the Hamiltonian

$$h_{\text{RD}} = -\frac{1}{\nu} p^x q^x + v(V'(u) + \lambda u).$$

Since h_{RD} is independent of z^t and z^x , one has $R_t h_{\text{RD}} = R_x h_{\text{RD}} = 0$. Hence, the two-contact Hamilton–De Donder–Weyl equations reduce to

$$\partial_t v - \partial_x p^x = \frac{\partial h_{\text{RD}}}{\partial u}, \quad -\partial_t u - \partial_x q^x = \frac{\partial h_{\text{RD}}}{\partial v}, \quad \partial_x u = \frac{\partial h_{\text{RD}}}{\partial p^x}, \quad \partial_x v = \frac{\partial h_{\text{RD}}}{\partial q^x}.$$

A direct computation gives

$$\begin{aligned} \frac{\partial h_{\text{RD}}}{\partial u} &= v(V''(u) + \lambda), & \frac{\partial h_{\text{RD}}}{\partial v} &= V'(u) + \lambda u, \\ \frac{\partial h_{\text{RD}}}{\partial p^x} &= -\frac{1}{\nu} q^x, & \frac{\partial h_{\text{RD}}}{\partial q^x} &= -\frac{1}{\nu} p^x. \end{aligned}$$

Therefore, $q^x = -\nu u_x$ and $p^x = -\nu v_x$. Substituting these expressions into the previous equations, one obtains

$$u_t - \nu u_{xx} + V'(u) + \lambda u = 0, \quad v_t + \nu v_{xx} = v(V''(u) + \lambda).$$

Hence the first equation is precisely (4.3.1), while the second one describes the complementary evolution of the auxiliary field v .

In particular, $v = 0$ is a consistent reduction, and every solution u of (4.3.1) defines a solution of the full Hamiltonian system by taking $v = 0$, $q^x = -\nu u_x$, and $p^x = 0$.

Thus (4.3.1) admits a natural two-contact Hamilton–De Donder–Weyl description on a reduced phase space. Since h_{RD} is independent of the dissipative variables, there is no explicit dissipation through the Reeb directions in this model.

A family of generalised Burgers-type equations in a two-contact setting

We now show that the previous two-contact construction can be modified so as to cover a wider class of Burgers-type equations. Consider the reduced phase space $P_0 = T^*\mathbb{R}^2$ with coordinates (u, v, p_u^x, p_v^x) , endowed with the one-form with values in $\mathbb{R}^2$ given by

$$\theta = \theta_e \otimes e_1 + \theta_{\mathbb{R}^2} \otimes e_2,$$

where $\theta_e = \frac{1}{2}(-v du + u dv)$ and $\theta_{\mathbb{R}^2} = p_u^x du + p_v^x dv$. Then,

$$\omega = -d\theta = -d\theta_e \otimes e_1 - d\theta_{\mathbb{R}^2} \otimes e_2 = -du \wedge dv \otimes e_1 + (du \wedge dp_u^x + dv \wedge dp_v^x) \otimes e_2.$$

We consider the two-contactification $M = P_0 \times \mathbb{R}^2$ of $P_0 = T^*\mathbb{R}^2$ with coordinates $(u, v, p_u^x, p_v^x, z^t, z^x)$ and canonical two-contact forms $\eta = (dz^t - \rho^* \theta_e) \otimes e_1 + (dz^x - \rho^* \theta_{\mathbb{R}^2}) \otimes e_2$, where $\rho : M \rightarrow P_0$ is the standard projection.

Let $D, B, C \in C^\infty(\mathbb{R})$, with $D(u) \neq 0$, and consider the Hamiltonian

$$h = -\frac{1}{D(u)} p_u^x p_v^x - \frac{B(u)}{D(u)} z^x + v C(u).$$

The associated two-contact Hamilton–De Donder–Weyl equations read

$$\partial_t v - \partial_x p_u^x = \frac{\partial h}{\partial u} + p_u^x \frac{\partial h}{\partial z^x}, \quad -\partial_t u - \partial_x p_v^x = \frac{\partial h}{\partial v} + p_v^x \frac{\partial h}{\partial z^x},$$

together with $\partial_x u = \frac{\partial h}{\partial p_u^x}$ and $\partial_x v = \frac{\partial h}{\partial p_v^x}$. Since

$$\frac{\partial h}{\partial v} = C(u), \quad \frac{\partial h}{\partial z^x} = -\frac{B(u)}{D(u)}, \quad \frac{\partial h}{\partial p_u^x} = -\frac{1}{D(u)} p_v^x, \quad \frac{\partial h}{\partial p_v^x} = -\frac{1}{D(u)} p_u^x,$$

the last two equations yield $p_v^x = -D(u) u_x$ and $p_u^x = -D(u) v_x$. Moreover, the equation for u becomes

$$-\partial_t u - \partial_x p_v^x = C(u) - \frac{B(u)}{D(u)} p_v^x.$$

Substituting $p_v^x = -D(u)u_x$, we obtain

$$u_t - \partial_x(D(u)u_x) + B(u)u_x + C(u) = 0. \quad (4.3.2)$$

Hence, this two-contact Hamiltonian system induces a whole family of generalised Burgers-type equations of convection–diffusion–reaction type [104].

Several relevant cases are recovered immediately. If $D(u) = \nu > 0$ is constant and $C(u) = 0$, then (4.3.2) reduces to $u_t + B(u)u_x = \nu u_{xx}$. In particular, for $B(u) = u$ one recovers the viscous Burgers equation $u_t + u u_x = \nu u_{xx}$ (see for instance [119]). More generally, if $D(u) = \nu$ is constant, then (4.3.2) becomes $u_t + B(u)u_x = \nu u_{xx} - C(u)$, which includes Burgers-type equations with reaction terms, namely with $C(u) \neq 0$ (cf. [51]).

This shows that the two-contact approach is flexible enough to accommodate a nontrivial family of nonlinear scalar equations while keeping the underlying geometric structure fixed and modifying only the Hamiltonian.

In this construction, the convection term arises directly from the dependence of the Hamiltonian on the spatial dissipative variable z^x . Indeed,

$$\frac{\partial h}{\partial z^x} = -\frac{B(u)}{D(u)},$$

and, after eliminating the momentum $p_v^x = -D(u)u_x$, the corresponding term in the reduced HdDW equation becomes $B(u)u_x$. Thus, this family provides a direct illustration of the z^x -dependent reduced models described in Remark 4.2.7.

4.3.3 Porous medium equation with linear absorption

Consider the porous medium equation with linear absorption

$$u_t = (u^m)_{xx} - bu, \quad m > 1, \quad b \geq 0, \quad (4.3.3)$$

which is a standard model of nonlinear diffusion in porous media with absorption (see, for instance, [8, 12, 13, 57, 126]). For simplicity, we restrict ourselves to $u > 0$.

Let us describe (4.3.3) on a two-contactification. Consider $P_0 = \mathbb{R}^4 \simeq T^*\mathbb{R}^2$ with global coordinates (u, v, p^x, q^x) , and define

$$\theta = \frac{1}{2}(-v du + u dv) \otimes e_1 + (p^x du + q^x dv) \otimes e_2.$$

This gives rise to an exact two-symplectic manifold $(P_0, \omega = -d\theta)$ with

$$\omega = (-du \wedge dv) \otimes e_1 + (du \wedge dp^x + dv \wedge dq^x) \otimes e_2.$$

Consider the contactification $M = P_0 \times \mathbb{R}^2$ with coordinates $(u, v, p^x, q^x, z^t, z^x)$ and two-contact form

$$\eta = (dz^t - \rho^* \theta_e) \otimes e_1 + (dz^x - \rho^* \theta_{\mathbb{R}^2}) \otimes e_2,$$

where $\rho : M \rightarrow P_0$ is the canonical projection. Define the Hamiltonian $h_{\text{PME}} \in C^\infty(P_0 \times \mathbb{R}^2)$ by

$$h_{\text{PME}} = -\frac{1}{mu^{m-1}} p^x q^x + 2bz^t.$$

Then, $R_t h_{\text{PME}} = 2b$ and $R_x h_{\text{PME}} = 0$. Hence, the associated Hamilton–De Donder–Weyl equations are

$$\partial_t v - \partial_x p^x = \frac{\partial h_{\text{PME}}}{\partial u} - bv, \quad -\partial_t u - \partial_x q^x = \frac{\partial h_{\text{PME}}}{\partial v} + bu, \quad \partial_x u = \frac{\partial h_{\text{PME}}}{\partial p^x}, \quad \partial_x v = \frac{\partial h_{\text{PME}}}{\partial q^x}.$$

A direct computation gives

$$\frac{\partial h_{\text{PME}}}{\partial u} = \frac{m-1}{mu^m} p^x q^x, \quad \frac{\partial h_{\text{PME}}}{\partial v} = 0, \quad \frac{\partial h_{\text{PME}}}{\partial p^x} = -\frac{1}{mu^{m-1}} q^x, \quad \frac{\partial h_{\text{PME}}}{\partial q^x} = -\frac{1}{mu^{m-1}} p^x.$$

Therefore, $q^x = -mu^{m-1}u_x = -(u^m)_x$, while the second balance equation becomes $-u_t - \partial_x q^x = bu$. Substituting the expression for q^x , one recovers precisely (4.3.3). Hence, (4.3.3) admits a natural two-contact Hamilton–De Donder–Weyl description on the region where $u > 0$ since $m > 1$.

The first equation determines the complementary evolution of the auxiliary field v , namely $\partial_t v - \partial_x p^x = \frac{m-1}{mu^m} p^x q^x - bv$. In particular, $v = 0$ and $p^x = 0$ define a consistent reduction, and every solution u of (4.3.3) lifts to a solution of the full Hamiltonian system by taking $q^x = -(u^m)_x$, $v = 0$, and $p^x = 0$.

Moreover, the dissipative variables satisfy

$$\partial_t z^t + \partial_x z^x = \theta_e(\phi'_t) + \theta_{\mathbb{R}^2}(\phi'_x) - h_{\text{PME}} = \frac{1}{2}(-vu_t + uv_t) + p^x u_x + q^x v_x - h_{\text{PME}}.$$

Using the constitutive relations for u_x and v_x , this becomes

$$\partial_t z^t + \partial_x z^x = \frac{1}{2}(-vu_t + uv_t) - \frac{1}{mu^{m-1}} p^x q^x - 2bz^t.$$

Under the consistent reduction $v = 0$ and $p^x = 0$, this becomes $\partial_t z^t + \partial_x z^x = -2bz^t$. Thus, the linear absorption term is encoded through the explicit dependence of the Hamiltonian on z^t .

4.3.4 Complex Ginzburg–Landau equation

Consider the complex Ginzburg–Landau equation

$$\psi_t = (1 + i\alpha)\psi_{xx} + (1 - \gamma)\psi - (1 + i\beta)|\psi|^2\psi, \quad \alpha, \beta, \gamma \in \mathbb{R}, \quad (4.3.4)$$

which is a standard model for nonlinear waves and pattern formation in dissipative media [5, 22, 55], and its variants with an additional linear gain/loss term also arise naturally in dissipative nonlinear optics and mode-locked laser theory [2, 54]. Writing $\psi = a + ib$, one obtains

$$a_t = a_{xx} - \alpha b_{xx} + R_a(a, b) - \gamma a, \quad b_t = \alpha a_{xx} + b_{xx} + R_b(a, b) - \gamma b,$$

where $R_a(a, b) = a - (a^2 + b^2)a + \beta(a^2 + b^2)b$ and $R_b(a, b) = b - \beta(a^2 + b^2)a - (a^2 + b^2)b$. It is convenient to write the diffusion part in matrix form as

$$\begin{pmatrix} a_t \\ b_t \end{pmatrix} = D \begin{pmatrix} a_{xx} \\ b_{xx} \end{pmatrix} + \begin{pmatrix} R_a(a, b) - \gamma a \\ R_b(a, b) - \gamma b \end{pmatrix}, \quad D = \begin{pmatrix} 1 & -\alpha \\ \alpha & 1 \end{pmatrix},$$

so that

$$D^{-1} = \frac{1}{1 + \alpha^2} \begin{pmatrix} 1 & \alpha \\ -\alpha & 1 \end{pmatrix}.$$

Instead of introducing spatial derivatives as independent constrained variables, we consider a reduced phase space $P_0 = T^*\mathbb{R}^4$ with coordinates $(a, b, c, d, p_a^x, p_b^x, q_a^x, q_b^x)$, where c, d are auxiliary fields. On P_0 define

$$\theta = \frac{1}{2}(-c da + a dc - d db + b dd) \otimes e_1 + (p_a^x da + p_b^x db + q_a^x dc + q_b^x dd) \otimes e_2.$$

Then, one has that

$$\omega = -d\theta = (-da \wedge dc - db \wedge dd) \otimes e_1 + (da \wedge dp_a^x + db \wedge dp_b^x + dc \wedge dq_a^x + dd \wedge dq_b^x) \otimes e_2.$$

Consider the two-contactification $M = P_0 \times \mathbb{R}^2$ with coordinates $(a, b, c, d, p_a^x, p_b^x, q_a^x, q_b^x, z^t, z^x)$ and canonical two-contact form given by $\eta = (dz^t - \rho^* \theta_e) \otimes e_1 + (dz^x - \rho^* \theta_{\mathbb{R}^4}) \otimes e_2$, where $\rho : M \rightarrow P_0$ is the standard projection. We take the Hamiltonian

$$h_{\text{CGL}} = -\frac{1}{1 + \alpha^2} \left(p_a^x (q_a^x + \alpha q_b^x) + p_b^x (-\alpha q_a^x + q_b^x) \right) - c R_a(a, b) - d R_b(a, b) + 2\gamma z^t.$$

Since $\partial h_{\text{CGL}} / \partial z^t = 2\gamma$ and $\partial h_{\text{CGL}} / \partial z^x = 0$, the two-contact Hamilton–De Donder–Weyl equations become

$$\partial_t c - \partial_x p_a^x = \frac{\partial h_{\text{CGL}}}{\partial a} - \gamma c, \quad -\partial_t a - \partial_x q_a^x = \frac{\partial h_{\text{CGL}}}{\partial c} + \gamma a,$$

$$\partial_t d - \partial_x p_b^x = \frac{\partial h_{\text{CGL}}}{\partial b} - \gamma d, \quad -\partial_t b - \partial_x q_b^x = \frac{\partial h_{\text{CGL}}}{\partial d} + \gamma b,$$

together with

$$\partial_x a = \frac{\partial h_{\text{CGL}}}{\partial p_a^x}, \quad \partial_x b = \frac{\partial h_{\text{CGL}}}{\partial p_b^x}, \quad \partial_x c = \frac{\partial h_{\text{CGL}}}{\partial q_a^x}, \quad \partial_x d = \frac{\partial h_{\text{CGL}}}{\partial q_b^x}.$$

A direct computation gives $\partial h_{\text{CGL}}/\partial c = -R_a(a, b)$, $\partial h_{\text{CGL}}/\partial d = -R_b(a, b)$, and

$$\partial_x a = -\frac{1}{1 + \alpha^2}(q_a^x + \alpha q_b^x), \quad \partial_x b = -\frac{1}{1 + \alpha^2}(-\alpha q_a^x + q_b^x).$$

Solving for q_a^x and q_b^x , one obtains

$$q_a^x = -a_x + \alpha b_x, \quad q_b^x = -\alpha a_x - b_x.$$

Substituting into the equations for a and b , we get

$$a_t = -\partial_x q_a^x + R_a(a, b) - \gamma a, \quad b_t = -\partial_x q_b^x + R_b(a, b) - \gamma b,$$

which reproduces precisely the real form of (4.3.4). Therefore, this complex Ginzburg–Landau equation with linear gain/loss admits a natural two-contact Hamilton–De Donder–Weyl description on a reduced phase space, now with a Hamiltonian affine in the dissipative variable z^t . The remaining equations describe the evolution of the auxiliary fields c and d . In particular, $c = d = p_a^x = p_b^x = 0$ is a consistent reduction, and every solution (a, b) of (4.3.4) determines a solution of the full Hamiltonian system by taking $q_a^x = -a_x + \alpha b_x$, $q_b^x = -\alpha a_x - b_x$.

4.3.5 Damped nonlinear Schrödinger equation in real form

Consider the damped nonlinear Schrödinger equation

$$i\psi_t + \psi_{xx} + |\psi|^2\psi + i\gamma\psi = 0, \quad \gamma \geq 0, \quad (4.3.5)$$

which may be regarded as a dissipative variant of the nonlinear Schrödinger equation, a standard model in nonlinear optics, while damped and driven versions play a central role in dissipative cavity optics [53, 87, 123]. Writing $\psi = a + ib$, one obtains

$$a_t = -b_{xx} - (a^2 + b^2)b - \gamma a, \quad b_t = a_{xx} + (a^2 + b^2)a - \gamma b.$$

Set $R_a(a, b) = -(a^2 + b^2)b - \gamma a$ and $R_b(a, b) = (a^2 + b^2)a - \gamma b$. Then the system reads $a_t = -b_{xx} + R_a(a, b)$, $b_t = a_{xx} + R_b(a, b)$. Instead of introducing derivative constraints by hand, we consider a reduced phase space $P_0 = T^*\mathbb{R}^4$ with coordinates $(a, b, c, d, p_a^x, p_b^x, q_a^x, q_b^x)$, endowed with the one-form taking values in $\mathbb{R}^2$ of the form

$$\theta = \frac{1}{2}(-c da + a dc - d db + b dd) \otimes e_1 + (p_a^x da + p_b^x db + q_a^x dc + q_b^x dd) \otimes e_2.$$

Then,

$$\omega = -(da \wedge dc + db \wedge dd) \otimes e_1 + (da \wedge dp_a^x + db \wedge dp_b^x + dc \wedge dq_a^x + dd \wedge dq_b^x) \otimes e_2.$$

Define the two-contactification $M = P_0 \times \mathbb{R}^2$ with coordinates $(a, b, c, d, p_a^x, p_b^x, q_a^x, q_b^x, z^t, z^x)$ and two-contact form $\eta = (dz^t - \rho^*\theta_e) \otimes e_1 + (dz^x - \rho^*\theta_{P_0}) \otimes e_2$, where $\rho : M \rightarrow P_0$ is the canonical projection. We take the Hamiltonian

$$h_{\text{dNLS}} = -p_a^x q_b^x + p_b^x q_a^x - c R_a(a, b) - d R_b(a, b).$$

Since h_{dNLS} is independent of z^t and z^x , one has $R_t h_{\text{dNLS}} = R_x h_{\text{dNLS}} = 0$. Hence, the two-contact Hamilton–De Donder–Weyl equations reduce to

$$\partial_t c - \partial_x p_a^x = \frac{\partial h_{\text{dNLS}}}{\partial a}, \quad -\partial_t a - \partial_x q_a^x = \frac{\partial h_{\text{dNLS}}}{\partial c},$$

$$\partial_t d - \partial_x p_b^x = \frac{\partial h_{\text{dNLS}}}{\partial b}, \quad -\partial_t b - \partial_x q_b^x = \frac{\partial h_{\text{dNLS}}}{\partial d},$$

together with

$$\partial_x a = \frac{\partial h_{\text{dNLS}}}{\partial p_a^x} = -q_b^x, \quad \partial_x b = \frac{\partial h_{\text{dNLS}}}{\partial p_b^x} = q_a^x, \quad \partial_x c = \frac{\partial h_{\text{dNLS}}}{\partial q_a^x} = p_b^x, \quad \partial_x d = \frac{\partial h_{\text{dNLS}}}{\partial q_b^x} = -p_a^x.$$

Therefore, $q_a^x = b_x$ and $q_b^x = -a_x$. Moreover, $\partial h_{\text{dNLS}}/\partial c = -R_a(a, b)$ and $\partial h_{\text{dNLS}}/\partial d = -R_b(a, b)$, so the equations for a and b become

$$a_t = -\partial_x q_a^x + R_a(a, b), \quad b_t = -\partial_x q_b^x + R_b(a, b).$$

Substituting the expressions for q_a^x and q_b^x , one recovers precisely the real form of (4.3.5). Hence the damped nonlinear Schrödinger equation admits a natural two-contact Hamilton–De Donder–Weyl description on a reduced phase space. The remaining equations determine the auxiliary fields c and d . In particular, $c = d = p_a^x = p_b^x = 0$ is a consistent reduction, and every solution (a, b) of (4.3.5) lifts to a solution of the full Hamiltonian system by taking $q_a^x = b_x$, $q_b^x = -a_x$, with z^t and z^x constant. The damping parameter γ is already incorporated into the reaction terms R_a and R_b . Therefore, the present construction does not use the dissipative variables to generate dissipation through the Reeb directions. If one wishes to introduce genuinely two-contact dissipation laws, one should allow the Hamiltonian to depend explicitly on z^t or z^x , which would produce a different non-conservative deformation of (4.3.5).

4.3.6 Fisher–KPP equation with linear loss

Consider the reaction–diffusion equation

$$u_t = \mu u_{xx} + ru(1 - u) - \lambda u, \quad \mu > 0, \quad r \geq 0, \quad \lambda \geq 0, \quad (4.3.6)$$

which may be regarded as a dissipative variant of the classical Fisher–KPP equation [27, 56, 102, 115, 129]. For particular choices of the coefficients, this equation describes several natural processes, such as the expansion of advantageous genes [27].

Let us describe (4.3.6) on the k -symplectic part of its k -contactification. Let $P_0 = T^*\mathbb{R}^2$ be the manifold with coordinates (u, v, p^x, q^x) , endowed with the one-form $\theta = \frac{1}{2}(-v du + u dv) \otimes e_1 + (p^x du + q^x dv) \otimes e_2$. Then, $\omega = -d\theta = -du \wedge dv \otimes e_1 + (du \wedge dp^x + dv \wedge dq^x) \otimes e_2$. We consider the two-contactification $M = P_0 \times \mathbb{R}^2$ with coordinates $(u, v, p^x, q^x, z^t, z^x)$ and the canonical two-contact form $\eta = (dz^t - \rho^* \theta_e) \otimes e_1 + (dz^x - \rho^* \theta_{P_0}) \otimes e_2$, where $\rho : P_0 \times \mathbb{R}^2 \rightarrow P_0$ is the natural projection.

We take the Hamiltonian

$$h_{\text{FKPP}} = -\frac{1}{\mu} p^x q^x + v(-ru(1 - u) + \lambda u).$$

Since h_{FKPP} is independent of z^t and z^x , one has $R_t h_{\text{FKPP}} = R_x h_{\text{FKPP}} = 0$. Hence, the two-contact Hamilton–De Donder–Weyl equations reduce to

$$\partial_t v - \partial_x p^x = \frac{\partial h_{\text{FKPP}}}{\partial u}, \quad -\partial_t u - \partial_x q^x = \frac{\partial h_{\text{FKPP}}}{\partial v}, \quad \partial_x u = \frac{\partial h_{\text{FKPP}}}{\partial p^x}, \quad \partial_x v = \frac{\partial h_{\text{FKPP}}}{\partial q^x}.$$

A direct computation gives $\partial h_{\text{FKPP}}/\partial v = -ru(1 - u) + \lambda u$, $\partial h_{\text{FKPP}}/\partial p^x = -q^x/\mu$, and $\partial h_{\text{FKPP}}/\partial q^x = -p^x/\mu$. Therefore $q^x = -\mu u_x$ and $p^x = -\mu v_x$, while the equation for u becomes

$$-\partial_t u - \partial_x q^x = -ru(1 - u) + \lambda u.$$

Substituting $q^x = -\mu u_x$, one recovers precisely (4.3.6). Hence, the Fisher–KPP equation with linear loss admits a natural two-contact Hamilton–De Donder–Weyl description on a reduced phase space.

The remaining equation determines the evolution of the auxiliary field v . In particular, $v = 0$ and $p^x = 0$ define a consistent reduction, and every solution u of (4.3.6) lifts to a solution of the full Hamiltonian system by taking $q^x = -\mu u_x$, with z^t and z^x constant.

Thus, the linear loss term is incorporated directly into the reaction part of the Hamiltonian, without requiring any explicit dependence on the dissipative variables.

4.3.7 A damped quartic field on a Minkowski space

Let $x^0 = t$ and x^1, x^2, x^3 be the standard coordinates on $\mathbb{R}^{3,1}$. Consider the damped ϕ^4 equation

$$\phi_{tt} - \Delta\phi + m^2\phi + g\phi^3 + \lambda\phi_t = 0, \quad m > 0, \quad g > 0, \quad \lambda \geq 0, \quad (4.3.7)$$

which may be regarded as a dissipative nonlinear Klein–Gordon model with quartic potential [79, 106, 117, 130].

We describe (4.3.7) on the canonical four-contact manifold $\mathcal{J}_{\mathbb{R},4} = \bigoplus^4 T^*\mathbb{R} \times \mathbb{R}^4$ with coordinates $(\phi, p^0, p^1, p^2, p^3, z^0, z^1, z^2, z^3)$ and canonical four-contact form $\eta_{\mathbb{R},4} = \sum_{\alpha=0}^3 (dz^\alpha - p^\alpha d\phi) \otimes e_\alpha$. Define the Hamiltonian

$$h_{\phi^4} = \frac{1}{2} \left((p^0)^2 - (p^1)^2 - (p^2)^2 - (p^3)^2 \right) + \frac{m^2}{2} \phi^2 + \frac{g}{4} \phi^4 + \lambda z^0.$$

The associated Hamilton–De Donder–Weyl equations are

$$\partial_0 \phi = \frac{\partial h_{\phi^4}}{\partial p^0} = p^0, \quad \partial_i \phi = \frac{\partial h_{\phi^4}}{\partial p^i} = -p^i, \quad i = 1, 2, 3,$$

$$\sum_{\mu=0}^3 \partial_\mu p^\mu = - \left(\frac{\partial h_{\phi^4}}{\partial \phi} + \sum_{\mu=0}^3 p^\mu \frac{\partial h_{\phi^4}}{\partial z^\mu} \right) = -(m^2 \phi + g\phi^3 + \lambda p^0),$$

and

$$\sum_{\mu=0}^3 \partial_\mu z^\mu = \sum_{\mu=0}^3 p^\mu \frac{\partial h_{\phi^4}}{\partial p^\mu} - h_{\phi^4} = \frac{1}{2} \left((p^0)^2 - (p^1)^2 - (p^2)^2 - (p^3)^2 \right) - \frac{m^2}{2} \phi^2 - \frac{g}{4} \phi^4 - \lambda z^0.$$

The regularity of h_{ϕ^4} allows us to describe the momenta in terms of the first derivatives of ϕ and relate h_{ϕ^4} to a second-order system of PDEs on the variable ϕ . Indeed, using $p^0 = \phi_t$ and $p^i = -\partial_i \phi$, $i = 1, 2, 3$, the momentum balance equation yields precisely (4.3.7). Hence, h_{ϕ^4} provides a natural 4-contact Hamilton–De Donder–Weyl description of the damped ϕ^4 field.

Moreover, the dissipative variables satisfy

$$\sum_{\mu=0}^3 \partial_\mu z^\mu = \frac{1}{2} (\phi_t^2 - |\nabla \phi|^2) - \frac{m^2}{2} \phi^2 - \frac{g}{4} \phi^4 - \lambda z^0.$$

Thus, the dissipation is encoded through the explicit dependence of the Hamiltonian on z^0 .

4.3.8 Damped sine–Gordon equation

Consider the damped sine–Gordon equation

$$u_{tt} - c^2 u_{xx} + \sin u + \lambda u_t = 0, \quad c > 0, \quad \lambda \geq 0, \quad (4.3.8)$$

which arises naturally in dissipative models of long Josephson junctions and, more broadly, in applied versions of the sine–Gordon model [21, 23, 92].

We describe (4.3.8) on the canonical two-contact manifold $\mathcal{J}_{\mathbb{R},2} = \bigoplus^2 T^*\mathbb{R} \times \mathbb{R}^2$ with coordinates (u, p^t, p^x, z^t, z^x) and the standard two-contact form $\eta_{\mathbb{R},2}$. Consider the Hamiltonian

$$h_{\text{sG}} = \frac{1}{2} \left((p^t)^2 - \frac{1}{c^2} (p^x)^2 \right) + (1 - \cos u) + \lambda z^t.$$

The Hamilton–De Donder–Weyl equations read

$$u_t = \frac{\partial h_{\text{sG}}}{\partial p^t} = p^t, \quad u_x = \frac{\partial h_{\text{sG}}}{\partial p^x} = -\frac{1}{c^2} p^x,$$

and

$$\partial_t p^t + \partial_x p^x = - \left(\frac{\partial h_{\text{sG}}}{\partial u} + p^t \frac{\partial h_{\text{sG}}}{\partial z^t} + p^x \frac{\partial h_{\text{sG}}}{\partial z^x} \right) = -(\sin u + \lambda p^t).$$

Moreover, the balance equation for the dissipative variables is

$$\partial_t z^t + \partial_x z^x = p^t \frac{\partial h_{\text{sG}}}{\partial p^t} + p^x \frac{\partial h_{\text{sG}}}{\partial p^x} - h_{\text{sG}} = \frac{1}{2} \left((p^t)^2 - \frac{1}{c^2} (p^x)^2 \right) - (1 - \cos u) - \lambda z^t.$$

Since h_{sG} is regular, one can describe the momenta by means of the derivatives u_t, u_x and remaining coordinates. Using $p^t = u_t$ and $p^x = -c^2 u_x$, the momentum balance equation yields precisely (4.3.8). Hence, h_{sG} provides a two-contact Hamilton–De Donder–Weyl formulation of the damped sine–Gordon equation. Furthermore, the dissipative variables satisfy

$$\partial_t z^t + \partial_x z^x = \frac{1}{2} (u_t^2 - c^2 u_x^2) - (1 - \cos u) - \lambda z^t.$$

One could use a more general approach to include a damped sine–Gordon equation with dissipation modelled by a function $\lambda(t, x)$. Indeed, let us define the Hamiltonian

$$h_{\text{sG}}^{\text{quad}} = \frac{1}{2} \left((p^t)^2 - \frac{1}{c^2} (p^x)^2 \right) + (1 - \cos u) + \frac{1}{2} (z^t)^2.$$

Then, the Hamilton–De Donder–Weyl equations become

$$u_t = p^t, \quad u_x = -\frac{1}{c^2} p^x, \quad \partial_t p^t + \partial_x p^x = -(\sin u + z^t p^t),$$

and

$$\partial_t z^t + \partial_x z^x = \frac{1}{2} \left((p^t)^2 - \frac{1}{c^2} (p^x)^2 \right) - (1 - \cos u) - \frac{1}{2} (z^t)^2.$$

Hence, after eliminating the momenta, which is possible due to the regularity of $h_{\text{sG}}^{\text{quad}}$, one obtains

$$u_{tt} - c^2 u_{xx} + \sin u + z^t u_t = 0.$$

Moreover, the form of $h_{\text{sG}}^{\text{quad}}$ shows that the equation is hyperbolic.

This formulation is useful if one wishes to realise a prescribed damping profile $\lambda = \lambda(x, t)$. Indeed, imposing $z^t = \lambda(x, t)$, the field equation becomes

$$u_{tt} - c^2 u_{xx} + \sin u + \lambda(x, t) u_t = 0,$$

while the balance equation for the dissipative variables reduces to

$$\partial_x z^x = \frac{1}{2} (u_t^2 - c^2 u_x^2) - (1 - \cos u) - \frac{1}{2} \lambda(x, t)^2 - \partial_t \lambda(x, t).$$

Therefore, once u and $\lambda(x, t)$ are fixed, the variable z^x can be chosen locally so that the above identity holds. In this sense, the quadratic dependence on z^t allows one to accommodate nonconstant damping profiles by adjusting the second dissipative variable z^x .

4.3.9 FitzHugh–Nagumo reaction–diffusion system

Consider the FitzHugh–Nagumo system for two dependent variables $u = u(t, x)$ and $v = v(t, x)$ of the form

$$u_t = D_u u_{xx} + f(u) - v + I, \quad v_t = D_v v_{xx} + \varepsilon(u - av), \quad (4.3.9)$$

where $D_u, D_v > 0$, $\varepsilon > 0$, $a > 0$, $f(u)$ is a certain function, and $I \in \mathbb{R}$. This system was originally introduced as a simplification of the Hodgkin–Huxley model and is a standard prototype for excitable media [28, 43].

Let us describe (4.3.9) on a two-contactification. Consider $P_0 = \mathbb{R}^8 \simeq T^*\mathbb{R}^4$ with global coordinates $(u, v, r, s, p_u^x, p_v^x, q_u^x, q_v^x)$, and define

$$\theta = \frac{1}{2} (-r du + u dr - s dv + v ds) \otimes e_1 + (p_u^x du + p_v^x dv + q_u^x dr + q_v^x ds) \otimes e_2.$$

This gives rise to an exact two-symplectic manifold $(P_0, \omega = -d\theta)$ with

$$\omega = (-du \wedge dr - dv \wedge ds) \otimes e_1 + (du \wedge dp_u^x + dv \wedge dp_v^x + dr \wedge dq_u^x + ds \wedge dq_v^x) \otimes e_2.$$

Consider the contactification $M = P_0 \times \mathbb{R}^2$ with coordinates $(u, v, r, s, p_u^x, p_v^x, q_u^x, q_v^x, z^t, z^x)$ and two-contact form

$$\eta = (dz^t - \rho^* \theta_e) \otimes e_1 + (dz^x - \rho^* \theta_{P_0}) \otimes e_2$$

with the projection $\rho : M \rightarrow P_0$. Define the Hamiltonian $h_{\text{FHN}} \in C^\infty(P_0 \times \mathbb{R}^2)$ of the form

$$h_{\text{FHN}} = -\frac{1}{D_u} p_u^x q_u^x - \frac{1}{D_v} p_v^x q_v^x - r(f(u) - v + I) - s\varepsilon(u - av).$$

Since h_{FHN} is independent of z^t and z^x , the Hamilton–De Donder–Weyl equations reduce to

$$\partial_t r - \partial_x p_u^x = \frac{\partial h_{\text{FHN}}}{\partial u}, \quad -\partial_t u - \partial_x q_u^x = \frac{\partial h_{\text{FHN}}}{\partial r},$$

$$\partial_t s - \partial_x p_v^x = \frac{\partial h_{\text{FHN}}}{\partial v}, \quad -\partial_t v - \partial_x q_v^x = \frac{\partial h_{\text{FHN}}}{\partial s},$$

together with $u_x = \partial h_{\text{FHN}} / \partial p_u^x = -q_u^x / D_u$, $v_x = \partial h_{\text{FHN}} / \partial p_v^x = -q_v^x / D_v$, $r_x = \partial h_{\text{FHN}} / \partial q_u^x = -p_u^x / D_u$, and $s_x = \partial h_{\text{FHN}} / \partial q_v^x = -p_v^x / D_v$. Hence, $q_u^x = -D_u u_x$ and $q_v^x = -D_v v_x$. Moreover,

$$\frac{\partial h_{\text{FHN}}}{\partial r} = -(f(u) - v + I), \quad \frac{\partial h_{\text{FHN}}}{\partial s} = -\varepsilon(u - av),$$

and the equations for u and v become

$$u_t = -\partial_x q_u^x + f(u) - v + I, \quad v_t = -\partial_x q_v^x + \varepsilon(u - av).$$

Substituting the expressions for q_u^x and q_v^x , one recovers precisely (4.3.9). Therefore, the FitzHugh–Nagumo system admits a natural two-contact Hamilton–De Donder–Weyl description on a reduced phase space.

The remaining equations determine the auxiliary fields r and s . In particular, $r = s = p_u^x = p_v^x = 0$ is a consistent reduction, and every solution (u, v) of (4.3.9) lifts to a solution of the full two-contact Hamiltonian system by taking $q_u^x = -D_u u_x$, $q_v^x = -D_v v_x$, with z^t and z^x constant.

Let us finally describe a simple infinitesimal two-contact symmetry in a particular affine FitzHugh–Nagumo-type case. Assume that $f(u) = a^{-1}u + c$, with $c \in \mathbb{R}$. Then

$$h = -\frac{1}{D_u} p_u^x q_u^x - \frac{1}{D_v} p_v^x q_v^x - r \left(\frac{1}{a} u + c - v + I \right) - s\varepsilon(u - av).$$

Consider the vector field

$$\Gamma = \frac{\partial}{\partial u} + \frac{1}{a} \frac{\partial}{\partial v} + \frac{1}{2} \left(r + \frac{s}{a} \right) \frac{\partial}{\partial z^t}.$$

Since

$$\mathcal{L}_{\partial_u + a^{-1}\partial_v} \theta_e = d \left[\frac{1}{2} \left(r + \frac{s}{a} \right) \right], \quad \mathcal{L}_{\partial_u + a^{-1}\partial_v} \theta_{P_0} = 0,$$

one has $\mathcal{L}_\Gamma \eta = 0$. Moreover, a short calculation shows that $\Gamma(h) = 0$. The corresponding current $\mathbf{F} = -\iota_\Gamma \eta$ is

$$\mathbf{F} = - \left(r + \frac{s}{a} \right) \otimes e_1 + \left(p_u^x + \frac{1}{a} p_v^x \right) \otimes e_2.$$

Since $R_t h = R_x h = 0$, this gives the conservation law

$$\partial_t \left[- \left(r + \frac{s}{a} \right) \right] + \partial_x \left(p_u^x + \frac{1}{a} p_v^x \right) = 0$$

along every solution of the associated two-contact Hamilton–De Donder–Weyl equations.

4.3.10 A simple Hamilton–Jacobi reconstruction

We conclude this chapter with a simple application of the Hamilton–Jacobi theory developed in the previous chapter to one of the models introduced above. Consider the damped Klein–Gordon model of Subsection 4.3.1 in the particular case $\epsilon = 0$. Its Hamiltonian is

$$h(u, p^t, p^x, z^t, z^x) = \frac{1}{2} \left((p^t)^2 - \frac{1}{c^2} (p^x)^2 \right) + \frac{1}{2} (z^t)^2, \quad c > 0.$$

The corresponding equation for the field u is

$$u_{tt} - c^2 u_{xx} + z^t u_t = 0.$$

We apply the z -independent Hamilton–Jacobi theorem, Theorem 3.2.4. Since $Q = \mathbb{R}$, a holonomic section $\gamma: \mathbb{R} \rightarrow \mathcal{J}_{\mathbb{R},2}$ is determined by two functions $\gamma^t, \gamma^x \in C^\infty(\mathbb{R})$ and takes the form

$$\gamma(u) = \left(u, \frac{d\gamma^t}{du}, \frac{d\gamma^x}{du}, \gamma^t(u), \gamma^x(u) \right).$$

The two-contact Hamilton–Jacobi equation is simply $h \circ \gamma = 0$, namely,

$$\frac{1}{2} \left(\frac{d\gamma^t}{du} \right)^2 - \frac{1}{2c^2} \left(\frac{d\gamma^x}{du} \right)^2 + \frac{1}{2} (\gamma^t)^2 = 0. \quad (4.3.10)$$

A particularly simple solution is

$$\gamma^t(u) = e^u, \quad \gamma^x(u) = c\sqrt{2}e^u + b, \quad b \in \mathbb{R}.$$

Indeed,

$$\frac{d\gamma^t}{du} = e^u, \quad \frac{d\gamma^x}{du} = c\sqrt{2}e^u,$$

and therefore

$$h \circ \gamma = \frac{1}{2} e^{2u} - \frac{1}{2c^2} (c^2 2e^{2u}) + \frac{1}{2} e^{2u} = 0.$$

The projected two-vector field along γ is

$$(X_h^\gamma)_t = \left(\frac{\partial h}{\partial p^t} \circ \gamma \right) \frac{\partial}{\partial u} = e^u \frac{\partial}{\partial u},$$

and

$$(X_h^\gamma)_x = \left(\frac{\partial h}{\partial p^x} \circ \gamma \right) \frac{\partial}{\partial u} = -\frac{\sqrt{2}}{c} e^u \frac{\partial}{\partial u}.$$

The two vector fields commute and hence define an integrable two-vector field. An integral section $u = u(t, x)$ therefore satisfies

$$u_t = e^u, \quad u_x = -\frac{\sqrt{2}}{c} e^u.$$

Equivalently,

$$\partial_t(e^{-u}) = -1, \quad \partial_x(e^{-u}) = \frac{\sqrt{2}}{c},$$

and hence

$$u(t, x) = -\log \left(C - t + \frac{\sqrt{2}}{c} x \right),$$

where $C \in \mathbb{R}$ and the solution is defined on every domain on which the argument of the logarithm is positive.

The corresponding lifted solution is

$$p^t = e^u, \quad p^x = c\sqrt{2}e^u, \quad z^t = e^u, \quad z^x = c\sqrt{2}e^u + b.$$

In particular,

$$u_{tt} = e^{2u}, \quad u_{xx} = \frac{2}{c^2} e^{2u}, \quad z^t u_t = e^{2u},$$

and consequently

$$u_{tt} - c^2 u_{xx} + z^t u_t = e^{2u} - 2e^{2u} + e^{2u} = 0.$$

Thus, the Hamilton–Jacobi construction produces directly an explicit solution of the dissipative field equation.

It is also straightforward to verify the balance equation for the dissipative variables. Indeed,

$$\partial_t z^t + \partial_x z^x = e^{2u} - 2e^{2u} = -e^{2u},$$

whereas

$$\frac{1}{2}(p^t)^2 - \frac{1}{2c^2}(p^x)^2 - \frac{1}{2}(z^t)^2 = -e^{2u}.$$

Hence, the full Hamilton–De Donder–Weyl system is satisfied.

This elementary example illustrates explicitly the role of the k -contact Hamilton–Jacobi theory: instead of solving the original second-order dissipative field equation directly, one first solves the first-order Hamilton–Jacobi equation (4.3.10) and subsequently integrates the lower-dimensional projected two-vector field.

Model	Field equation	Hamiltonian / geometric realisation
Damped wave	$u_{tt} - c^2 u_{xx} + k u_t = 0$	On $\mathcal{J}_{\mathbb{R},2}$: $h = \frac{1}{2\rho}(p^t)^2 - \frac{1}{2\tau}(p^x)^2 + k z^t$, with $c^2 = \tau/\rho$.
Damped Klein-Gordon-type model	$u_{tt} - c^2 u_{xx} + z^t u_t + \epsilon u = 0$	On $\mathcal{J}_{\mathbb{R},2}$: $h = \frac{1}{2}((p^t)^2 - c^{-2}(p^x)^2) + \frac{1}{2}(z^t)^2 + \frac{\epsilon}{2}u^2$.
Allen-Cahn type / reaction-diffusion	$u_t - \nu u_{xx} + V'(u) + \lambda u = 0$	On the contactification of a reduced exact 2-symplectic phase space: $h_{\text{RD}} = -\nu^{-1} p^x q^x + v(V'(u) + \lambda u)$.
Generalised Burgers family	$u_t - \partial_x(D(u)u_x) + B(u)u_x + C(u) = 0$	Two-contactification of an exact two-symplectic phase space: $h = -D(u)^{-1} p_u^x p_v^x - \frac{B(u)}{D(u)} z^x + v C(u)$. It includes viscous Burgers for $D(u) = \nu$, $B(u) = u$, $C(u) = 0$.
Porous medium with linear absorption	$u_t = (u^m)_{xx} - bu$, $m > 1$	Two-contactification of an exact two-symplectic phase space: $h_{\text{PME}} = -(mu^{m-1})^{-1} p^x q^x + 2bz^t$. This realisation is local on the region where $u > 0$.
Complex Ginzburg-Landau	$\psi_t = (1 + i\alpha)\psi_{xx} + (1 - \gamma)\psi - (1 + i\beta) \psi ^2\psi$	In real form, on the two-contactification of an exact two-symplectic phase space: $h_{\text{CGL}} = -\frac{p_a^x(q_a^x + \alpha q_b^x) + p_b^x(-\alpha q_a^x + q_b^x)}{1 + \alpha^2} - c R_a - d R_b + 2\gamma z^t$.
Damped nonlinear Schrödinger	$i\psi_t + \psi_{xx} + \psi ^2\psi + i\gamma\psi = 0$	In real form, on the two-contactification of an exact two-symplectic phase space: $h_{\text{dNLS}} = -p_a^x q_b^x + p_b^x q_a^x - c R_a(a, b) - d R_b(a, b)$.
Fisher-KPP with linear loss	$u_t = \mu u_{xx} + ru(1 - u) - \lambda u$	Two-contactification of an exact two-symplectic phase space: $h_{\text{FKPP}} = -\mu^{-1} p^x q^x + v(-ru(1 - u) + \lambda u)$.
Damped ϕ^4 field	$\phi_{tt} - \Delta\phi + m^2\phi + g\phi^3 + \lambda\phi_t = 0$	On $\mathcal{J}_{\mathbb{R},4}$: $h_{\phi^4} = \frac{1}{2}((p^0)^2 - \sum_{i=1}^3 (p^i)^2) + \frac{m^2}{2}\phi^2 + \frac{g}{4}\phi^4 + \lambda z^0$.
Damped sine-Gordon	$u_{tt} - c^2 u_{xx} + \sin u + \lambda u_t = 0$	On $\mathcal{J}_{\mathbb{R},2}$: $h_{\text{sG}} = \frac{1}{2}((p^t)^2 - c^{-2}(p^x)^2) + (1 - \cos u) + \lambda z^t$.
FitzHugh-Nagumo	$u_t = D_u u_{xx} + f(u) - v + I,$ $v_t = D_v v_{xx} + \varepsilon(u - av)$	Two-contactification of an exact two-symplectic phase space: $h_{\text{FHN}} = -D_u^{-1} p_u^x q_u^x - D_v^{-1} p_v^x q_v^x - r(f(u) - v + I) - s\varepsilon(u - av)$.

Table 4.1: Examples developed in Chapter 4 of this dissertation. Canonical k -contact models coexist with realisations on contactifications of reduced exact k -symplectic phase spaces, depending on the structure of the equation under consideration.

Model	Configuration space Q	Representative Hamiltonian h	Target equation
Damped sine-Gordon field	$\mathbb{R}$, coordinate u	$h = \frac{1}{2}((p_u^t)^2 - c^{-2}(p_u^x)^2) + (1 - \cos u) + \lambda z^t$	$u_{tt} - c^2 u_{xx} + \sin u + \lambda u_t = 0.$
Damped double sine-Gordon field	$\mathbb{R}$, coordinate u	$h = \frac{1}{2}((p_u^t)^2 - c^{-2}(p_u^x)^2) + a(1 - \cos u) + b(1 - \cos 2u) + \lambda z^t$	$u_{tt} - c^2 u_{xx} + a \sin u + 2b \sin(2u) + \lambda u_t = 0.$
Damped ϕ^4 field	$\mathbb{R}$, coordinate u	$h = \frac{1}{2}((p_u^t)^2 - c^{-2}(p_u^x)^2) + \frac{m^2}{2}u^2 + \frac{\beta}{4}u^4 + \lambda z^t$	$u_{tt} - c^2 u_{xx} + m^2 u + \beta u^3 + \lambda u_t = 0.$
Coupled dissipative scalar fields	$\mathbb{R}^2$, coordinates (u, v)	$h = \frac{1}{2}((p_u^t)^2 - c_u^{-2}(p_u^x)^2 + (p_v^t)^2 - c_v^{-2}(p_v^x)^2) + V(u, v) + \lambda z^t$	$u_{tt} - c_u^2 u_{xx} + \partial_u V + \lambda u_t = 0,$ $v_{tt} - c_v^2 v_{xx} + \partial_v V + \lambda v_t = 0.$
Driven Josephson-type junction	$\mathbb{R}$, coordinate u	$h = \frac{1}{2}((p_u^t)^2 - c^{-2}(p_u^x)^2) + (1 - \cos u) - Iu + \lambda z^t$	$u_{tt} - c^2 u_{xx} + \sin u + \lambda u_t = I.$
Dissipative nonlinear sigma model	A Riemannian manifold (N, g)	$h = \frac{1}{2}g^{AB}(u)(p_A^t p_B^t - c^{-2}p_A^x p_B^x) + U(u) + \lambda z^t$	Schematic damped wave-map equation with potential: $\square_g u + \nabla U(u) + \lambda u_t = 0.$
Damped massive Klein-Gordon system	$\mathbb{R}^N$, coordinates $(u^1, \dots, u^N)$	$h = \frac{1}{2} \sum_{A=1}^N ((p_A^t)^2 - c_A^{-2}(p_A^x)^2) + m_A^2 (u^A)^2 + \lambda z^t$	$u_{tt}^A - c_A^2 u_{xx}^A + m_A^2 u^A + \lambda u_t^A = 0, \quad A = 1, \dots, N.$
Coupled damped Klein-Gordon system	$\mathbb{R}^N$, coordinates $(u^1, \dots, u^N)$	$h = \frac{1}{2} \sum_{A=1}^N ((p_A^t)^2 - c_A^{-2}(p_A^x)^2) + V(u^1, \dots, u^N) + \lambda z^t$	$u_{tt}^A - c_A^2 u_{xx}^A + \partial_{u^A} V + \lambda u_t^A = 0, \quad A = 1, \dots, N.$
Damped anisotropic wave equation	$\mathbb{R}$, coordinate u	$h = \frac{1}{2}((p_u^t)^2 - \sum_{a=1}^d c_a^{-2}(p_u^{x^a})^2) + V(u) + \lambda z^t$	$u_{tt} - \sum_{a=1}^d c_a^2 u_{x^a x^a} + V'(u) + \lambda u_t = 0.$
Damped nonlinear telegraph equation with variable potential	$\mathbb{R}$, coordinate u	$h = \frac{1}{2}((p_u^t)^2 - c^{-2}(p_u^x)^2) + W(u) + \lambda z^t$	$u_{tt} - c^2 u_{xx} + W'(u) + \lambda u_t = 0,$ including damped telegrapher and nonlinear wave models.

Table 4.2: Possible additional applications of the k -contact formalism. All models in this table are intended as canonical k -contact models on $\mathcal{J}_{Q,k} = \bigoplus^k T^*Q \times \mathbb{R}^k$.

Conclusions

This doctoral dissertation has developed geometric Hamilton–Jacobi theories for non-conservative classical field theories within the framework of k -contact geometry. The main objective has been to extend the geometric Hamilton–Jacobi method to a setting capable of encoding dissipation, source terms, and balance laws while retaining the geometric structures required by field theory. In this way, the dissertation contributes both to the foundations of k -contact Hamiltonian dynamics and to its application to physically relevant partial differential equations.

After reviewing the symplectic, contact, k -polysymplectic, and k -polycosymplectic structures underlying the subsequent constructions, particular attention was devoted to k -contact Hamiltonian systems and their Hamilton–De Donder–Weyl equations. An important feature of this setting is that, when $k > 1$, a Hamiltonian function does not generally determine a unique Hamiltonian k -vector field. This non-uniqueness must be incorporated explicitly into the Hamilton–Jacobi problem and distinguishes the field-theoretic situation from ordinary contact Hamiltonian mechanics. The evolution k -contact formalism was also analysed, providing a counterpart of the evolution description used in contact dynamics and clarifying the role played by the Reeb vector fields and the additional contact variables.

Two complementary families of k -contact Hamilton–Jacobi theories were developed. In the z -independent formulation, the dissipative variables are not retained in the base of the projected dynamics. This approach is naturally related, in the canonical k -contact setting, to families of Legendrian submanifolds and provides a direct extension of geometric Hamilton–Jacobi constructions familiar from symplectic, k -symplectic, and contact geometry. In the z -dependent formulation, the contact variables are incorporated into the base manifold. The resulting geometric conditions involve maximally coisotropic submanifolds and allow the dependence of the dynamics on the dissipative variables to be treated intrinsically. Both formulations were studied for the Hamiltonian and evolution Hamilton–De Donder–Weyl equations. Consequently, the dissertation does not produce a single Hamilton–Jacobi equation, but a collection of related theories adapted to different choices of projection and different descriptions of the k -contact dynamics.

Complete solutions were introduced in this setting and related to suitable notions of integrability. These constructions extend the classical interpretation of a complete Hamilton–Jacobi solution as a family of invariant geometric objects that locally foliates the phase space. They also show that the Hamilton–Jacobi method retains its geometric meaning in the presence of non-conservative effects, although the non-uniqueness of Hamiltonian k -vector fields and the role of the contact variables require additional compatibility conditions. The contact and k -symplectic theories are recovered in the corresponding particular cases, providing a consistency check for the proposed framework.

The formalism was applied to several non-conservative field equations. Detailed Hamilton–Jacobi analyses were carried out for damped telegrapher and Klein–Gordon-type equations, a dissipative Hunter–Saxton equation, and a dissipative wave equation. These examples illustrate the differences between the Hamiltonian and evolution approaches and between the z -independent and z -dependent formulations. They also provide explicit complete solutions in situations where the relevant Hamilton–Jacobi equations can be solved globally. A covariant model inspired by extended irreversible thermodynamics was studied as well. In that example, the dissipative variables acquire the physical interpretation of components of an entropy current, and the Hamilton–Jacobi formalism yields a geometric lift of the Maxwell–Cattaneo

dynamics together with its entropy production law. This shows that the contact variables need not be regarded merely as auxiliary coordinates: in suitable models, they carry direct physical information.

A further collection of k -contact Hamilton–De Donder–Weyl models was constructed to demonstrate the range of partial differential equations that can be accommodated within the formalism. These include damped Klein–Gordon, Allen–Cahn-type, porous-medium, complex Ginzburg–Landau, damped nonlinear Schrödinger, Fisher–KPP, damped ϕ^4 , sine–Gordon, and FitzHugh–Nagumo equations. The examples exhibit two related but conceptually distinct mechanisms. In some cases, non-conservative effects are generated intrinsically through explicit dependence of the Hamiltonian on the dissipative variables, and therefore through the corresponding Reeb terms. In others, dissipative or reaction terms are incorporated into an enlarged Hamiltonian system whose exact k -polysymplectic structure is subsequently k -contactified. Although the latter constructions do not use the dissipative variables themselves to generate dissipation, they provide systematic k -contact Hamilton–De Donder–Weyl embeddings of the original equations and identify phase spaces on which the Hamilton–Jacobi methods developed in this dissertation may subsequently be investigated.

These applications also show that dependence on the dissipative variables need not be affine. Nonlinear dependence can produce field equations with variable coefficients or non-autonomous dissipative terms, thereby enlarging the family of systems accessible to k -contact geometry. At the same time, the examples reveal limitations that should be kept explicit. A k -contact Hamiltonian representation is generally not unique, auxiliary variables may satisfy additional balance equations, and the existence of a Hamilton–De Donder–Weyl formulation does not by itself guarantee the existence of a complete Hamilton–Jacobi solution. Each model therefore requires a separate analysis of regularity, integrability, and the geometric meaning of its dissipative variables.

Several directions remain open. A natural first problem is to develop systematic criteria for the existence of complete solutions and to clarify their relation to integrability for $k > 1$. It would also be useful to determine when different k -contact Hamiltonian representations of the same partial differential equation are geometrically equivalent and when the dissipative variables admit a distinguished physical interpretation. Further work may extend the theory to higher-order models as well as study reduction and symmetry methods in conjunction with the Hamilton–Jacobi formalism. The new models presented here provide a broad collection of examples on which these questions can be tested.

In conclusion, this dissertation establishes a geometric Hamilton–Jacobi framework for k -contact Hamiltonian systems and demonstrates its relevance to non-conservative field theory. The resulting picture combines projected dynamics, Legendrian and coisotropic geometry, complete solutions, integrability, and balance laws within a common formalism. The applications show both the flexibility of k -contact geometry and the need to distinguish carefully between different mechanisms by which non-conservative effects enter a Hamiltonian description. Taken together, these results provide a foundation for the further development of Hamilton–Jacobi methods in dissipative and non-conservative classical field theories.

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